



Dipartimento di Fisica “A. Volta”

Corso di laurea in Fisica

Hamiltonian Structures in the C^* -Algebraic Reconstruction of Classical and Quantum Mechanics

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Introduction

It is wrong to think that the task of physics is to find out how nature is. Physics concerns what we can say about Nature.

Niels Bohr, [Pet63]

The aim of the quantum reconstruction program is to derive quantum mechanics from accessible postulates instead of taking Hilbert space as a primitive structure [Har01]. Reconstructions uncover why quantum theory assumes its characteristic form, addressing Wheeler’s call to explain “Why the quantum?”, whereas interpretational approaches take the mathematical structure as given. Von Neumann’s Hilbert space axiomatization supplies a powerful formalism but leaves most foundational questions unanswered [Von18]. This drove many scientists toward operationalism, which seeks the minimal structure encapsulating laboratory operations. This approach was started by Peres [Per02] but we can track its roots from Bohr’s epistemological philosophy that quantum formalism expresses communicable outcomes of macroscopic devices. Constructing the theory on these procedures selects bounded observables as primitive operational objects. At the same time, the algebraic formulation arose from Segal’s observation that not every Hilbert space operator corresponds to a measurable quantity [Seg47]. In fact, Segal identified the C^* -algebra studied by Gelfand and Naimark [GN94] as the appropriate setting. Gelfand-Naimark-Segal construction associates a Hilbert space representation to each state without assuming it beforehand, deriving the familiar formalism from the operational content of the algebra’s relations.

An alternative approach used to understand quantum mechanics relies on finding analogies between the quantum formalism and the classical one. Kibble discovered that a projective Hilbert space supplies a symplectic structure whose Hamiltonian flow generates Schrödinger evolution [Kib79]. Then, Ashtekar and Schilling showed that there is a full Kähler structure that embraces a symplectic form, the Fubini-Study metric, and a compatible complex structure [AS97]. To conclude, Brody and Hughston formalised that expectation values and quantum uncertainties admit geometric expressions on a Kähler manifold [BH01]. This formulation follows from the inner-product geometry, providing a unified description where dynamics, symmetries, and measurements arise from the manifold’s geometry without assuming separately the Schrödinger equation.

Motivated by this geometric perspective, the central problem of this thesis is the investigation of the structural correspondence between classical and quantum mechanics. In fact, both frameworks share multiple Hamiltonian structures. Classical mechanics evolves on a symplectic manifold $(\mathcal{M}, \omega)$ via Hamiltonian vector fields generated by smooth functions. Analogously, the projective Hilbert space of quantum pure states carries a symplectic form whose associated Hamiltonian flow generates the Schrödinger evolution [Kib79]. Another correspondence appears in the Poisson bracket to commutator analogy first recognised by Dirac when he proposed that the quantum commutator corresponds to $i\hbar$ times the classical Poisson bracket. Moreover, Noether’s theorem in both theories associates continuous symmetries with conserved quantities, with the quantum geometric version using a momentum map on the Kähler state space [AS97]. These parallels raise whether these are distinct theories or two realisations of a common framework. This thesis reconstructs both from a shared algebraic starting point examining where their geometries diverge.

Understanding shared algebraic and geometric structures is necessary for several reasons. Within this framework, the problem of quantization reduces to deforming a commutative classical observable algebra into a noncommutative one, with the Poisson bracket arising as the first-order term in the commutator expansion. This approach formalizes also the classical limit, that requires a rigorous framework such

as Landsman's continuous fields of C^* -algebras in [Lan98]. Moreover, formulating both theories within a shared algebraic language help us understand better the boundary between a quantum system and a classical measuring apparatus.

The literature contains many classical quantum correspondences. Dirac's canonical quantization suffers from ordering ambiguity and the Groenewold van Hove obstruction [Gro46], while Weyl's systematic map from phase space functions to operators does not preserve the Poisson brackets [WW09]. For these reasons, we present Segal's algebraic formulation that uses C^* -algebras treating classical and quantum mechanics on the same footing [Seg47]. In addition, Strocchi showed that the GNS construction yields a symplectic structure on the state space [Str66] while Landsman provided the most comprehensive algebraic treatment of the classical limit [Lan98]. Separately, the geometric formulation, initiated by Kibble, showed that a projective Hilbert space carries a symplectic form whose Hamiltonian flow is Schrödinger evolution [Kib79]. While Heslot extended this by proving that the geometry of projective space determines the quantum formalism [Hes85]. Ashtekar and Schilling completed the picture with a canonical Kähler structure combining symplectic form, Fubini Study metric, and complex structure, with the metric's geodesic distance matching Born rule transition probabilities [AS97]. Moreover, Cirelli, Manià and Pizzocchero showed that expectation values, uncertainties, and the commutator structure have direct algebraic expressions on the state manifold [CMP90]. These approaches have achieved significant successes including the Poisson bracket commutator link, the unified C^* -algebraic framework, and the Kähler structure. However the canonical quantization is not unique. The algebraic formulation does not explain why quantum mechanics requires noncommutativity or how the classical limit operates operationally.

A synopsis of the thesis is the following.

Chapter one builds the classical picture from the Gelfand-Naimark theorem. This identifies a unital commutative C^* -algebra with continuous functions on its spectrum, so phase space is recovered as a topological space. Since that recovery gives only topology, the chapter introduces a dense smooth subalgebra with a Poisson bracket. This extra structure is not intrinsic to the C^* -algebra but it is needed to give the spectrum a differentiable symplectic structure. We define states as normalized positive functionals while their probabilistic interpretation comes from the Riesz-Markov theorem. In addition, we state that time evolution can be algebraically represented by a one-parameter automorphism group. A realization of this formalism is presented as the classical spin system derived from the group $\mathfrak{so}(3)$. Its Poisson bracket, via Gelfand's theorem, yields the sphere $\mathbb{S}^2$ as the phase space, where the symplectic form, the $SO(3)$ moment map, and angular momentum conservation along the symmetry axis are built.

Chapter two turns to the algebraic formulation of quantum mechanics using a non-commutative observable algebra. The GNS construction shows that every state on a C^* -algebra yields a Hilbert space and a representation by bounded operators, with the state recovered as an expectation on a cyclic vector. In the non-commutative case, GNS serves the same purpose as Gelfand's theorem in the commutative one. It moves from an abstract algebra to a Hilbert space representation without assuming it beforehand. It also connects pure states to irreducible representations. Because only bounded operators can lie in a C^* -algebra, we introduce the Weyl algebra to handle the unboundedness of position and momentum. Consequently, we show that representing the Weyl C^* -algebra leads to Schrödinger wave functions. The chapter ends with a systematic comparison of the two reconstructions. In the non-commutative setting, spectrum points are replaced by Hilbert space representations, and the Poisson bracket, restricted to the smooth subalgebra, is replaced by a Lie bracket on the whole algebra. The two-level system based on $\mathfrak{su}(2)$ provides an example. GNS applied to the $SO(3)$ -invariant state yields the Pauli algebra as an irreducible representation. Comparing the Bloch ball with the simplex of classical states on the sphere shows how the same Lie algebra generates different state geometries depending on whether it uses a Poisson bracket or a commutator.

Chapter three gives a geometric reformulation of quantum mechanics. It starts with the pure state space, identified with projective Hilbert space, without assuming the Schrödinger equation or symmetries as extra ingredients. We show explicitly that this projective space carries a Kähler structure combining a symplectic form, a Riemannian metric, and a complex structure. As in classical mechanics, the symplectic form defines a Poisson bracket that drives time evolution. In fact, given the energy selfadjoint operator H , its expectation value is a smooth function on the state space, while the corresponding Hamiltonian vector field generates a flow whose integral curves satisfy the Schrödinger equation. The Riemannian metric

accounts for statistical and probabilistic features defining a distance between pure states whose geodesic separation matches the Born's rule transition probability. Continuous symmetries appear as flows that preserve the Kähler structure, and the geometry provides a moment map pairing each symmetry generator with a conserved quantity, a geometric Noether theorem. Dynamics and symmetries all follow from the state manifold's geometry, providing a geometric formulation of quantum mechanics. In an example, we extend the previous two-level quantum system to a Kähler manifold showing the dynamical analogies with the first classical example.

Chapter 1

Algebraic Construction of Classical Mechanics

The reader will find no figures in this work. The methods which I set forth do not require either constructions or geometrical or mechanical reasonings: but only algebraic operations, subject to a regular and uniform rule of procedure.

Giuseppe Luigi Lagrangia, Preface to *Mécanique Analytique* [Lag09]

This chapter establishes the geometric and algebraic language of classical mechanics in an invariant form. Moving beyond the traditional Newtonian approach, we adopt a coordinate-free framework on smooth manifolds. The logical sequence of the chapter is as follows: First, we introduce Lie algebras, relying on Humphreys [Hum10] and Lee [Lee13], to provide the formal infrastructure for the algebraic brackets. Second, using the foundations of symplectic geometry, following Abraham and Marsden [AM08], we identify $C^\infty(M)$ as the natural space for observables. In this context, Poisson brackets endow the space with a Lie algebra structure. Finally, by applying the Gelfand theorem, we recognize $C_0(M)$ as a commutative C^* -algebra. This leads to our primary structural result, namely classical mechanics formally coincides with the theory of commutative C^* -algebras.

1.1 A Brief Introduction to Differential Geometry

The geometric formulation of classical mechanics replaces the traditional Euclidean space with a coordinate-free framework on smooth manifolds. This abstraction is physically justified by holonomic constraints, which deform the flat configuration space and remove any global vector space structure. This section introduces the topological and differential tools required to extend ordinary calculus to such curved spaces.

Definition 1.1.1 (Topological manifold). *A topological manifold of dimension n is a topological space M that is Hausdorff, second countable, and locally Euclidean of dimension n . Every point $p \in M$ has an open neighborhood homeomorphic to an open subset of $\mathbb{R}^n$. We write $\dim M = n$.*

In physics, configuration spaces, the space formed by all the possible spatial configurations of a system of particles, provide the most natural examples of such topological spaces. For instance, a simple pendulum constrained to move in a plane has the unit circle $\mathbb{S}^1$ as its configuration space, while the configurations of a free rigid body, in a frame of reference comoving with the centre of mass, are described by the special orthogonal group $SO(3)$. Neither of these spaces is globally homeomorphic to a Euclidean space. Hence, to perform calculus, a way of circumnavigate this consist of covering the manifold with open sets that locally resemble flat space.

Definition 1.1.2 (Chart, atlas). A chart on M is a pair (U, φ) , with $U \subseteq M$ open and $\varphi : U \rightarrow \varphi(U) \subseteq \mathbb{R}^n$ a homeomorphism. Two charts $(U, \varphi), (V, \psi)$ are C^∞ -compatible if the transition map

$$\psi \circ \varphi^{-1} : \varphi(U \cap V) \rightarrow \psi(U \cap V)$$

is a C^∞ -diffeomorphism whenever $U \cap V \neq \emptyset$. An atlas is a collection of pairwise compatible charts covering M . The union of all atlases compatible with a particular one on M is called the maximal atlas.

Remark 1.1.3. An equivalent formulation of Definition 1.1.2 could have been stated using $\varphi \circ \psi^{-1}$ that is a diffeomorphism being the inverse of $\psi \circ \varphi^{-1}$.

The compatibility condition ensures that if two observers describe the same physical region using different local coordinates, the transformation between their descriptions is smooth. This allows for a globally consistent notion of differentiability.

Definition 1.1.4 (Smooth manifold). A smooth manifold is a pair $M \equiv (M, \mathcal{A})$, where M is a n -dimensional topological manifold and $\mathcal{A}$ is a maximal atlas on M referred to as a smooth structure.

The configuration space is a smooth manifold. Observables are scalar functions on it, while symmetries and coordinate transformations are mappings between manifolds. Since these spaces generally lack a global Euclidean structure, smoothness must be checked locally, pulling the map back through charts into Euclidean space, where ordinary multivariable calculus applies.

Definition 1.1.5 (Smooth map, diffeomorphism). Let M, N be smooth manifolds, respectively of dimension $m, n \geq 1$. A map $f : M \rightarrow N$ is smooth at $p \in M$ if there exist charts (U, φ) around p and (V, ψ) around $f(p)$, with $f(U) \subseteq V$, such that

$$\psi \circ f \circ \varphi^{-1} : \varphi(U) \rightarrow \psi(V)$$

is C^∞ in the Euclidean topology. A function f is smooth if it is such at every point, while it is a diffeomorphism if it is a smooth bijection with smooth inverse. We write $C^\infty(M) := C^\infty(M; \mathbb{R})$.

The concept of manifold itself does not carry a vector space structure. Despite that, we can always fix a point and define a tangent vector as a derivation. This creates a vector space at every point of M called *tangent space*.

Definition 1.1.6 (Tangent vector). Let M be a smooth manifold as in Definition 1.1.4 and let $p \in M$. A tangent vector at p is a linear map $v : C^\infty(M) \rightarrow \mathbb{R}$ satisfying the Leibniz rule

$$v(fg) = f(p)v(g) + g(p)v(f), \quad f, g \in C^\infty(M).$$

The set of these tangent vectors is the tangent space at $p \in M$ denoted as $T_p M$.

Remark 1.1.7. Equivalently, a smooth curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow M$ with $\varepsilon > 0$ and $\gamma(0) = p$ induces the derivation $v_\gamma(f) := \left. \frac{d}{dt} \right|_{t=0} f(\gamma(t))$, as per Definition 1.1.6 corresponds to one of such maps. It can be proved that two curves induce the same tangent vector if and only if their local representatives agree to first order at p . This equivalent identification $T_p M \cong \{\text{curves through } p\} / \sim$ is used at a later stage.

Most of the times it is convenient to write statements locally on a specific chart. This allows one to use the rules of calculus on manifolds in order to simplify computations.

Definition 1.1.8 (Coordinate basis). Let (U, φ) be a chart around $p \in M$, an n -dimensional smooth manifold, with $\varphi : U \rightarrow V \subseteq \mathbb{R}^n$ a homeomorphism. Let $r^i : \mathbb{R}^n \rightarrow \mathbb{R}$ with $i = 1, \dots, n$ be the standard coordinates of $\mathbb{R}^n$. Define the coordinate basis as

$$x^i := r^i \circ \varphi : U \rightarrow \mathbb{R} \quad i = 1, \dots, n$$

the vectors

$$\left. \frac{\partial}{\partial x^i} \right|_p \in T_p M, \quad \left. \frac{\partial}{\partial x^i} \right|_p (f) := \left. \frac{\partial (f \circ \varphi^{-1})}{\partial r^i} \right|_{\varphi(p)}$$

Proposition 1.1.9. *A coordinate basis as in Definition 1.1.8, forms a basis of $T_p M$. Hence $\dim T_p M = n$, and $v = v^i \partial_i|_p$ using the Einstein summation convention.*

A proof of this proposition can be found in [Lee13, Proposition 3.15]. We can now assemble the tangent spaces at every point of M into a single geometric object, endowed with its own smooth structure. In literature, it is used as the natural stage for the Lagrangian description of a mechanical system.

Definition 1.1.10 (Tangent space). *Let M be a smooth manifold as in Definition 1.1.4. The tangent space of M is*

$$TM := \bigsqcup_{p \in M} T_p M.$$

It is endowed with the canonical projection $\pi : TM \rightarrow M$, $\pi(v) = p$ for $v \in T_p M$.

A single tangent vector specifies a generalized velocity at a specific configuration. To model global dynamical systems, we must assign smoothly a velocity vector to every point of the manifold, which leads to the notion of a vector field.

Definition 1.1.11 (Vector field). *Let M be a smooth manifold as in Definition 1.1.4. A vector field on M is a smooth map $X : M \rightarrow TM$ with $\pi \circ X = \text{id}_M$. We denote by $\mathfrak{X}(M)$ the $C^\infty(M)$ -module of vector fields on M .*

The tangent space captures the kinematics of a system. It is the space of generalized velocities. The Hamiltonian framework also requires generalized momenta, which are dual to velocities and are modeled as linear functionals on the tangent space.

Definition 1.1.12 (Covector). *Let M be a smooth manifold. The cotangent space at $p \in M$ is the dual vector space $T_p^* M := (T_p M)^*$. For $f \in C^\infty(M)$, its differential at p is the covector*

$$df_p \in T_p^* M, \quad df_p(v) := v(f), \quad v \in T_p M.$$

Remark 1.1.13. In the coordinate basis, $\{dx^1|_p, \dots, dx^n|_p\}$ is dual to $\{\partial/\partial x^i|_p\}$, and $df_p = \frac{\partial f}{\partial x^i} \Big|_p dx^i|_p$.

The same construction that produced the tangent space from the individual tangent spaces can be repeated for their duals.

Definition 1.1.14 (Cotangent space). *Let M be a smooth manifold, the cotangent space is*

$$T^*M := \bigsqcup_{p \in M} T_p^* M,$$

*endowed with the canonical projection $\pi : T^*M \rightarrow M$.*

The next proposition follows from the Vector Bundle Construction Lemma [Lee13, Lemma 10.6] that will not be proved here. In short, it states that given a surjective local homeomorphism $\pi : X \rightarrow Y$ from a topological space X and a smooth manifold Y , there exists a unique smooth structure on X , such that π becomes a local diffeomorphism.

Proposition 1.1.15 (TM and T^*M as smooth manifolds). *Let M be a smooth manifold of dimension n . Then TM and T^*M , endowed with the natural projections $\pi_{TM} : TM \rightarrow M$ and $\pi_{T^*M} : T^*M \rightarrow M$, are smooth manifolds of dimension $2n$.*

It descends from Proposition 1.1.15 that for TM a coordinate basis $(U, \varphi = (x^1, \dots, x^n))$ on M induces a counterpart on $\pi^{-1}(U)$ via $v = v^i \partial_i|_p \mapsto (x^1(p), \dots, x^n(p), v^1, \dots, v^n)$. A complete proof of this statement is shown in [Lee13, Proposition 3.18]. The same construction can be extended to T^*M and it can be found in [Lee13, Proposition 11.9].

Now, we generalize the pointwise notion of covector to multilinear objects of arbitrary order, the building blocks from which differential forms are constructed.

Definition 1.1.16 (Differential k -covectors). Let M be a smooth manifold of dimension n and let $k \geq 0$. For each $p \in M$, we denote by $\Lambda^k(T_p^*M)$ the vector space of alternating k -covectors at p that are k -linear alternating maps on T_pM . The disjoint union

$$\Lambda^k(T^*M) := \bigsqcup_{p \in M} \Lambda^k(T_p^*M),$$

is the space of alternating k -covectors.

Using the same construction seen above in Proposition 1.1.15, we can state that $\Lambda^k(T^*M)$ is equipped with a smooth structure over M via the canonical projection $\pi : \Lambda^k(T^*M) \rightarrow M$. A smooth choice of one such alternating k -covector at every point of M defines a differential k -form on the manifold.

Definition 1.1.17 (Differential k -form). A differential k -form on M is a smooth section $\omega : M \rightarrow \Lambda^k(T^*M)$ of this space. We denote the $C^\infty(M)$ -module of differential k -forms on M by $\Omega^k(M)$.

Remark 1.1.18. By convention, for $k = 0$ we set $\Lambda^0(T_p^*M) := \mathbb{R}$, which yields the identity $\Omega^0(M) = C^\infty(M)$. For $k = 1$, Definition 1.1.17 naturally recovers the space of 1-forms $\Omega^1(M)$ introduced in Definition 1.1.14. Furthermore, if $k > n = \dim M$, the alternating property implies that $\Lambda^k(T_p^*M) = \{0\}$, and consequently $\Omega^k(M) = \{0\}$.

Operators on vector fields and forms

With the structure of manifolds, tangent spaces, and the algebra of differential forms in place, we introduce the operators that act on them. In the geometric theory of mechanics, these operators map physical quantities between different spaces, through symmetries or coordinate transformations and generate the dynamical equations. The pushforward maps velocity vectors, its dual, the pullback, acts on configuration constraints and momenta.

First we see that there is a unique way to compute the product between vectors and covectors.

Definition 1.1.19 (Interior product). Let M be a smooth manifold, let $V \in \mathfrak{X}(M)$ be a smooth vector field, and let $\omega \in \Omega^k(M)$ be a smooth k -form. The interior product of ω by V is the $(k-1)$ -form $\iota_V \omega \in \Omega^{k-1}(M)$ defined by

$$(\iota_V \omega)(W_1, \dots, W_{k-1}) = \omega(V, W_1, \dots, W_{k-1}),$$

for all vector fields $W_1, \dots, W_{k-1}$. For $k = 0$, we set $\iota_V \omega = 0$.

While the interior product acts as an algebraic contraction that reduces the degree of a form, we conversely require an operation to build higher-degree forms from lower-degree ones. To construct the full exterior algebra, we introduce an intrinsic product that combines a k and an l -covector into a $(k+l)$ -covector, preserving the alternating property.

Definition 1.1.20 (Wedge product). Let M be a smooth manifold, and let $\omega \in \Lambda^k(T_p^*M)$ and $\eta \in \Lambda^l(T_p^*M)$ as per Definition in 1.1.16. Their wedge product, or exterior product, is the alternating $(k+l)$ -covector $\omega \wedge \eta \in \Lambda^{k+l}(T_p^*M)$ defined by its evaluation on tangent vectors $v_1, \dots, v_{k+l} \in T_pM$:

$$(\omega \wedge \eta)(v_1, \dots, v_{k+l}) := \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) \omega(v_{\sigma(1)}, \dots, v_{\sigma(k)}) \eta(v_{\sigma(k+1)}, \dots, v_{\sigma(k+l)}),$$

where S_{k+l} is the symmetric group of permutations of $\{1, \dots, k+l\}$ and $\text{sgn}(\sigma)$ denotes the signature of the permutation.

This algebraic operation naturally extends pointwise to smooth sections.

Remark 1.1.21. A fundamental algebraic consequence of Definition 1.1.20 is its graded commutativity: for any $\alpha \in \Omega^k(M)$ and $\beta \in \Omega^l(M)$, the relation

$$\alpha \wedge \beta = (-1)^{kl} \beta \wedge \alpha,$$

holds identically. This entails that the wedge product of any 1-form with itself strictly vanishes.

We now examine the differential operators that map kinematic quantities between different manifolds. The most fundamental of such mappings actively transports tangent vectors along a smooth function.

Definition 1.1.22 (Pushforward / Differential). *Let $F : M \rightarrow N$ be a smooth map between smooth manifolds. For each $p \in M$, the pushforward (or differential) of F at p is the linear map*

$$F_* : T_p M \rightarrow T_{F(p)} N,$$

such that, for any tangent vector $X \in T_p M$ and for any smooth function $f \in C^\infty(N)$

$$(F_* X)(f) := X(f \circ F).$$

The pushforward map is also denoted by dF_p .

Remark 1.1.23. We can state an alternative of Definition 1.1.22 using the geometric description of tangent vectors as equivalence classes of curves. Let $\gamma \in C^\infty((-\epsilon, \epsilon); M)$ such that $\gamma(0) = p$ and $\gamma'(0) = v \in T_p M$. Then the pushforward $F_* v$ is the velocity vector of the curve $F \circ \gamma$ at $t = 0$:

$$F_* v = \left. \frac{d}{dt} \right|_{t=0} (F \circ \gamma)(t).$$

The differential of a function yields the linear approximation of a scalar observable. To build the geometric machinery of classical mechanics, in particular the canonical symplectic form on the cotangent space, we need to extend this operation to higher-degree forms. The exterior derivative serves the job.

Definition 1.1.24 (Exterior derivative). *Let M be a smooth manifold, define a family of mappings $d^k(U) : \Omega^k(U) \rightarrow \Omega^{k+1}(U)$ with $k = 0, 1, 2, \dots, n$, and U is open in M , which we denote by d , called the exterior derivative on M , such that*

(i) *d is a $\wedge$ -antiderivation, that is, d is $\mathbb{R}$ linear and for $\alpha \in \Omega^k(U)$, $\beta \in \Omega^l(U)$,*

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta,$$

where $\wedge$ denotes the wedge-product of forms.

(ii) *If $f \in C^\infty(U)$, $d^0 f = df$ is the differential as in Definition 1.1.22.*

(iii) *$d \circ d = 0$, that is, $d^{k+1}(U) \circ d^k(U) = 0$.*

(iv) *d is natural with respect to restrictions, that is, if $U \subset V \subset M$ are open and $\alpha \in \Omega^k(V)$, then $d(\alpha|_U) = (d\alpha)|_U$.*

Remark 1.1.25. It can be proved that a family of mappings as in Definition 1.1.24 exist and it is unique.

A crucial feature of the exterior derivative is its commutation with the pullback, a property that ensures the geometric consistency of coordinate transformations and symmetries. While the pushforward linearly transports velocities along a map, the pullback naturally acts in the opposite direction, transporting observables and differential forms.

Definition 1.1.26 (Pullback). *Let $F : M \rightarrow N$ be a smooth map between smooth manifolds. For $k \geq 0$, let $\Omega^k(N)$ denote the space of smooth differential k -forms on N . The pullback is the map*

$$F^* : \Omega^k(N) \rightarrow \Omega^k(M)$$

defined as follows.

1. *If $k = 0$, given $f \in C^\infty(N) \equiv \Omega^0(N)$, the pullback $F^* f \in C^\infty(M)$ is defined by precomposition:*

$$F^* f = f \circ F.$$

2. If $k \geq 1$, for $\Lambda \in \Omega^k(N)$, the pullback $F^*\Lambda \in \Omega^k(M)$ is defined pointwise. For each $p \in M$ and all tangent vectors $X_1, \dots, X_k \in T_p M$,

$$(F^*\Lambda)_p(X_1, \dots, X_k) = \Lambda_{F(p)}(F_*X_1, \dots, F_*X_k).$$

Remark 1.1.27. The case $k = 1$ recovers the pullback of covectors: for $\omega \in T_{F(p)}^*N$ and $v \in T_p M$,

$$(F^*\omega)(v) = \omega(F_*v).$$

Having described how differential forms and covectors transform under smooth maps, we now turn to the dynamical content of a vector field. In fact, the latter generates, through its integral curves, a continuous family of diffeomorphisms of the manifold. This flow is the fundamental geometric object underlying the notion of time evolution, and it will later allow us to identify the trajectories of a Hamiltonian system with the flow of the corresponding Hamiltonian vector field.

Definition 1.1.28 (Flow of a Vector Field). *Let X be a smooth vector field on a smooth manifold M . An integral curve of X is a smooth curve $\gamma : I \rightarrow M$, where $I \subseteq \mathbb{R}$ is an open interval, such that $\gamma'(t) = X(\gamma(t))$ for all $t \in I$.*

The (local) flow of X is a smooth map $\theta : \mathcal{D} \rightarrow M$, where $\mathcal{D} \subset \mathbb{R} \times M$ is an open neighborhood of $\{0\} \times M$ known as the flow domain, satisfying the following properties

1. *For each $p \in M$, the slice $\mathcal{D}^{(p)} = \{t \in \mathbb{R} : (t, p) \in \mathcal{D}\}$ is an open interval containing 0, and the curve $\theta^{(p)} : \mathcal{D}^{(p)} \rightarrow M$ defined by $\theta^{(p)}(t) = \theta(t, p)$ is the unique maximal integral curve of X satisfying the initial condition $\theta(0, p) = p$, namely*

$$\frac{\partial \theta}{\partial t}(t, p) = X(\theta(t, p)).$$

2. **One-Parameter Group Property:** *Denoted by $\theta_t(p) = \theta(t, p)$, then for any $s \in \mathcal{D}^{(p)}$ and $t \in \mathcal{D}^{(\theta_s(p))}$ such that $t + s \in \mathcal{D}^{(p)}$, we have*

$$\theta_t(\theta_s(p)) = \theta_{t+s}(p).$$

Proposition 1.1.29. *Let M be a smooth manifold and let X be a vector field with $\theta : \mathcal{D} \rightarrow M$ its flow as in Definition 1.1.28. If $\mathcal{D} = \mathbb{R} \times M$, the vector field X is called complete and the maps $\theta_t : M \rightarrow M$ constitute a global one-parameter group of diffeomorphisms.*

A proof of this statement can be constructed merging [Lee13, Proposition 9.7] and [Lee13, Theorem 9.12]. The concept of flow translates the analytical problem of solving systems of differential equations into the geometric action of a one-parameter group of diffeomorphisms. In the context of dynamics, if X is the Hamiltonian vector field governing a mechanical system, its integral curves represent the physical trajectories in phase space, and the associated flow θ_t corresponds to the time evolution operator, which inherently preserves the symplectic structure. This will be analysed later in 1.4

Lie derivative

Directional derivatives of smooth functions on a manifold are defined via tangent vector acting as a derivation when applied to a real-valued function as shown in 1.1.7. In Euclidean spaces, the directional derivative of a vector field relies on the vector space structure of $\mathbb{R}^n$ and the identification of tangent vectors with elements of $\mathbb{R}^n$. On a general manifold, this identification fails. Replacing $v \in T_p M$ with $V \in \mathfrak{X}(M)$ overcomes this, as the flow of V enables the pullback of W to p for differentiation. Hence we adopt the following definition.

Definition 1.1.30 (Lie derivative). *Let M be a smooth manifold, $V \in \mathfrak{X}(M)$, and let θ be the flow of V as in Definition 1.1.28. For $W \in \mathfrak{X}(M)$, the Lie derivative of W with respect to V is the vector field $\mathcal{L}_V W \in \mathfrak{X}(M)$ such that*

$$(\mathcal{L}_V W)_p = \left. \frac{d}{dt} \right|_{t=0} d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)}) = \lim_{t \rightarrow 0} \frac{d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)}) - W_p}{t}. \quad (1.1)$$

If $\partial M \neq \emptyset$, the definition still goes through as long as V is tangent to ∂M , since this entails that the flow of V exists on all of M .

Lemma 1.1.31. *Suppose M is a smooth manifold with or without boundary, and $V, W \in \mathfrak{X}(M)$. If $\partial M \neq \emptyset$, assume in addition that V is tangent to ∂M . Then $(\mathcal{L}_V W)_p$ exists for every $p \in M$, and $\mathcal{L}_V W$ is a smooth vector field.*

The proof of this result is stated in [Lee13, Lemma 9.36]. The next theorem is crucial to understand the role of this construction throughout the thesis.

Theorem 1.1 (Lie derivative of vector fields). *If M is a smooth manifold and $V, W \in \mathfrak{X}(M)$, then $\mathcal{L}_V W = [V, W]$, the commutator between V and W .*

This theorem, proved in [Lee13, Theorem 9.38], provides a direct geometric interpretation of the Lie bracket. It establishes that the commutator $[V, W]$ measures the failure of the flows of V and W to commute. Equivalently, it quantifies the rate at which the vector field W changes when observed from a point moving along the flow of V . This identity will be a cornerstone in the following sections, as it links the differential structure of the manifold to the algebraic properties of vector fields, and, later, to the Poisson bracket of observables.

Corollary 1.1.32. *Suppose M is a smooth manifold with or without boundary, and $V, W, X \in \mathfrak{X}(M)$.*

- (a) $\mathcal{L}_V W = -\mathcal{L}_W V$.
- (b) $\mathcal{L}_V [W, X] = [\mathcal{L}_V W, X] + [W, \mathcal{L}_V X]$.
- (c) $\mathcal{L}_{[V, W]} X = \mathcal{L}_V \mathcal{L}_W X - \mathcal{L}_W \mathcal{L}_V X$.
- (d) If $h \in C^\infty(M)$, then $\mathcal{L}_V (hW) = (Vh)W + h\mathcal{L}_V W$.
- (e) If $F : M \rightarrow N$ is a diffeomorphism, then $F_*(\mathcal{L}_V X) = \mathcal{L}_{F_*V} F_*X$.

As established by [Lee13, Corollary 9.39], these properties highlight the algebraic role of the Lie derivative. In particular, (b) and (c) show that $\mathcal{L}_V$ acts as a derivation on the Lie algebra of vector fields, while (d) confirms its behavior as a derivation with respect to multiplication by smooth functions. Property (e) establishes the naturality of the Lie derivative under diffeomorphisms, ensuring that the construction is intrinsic to the manifold structure. Now, we extend Definition 1.1.30 to differential forms.

Definition 1.1.33 (Lie derivative of a differential form). *Let M be a smooth manifold, let $V \in \mathfrak{X}(M)$ have flow θ , and let $\omega \in \Omega^k(M)$ be a smooth differential form on M . The Lie derivative of ω with respect to V is $\mathcal{L}_V \omega \in \Omega^k(M)$ whose value at $p \in M$ is*

$$(\mathcal{L}_V \omega)_p = \frac{d}{dt} \Big|_{t=0} (\theta_t^* \omega)_p = \lim_{t \rightarrow 0} \frac{(\theta_t^* \omega)_p - \omega_p}{t},$$

wherever this derivative exists, where $\theta_t^* \omega$ is the pullback of ω by the time- t flow map of V .

Lie differentiation satisfies a product rule with respect to wedge products.

Proposition 1.1.34. *Suppose M is a smooth manifold, $V \in \mathfrak{X}(M)$, and $\omega, \eta \in \Omega^k(M)$. Then*

$$\mathcal{L}_V (\omega \wedge \eta) = (\mathcal{L}_V \omega) \wedge \eta + \omega \wedge (\mathcal{L}_V \eta).$$

This product rule, as proved in [Lee13, Proposition 14.33], together with the fact that the Lie derivative commutes with the exterior derivative d , makes $\mathcal{L}_V$ a derivation of the algebra of differential forms. This algebraic compatibility is essential for deriving the fundamental identity that links Lie derivatives, exterior derivatives, and interior products, commonly known as Cartan's Magic Formula.

Theorem 1.2 (Cartan's Magic Formula). *On a smooth manifold M , for any smooth vector field V and any smooth differential form ω ,*

$$\mathcal{L}_V \omega = \iota_V (d\omega) + d(\iota_V \omega). \quad (1.2)$$

For a complete proof of this formula, see [Lee13, Theorem 14.35]. As we will show, it is necessary for characterizing symplectic vector fields, and it will serve as a key tool in the proof of Liouville's Theorem and the construction of the momentum map.

1.2 Lie Algebras and Representations

In this section we develop the algebraic component of classical mechanics. We first introduce Lie algebras abstractly. We then specialize to Lie groups, smooth manifolds carrying a compatible group structure, and show how every Lie group determines a Lie algebra through its left-invariant vector fields, with the exponential map provides the bridge back from the algebra to the group. Finally, we introduce the notion of a representation, which realizes the abstract elements of a Lie algebra or Lie group as concrete linear operators on a vector space. This last step is what will allow us, in later chapters, to identify the continuous symmetries of a quantum system with observables that can actually be measured.

Note that in this chapter, $\mathbb{K}$ denotes an arbitrary commutative field with characteristic zero, following [Hum10].

Definition 1.2.1 (Lie algebra). A Lie algebra over $\mathbb{K}$ is a $\mathbb{K}$ -vector space $\mathfrak{g}$ equipped with a bilinear map $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ satisfying

$$[X, Y] = -[Y, X], \quad \text{antisymmetry} \quad (1.3)$$

$$[[X, Y], Z] + [[Y, Z], X] + [[Z, X], Y] = 0, \quad \text{Jacobi identity} \quad (1.4)$$

for all $X, Y, Z \in \mathfrak{g}$.

Remark 1.2.2. Every associative algebra $(\mathcal{A}, \cdot)$ carries a canonical Lie algebra structure via the commutator $[a, b] := ab - ba$. It satisfies identically the antisymmetry and the Jacobi identity:

$$[a, b], c] = (abc - bac) - c(ab - ba) = -[[b, c], a] - [[c, a], b].$$

The Jacobi identity is used in a Lie algebra to mimic the chain property of a derivation. For example, this allows $\mathfrak{g}$ to codify infinitesimal calculus on manifolds.

Example 1.2.3 (Vector fields as a Lie algebra). Let M be a smooth manifold. The space of smooth vector fields $\mathfrak{X}(M)$, introduced in Definition 1.1.11, forms an infinite-dimensional Lie algebra over $\mathbb{R}$. Since vector fields act as derivations on the algebra of smooth functions $C^\infty(M)$, we can define the Lie bracket between two vector fields $X, Y \in \mathfrak{X}(M)$ as their commutator:

$$[X, Y](f) := X(Y(f)) - Y(X(f)), \quad \forall f \in C^\infty(M).$$

While the composition of two derivations XY is not generally a derivation, their commutator inherently is, ensuring that $[X, Y] \in \mathfrak{X}(M)$ is again a well-defined vector field. The bilinearity and antisymmetry are manifest, while the Jacobi identity follows directly from the associative nature of operator composition.

Example 1.2.4. One of the most important examples of Lie algebra, as per Definition 1.2.1, is $\mathfrak{gl}(V)$, the space of all linear endomorphisms of a finite dimensional vector space V , equipped with the commutator $[A, B] = AB - BA$ as Lie bracket. When $V = \mathbb{C}^n$, $\mathfrak{gl}(V) \cong \mathfrak{gl}(n, \mathbb{C})$ can be identified with $M(n, \mathbb{C})$. A natural basis, denoted E_{ij} for $i, j = 1, \dots, n$, is given by the *matrix units*, i.e., the matrix with entry 1 in position (i, j) and 0 elsewhere, satisfying

$$[E_{ij}, E_{kl}] = \delta_{jk}E_{il} - \delta_{li}E_{kj}.$$

Example 1.2.5. Another notable example of Lie algebra, as per Definition 1.2.1, is $\mathfrak{su}(n)$. It consists of the matrices in $M(n, \mathbb{C})$ which are traceless and anti-Hermitian. The dimension of the resulting Lie algebra is $n^2 - 1$. With $n = 2$, the basis of $\mathfrak{su}(2)$ is $\{T_j = -i\sigma_j/2\}_{j=x,y,z}$, where σ_j are the Pauli operators which are used for representing the spin of a fermion in quantum physics. Instead, with $n = 3$, the basis is $\{-i\lambda_a/2\}_{a=1,\dots,8}$ which is composed by the eight Gell-Mann matrices λ_a used in quantum chromodynamics.

We recall some basic concepts of the theory of Lie algebras. A *homomorphism* is a map between the same mathematical structures that preserves the inner operations. In particular, when the structure is a Lie algebra we can use the following definition.

Definition 1.2.6 (Homomorphism of Lie Algebras). Let $\mathfrak{g}$ and $\mathfrak{h}$ be Lie algebras over the same field $\mathbb{K}$. A linear map $\phi : \mathfrak{g} \rightarrow \mathfrak{h}$ is a Lie algebra homomorphism if it preserves the Lie bracket:

$$\phi([X, Y]) = [\phi(X), \phi(Y)] \quad \forall X, Y \in \mathfrak{g}.$$

When such a homomorphism is invertible, it identifies the two Lie algebras as abstractly the same structure.

Definition 1.2.7 (Isomorphism of Lie Algebras). An isomorphism is a Lie algebra homomorphism $\phi : \mathfrak{g} \rightarrow \mathfrak{h}$ that is bijective.

Lie Groups

To benefit from the algebraic structure of Lie algebras within the framework of differential geometry, we must introduce manifolds equipped with compatible smooth group operations. In order to canonically endow the tangent space at the identity element with a non-trivial Lie algebra structure, the manifold must globally be a Lie group.

To formalize this relationship, we first define a Lie group by requiring the group operations to be smooth.

Definition 1.2.8 (Lie Group). *A Lie group is a smooth manifold G together with a smooth map from $G \times G \rightarrow G$ that makes G into a group. Furthermore the inverse map $g \mapsto g^{-1}$ is smooth.*

We can specialize Definition 1.2.6 to the case of Lie groups.

Definition 1.2.9 (Homomorphism of Lie Groups). *Let G be a Lie group as in Definition 1.2.8, the map $F : G \rightarrow H$ where G and H are Lie groups is called a homomorphism if it is a smooth map and a homomorphism between groups.*

Consider the tangent space of a Lie group. For every $g \in G$, we can define a map $L_g \in C^\infty(G; G)$ by $L_g(h) = gh$ called the "left multiplication". The differential dL_g at a point $h \in G$ is a linear map from $T_h G$ to $T_{gh} G$ as stated in Definition 1.1.22. A vector field X on G is called *left-invariant* if

$$dL_g(X_h) = X_{gh} \quad \forall g, h \in G.$$

It is also true that, for any given vector $v \in T_e G$, there exists a unique smooth left-invariant vector field X^v . It can be constructed starting from the vector tangent to the identity by setting $X_e^v = v$. We can then use L_g to define

$$X_g^v = dL_g(X_e^v) = dL_g(v).$$

By construction, this is left-invariant and smooth. One can show that the commutator of two left-invariant vector fields is, again, a left-invariant vector field. Also, it is worth noting that left-invariant vector fields form a linear subspace of $\mathfrak{X}(G)$ closed respect to the commutator, called a *Lie subalgebra* and thus they inherit the property of a Lie algebra.

Definition 1.2.10 (Lie algebra of a Lie group). *Let G be a Lie group, the Lie algebra $\mathfrak{g}$ of G is the tangent space at the identity with the bracket operation defined by*

$$[v, w] := [X^v, X^w]_e.$$

On a general smooth manifold, vector fields typically admit only local flows, potentially developing singularities in finite time. However, the underlying homogeneous structure of a Lie group circumvents this limitation. Local solutions to the associated differential equations can be indefinitely translated across the manifold via the group multiplication. This mechanism prevents finite-time divergences and guarantees the existence of global flows, ensuring that infinitesimal symmetries seamlessly integrate into one-parameter subgroups of finite transformations.

Proposition 1.2.11 (Completeness of left-invariant vector fields). *Let $X^v \in \mathfrak{X}(G)$ be the left-invariant vector field associated to $v \in \mathfrak{g} = T_e G$. Then X^v is complete: its maximal integral curve through e is defined for all $t \in \mathbb{R}$.*

The complete proof can be found in [[Lee13], Theorem 9.18]. Completeness alone does not yet reveal the full algebraic content of the flow. The following proposition shows that the integral curve of a left-invariant vector field is in fact a group homomorphism.

Proposition 1.2.12 (One-parameter subgroup). *Let $\gamma_v : \mathbb{R} \rightarrow G$ be the integral curve of X^v through e . Then*

$$\gamma_v(t+s) = \gamma_v(t)\gamma_v(s) \quad \forall t, s \in \mathbb{R},$$

$\gamma_v : \mathbb{R} \rightarrow G$ is a Lie group homomorphism called a one-parameter subgroup.

This is proved in [Lee13, Theorem 20.1]. These concepts naturally converge to establish a canonical correspondence between a Lie algebra and its associated Lie group. The mathematical tool enacting this exact correspondence is the exponential map.

Definition 1.2.13 (Exponential map). *Let G be a Lie group and let $\gamma_v : \mathbb{R} \rightarrow G$ be the integral curve of X^v through e . The exponential map of G is*

$$\exp : \mathfrak{g} \rightarrow G, \quad \exp(v) := \gamma_v(1).$$

Thanks to the completeness granted by Proposition 1.2.11, the integral curve γ_v is extended globally $\forall t \in \mathbb{R}$. This justifies Definition 1.2.13 and it is at the heart of the next properties we discuss.

Proposition 1.2.14 (Properties of exponential map). *Let G be a Lie group, it can be shown that*

1. $\exp(tv) = \gamma_v(t) \forall t \in \mathbb{R}$.
2. $\exp(0) = e$ and $d(\exp)_0 = \text{id}_{\mathfrak{g}}$.
3. *Naturality: if $F : G \rightarrow H$ is a homomorphism of Lie groups with differential $dF_e : \mathfrak{g} \rightarrow \mathfrak{h}$, then*

$$F(\exp_G(v)) = \exp_H(dF_e(v)), \quad v \in \mathfrak{g}.$$

This statement is proved in [Lee13, Proposition 20.8]. The naturality property is particularly restrictive. It forces the exponential map to commute with any homomorphism of Lie groups. Before exploiting this, it is worth noting a subtlety in the global behaviour of $\exp$.

Remark 1.2.15. Global surjectivity of $\exp$ is not automatic in general, it fails ,e.g., for $SL(2, \mathbb{R})$. Yet it is verified for G compact and connected. This covers the classical groups used in this thesis ($SU(2)$, $U(n)$).

Remark 1.2.16 (Matrix Lie groups). For $G \subseteq GL(n, \mathbb{C})$ a closed subgroup, $\mathfrak{g} \subset \mathfrak{gl}(n, \mathbb{C}) = M_n(\mathbb{C})$ is identified with a linear subspace of matrices via $T_e G \subset T_e GL(n, \mathbb{C}) \cong M_n(\mathbb{C})$. Left multiplication is matrix multiplication: $L_g(h) = gh$, so $dL_g(v) = gv$ for $v \in \mathfrak{g}$. The left-invariant vector field associated to v is therefore $X_g^v = gv$, and its integral curve through e solves the linear matrix ODE

$$\dot{\gamma}(t) = \gamma(t)v, \quad \gamma(0) = \mathbb{1}.$$

The unique solution is the matrix exponential series

$$\gamma(t) = e^{tv} := \sum_{k=0}^{\infty} \frac{(tv)^k}{k!}.$$

Hence $\exp(v) = e^v$. The abstract exponential map of Definition 1.2.13 coincides with the ordinary matrix exponential whenever G is realized as a matrix group. In particular

$$\mathfrak{g} = \{v \in M_n(\mathbb{C}) \mid e^{tv} \in G \forall t \in \mathbb{R}\}.$$

Example 1.2.17. Let $G = SU(2) = \{g \in GL(2, \mathbb{C}) : g^\dagger g = \mathbb{1}, \det g = 1\}$, differentiating both defining conditions along $\gamma(t) = e^{tv}$, $\gamma(0) = \mathbb{1}$, yields the linearized conditions on v :

$$\left. \frac{d}{dt} \right|_{t=0} (e^{tv})^\dagger e^{tv} = v^\dagger + v = 0, \quad \left. \frac{d}{dt} \right|_{t=0} \det e^{tv} = \text{tr}(v) = 0,$$

using $\det e^{tv} = e^{t \text{tr} v}$. It follows that

$$\mathfrak{su}(2) = \{v \in M_2(\mathbb{C}) \mid v^\dagger = -v, \text{tr}(v) = 0\},$$

a 3-dimensional real vector space. Take $v = -\frac{i\theta}{2}\sigma_3$, $\theta \in \mathbb{R}$, $\sigma_3 = \text{diag}(1, -1)$. Since $\sigma_3^2 = \mathbb{1}$, the series splits into even and odd powers:

$$\exp(v) = e^{-i\theta\sigma_3/2} = \cos\left(\frac{\theta}{2}\right)\mathbb{1} - i\sin\left(\frac{\theta}{2}\right)\sigma_3 = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}.$$

This matrix is unitary with determinant 1 for every θ , confirming $\exp(v) \in SU(2)$ and $\theta \mapsto \exp(v)$ satisfies the one-parameter subgroup law, consistently with Proposition 1.2.12.

Representations of groups

Lie groups and Lie algebras are abstract mathematical structures. To use them in physics, we need to identify their elements with linear operators acting on a vector space. This identification is called a representation. In quantum mechanics and quantum information theory the relevant state space is a Hilbert space, and representation theory gives us the dictionary between abstract symmetry generators and concrete linear operators. This is what turns continuous symmetries into observables we can actually measure.

In this section, we start with the general definition of a representation for arbitrary groups, then specialize to Lie groups and Lie algebras, following [Hal03].

Definition 1.2.18 (Representation of a group). *Let G be a group and V a vector space over $\mathbb{K}$. A representation of G on V is a homomorphism between groups:*

$$\rho : G \rightarrow GL(V),$$

where $GL(V)$ denotes the general linear group, the set of all the automorphisms $\psi : V \rightarrow V$. A representation is often denoted by (V, ρ) .

In the case of Lie algebras this can be transported as the following.

Definition 1.2.19 (Representation of a Lie Algebra). *A representation of a Lie Algebra $\mathfrak{g}$ is a homomorphism $\phi : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ where V is a $\mathbb{K}$ -vector space.*

Remark 1.2.20. It is worth noting that this definition is slightly different from the previous one. In the case of a group, the range of the homomorphism ρ is the set of all automorphisms. Instead in Definition 1.2.19, the maps onto $\mathfrak{gl}(V)$ are not all isomorphisms. Observe that $\mathfrak{gl}(V)$ is the *tangent space* to $GL(V)$ at the identity. Defining an exponential map $g(t) = e^{tX}$ one has $g'(0) = X$, which is generically non-invertible even though every $g(t)$ is invertible. Algebraically, this reflects the fact that a Lie algebra homomorphism $\rho : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ needs only preserve the bracket, $\rho([X, Y]) = [\rho(X), \rho(Y)]$, whereas a group homomorphism $\rho : G \rightarrow GL(V)$ must preserve inverses. This forces $\rho(g) \in GL(V)$ for every $g \in G$.

Example 1.2.21. We define the *adjoint representation* as the homomorphism $\text{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ which sends $x \mapsto \text{ad } x$ with $\text{ad } x$ being an endomorphism on $\mathfrak{g}$ defined as

$$\text{ad } x(y) := [x, y].$$

In addition, it can be easily proved that

$$\text{ad}[x, y] = [\text{ad } x, \text{ad } y],$$

showing the clear preservation of Lie brackets.

One of the most relevant properties of a representation is faithfulness.

Definition 1.2.22 (Faithful Representation). *A representation (V, ρ) is said to be faithful if ρ is injective that is*

$$\ker \rho = \{g \in G \mid \rho(g) = \mathbb{1}_V\} = \{e\},$$

where e is the identity element of the group.

Establishing whether a representation is faithful determines if it captures the full algebraic structure of the original group without redundancy. However, physical systems may realize the identical abstract symmetry through fundamentally different matrix formulations, reflecting a different choice of basis for the underlying state space. Therefore, before attempting to systematically decompose these spaces, it is strictly necessary to formulate a rule to identify when two representations are mathematically equivalent and physically indistinguishable.

Definition 1.2.23 (Equivalent representations). *Let (V, ρ) and (W, π) be two representations of a group G , with V and W two $\mathbb{K}$ -vector spaces. The representations are said to be equivalent if there exists a vector space isomorphism $T : V \rightarrow W$ such that:*

$$T\rho(g) = \pi(g)T \quad \forall g \in G.$$

Remark 1.2.24. Another way to express Definition 1.2.23 is the following. In order to define a representation equivalent to (V, ρ) on a $\mathbb{K}$ -vector space W , given a vector space isomorphism T we define the homomorphism of the equivalent representation as

$$\pi := T \circ \rho \circ T^{-1}.$$

Now we are ready to define what it means to decompose representations into the so called *irreducible representations*.

Definition 1.2.25 (Irreducibility of a representation). *Let (V, π) be a finite-dimensional representation of a group G , acting on a space V . We say that:*

- *A subspace W of V is invariant if $\pi(g)(w) \in W \quad \forall w \in W, \forall g \in G$.*
- *An invariant subspace W is nontrivial if $W \neq \{0\}$ and $W \neq V$.*
- *A representation with no nontrivial invariant subspaces is irreducible.*

The terms invariant, nontrivial, and irreducible are defined analogously for representations of Lie algebras.

Thanks to such a definition we can state a notable result.

Lemma 1.2.26 (Schur's Lemma). *Let $\pi : G \rightarrow GL(V)$ be a finite-dimensional irreducible representation over $\mathbb{C}$.*

1. *If $T : V \rightarrow V$ is a linear map commuting with $\pi(g)$ for every g , then $T = \lambda \text{id}_V$ for some $\lambda \in \mathbb{C}$.*
2. *More generally, if $\pi_1 : G \rightarrow GL(V_1)$, $\pi_2 : G \rightarrow GL(V_2)$ are both irreducible and $T : V_1 \rightarrow V_2$ satisfies $T\pi_1(g) = \pi_2(g)T$ for all g , then either $T = 0$ or T is an isomorphism.*

We will not show the proof of this lemma. The reader is referred to [Hal03, Theorem 4.26]. The physical implication of Schur's Lemma manifests prominently in the study of Casimir operators. Those are possibly non-linear algebraic invariants constructed from the generators of the Lie algebra that, by definition, commute with all elements of the algebra itself. Since a Casimir operator commutes with the entire representation, Schur's Lemma dictates that it must act as a scalar multiple of the identity on any irreducible representation space. This scalar value yields the exact quantum numbers necessary to classify the fundamental multiplets of a quantum system. This lemma can be extended to Lie algebras as well. We can show an example of its application on $\mathfrak{su}(2)$.

Example 1.2.27 (Schur's Lemma on $\mathfrak{su}(2)$). Let $\pi : \mathfrak{su}(2) \rightarrow \text{End}(V)$ be a finite-dimensional irreducible representation, let $J_i := iT_i$ for $i = x, y, z$ as shown in Example 1.2.5 and define:

$$\hat{J}^2 := \hat{J}_x^2 + \hat{J}_y^2 + \hat{J}_z^2$$

be the total angular momentum squared operator. Using $[\hat{J}_i, \hat{J}_j] = i\varepsilon_{ijk}\hat{J}_k$, a direct computation yields $[\hat{J}^2, \hat{J}_i] = 0$ for every i . Hence $\hat{J}^2$ commutes with the entire representation.

By Schur's Lemma 1.2.26 item 1., any operator commuting with an irreducible representation is a scalar multiple of the identity:

$$\hat{J}^2 = \lambda \mathbb{1}, \quad \text{for some } \lambda \in \mathbb{R}.$$

In this simple case, we see how Schur's Lemma guarantees that $\hat{J}^2$ takes one fixed value on the whole representation, not just on individual states.

To conclude, we state an important definition that will be needed in the next chapters.

Definition 1.2.28 (Strongly continuous unitary representation). *Let $\mathcal{H}$ be a Hilbert space, and let $U(\mathcal{H})$ denote the group of unitary operators on $\mathcal{H}$. Then, a homomorphism $\Pi : G \rightarrow U(\mathcal{H})$, where G is a Lie group, is called a strongly continuous unitary representation of G if Π satisfies the following continuity condition called strong continuity:*

$$A_n, A \in G \quad \text{and} \quad A_n \rightarrow A \quad \implies \quad \Pi(A_n)(v) \rightarrow \Pi(A)(v) \quad \forall v \in \mathcal{H}.$$

1.3 Elements of symplectic geometry

In the previous sections, we established the formalism of differential geometry and the theory of Lie groups, which are essential tools for describing the global symmetries and the topology of physical systems. However, the configuration manifold turns out to be insufficient. Dynamical evolution naturally arises from the phase space, mathematically identified with the cotangent space of the base manifold. This structure is complete enough to yield a way to compute an isomorphism between covectors and vector fields. The framework provided is called symplectic geometry.

In this section, we will introduce the foundations of symplectic manifolds, starting from the fundamental notion of a closed and non-degenerate 2-form, which constitutes the mathematical core required to define canonical transformations and govern Hamiltonian dynamics.

Definition 1.3.1 (Non-degenerate 2-form on a manifold). *Let M be a smooth manifold and $\omega \in \Omega^2(M)$. We say that ω is non-degenerate if, for every $p \in M$, the bilinear form ω_p is non-degenerate on $T_p M$: for every $X \in T_p M$, if $\forall Y \in T_p M$*

$$\omega_p(X, Y) = 0 \implies X = 0.$$

The next proposition shows that any skew-symmetric two-form, degenerate or not, admits a pointwise normal form, a purely algebraic fact from which the non-degenerate case will follow as an immediate specialization.

Proposition 1.3.2. *Let $\omega \in \Omega^2(M)$ on a smooth manifold M , and for $p \in M$ set $r_p := \text{rank}(\omega_p)$. Then for every $p \in M$ there exists an integer n_p with $r_p = 2n_p$, and an ordered basis $\hat{e} = (e_i)$ of $T_p M$, with dual ordered basis (α^i) as per Remark 1.1.13, such that*

$$\omega_p = \sum_{i=1}^{n_p} \alpha^i \wedge \alpha^{i+n_p},$$

or, equivalently, the matrix of ω_p in this basis is

$$\begin{pmatrix} 0 & \mathbb{1} & 0 \\ -\mathbb{1} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

where $\mathbb{1}$ is the $n_p \times n_p$ identity matrix and the bottom-right block has size $(d-2n_p) \times (d-2n_p)$, $d := \dim T_p M$.

A proof of this statement can be found in [AM08, Proposition 3.1.2].

Remark 1.3.3. Observe that $(e_i)_{i=1, \dots, 2n}$ is a basis of $T_p M$ alone, chosen independently at each p , with no claim that $p \mapsto e_i(p)$ varies smoothly, or even continuously.

Specializing the previous proposition to the non-degenerate case removes the degenerate block entirely and it fixes the rank at every point, with the following immediate consequences.

Corollary 1.3.4. *Suppose ω is non-degenerate by Definition 1.3.1. Then $r_p = \dim T_p M =: d$ for every $p \in M$, and $n_p \equiv n$ is the same integer at every point. In particular:*

- d must be even, so a skew-symmetric non-degenerate 2-form can exist only on an even-dimensional manifold.
- we can define $\omega^n := \omega \wedge \dots \wedge \omega$ with $n = d/2$ factors, which is nowhere-vanishing and, in the basis used above, it can be written as $\omega_p^n = n! \alpha^1 \wedge \alpha^{n+1} \wedge \dots \wedge \alpha^n \wedge \alpha^{2n} \neq 0$. So, ω^n is a volume form on M , that can be normalized as

$$\Omega_\omega := \frac{(-1)^{n(n-1)/2}}{n!} \omega^n.$$

This is proved in [AM08, Proposition 3.1.5]. Beyond restricting the dimension and providing a canonical volume measure, the non-degeneracy of ω plays a far more active role in the formulation of mechanics. Specifically, a non-degenerate bilinear form naturally induces a point-wise vector space isomorphism between the tangent and cotangent spaces. This duality is the fundamental mathematical machinery required to translate the differential of an observable into a unique vector field that generates the dynamics.

Definition 1.3.5 (Musical isomorphisms). Let M be a smooth manifold and $\omega \in \Omega^2(M)$ be non-degenerate by Definition 1.3.1. We define $\flat : \mathfrak{X}(M) \rightarrow \Omega^1(M)$ so that a vector field

$$X \mapsto X^\flat = \iota_X \omega = \omega(X, \cdot).$$

The inverse map is $\sharp : \Omega^1(M) \rightarrow \mathfrak{X}(M)$ so that $\alpha \mapsto \alpha^\sharp$ where $\alpha^\sharp$ is the only vector field X such that $\iota_X \omega = \alpha$.

Remark 1.3.6. Note that the inverse map $\sharp$ is well-defined because $\flat$ is bijective being ω non-degenerate as in Definition 1.3.1. We can infer that $(X^\flat)^\sharp = X$ and $(\alpha^\sharp)^\flat = \alpha$ due to the non-degeneracy of ω .

With the musical isomorphisms in hand, we are ready to state a central theorem that underlines the importance of the closure of the symplectic form.

Theorem 1.3 (Darboux). Suppose ω is a non-degenerate two-form on a $2n$ -manifold M by Definition 1.3.1. Then ω is closed, meaning $d\omega = 0$, where d denotes the exterior derivative, if and only if there exists a chart (U, ϕ) at each $m \in M$ such that $\phi(m) = \mathbf{0}$ and $\phi(u) = (x^1(u), \dots, x^n(u), y^1(u), \dots, y^n(u))$ with

$$\omega|_U = \sum_{i=1}^n dx^i \wedge dy^i.$$

We will not show the proof of this theorem. We refer the reader to [AM08, Theorem 3.2.2].

Remark 1.3.7. One direction of the equivalence descend by direct inspection. If we start from the fact that the canonical local expression of the form is exact, then it is automatically closed. The true analytical depth of Darboux's Theorem, however, lies in its sufficiency. It proves that the closure of ω is the only requirement needed to guarantee the existence of a local canonical frame. Consequently, any two symplectic manifolds of the same dimension are locally diffeomorphic.

Finally, we can state the next important definition that encodes all the structure to derive Hamiltonian mechanics in a natural way.

Definition 1.3.8 (Symplectic manifold). A symplectic form ω on a smooth manifold M is a non-degenerate, closed, two-form on M . A symplectic manifold (M, ω) is a smooth manifold M together with a symplectic form ω on M . The charts as per the Darboux Theorem 1.3 are called symplectic charts and the associated coordinates are called canonical.

Having in hand the geometric framework of the phase space, it is natural to investigate what class of transformations preserves the structure of the symplectic manifold. Such mappings are fundamental as they generate all the symmetries within Hamiltonian mechanics.

Definition 1.3.9 (Canonical transformation). Let (M, ω) and (N, ρ) be symplectic manifolds. Then $F \in C^\infty(M; N)$ is called symplectic or canonical transformation if $F^* \rho = \omega$.

Since canonical transformations carry ρ exactly onto ω , they must preserve the associated volume form, which forces them to be local diffeomorphisms.

Proposition 1.3.10. If (M, ω) and (N, ρ) be symplectic manifolds with $\dim M = \dim N$ and $F : M \rightarrow N$ symplectic, then F is volume preserving,

$$F^* \Omega_\rho = \Omega_\omega,$$

with Ω_ω defined as in 1.3.4, and F is a local diffeomorphism.

This statement is proved in [AM08, Proposition 3.2.6]. The next theorem is notationally dense, but it is central to this introduction to symplectic geometry. The problem we are facing is the following. Constructing a 1-form on a manifold M , requires a rule that assigns a real number to any tangent vector. When M is already a cotangent space of another manifold Q , $M = T^*Q$, as in the case of the phase space in Hamiltonian mechanics, there is a natural way to accomplish this.

Fixing a $q \in Q$, from the perspective of Q , every point in M is a 1-form, which we denote as α_q . From the perspective of M , we can instead consider the tangent vector at the point α_q , calling it w_{α_q} . At this point, the natural way to create a 1-form is to use α_q viewed as a form acting on tangent vectors in TQ . Thus, we can apply a pushforward to w_{α_q} in order to move it to the tangent space of Q , pairing it with α_q . This is how the canonical form θ_0 works.

Theorem 1.4 (Canonical forms). *Let Q be an n -manifold and $M = T^*Q$. Consider now the maps $\tau_Q^* : T^*Q \rightarrow Q$ and $d\tau_Q^* : TM \rightarrow TQ$, which is the pushforward of τ_Q^* . Let α_q denote a point in T^*Q , with $q \in Q$, and $w_{\alpha_q} \in TM$ in the fiber above α_q . Define*

$$\theta_{\alpha_q} : T_{\alpha_q}M \rightarrow \mathbb{R} \quad w_{\alpha_q} \mapsto \alpha_q(d\tau_Q^*(w_{\alpha_q})),$$

and the map $\theta_0 : \alpha_q \rightarrow \theta_{\alpha_q}$. In this case $\theta_0 \in \Omega^1(M)$ and $\omega_0 = -d\theta_0$ is a symplectic form on M . The forms θ_0 and ω_0 are called the canonical forms on M .

This theorem is proved in [AM08, Theorem 3.2.10].

Remark 1.3.11. The superscript $*$ in τ_Q^* labels it as the *cotangent-space* projection, as opposed to $\tau_Q : TQ \rightarrow Q$. It should not be confused with the pullback operator.

Corollary 1.3.12 (Canonical forms in coordinates). *From Theorem 1.4, in finite-dimensional smooth manifolds written in local coordinates of M ($q^1, \dots, q^n, p_1, \dots, p_n$), it follows that*

$$\theta_0 = \sum_{i=1}^n p_i dq^i \quad \omega_0 = \sum_{i=1}^n dq^i \wedge dp_i.$$

The full construction of the canonical forms in local coordinates can be found in [AM08, p.179]. We can also prove that these canonical forms are not changed by any diffeomorphism on the manifold.

Theorem 1.5 (Invariance of the canonical forms). *Let Q be a smooth manifold and $f : Q \rightarrow Q$ a diffeomorphism. Define the lift of f as*

$$T^*f : T^*Q \rightarrow T^*Q, \quad (T^*f(\alpha_q))(v) = \alpha_q(df(v)) \quad \forall v \in T_qQ.$$

*Then T^*f is symplectic and the canonical form θ_0 , defined in Theorem 1.4, is invariant under the application of the pulled-back lift: $(T^*f)^*\theta_0 = \theta_0$.*

We will not prove this theorem, instead, we refer the reader to [AM08, Theorem 3.2.12].

Remark 1.3.13. In coordinates, if $f(q^1, \dots, q^n) = (Q^1, \dots, Q^n)$, then the lift of f is a map that

$$(Q^1, \dots, Q^n, P_1, \dots, P_n) \mapsto (q^1, \dots, q^n, p_1, \dots, p_n),$$

where

$$p_j = \sum_{i=1}^n \frac{\partial Q^i}{\partial q^j} P_i.$$

1.4 Hamiltonian Vector Fields and Poisson Brackets

Having built the geometric picture of phase space, this section focuses on dynamics and symmetries in Hamiltonian mechanics, where the geometric and algebraic structures introduced in the previous chapters come together. We first show how a single observable, the Hamiltonian function, representing the total energy of a physical system, generates a flow on phase space via the symplectic form. This flow governs the time evolution of all observables. The Poisson bracket, which endows the space of observables with a Lie algebra structure, encodes the compatibility between the symplectic geometry and the algebra of functions. In the following sections, this algebraic framework will be essential for the systematic analysis of symmetries and their associated conserved quantities through Noether's theorem.

Definition 1.4.1 (Hamiltonian Vector Field). *Let (M, ω) be a symplectic manifold and $H : M \rightarrow \mathbb{R}$ a given C^∞ -function. The vector field X_H determined by*

$$\iota_{X_H} \omega = dH, \tag{1.5}$$

called the Hamiltonian vector field where H is called the energy function. (M, ω, H) is a Hamiltonian system.

This definition selects the vector fields that arise from a global energy function H . It is useful, however, to isolate a weaker and more geometric requirement, namely that the flow of X preserves the symplectic form ω without necessarily deriving from a globally defined Hamiltonian. The next definition makes this distinction precise, and the remark that follows shows exactly how the two notions differ.

Definition 1.4.2 (Symplectic and Globally Hamiltonian Vector Fields). *Let (M, ω) be a symplectic manifold as in Definition 1.3.8. A vector field $X \in \mathfrak{X}(M)$ is called symplectic or locally Hamiltonian if $\mathcal{L}_X \omega = 0$.*

The vector field X is called globally Hamiltonian if there exists a smooth function $H \in C^\infty(M)$ such that the contraction satisfies $\iota_X \omega = dH$.

Remark 1.4.3 (Closedness and exactness of Hamiltonian vector fields). The condition for a vector field X to be locally Hamiltonian, namely $\mathcal{L}_X \omega = 0$, means that it preserves the symplectic form. We can state an equivalent formulation using Cartan's magic formula as per Equation (1.2), the Lie derivative of the symplectic form is

$$\mathcal{L}_X \omega = \iota_X(d\omega) + d(\iota_X \omega).$$

Since ω is a symplectic form, it is automatically closed $d\omega = 0$. The equation reduces to:

$$\mathcal{L}_X \omega = d(\iota_X \omega).$$

Consequently the invariance of the symplectic structure stands if and only if $d(\iota_X \omega) = 0$, that is requiring the closedness of $\iota_X \omega$. Analogously, the condition for X to be globally Hamiltonian postulates the existence of H such that $\iota_X \omega = dH$. That is, by definition, requiring the exactness of $\iota_X \omega$.

The main advantage of this algebraic definition is its reliance on the symplectic structure, which maps dH into the corresponding vector field. When evaluated in local canonical coordinates, whose existence is guaranteed by Darboux's Theorem 1.3, this geometric mapping recovers the classical Hamilton's equations of motion.

Theorem 1.6 (Hamilton Equations). *Let $(q^1, \dots, q^n, p_1, \dots, p_n)$ be canonical coordinates for ω , so $\omega = \sum_{i=1}^n dq^i \wedge dp_i$. Then in these coordinates and dropping base points,*

$$X_H = \left(\frac{\partial H}{\partial p_i}, -\frac{\partial H}{\partial q^i} \right) = J \cdot dH, \quad (1.6)$$

here dH denotes the column vector of its coordinate partial derivatives and $J = \begin{pmatrix} 0 & \mathbb{1} \\ -\mathbb{1} & 0 \end{pmatrix}$. Thus $(q(t), p(t))$ is an integral curve of X_H if and only if Hamilton's equations hold:

$$\boxed{\dot{q}^i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q^i}, \quad i = 1, \dots, n.}$$

Proof. Let Y be defined by Equation (1.6). We then have to verify 1.5. Since by construction $\iota_Y dq^i = \partial H / \partial p_i$, $\iota_Y dp_i = -\partial H / \partial q^i$, it descends

$$\iota_Y \omega = \sum \iota_Y (dq^i \wedge dp_i) = \sum (\iota_Y dq^i) \wedge dp_i - \sum dq^i \wedge \iota_Y dp_i = \sum \frac{\partial H}{\partial p_i} dp_i + \frac{\partial H}{\partial q^i} dq^i = dH.$$

By the non-degeneracy of ω , $Y = X_H$. □

A direct physical consequence of the anti-symmetry inherent to the symplectic form is the emergence of conserved quantities which we will study in the next section. Here, we can show the main example, the Hamiltonian itself. Evaluating the time derivative of the energy along its own integral curves demonstrates that the dynamics intrinsically preserves the total energy of the system.

Proposition 1.4.4 (Conservation of energy). *Let (M, ω, H) be a Hamiltonian system through Definition 1.4.1 and let $c(t)$ be an integral curve for X_H . Then $H(c(t))$ is constant in as a function of $t \in \mathbb{R}$.*

Proof. By the chain rule and Equation (1.5),

$$\begin{aligned}\frac{d}{dt}H(c(t)) &= dH(c(t)) \cdot c'(t) \\ &= \langle dH(c(t)), X_H(c(t)) \rangle \\ &= \omega(X_H(c(t)), X_H(c(t))) \\ &= 0\end{aligned}$$

since ω is skew-symmetric. Here $\langle \cdot, \cdot \rangle$ denotes the classical dual pairing between forms and vectors. $\square$

This conservation law suggests a more general principle, namely, any symmetry of the Hamiltonian system produces a conserved quantity. It will be formulated algebraically later that the Poisson bracket structure provides the universal mechanism for generating conserved quantities from symmetries. That is the content of Noether's Theorem 1.9.

Another important statement is that the phase flow actively preserves the underlying symplectic structure. Consequently, time evolution in classical mechanics unfolds as a continuous one-parameter group of canonical transformations, a structural feature presented by Liouville's Theorem.

Proposition 1.4.5 (Liouville's Theorem). *Let (M, ω, X_H) be a Hamiltonian system, and F_t be the flow of X_H . Then for each t , $F_t^* \omega = \omega$, that is, F_t is symplectic on its domain. Thus F_t also preserves the volume form Ω_ω .*

Proof. We have

$$\begin{aligned}\frac{d}{dt}F_t^* \omega &= \lim_{h \rightarrow 0} \frac{F_{t+h}^* \omega - F_t^* \omega}{h} \\ &= F_t^* \left(\lim_{h \rightarrow 0} \frac{F_h^* \omega - \omega}{h} \right) \\ &= F_t^* \mathcal{L}_{X_H} \omega \\ &= F_t^* (i_{X_H} d\omega + di_{X_H} \omega) \\ &= F_t^* (0 + ddH) \quad (\text{since } d\omega = 0 \text{ and } i_{X_H} \omega = dH) \\ &= 0\end{aligned}$$

Thus $F_t^* \omega$ is constant in t . Since $F_0 = \mathbb{1}$, it follows that $F_t^* \omega = \omega$. $\square$

Poisson Bracket

In this section, we see that the construction above has a much deeper significance enclosed in the geometrical formulation. In fact, analysing the structure that lies beyond classical observables, we can find a Lie product given by the Poisson brackets. To show this, we start from enriching $\Omega^1(M)$ with a specific Lie bracket.

Definition 1.4.6 (Poisson Bracket of forms). *Suppose (M, ω) is a symplectic manifold as in Definition 1.3.8 and $\alpha, \beta \in \Omega^1(M)$. The Poisson bracket is the one-form*

$$\{\alpha, \beta\} = -[\alpha^\sharp, \beta^\sharp]^\flat,$$

where $\flat$ and $\sharp$ are the musical isomorphisms as in Definition 1.3.5.

Note that we have the following commutative diagram:

$$\begin{array}{ccc} \mathfrak{X} \times \mathfrak{X} & \xrightarrow{[\cdot, \cdot]} & \mathfrak{X} \\ \downarrow \flat \times \flat & & \downarrow \flat \\ \Omega^1 \times \Omega^1 & \xrightarrow{\{\cdot, \cdot\}} & \Omega^1 \end{array}$$

As we showed in Example 1.2.3, the commutator $[\cdot, \cdot]$ makes $\mathfrak{X}(M)$ a Lie algebra. Since $\flat$ is linear, then $\Omega^1(M)$ as a real vector space with the composition $\{\cdot, \cdot\}$ is a Lie algebra.

Our aim is to equip the observable algebra $C^\infty(M)$ with a Lie bracket. To this end, we will define a binary operation on smooth functions whose image under the exterior derivative $d : C^\infty(M) \rightarrow \Omega^1(M)$ is compatible with the Poisson bracket on forms. Establishing this compatibility will induce a Lie algebra homomorphism from $\mathfrak{X}(M)$ to the observables.

Definition 1.4.7 (Poisson Bracket of functions). *Let (M, ω) be a symplectic manifold and $f, h \in C^\infty(M)$, with $X_f = (df)^\sharp \in \mathfrak{X}(M)$ as in Definition 1.3.5, analogously for h . The Poisson bracket between f and g is*

$$\{f, h\} = \omega(X_f, X_h) \in C^\infty(M).$$

The following properties confirm that $\{\cdot, \cdot\}$ behaves as expected, namely the Poisson bracket between f and h , measures the rate of change of f along the flow of X_h . It vanishes exactly when f is conserved along such flow, giving the bracket its direct dynamical meaning.

Proposition 1.4.8. *Let (M, ω) be a symplectic manifold and $f, h \in C^\infty(M)$. Then*

$$\{f, h\} = -\mathcal{L}_{X_f} h = \mathcal{L}_{X_h} f.$$

In addition, we have that

(i) For $f_0 \in C^\infty(M)$, the map $h \mapsto \{f_0, h\}$ is a derivation.

(ii) f is constant along the orbits of $X_h \iff h$ is constant on the orbits of $X_f \iff \{f, h\} = 0$.

Proof. It holds that $\mathcal{L}_{X_f} h = i_{X_f} dh = i_{X_f} i_{X_h} \omega$ since $dh = i_{X_h} \omega$. Since $i_X i_Y \omega = -i_Y i_X \omega = \omega(Y, X)$, the last equality follows. Now, we prove the two statements below.

(i) $h \mapsto \{f_0, h\} = -\mathcal{L}_{X_{f_0}} h$ is clearly a derivation.

(ii) If F_t is the flow of X_f ,

$$\frac{d}{dt}(h \circ F_t) = F_t^* \mathcal{L}_{X_f} h = -F_t^* \{f, h\},$$

which vanishes if and only if $\{f, h\} = 0$. Since $\{f, h\} = -\{h, f\}$, this is equivalent to $(d/dt)(f \circ H_t) = 0$, where H_t is the flow of X_h .

□

Observe that the identity $\{H, H\} = 0$ corresponds to conservation of energy by (ii) already proved by Proposition 1.4.4.

In this standard representation, the abstract symplectic structure reduces to its canonical form, yielding the familiar coordinate expression for the Poisson bracket.

Corollary 1.4.9. *In canonical coordinates $(q^1, \dots, q^n, p_1, \dots, p_n)$, we have*

$$\{f, g\} = \sum_{i=1}^n \left(\frac{\partial f}{\partial q^i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q^i} \right).$$

Remark 1.4.10. It follows from Corollary 1.4.9 that

$$\{q^i, q^j\} = 0 \quad \{p_i, p_j\} = 0 \quad \{q^i, p_j\} = \delta_j^i.$$

In addition to all these important properties, we can finally observe that the exterior derivative d acts as a compatible morphism, mapping the bracket of scalar observables directly to the bracket of their corresponding 1-forms on the cotangent space.

Proposition 1.4.11. *Let (M, ω) be a symplectic manifold and $f, g \in C^\infty(M)$. Then $d\{f, g\} = \{df, dg\}$.*

For a detailed proof, we refer the reader to [AM08, Proposition 3.3.16]. Furthermore, we can prove that the Poisson brackets are an operation suitable for making the classical observable space a Lie algebra.

Proposition 1.4.12. *The real vector space $C^\infty(M)$, together with the Poisson bracket $\{\cdot, \cdot\}$, forms a Lie algebra.*

Proof. Since d and $\omega^\sharp$ are $\mathbb{R}$ linear, the map $f \mapsto X_f$ is $\mathbb{R}$ linear. Hence $\{f, g\} = \omega(X_f, X_g)$, which entails $\{f, f\} = 0$. For Jacobi's identity, we have

$$\begin{aligned}\{f, \{g, h\}\} &= -\mathcal{L}_{X_f}\{g, h\} = \mathcal{L}_{X_f}(\mathcal{L}_{X_g}h) \\ \{g, \{h, f\}\} &= \mathcal{L}_{X_g}(\mathcal{L}_{X_h}f) = -\mathcal{L}_{X_g}(\mathcal{L}_{X_f}h) \\ \{h, \{f, g\}\} &= \mathcal{L}_{X_{\{f, g\}}}h\end{aligned}$$

However, $X_{\{f, g\}} = (d\{f, g\})^\sharp = \{df, dg\}^\sharp = -[(df)^\sharp, (dg)^\sharp]$ by Proposition 1.4.11. Hence $X_{\{f, g\}} = -[X_f, X_g]$ and the result follows. $\square$

The Jacobi identity just proved a structural link between the algebra of observables and the algebra of vector fields. Specifically, the bracket of the Hamiltonian vector fields corresponding to two functions should correspond to the Hamiltonian vector field of their Poisson bracket. The following theorem formalizes this correspondence, showing that the mapping from functions to vector fields is a Lie algebra homomorphism up to a sign.

Theorem 1.7 (Lie Algebra Homomorphism of Hamiltonian Vector Fields). *Let (M, ω) be a symplectic manifold. The map $\mathcal{H} : C^\infty(M) \rightarrow \mathfrak{X}(M)$ defined by $f \mapsto \mathcal{H}_f = -X_f$, where $\iota_{\mathcal{H}_f}\omega = -df$, relates the Poisson algebra of smooth functions to the Lie algebra of vector fields via the identity*

$$\mathcal{H}_{\{f, g\}} = [\mathcal{H}_f, \mathcal{H}_g] \quad \forall f, g \in C^\infty(M).$$

Consequently, the map $f \mapsto -X_f$ constitutes a strict Lie algebra homomorphism from $(C^\infty(M), \{\cdot, \cdot\})$ to $(\mathfrak{X}(M), [\cdot, \cdot])$.

This follows from the work of [AM08, Corollary 3.3.18]. It guarantees that the algebra of classical observables is intrinsically tied to the geometry of infinitesimal canonical transformations. Physically, it implies that the algebraic structure of symmetries, encoded in the Poisson bracket, is faithfully represented by the commutation of their dynamical flows on the phase space.

Corollary 1.4.13 (Lie Subalgebra of Globally Hamiltonian Vector Fields). *The space of globally Hamiltonian vector fields on a symplectic manifold (M, ω) , introduced in Definition 1.4.2 as $\mathfrak{X}_{\mathcal{H}}(M)$, forms a linear subspace of $\mathfrak{X}(M)$ that is closed under the Lie bracket operation.*

For the full proof, see [AM08, Corollary 3.3.18].

Remark 1.4.14. The structural rigidity established in Corollary 1.4.13 follows as a direct algebraic consequence of Theorem 1.7. By definition, the space $\mathfrak{X}_{\mathcal{H}}(M)$ is the image of the map $\mathcal{H} : C^\infty(M) \rightarrow \mathfrak{X}(M)$. Since the homomorphic image of any Lie algebra necessarily forms a Lie subalgebra of the codomain, the closure of $\mathfrak{X}_{\mathcal{H}}(M)$ under the Lie bracket is automatically ensured. Such a subspace is called a *Lie subalgebra*. Furthermore, the kernel of this homomorphism consists of functions $f \in C^\infty(M)$ satisfying $df = 0$, which correspond to locally constant functions on the manifold M .

Beyond its structural role in defining symmetries, the primary physical utility of the Poisson bracket lies in its dynamical capacity to generate time evolution on the phase space.

Proposition 1.4.15. *Let X_H be a Hamiltonian vector field on a symplectic manifold (M, ω) with Hamiltonian $H \in C^\infty(M)$ and flow F_t . Then for $f \in C^\infty(M)$ we have*

$$\frac{d}{dt}(f \circ F_t) = \{f \circ F_t, H\}.$$

Proof. It follows from this calculation:

$$\begin{aligned}\frac{d}{dt}(f \circ F_t) &= \frac{d}{dt}F_t^*f = F_t^*\mathcal{L}_{X_H}f \\ &= L_{X_H}(f \circ F_t) = \{f \circ F_t, H\}.\end{aligned}$$

$\square$

1.5 Symmetries and Geometrical Noether's Theorem

The relation between continuous symmetries and conserved quantities constitutes a key principle in theoretical physics. To analyse this correspondence formally, it is necessary to formalize the concept of physical symmetry through the Lie group actions on smooth manifolds.

In this section, we transition from the local dynamics generated by a single Hamiltonian observable to the global symmetry transformations induced by Lie groups. We shall establish how a symplectic group action on a phase space naturally generates a set of fundamental conserved observables, an algebraic and geometric synthesis culminating in the modern geometrical formulation of Noether's Theorem.

First we formalise how a Lie group can act on the points of the underlying phase space.

Definition 1.5.1 (Left Action). *Let M be a smooth manifold. A left action, which we will refer as action for simplicity, of a Lie group G on M is a smooth mapping $\Phi : G \times M \rightarrow M$ such that*

- $\Phi(e, x) = x \ \forall x \in M$,
- given $g, h \in G$ we have $\Phi(g, \Phi(h, x)) = \Phi(gh, x) \ \forall x \in M$.

The action $\Phi(g, \cdot)$ will be also denoted by $\Phi_g(\cdot)$.

The action of a group on the manifold identifies an orbit.

Definition 1.5.2 (Orbit). *Let M be a smooth manifold and Φ be an action of G on M . For $x \in M$ the orbit is*

$$G \cdot x = \{\Phi_g(x) \mid g \in G\}.$$

Not every action explores the manifold, or the group itself, in the same way. Some collapse the whole space into a single orbit, some fail to distinguish between different group elements acting identically, and some may fix certain points while moving others freely. The next definition isolates these three qualitatively different behaviours, each of which constrains what geometric structures can later be built out of the action.

Definition 1.5.3 (Transitive, effective and free action). *Let M be a smooth manifold.*

- *An action of a Lie group G on M is transitive if there is just one orbit.*
- *It is effective if $\Phi_g = \mathbb{1}$ implies $g = e$, that is $g \mapsto \Phi_g$ is one-to-one.*
- *An action is free if, for each $x \in M$, $g \mapsto \Phi_g(x)$ is one-to-one.*

While the orbit characterizes the global, geometric footprint of a group action on a manifold, its local, differential structure is governed by the continuous flow of the symmetry group. Hence, the tangent space to the orbit at any point $x \in M$ is spanned by induced velocity vector fields evaluated at x . This localized, differential perspective leads naturally to the formal definition of the infinitesimal generator.

Definition 1.5.4 (Infinitesimal generator). *Let M be a smooth manifold and let G be a Lie group. Suppose $\Phi : G \times M \rightarrow M$ is a smooth action. If $\xi \in T_e G \cong \mathfrak{g}$, then*

$$\Phi^\xi : \mathbb{R} \times M \rightarrow M, \quad (t, x) \mapsto \Phi(\exp t\xi, x)$$

is an $\mathbb{R}$ -action on M , that is, Φ^ξ is a flow on M . The corresponding vector field on M given by

$$\xi_M(x) = \left. \frac{d}{dt} \Phi(\exp t\xi, x) \right|_{t=0}$$

is called the infinitesimal generator of the action corresponding to ξ .

Remark 1.5.5. The role of the exponential map in this definition serves as the bridge between abstract symmetries and physical dynamics on the manifold. Intuitively, an element $\xi \in \mathfrak{g}$ represents an infinitesimal transformation, meaning a direction of symmetry at the identity of the group. The exponential map $\exp(t\xi)$ integrates this infinitesimal tangent vector into a one-parameter subgroup, within the Lie group G , parametrized by $t \in \mathbb{R}$. By feeding this trajectory into the action Φ , we translate the abstract symmetry into a concrete flow on the physical space M . Taking the temporal derivative at $t = 0$ extracts the physical velocity field. This is the infinitesimal generator, that pushes the points of the manifold along that specific direction of symmetry.

Example 1.5.6 (Infinitesimal generator of planar rotations). Consider the canonical action of the rotation group $G = SO(2)$ on the configuration manifold $M = \mathbb{R}^2$. The Lie algebra $\mathfrak{so}(2)$ is spanned by the basis element $\xi = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. The exponential map yields the standard rotation matrix $\exp(t\xi) = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}$. Given a point $\mathbf{x} = (x, y)^T \in \mathbb{R}^2$, the infinitesimal generator associated with ξ is computed directly from the flow:

$$\xi_M(x, y) = \left. \frac{d}{dt} \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \right|_{t=0} = \begin{pmatrix} -y \\ x \end{pmatrix}.$$

Dynamically, this yields the vector field $\xi_M = -y \frac{\partial}{\partial x} + x \frac{\partial}{\partial y}$, which is the standard generator of angular momentum and continuous rotational symmetries in the plane.

The concrete derivation of the rotational generator in the preceding example underlines a systematic correspondence. More generally, on a symplectic manifold (M, ω) , the mapping $\mathfrak{g} \rightarrow \mathfrak{X}(M)$ that assigns $\xi \mapsto \xi_M$ acts as a Lie algebra anti-homomorphism, satisfying the fundamental bracket relation:

$$[\xi_M, \eta_M] = -[\xi, \eta]_M \quad \forall \xi, \eta \in \mathfrak{g}. \quad (1.7)$$

This allows us to classify and study continuous symmetries through the local generators of the Lie algebra $\mathfrak{g}$ as shown in detail by [AM08, Construction on p.271]. Choosing a basis $\{e_1, \dots, e_k\}$ of $\mathfrak{g}$ defines the structure constants $[e_\alpha, e_\beta] = c_{\alpha\beta}^\gamma e_\gamma$. The abstract commutation relations are mapped directly to the physical velocity fields on the manifold:

$$[e_{\alpha,M}, e_{\beta,M}] = -c_{\alpha\beta}^\gamma e_{\gamma,M}. \quad (1.8)$$

From this, we can infer that when the group action preserves the symplectic form, these infinitesimal generators $e_{\alpha,M}$ are necessarily locally Hamiltonian vector fields. Assuming they are globally Hamiltonian, each basis generator is constructed by a smooth scalar observable $J_\alpha \in C^\infty(M)$ such that $e_{\alpha,M} = X_{J_\alpha}$. Imposing that these Hamiltonian functions faithfully preserve the commutation relations of the symmetry group constitutes the exact geometric requirement that motivates the definition of the momentum map.

Momentum Map

We can now focus more deeply on a concept briefly outlined in the previous construction, the momentum map. This mathematical object encodes the infinitesimal generators of a group action into a single mapping, taking values in the dual of the Lie algebra. In the following, we first give the formal definition of the momentum map. Subsequently we illustrate its meaning with examples, and eventually we prove the geometric version of Noether's theorem.

Definition 1.5.7 (Momentum map). Let (P, ω) be a symplectic manifold and $\Phi : G \times P \rightarrow P$ a symplectic action of the connected Lie group G on P , that is, for each $g \in G$, the map $\Phi_g : P \rightarrow P$ with $x \mapsto \Phi(g, x)$ is symplectic. We say that a map

$$J : P \rightarrow \mathfrak{g}^*,$$

where $\mathfrak{g}^*$ is the dual of the Lie algebra of G , is a momentum map provided that, for every $\xi \in \mathfrak{g}$,

$$d\hat{J}(\xi) = \iota_{\xi_P} \omega, \quad (1.9)$$

where $\hat{J}(\xi) : P \rightarrow \mathbb{R}$ is defined by $\hat{J}(\xi)(x) = \langle J(x), \xi \rangle$ while ξ_P is the infinitesimal generator of the action corresponding to ξ . The quadruple (P, ω, Φ, J) is called a Hamiltonian G -space.

Remark 1.5.8 (Equivalent definition of the momentum map). In other words, the request presented in Equation (1.9) can be stated as

$$X_{\hat{J}(\xi)} = \xi_P,$$

for all $\xi \in \mathfrak{g}$. This means that the fundamental vector field ξ_P associated with the Lie algebra element ξ coincides with the Hamiltonian vector field generated by the corresponding component $\hat{J}(\xi) = \langle J, \xi \rangle$ of the momentum map. Consequently, the components of the momentum map serve as the Hamiltonian functions that generate the infinitesimal action of the symmetry group on the symplectic manifold.

Before turning to concrete examples, it is worth covering briefly why the dual Lie algebra $\mathfrak{g}^*$ is the natural target space for the momentum map, rather than an arbitrary construction.

Remark 1.5.9 (Geometric Intuition of the Momentum Map). The abstract codomain of J , namely the dual of the Lie algebra $\mathfrak{g}^*$, can initially appear counterintuitive. In classical mechanics, an observable is inherently a real-valued function on the phase space. A continuous symmetry is generated by an element of the Lie algebra, $\xi \in \mathfrak{g}$, per Definition 1.5.4. To systematically assign a scalar conserved quantity to each possible symmetry direction ξ , we require a map that linearly takes a vector in $\mathfrak{g}$ and yields a real number at each point $x \in P$. The mathematical space is exactly the dual space $\mathfrak{g}^*$. Therefore, evaluating $J(x) \in \mathfrak{g}^*$ on a generator $\xi \in \mathfrak{g}$ through the natural pairing $\hat{J}(\xi)(x) = \langle J(x), \xi \rangle$ yields the observable $\hat{J}(\xi)(x)$. Geometrically, this translates to taking the scalar product between an element representing the generalized momentum and a tangent vector representing the infinitesimal symmetry transformation.

With this interpretation in hand, we make the abstract construction concrete in the simplest possible setting, a free particle.

Example 1.5.10 (Translational Symmetry and Linear Momentum). Consider the phase space of a free particle, $P = T^*\mathbb{R}^3 \cong \mathbb{R}^3 \times \mathbb{R}^3$, with canonical coordinates (q, p) and symplectic form $\omega = \sum dq^i \wedge dp_i$. Let $G = \mathbb{R}^3$ act on P via spatial translations: $\Phi_a(q, p) = (q + a, p)$. The Lie algebra $\mathfrak{g}$ is isomorphic to $\mathbb{R}^3$, and its elements ξ represent constant translation vectors.

The infinitesimal generator on P is the constant vector field $\xi_P = \sum \xi^i \frac{\partial}{\partial q^i}$. By solving the defining relation $d\hat{J}(\xi) = \iota_{\xi_P} \omega$, one finds $\hat{J}(\xi)(q, p) = \langle p, \xi \rangle$. Consequently, the momentum map $J : P \rightarrow (\mathbb{R}^3)^*$ is $J(q, p) = p$, which is the standard classical linear momentum.

The translational example above relied on the global exactness of the symplectic form, a convenient generic circumstance. It is natural to ask what happens when this hypothesis fails, or more generally when the infinitesimal generators of the action are only locally, rather than globally, Hamiltonian.

Remark 1.5.11 (Limitations for momentum map). If the infinitesimal generators ξ_P associated with the action are only locally Hamiltonian, we cannot define a global momentum map. According to Definition 1.5.7, the momentum map requires the existence of a family of global smooth functions $\hat{J}(\xi)$ on P whose differential satisfies $d\hat{J}(\xi) = \iota_{\xi_P} \omega$. If the 1-form $\iota_{\xi_P} \omega$ is closed but not exact, such global functions do not exist on the manifold. This prevents the formulation of the momentum map.

The following theorem, proved in [AM08, Theorem 4.2.10], establishes a systematic procedure to compute the momentum map directly, relying upon the critical hypothesis that the symplectic manifold admits an exact symplectic form whose globally defined primitive is strictly invariant under the designated group action.

Theorem 1.8. *Let Φ be a symplectic action of a connected Lie group G on a symplectic manifold (P, ω) . Assume ω on P is exact, $\omega = -d\theta$ and the action leaves θ invariant. That is, $\Phi_g^* \theta = \theta$ for all $g \in G$. Then $J : P \rightarrow \mathfrak{g}^*$ defined by*

$$\langle J(x), \xi \rangle = (\iota_{\xi_P} \theta)(x),$$

is a momentum map for that action.

Example 1.5.12 (Momentum map for translational symmetry via exact forms). The preceding theorem provides an algorithmic method to compute the momentum map in the case of cotangent spaces $P = T^*Q$, where the canonical symplectic form is structurally exact. We consider again the phase space of a free particle $P = T^*\mathbb{R}^3 \cong \mathbb{R}^3 \times \mathbb{R}^3$, equipped with canonical coordinates (q, p) . By defining the canonical 1-form as $\theta = \sum_{i=1}^3 p_i dq^i$, its exterior derivative is given by:

$$d\theta = \sum_{i=1}^3 dp_i \wedge dq^i = - \sum_{i=1}^3 dq^i \wedge dp_i = -\omega. \quad (1.10)$$

Thus, we recover the required condition $\omega = -d\theta$.

The action of the translation group $G = \mathbb{R}^3$ operates on the phase space via $\Phi_a(q, p) = (q + a, p)$. We first verify that this action preserves the 1-form θ :

$$\Phi_a^* \theta = \sum_{i=1}^3 (p_i \circ \Phi_a) d(q^i \circ \Phi_a) = \sum_{i=1}^3 p_i d(q^i + a^i) = \sum_{i=1}^3 p_i dq^i = \theta. \quad (1.11)$$

Since θ is strictly invariant under the global action, the hypotheses of the theorem are fully satisfied. The infinitesimal generator associated with the Lie algebra element $\xi \in \mathbb{R}^3$ is the constant vector field $\xi_P = \sum_{i=1}^3 \xi^i \frac{\partial}{\partial q^i}$. Evaluating the contraction of θ along ξ_P directly yields the momentum observable:

$$\langle J(q, p), \xi \rangle = (\iota_{\xi_P} \theta)(q, p) = \left(\sum_{i=1}^3 p_i dq^i \right) \left(\sum_{j=1}^3 \xi^j \frac{\partial}{\partial q^j} \right) = \sum_{i=1}^3 p_i \xi^i = \langle p, \xi \rangle. \quad (1.12)$$

This algebraic procedure cleanly and instantaneously yields the linear momentum map $J(q, p) = p$.

Noether's Theorem

The momentum map is the structure that assigns to each infinitesimal symmetry generator a scalar observable whose Hamiltonian vector field coincides with that generator. In other words, the momentum map lifts the abstract Lie algebra of symmetries into the concrete Poisson algebra of observables on phase space. This lift is precisely what allows us to translate invariance of the Hamiltonian under the group action into conservation laws. The modern geometric formulation of Noether's Theorem, presented below, establishes this correspondence rigorously and shows that the momentum map is the central unifying object linking symmetries, conserved quantities, and the algebraic structure of classical mechanics.

Theorem 1.9 (Geometric Noether Theorem). *Let $\Phi : G \times P \rightarrow P$ be a symplectic action of a connected Lie group G on a symplectic manifold (P, ω) admitting a momentum map $J : P \rightarrow \mathfrak{g}^*$ according to Definition 1.5.7. A Hamiltonian function $H : P \rightarrow \mathbb{R}$ is invariant under the action of G , that is*

$$H(\Phi_g(x)) = H(x) \quad \forall x \in P, \forall g \in G,$$

if and only if the momentum map J is an integral of the motion for the Hamiltonian vector field X_H such that

$$J(F_t(x)) = J(x) \quad \forall x \in P, \forall t \in \mathbb{R},$$

where F_t denotes the flow of X_H .

Proof. Both invariance statements are equivalent to the pointwise vanishing of the bracket $\{H, \hat{J}(\xi)\}$ for every $\xi \in \mathfrak{g}$, where $\hat{J}(\xi) := \langle J(\cdot), \xi \rangle$. We show the chain of equivalences.

Since G is connected, it is generated by $\exp(\mathfrak{g})$: Every $g \in G$ is a finite product $\exp(\xi_1) \cdots \exp(\xi_k)$. Hence

$$H(\Phi_g(x)) = H(x) \quad \forall g \in G \iff H(\Phi_{\exp(t\xi)}(x)) = H(x) \quad \forall \xi \in \mathfrak{g}, \forall t \in \mathbb{R},$$

the backward implication holding because invariance under each individual flow $(\xi_i)_P$ composes to invariance under the product g . Differentiating the expression on the RHS in t at $t = 0$ yields

$$\langle dH(x), \xi_P(x) \rangle = 0 \implies \mathcal{L}_{\xi_P} H = 0 \quad \forall \xi \in \mathfrak{g}.$$

By the equivalent definition of the momentum map in Remark 1.5.8, $\xi_P = X_{\hat{J}(\xi)}$, hence

$$\mathcal{L}_{\xi_P} H = \mathcal{L}_{X_{\hat{J}(\xi)}} H = \{H, \hat{J}(\xi)\}.$$

By skew-symmetry of the Poisson bracket it descends that

$$\{H, \hat{J}(\xi)\} = 0 \iff \{\hat{J}(\xi), H\} = \mathcal{L}_{X_H} \hat{J}(\xi) = 0.$$

At last, $\mathcal{L}_{X_H} \hat{J}(\xi) = 0$ for all t is equivalent to $\hat{J}(\xi)$ being constant along the flow F_t of X_H ,

$$\hat{J}(\xi)(F_t(x)) = \hat{J}(\xi)(x) \quad \forall t \in \mathbb{R}, \forall \xi \in \mathfrak{g},$$

and, since the pairing between $\mathfrak{g}^*$ and $\mathfrak{g}$ is non-degenerate, this holds for all $\xi \in \mathfrak{g}$ if and only if

$$J(F_t(x)) = J(x) \quad \forall t \in \mathbb{R}.$$

Chaining all the equivalences together proves the theorem. $\square$

The proof of the geometric Noether theorem presented above reveals that the entire argument operates at the infinitesimal level. It uses only the Lie algebra $\mathfrak{g}$ and its action on phase space via the infinitesimal generators. Consequently, the conservation of the momentum map follows purely from the algebraic compatibility between the Poisson bracket and the Lie bracket of vector fields, as established in Theorem 1.7. This underscores that the theorem is fundamentally a statement about Lie algebras and their representation on the Poisson algebra of observables.

Remark 1.5.13 (Conservation along the Flow). The deeper geometrical meaning of this theorem can be seen from its proof. The derivation demonstrates that the directional derivative of H along the symmetry generator ξ_p vanishes, $\mathcal{L}_{\xi_p} H = 0$. In turn, it holds that $\mathcal{L}_{X_H} \hat{J}(\xi) = 0$. Thus, the observable $\hat{J}(\xi)$ remains strictly constant when transported along the dynamical orbits generated by X_H .

1.6 Classical Mechanics as a Commutative C^* -algebra

The standard description of a classical system establishes geometry as the primary source of the whole framework. The Gelfand theory of commutative C^* -algebras introduces a change of perspective for the foundation of Hamiltonian mechanics. By exploiting the Gelfand-Naimark representation theorem one can reverse this relation and construct the theory starting from an abstract algebra of observables. However, this framework reconstructs only a Hausdorff topological space through the Gelfand spectrum of the algebra. In fact, the full picture will be recovered adding missing structures to enrich the topological space with properties such as differentiability. This approach dictates that the algebra of physical observables can provide a self-contained axiomatic basis for the formulation of classical mechanics.

Basic topics in C^* -algebras theory

To reconstruct a classical theory algebraically, we first need to introduce C^* -algebras. This abstract mathematical structure recurs in both classical and quantum descriptions of observables. Specifically, we adopt this framework because involution naturally models the complex conjugation or the adjoint of physical quantities, while the underlying metric completeness ensures that the approximation of observables is rigorously well-defined. Here, we will present only the basic definitions to introduce the reader to the theory of commutative C^* -algebras following the approach of Bratteli and Robinson [BR02, Chapter 2]. We begin by outlining the notion of an algebra, the basic structure on which the involution defining a C^* -algebra will be built.

Definition 1.6.1 (Algebra). A complex algebra $\mathfrak{A}$ is a $\mathbb{C}$ -vector space equipped with an associative and distributive multiplication law which is compatible with the other vector space operations.

Definition 1.6.2 ($*$ -algebra). Let $\mathfrak{A}$ be an algebra as in Definition 1.6.1. An involution on $\mathfrak{A}$ is a map associating an element $A \mapsto A^*$ satisfying three distinct conditions:

- $A^{**} = A$,
- $(AB)^* = B^*A^*$,
- $(\alpha A + \beta B)^* = \bar{\alpha}A^* + \bar{\beta}B^* \quad \forall \alpha, \beta \in \mathbb{C}$.

An algebra equipped with an involution map is called a $*$ -algebra.

Example 1.6.3. One of the classical example of a $*$ -algebra is $C(X)$. That is the algebra of continuous functions on a compact Hausdorff space X with $f^*(x) := \overline{f(x)}$, and is commutative. A non commutative example is the matrix algebra $M_n(\mathbb{C})$ with $A^* := \bar{A}^T$ and $n \geq 2$.

Within a $*$ -algebra it is natural to isolate subsets that remain stable under the involution, since they will later correspond to physically meaningful collections of observables, as we will present in Definition 1.6.32. Therefore, we introduce the following notion of a selfadjoint subset.

Definition 1.6.4 (Selfadjoint subset). *Let $\mathfrak{A}$ be a $*$ -algebra in the sense of Definition 1.6.2. A subset $\mathfrak{B} \subset \mathfrak{A}$ is called selfadjoint if $A \in \mathfrak{B}$ implies that $A^* \in \mathfrak{B}$.*

Some distinguished elements of this subset are the adjoint of themselves. As we will see later in Definition 1.6.17, selfadjoint elements have a particular meaning in algebraic classical mechanics.

Definition 1.6.5 (Selfadjoint element). *Let $\mathfrak{A}$ be a $*$ -algebra in the sense of Definition 1.6.2. An element $A \in \mathfrak{A}$ is selfadjoint if $A = A^*$. The set of all selfadjoint elements will be referred to selfadjoint sector.*

Example 1.6.6. Referring to the previous Example 1.6.3, in the algebra $C(X)$ selfadjoint means real-valued. Instead, in $M_n(\mathbb{C})$, it means Hermitian. With $n = 2$, these matrices span a four-dimensional space generated by $\mathbb{1}$ and the three Pauli matrices.

We endow the algebra with an additional structure that measures the length of its elements, called a *norm*. This operation will let us control convergence and continuity within the algebra.

Definition 1.6.7 (Normed algebra). *An algebra $\mathfrak{A}$, as in Definition 1.6.1, endowed with a map $A \in \mathfrak{A} \mapsto \|A\| \in \mathbb{R}$ satisfying the requirements*

- $\|A\| \geq 0$ and $\|A\| = 0$, if and only if, $A = 0$,
- $\|\alpha A\| = |\alpha| \|A\|$,
- $\|A + B\| \leq \|A\| + \|B\|$, triangle inequality,
- $\|AB\| \leq \|A\| \|B\|$, product inequality,

is called a normed algebra, while the map is called a norm.

Normed algebras are relevant objects inside our construction, since the norm endows each of them with a notion of a distinguished topology called *uniform*.

Definition 1.6.8 (Uniform topology). *Let $\mathfrak{A}$ be a normed algebra as in Definition 1.6.7. The norm defines a metric topology on $\mathfrak{A}$ which is referred to as the uniform topology. The open neighborhoods of an element $A \in \mathfrak{A}$ are*

$$\mathcal{U}_{A,\varepsilon} = \{B \in \mathfrak{A} \mid \|B - A\| < \varepsilon\},$$

where $\varepsilon \in \mathbb{R}^+$.

Analogously, endowing a vector space with a norm makes it a *normed vector space*, and the uniform topology just introduced allows us to speak of convergent sequences within it. However, in a generic normed space, such sequences are not guaranteed to converge to an element within the space. Imposing completeness leads to the concept of a *Banach space*.

Definition 1.6.9 (Banach Space on $\mathbb{C}$). *A Banach space is a normed $\mathbb{C}$ -vector space $(X, \|\cdot\|)$ which is complete to the metric induced by its norm.*

A Banach space can be provided with a multiplication law and an involution creating a Banach $*$ -algebra.

Definition 1.6.10 (Banach $*$ -algebra). *A Banach $*$ -algebra is a normed algebra $\mathfrak{A}$ over a $\mathbb{C}$ -vector space, as in Definition 1.6.7, equipped with an involution, as in Definition 1.6.2, which is complete with respect to the metric induced by its norm, and such that the norm satisfies the isometric property*

$$\|A\| = \|A^*\|, \quad \forall A \in \mathfrak{A}.$$

From a physical perspective, completeness guarantees that if an idealized physical quantity can be approximated with arbitrary precision in norm by a sequence of valid measurements in $\mathfrak{A}$, the limiting observable is itself a well defined element of the algebra. In a Banach $*$ -algebra, the norm and involution are constrained by $\|A\| = \|A^*\|$. Imposing the stronger identity $\|A^*A\| = \|A\|^2$ makes the norm compatible with the algebraic structure and defines a C^* -algebra.

Definition 1.6.11 (C^* -algebra). A C^* -algebra is a Banach $*$ -algebra $\mathfrak{A}$, as in Definition 1.6.10, with the property $\|A^*A\| = \|A\|^2 \ \forall A \in \mathfrak{A}$.

Example 1.6.12 (The C^* -identity on $M_n(\mathbb{C})$). Equip $M_n(\mathbb{C})$ with the operator norm induced by the Euclidean norm on $\mathbb{C}^n$, namely

$$\|A\| := \sup_{\|v\|=1} \|Av\|, \quad v \in \mathbb{C}^n.$$

For every $A \in M_n(\mathbb{C})$ the matrix A^*A is positive semidefinite and Hermitian in the sense of Definition 1.6.5, so its operator norm equals its largest eigenvalue, and this eigenvalue coincides with the square of the largest singular value of A . This yields $\|A^*A\| = \|A\|^2$.

The abstraction of bounded operators leads, both in classical and in quantum physics, to the study of Banach algebras and their involutions. This approach allows the characterization of states and observables in both theories. A key class of C^* -algebras, particularly relevant for this reconstruction, consists of those possessing an identity element.

Definition 1.6.13 (Unital C^* -algebra). A unital C^* -algebra is a C^* -algebra $\mathfrak{A}$, as in Definition 1.6.11, containing an identity element $\mathbb{1}$ such that $A = \mathbb{1}A = A\mathbb{1}$ for all $A \in \mathfrak{A}$.

The identity plays the role of a distinguished element acting trivially on every other one. Before relying on this notion, it is worth verifying that, when it exists, it is uniquely determined.

Proposition 1.6.14. A C^* -algebra $\mathfrak{A}$ can possess at most one identity element.

Proof. Suppose a second identity element $\mathbb{1}'$ exists. It must satisfy $\mathbb{1}' = \mathbb{1}\mathbb{1}' = \mathbb{1}$. □

Algebraic description of Hamiltonian Mechanics

We apply the abstract framework developed above to the specific case of Hamiltonian mechanics. In the classical setting, physical intuition suggests starting from the phase space and only afterward considering functions defined on it. Our goal will be to establish that the reverse order is equally legitimate. We now carry out this reconstruction explicitly, showing how the building blocks of classical mechanics, observables, states, and dynamics, admit a counterpart in the framework of commutative C^* -algebras.

Observables. We begin by algebraically characterizing the observables used in Hamiltonian mechanics. Let Γ be a generic compact phase space, in the sense of Definition 1.3.8. For simplicity, we assume that there exist global coordinates (q, p) on Γ . Let $\mathcal{P}$ be the algebra of *polynomial observables* generated by the phase space coordinates q and p . Operationally, these correspond to the measurement of position or momentum of a physical system. Given that the coordinates take different values for different points on Γ , they separate points. $\mathcal{P}$ also includes constant functions, consequently it fulfills the hypotheses of Stone-Weierstrass Theorem [RS03, Theorem IV.9]. This establishes that $\mathcal{P}$ is dense in $C^0(\Gamma; \mathbb{R})$ under the uniform norm topology:

$$\|f\|_\infty = \sup_{p \in \Gamma} |f(p)| \quad \forall f \in \mathcal{P}. \quad (1.13)$$

Taking the closure with respect to it yields $\overline{\mathcal{P}}^{\|\cdot\|_\infty} = C^0(\Gamma; \mathbb{R})$.

In a general *classical scenario*, the phase space Γ is abstracted to a compact Hausdorff space X , replacing $C^0(\Gamma; \mathbb{R})$ with the classical continuous algebra in Definition 1.6.15. Physical observables are then directly recovered as the selfadjoint elements invariant under complex conjugation via Definition 1.6.17.

Definition 1.6.15 (Classical Continuous Algebra). Let X be a compact Hausdorff space. The classical continuous algebra is the commutative unital C^* -algebra $\mathfrak{A} = C^0(X; \mathbb{C})$, as in Definition 1.6.13.

In the case above, $X = \Gamma$ and $\mathfrak{A} = C^0(\Gamma)$.

Remark 1.6.16. The space $C^0(X; \mathbb{C})$ is a commutative unital C^* -algebra equipped with a pointwise multiplication $(fh)(x) = f(x)h(x)$, the identity element $\mathbb{1}(x) = 1$, the involution $f^*(x) = \overline{f(x)}$ and the supremum norm in Equation (1.13). The C^* -algebra condition as in Definition 1.6.11 follows from $\|f^*f\|_\infty = \|f\|_\infty^2$.

This construction establishes that the algebra of classical observables can be modeled by a commutative unital C^* -algebra. Later in this section, this will constitute a key component for the algebraic reconstruction of classical mechanics, as seen in Definition 1.6.35. Since complex-valued elements lack a direct physical interpretation, physical *observables* are strictly identified with the selfadjoint elements of the algebra.

Definition 1.6.17 (Classical Observable Algebra). *Let $\mathfrak{A}$ be a commutative unital C^* -algebra, as in Definition 1.6.13. The classical observable algebra $\mathfrak{A}_{sa}$ is the selfadjoint sector, as seen in Definition 1.6.5.*

A more detailed discussion about the physical quantities really measurable will be done in Subsection 1.6.

States. Having outlined the algebraic structure of observables, we now turn to states. From an operational point of view, if we prepare the system in a particular state ω and we measure an observable n times, then we can define its mean value as

$$\langle f \rangle_n^{(\omega)} := \frac{1}{n} \sum_{i=1}^n m_i^{(\omega)}(f) \quad \forall f \in \mathfrak{A}_{sa}.$$

Here $m_i^{(\omega)}(f)$ represents the i -th measurement and $\mathfrak{A}_{sa}$ is the classical observable algebra in Definition 1.6.17. The limit of this expression, assuming that it converges, can be defined dually as

$$\omega(f) := \lim_{n \rightarrow \infty} \langle f \rangle_n^{(\omega)} \quad \forall f \in \mathfrak{A}_{sa}.$$

In this picture, a state ω , acting on observables, yields the average of the measurements. This definition can however be extended to the whole unital commutative C^* -algebra $\mathfrak{A}$. The state ω becomes a linear functional that must preserve positivity $\omega(f^*f) \geq 0$ for all $f \in \mathfrak{A}$. In addition, it can be shown that positivity implies the Cauchy-Schwarz inequality, as proved in [BR02, Lemma 2.3.10], namely

$$|\omega(A^*B)| \leq (\omega(A^*A) \omega(B^*B))^{1/2} \quad \forall A, B \in \mathfrak{A}. \quad (1.14)$$

Therefore, if $\mathfrak{A}$ is unital, we can set $B = \mathbb{1}$ in Equation (1.14) to obtain $|\omega(A^*)|^2 \leq \omega(A^*A) \omega(\mathbb{1})$ for all $A \in \mathfrak{A}$. If $\omega(\mathbb{1})$ were zero, this inequality would imply $\omega(A^*) = 0$ for every $A \in \mathfrak{A}$, thus forcing the functional ω to be identically zero. Consequently, for any non-trivial positive linear functional, it must hold that $\omega(\mathbb{1}) > 0$. This ensures that one can always normalize such a functional to define a proper state via the map

$$\omega \mapsto \omega_{norm} = \omega(\mathbb{1})^{-1} \omega.$$

We can formalize this concept in the following definition.

Definition 1.6.18 (State on a C^* -Algebra). *Let $\mathfrak{A}$ be a unital C^* -algebra as in Definition 1.6.13. A state on $\mathfrak{A}$ is a linear functional $\omega : \mathfrak{A} \rightarrow \mathbb{C}$ such that*

1. $\omega(A^*A) \geq 0$ for all $A \in \mathfrak{A}$ *positivity,*
2. $\omega(\mathbb{1}) = 1$ *normalization.*

The set of all states is denoted by $\mathcal{S}(\mathfrak{A})$.

States on a C^* -algebra are not isolated objects, since a convex combination of two physically admissible preparations of a system is itself a physically admissible preparation. Thus, we can analyse the properties of this space.

Proposition 1.6.19 (Convexity of the State Space). *Let $\mathfrak{A}$ be a unital C^* -algebra as in Definition 1.6.13. The set $\mathcal{S}(\mathfrak{A})$ of all states on $\mathfrak{A}$ is convex.*

Proof. Let $\omega_1, \omega_2 \in \mathcal{S}(\mathfrak{A})$ and let $t \in [0, 1]$. Define $\omega = t\omega_1 + (1-t)\omega_2$ which is linear on $\mathfrak{A}$. Positivity follows because for any $A \in \mathfrak{A}$, $\omega(A^*A) = t\omega_1(A^*A) + (1-t)\omega_2(A^*A) \geq 0$. Normalization holds since $\omega(\mathbb{1}) = t\omega_1(\mathbb{1}) + (1-t)\omega_2(\mathbb{1}) = t + (1-t) = 1$. Thus $\omega \in \mathcal{S}(\mathfrak{A})$ and $\mathcal{S}(\mathfrak{A})$ is convex. $\square$

Probabilistic interpretation of states. One can ask which is the physical interpretation of such an abstract object. The solution comes from considering the classical scenario given by X , a compact Hausdorff space and $C^0(X)$ the associated classical continuous algebra as in Definition 1.6.15. It is proved in [BR02, Proposition 2.3.11] that a state $\omega \in \mathcal{S}(C^0(X))$ is continuous. Therefore, by the Riesz-Markov Representation Theorem [RS03, Theorem IV.14], it identifies a unique Borel measure μ_ω on X such that

$$\omega(f) = \int_X f d\mu_\omega, \quad \mu_\omega(X) = \omega(1) = 1. \quad (1.15)$$

Thus, we can interpret the state as a probability distribution on the state space and observables as random variables. This probabilistic interpretation represents the core of statistical mechanics.

In this description, Hamiltonian mechanics is recovered as a special case. In fact, we can take μ_{P_0} as the probability measure concentrated on the point $P_0 = \{q_0, p_0\}$ of a compact phase space Γ , in the sense of Definition 1.3.8. Then μ_{P_0} is a Dirac measure and the corresponding state ω_{P_0} is

$$\omega_{P_0}(f) = \int_\Gamma f d\mu_{P_0} = f(P_0). \quad (1.16)$$

These are also called *pure states* since they cannot be written as convex non-trivial linear combinations of other states. Measurements on pure states always yield the same result $f_\omega = \omega(f)$. Hence there is no dispersion and the variance of that state vanishes. We can formalize these concepts generally in the following way.

Definition 1.6.20 (Pure State). *Let $\mathfrak{A}$ be a commutative unital C^* -algebra. A state $\omega \in \mathcal{S}(\mathfrak{A})$, as in Definition 1.6.18, is called pure if it is an extreme point of the convex set $\mathcal{S}(\mathfrak{A})$. That is, whenever $\omega = t\omega_1 + (1-t)\omega_2$ with $0 < t < 1$ and $\omega_1, \omega_2 \in \mathcal{S}(\mathfrak{A})$, then $\omega_1 = \omega_2 = \omega$.*

We extend the statistical definition of variance to our algebraic framework. This notion will provide the precise sense in which pure states correspond to dispersion-free measurements.

Definition 1.6.21 (Variance of a State). *Let $\mathfrak{A}$ be a unital C^* -algebra, as in Definition 1.6.13, and let $\omega \in \mathcal{S}(\mathfrak{A})$ be a state, defined in Definition 1.6.18. The variance of an element $A \in \mathfrak{A}$ in the state ω is*

$$\text{Var}_\omega(A) := \omega(A^*A) - \omega(A^*)\omega(A).$$

The quantity is non-negative by the Cauchy-Schwarz inequality [BR02, Lemma 2.3.10]. In particular, since their measurements are dispersion-free, pure states show a strong form of this property.

Proposition 1.6.22 (Zero Variance for Pure States). *Let $\mathfrak{A}$ be a commutative unital C^* -algebra as in Definition 1.6.13 and let $\omega \in \mathcal{S}(\mathfrak{A})$ be a pure state, shown in Definition 1.6.20. Then*

$$\text{Var}_\omega(A) = 0 \quad \forall A \in \mathfrak{A}.$$

Proof. For commutative C^* -algebras, every pure state is multiplicative, that is, $\omega(AB) = \omega(A)\omega(B)$ for all $A, B \in \mathfrak{A}$, see [BR02, Proposition 2.3.27]. In particular, $\omega(A^*A) = \omega(A^*)\omega(A)$. Substituting into Definition 1.6.21 yields $\text{Var}_\omega(A) = 0$ for every $A \in \mathfrak{A}$. $\square$

Duality between states and observables. To conclude the excursus on observables and states, we establish an intrinsic duality between these two notions in classical scenarios. On the one hand, given $\mathfrak{A}$ the classical continuous C^* -algebra, defined in Definition 1.6.15, every state $\omega \in \mathcal{S}(\mathfrak{A})$ induces an evaluation. When that map is restricted to its classical observable algebra $\mathfrak{A}_{sa}$ introduced in Definition 1.6.17 it yields strictly real values. This property is mathematically guaranteed by the positivity of the state, ensuring that $A = A^* \in \mathfrak{A}_{sa}$ implies $\omega(A) \in \mathbb{R}$. Consequently, because every element $A \in \mathfrak{A}$ admits a unique decomposition into selfadjoint components as $A = A_1 + iA_2$, with $A_1, A_2 \in \mathfrak{A}_{sa}$, the action of a state on the entire complex C^* -algebra is completely and uniquely determined by its values on the physical classical observables.

Conversely, the uniqueness of a state given its observable values relies on the topological properties of the space X associated with the classical continuous algebra $\mathfrak{A}$. Since X is compact and Hausdorff, it satisfies the hypothesis of Urysohn's lemma [RS03, Theorem IV.7]. It states that for any two distinct points p and q in X , there exists a strictly real-valued function $f \in C^0(X; \mathbb{R}) \cong \mathfrak{A}_{sa}$ such that $f(p) = 0$ and $f(q) = 1$. In terms of pure states, this yields $\omega_p(f) = 0$ and $\omega_q(f) = 1$. Thus, the classical observable algebra $\mathfrak{A}_{sa}$ separates points on X .

Combining these findings yields a duality relation. A physical state is completely determined by the collection of its values on the selfadjoint observables, and the algebra of such observables is structurally sufficient to reconstruct the points of the underlying state space.

Time evolution. Having established observables and states, we algebraically investigate dynamics in Hamiltonian mechanics. Let Γ be a compact phase space, as in Definition 1.3.8. In this framework, time evolution of the system $\{q, p\} \rightarrow \{q_t, p_t\}$ is continuous with respect to $t \in \mathbb{R}$, with a continuous dependence on the initial data at time t_0 . Therefore it defines a one-parameter family of continuous invertible maps $\alpha_{t_0, t}$ of $C^0(\Gamma; \mathbb{R})$ into itself. These preserve all the algebraic relations, including the $*$ -operation. Such maps are called $*$ -automorphisms. Thus, the dynamics is encoded by a one-parameter group of automorphisms of the observable algebra.

Definition 1.6.23 (Time Evolution). *Let $\mathfrak{A}$ be a unital C^* -algebra as in Definition 1.6.13. A time evolution is a strongly continuous one-parameter group of $*$ -automorphisms $\alpha_t : \mathfrak{A} \rightarrow \mathfrak{A}$, $t \in \mathbb{R}$, such that*

1. $\alpha_0 = \text{id}$,
2. $\alpha_{s+t} = \alpha_s \circ \alpha_t$ for all $s, t \in \mathbb{R}$,
3. each α_t is linear, multiplicative, preserves the involution, and is norm-continuous,
4. for every $A \in \mathfrak{A}$, the map $t \mapsto \alpha_t(A)$ is continuous in the norm topology.

By duality, the time evolution of states is given by $(\alpha_t^* \omega)(A) = \omega(\alpha_t(A))$ for all $A \in \mathfrak{A}$. In addition, since α_t are $*$ -automorphisms, they preserve every selfadjoint sector $\mathfrak{A}_{sa}$, ensuring that real physical observables evolve into real physical observables.

Conclusion. *The physics of a classical system can be rewritten algebraically as the pair $(\mathfrak{A}, \alpha_t)$ where $\mathfrak{A}$ is a commutative unital C^* -algebra as in Definition 1.6.13 and α_t is a time evolution on it, as in Definition 1.6.23. To fix the initial conditions, it is given a state $\omega \in S(\mathfrak{A})$, as in Definition 1.6.18.*

Characters of a C^* -algebra and Gelfand-Naimark Theorem

In the previous discussion, we have concluded that Hamiltonian mechanics can be rewritten in algebraic terms. In this section, we discuss in detail the methods required to derive a state space from a C^* -algebra. We demonstrate how the definition of a commutative C^* -algebra can be exploited to obtain a topological space through the Gelfand-Naimark theorem, which will act as the state space of the system. We achieve this using a class of multiplicative linear functionals called characters. These characters coincide with the pure states and their space will coincide with the actual state space.

Definition 1.6.24 (Character of a commutative C^* -Algebra). *Let $\mathfrak{A}$ be a commutative C^* -algebra, as in Definition 1.6.11. A character ω of $\mathfrak{A}$ is a non vanishing linear functional $\omega : \mathfrak{A} \rightarrow \mathbb{C}$ that preserves algebraic multiplication, meaning*

$$\omega(AB) = \omega(A)\omega(B),$$

for all $A, B \in \mathfrak{A}$. The Gelfand spectrum $\sigma(\mathfrak{A})$ is the set of all characters on $\mathfrak{A}$.

Characters map observables to complex numbers and naturally belong to the dual space of the algebra. To analyse their geometric properties, we must equip this dual space with a metric structure.

Definition 1.6.25 (Dual Operator Norm). *Let $\mathfrak{A}$ be a C^* -algebra as in Definition 1.6.11. The dual space $\mathfrak{A}^*$ is a Banach space as in Definition 1.6.9 equipped with the dual operator norm:*

$$\|\alpha\| = \sup_{A \in \mathfrak{A}} \{ |\alpha(A)| \mid \|A\| \leq 1 \}, \quad \alpha \in \mathfrak{A}^*.$$

This definition is needed to understand the boundedness and continuity of characters that make the Gelfand's spectrum a unit ball.

Lemma 1.6.26 (Boundedness of characters). *Let $\mathfrak{A}$ be a commutative C^* -algebra, as in Definition 1.6.11, and its Gelfand spectrum $\sigma(\mathfrak{A})$ per Definition 1.6.24. We have $\sigma(\mathfrak{A}) \subseteq \mathfrak{A}^*$ and, for any $\omega \in \sigma(\mathfrak{A})$, $\|\omega\| = 1$. Furthermore, if $\mathfrak{A}$ is unital then $\omega(1_{\mathfrak{A}}) = 1$.*

For a complete proof of this lemma, we refer to [Lan17, Lemma C.9]. So far, characters have been defined only abstractly. We established that they are pure states, see in Definition 1.6.20. We need the following result proved in [BR02, Proposition 2.3.27].

Proposition 1.6.27 (Equivalence of Pure States and Characters). *Let ω be a nonzero linear functional over the commutative C^* -algebra $\mathfrak{A}$. Then ω is a pure state if and only if $\omega \in \sigma(\mathfrak{A})$.*

We need a topological structure for the spectrum to behave as a classical state space. The uniform topology induced by the dual norm is inadequate because in any infinite dimensional Banach space. The closed unit ball is never compact with respect to the uniform norm, see Riesz Lemma [RS03, Theorem II.4]. We therefore introduce a coarser topology designed to recover this required compactness.

Definition 1.6.28 (Weak* Topology). *Let $\mathfrak{A}^*$ be the topological dual space of a C^* -algebra $\mathfrak{A}$, as in Definition 1.6.11. The weak* topology on $\mathfrak{A}^*$ is the coarsest topology such that for every fixed $A \in \mathfrak{A}$ the evaluation map $\hat{A} : \mathfrak{A}^* \rightarrow \mathbb{C}$ defined by $\hat{A}(\alpha) = \alpha(A)$ is continuous $\forall \alpha \in \mathfrak{A}^*$.*

This topology controls pointwise convergence. A sequence of functionals converges if and only if the expectation value of every individual observable converges. By relaxing the strict requirement of uniform convergence across the entire algebra, this topology restores the geometric compactness essential for a well defined physical phase space.

Proposition 1.6.29 (Weak* Compactness of the Spectrum). *Let $\mathfrak{A}$ be a unital commutative C^* -algebra as in Definition 1.6.11. The Gelfand spectrum $\sigma(\mathfrak{A})$, defined in Definition 1.6.24, is a compact Hausdorff space with respect to the weak* topology.*

The complete proof of this result can be found in [Lan17, Lemma C.10].

We present the central result of this chapter. This theorem allows us to completely reconstruct the topological state space of a physical classical system directly from its abstract commutative algebra, establishing a correspondence between algebraic structures and geometry.

Theorem 1.10 (Gelfand-Naimark Representation Theorem). *Let $\mathfrak{A}$ be a commutative C^* -algebra and X be the Gelfand spectrum of $\mathfrak{A}$, as in Definition 1.6.24, equipped with the weak* topology induced by $\mathfrak{A}^*$ as in Definition 1.6.28. Then $\mathfrak{A}$ is isomorphic to the algebra $C_0(X)$ of continuous functions over X which vanish at infinity.*

A complete proof of this theorem is available in [BR02, Theorem 2.1.11]. As a special case we state the following corollary that is also proved in [BR02, Theorem 2.1.11A].

Corollary 1.6.30. *Under the hypotheses of Theorem 1.10, X is a locally compact Hausdorff space which is compact, if and only if, $\mathfrak{A}$ is unital, as in Definition 1.6.13. Thus, we have $\mathfrak{A} \cong C^0(X)$, the space of continuous functions over X .*

The Gelfand-Naimark Theorem 1.10 establishes the duality necessary to reconstruct the state space from any commutative C^* -algebra. It yields a locally compact Hausdorff space X that matches the Gelfand spectrum formed by the characters of the algebra. In addition, this spectrum is isomorphic to the classical continuous algebra, seen in Definition 1.6.15.

The topology of X directly reflects the algebraic structure of $\mathfrak{A}$. As formalized in Proposition 1.6.29, the existence of the identity element ensures that the Gelfand spectrum forms a weak* closed subset of the dual unit ball, which guarantees its strict compactness via the Banach-Alaoglu Theorem [RS03, Theorem IV.21]. Consequently, the Gelfand isomorphism maps the abstract algebra to the space $C_0(X)$ of continuous functions vanishing at infinity, which identically reduces to the full space of continuous functions $C^0(X)$ in this compact setting, as stated in Corollary 1.6.30.

Reconstruction of the Symplectic Manifold

The Gelfand isomorphism identifies the topological spectrum X purely as a Hausdorff space, discarding all differentiable and symplectic information. To recover the phase space of classical mechanics, the C^* -algebraic framework must be augmented with additional algebraic structures. These structures are captured by the following two definitions. The Poisson algebra formalizes the Lie bracket of observables onto the C^* -algebra itself, while the smooth dense subalgebra supplies the required differentiability in a subset of $\mathfrak{A}$. The compatibility between them, enforced by the Leibniz rule, ensures that the resulting geometry is coherent. However, the choice of such a subalgebra is not inherent to $\mathfrak{A}$ itself and constitutes the central external input of this reconstruction procedure.

We begin by defining the main structure that allows us to recover the Lie algebra characteristic of classical mechanics.

Definition 1.6.31 (Poisson Algebra). *A Poisson algebra is an associative commutative algebra $\mathfrak{P}$, as in Definition 1.6.1, equipped with a Lie bracket $\{\cdot, \cdot\}$, per Definition 1.2.1, acting as a derivation with respect to the associative product. Hence it satisfies the Leibniz rule*

$$\{f, lh\} = \{f, l\}h + l\{f, h\} \quad \forall f, h, l \in \mathfrak{P}.$$

In addition, to recover the smooth structure of the phase space, we need an isomorphism from a subalgebra of $\mathfrak{A}$ to a specific $C_0^\infty(M)$ where M is a smooth manifold. These strong requirements are coded inside the following definition.

Definition 1.6.32 (Smooth Dense Poisson Subalgebra). *Let $\mathfrak{A}$ be a commutative unital C^* -algebra as in Definition 1.6.13 with Gelfand spectrum X as in Definition 1.6.24. A smooth dense Poisson subalgebra is a Poisson subalgebra $(\mathfrak{A}^\infty, \{\cdot, \cdot\})$, seen in Definition 1.6.31, with $\mathfrak{A}^\infty \subset \mathfrak{A}$ satisfying*

1. $\mathfrak{A}^\infty$ is a selfadjoint subset as in Definition 1.6.4,
2. $\mathfrak{A}^\infty$ is dense in $\mathfrak{A}$ with respect to the uniform norm,
3. there exists a smooth manifold M such that $\mathfrak{A}^\infty \cong C_0^\infty(M)$ as algebras.

We can ask whether the Poisson and the smooth structure defined on $\mathfrak{A}^\infty$ can be transported and extended to the whole space of characters $\sigma(\mathfrak{A}) \equiv X$. The first answer is negative due to the behaviour of the Poisson brackets, as explained in Remark 1.6.33.

Remark 1.6.33 (Topological Obstruction to Poisson Structure). The Poisson bracket on the Poisson subalgebra, as seen in Definition 1.6.31, acts as a first-order differential operator. This makes it generally unbounded and discontinuous with respect to the uniform supremum norm of the C^* -algebra $\mathfrak{A}$. This obstruction restricts the Poisson algebra strictly to the incomplete dense subalgebra $\mathfrak{A}^\infty$, sacrificing Banach completeness to preserve the Leibniz rule.

At the same time, when we examine the smooth structure, we find that we can extend it. In fact, if there exists a smooth structure on M , as in Definition 1.6.32, and M is compact, we can prove the following result.

Proposition 1.6.34. *Let $\mathfrak{A}$ be a commutative unital C^* -algebra as in Definition 1.6.13 with Gelfand spectrum X as in Definition 1.6.24. Let $(\mathfrak{A}^\infty, \{\cdot, \cdot\})$ be the smooth dense Poisson subalgebra of $\mathfrak{A}$, as in Definition 1.6.32, with $\mathfrak{A}^\infty \cong C_0^\infty(M)$. If M is compact, then X is a smooth manifold.*

Proof. Since M is compact, $C_0^\infty(M) = C^\infty(M)$ contains the constant functions. By the isomorphism of Definition 1.6.32, it descends that $\mathbb{1}_{\mathfrak{A}} \in \mathfrak{A}^\infty$. In particular $\mathfrak{A}^\infty$ is a unital subalgebra of $\mathfrak{A}$. Let $\sigma(\mathfrak{A}^\infty)$ denote the set of linear multiplicative non-trivial functionals on $\mathfrak{A}^\infty$, and define an evaluation map as

$$\text{ev} : X \rightarrow \sigma(\mathfrak{A}^\infty), \quad \text{ev}(x) = \text{ev}_x := x|_{\mathfrak{A}^\infty}, \quad \text{ev}_x(f) = f(x) \quad \forall f \in \mathfrak{A}^\infty.$$

It is well defined and continuous by [Lan17, Construction p. 92] being dual to the inclusion $\mathfrak{A}^\infty \hookrightarrow \mathfrak{A}$. Since $\mathfrak{A}^\infty$ is dense in $\mathfrak{A}$ and multiplication is jointly continuous, two continuous maps out of $\mathfrak{A}$ that agree on $\mathfrak{A}^\infty$ must coincide. Likewise two continuous maps out of $\mathfrak{A} \times \mathfrak{A}$ that agree on $\mathfrak{A}^\infty \times \mathfrak{A}^\infty$ must coincide, via [Mun00, Exercise 13, §18].

Injectivity. If $x_1|_{\mathfrak{A}^\infty} = x_2|_{\mathfrak{A}^\infty}$, the above result yields $x_1 = x_2$ on $\mathfrak{A}$.

Surjectivity. Under $\mathfrak{A}^\infty \cong C_0^\infty(M)$, every $\omega \in \sigma(\mathfrak{A}^\infty)$ induces an evaluation by duality $\omega(f) = f(p)$ $\forall f \in C_0^\infty(M)$ for $p \in M$. Since all $f \in \mathfrak{A}^\infty$ are bounded by the supremum norm inherited from $\mathfrak{A}$, ω is bounded. Thus the B.L.T. Theorem [RS03, Theorem I.7] extends the functionals uniquely to a bounded linear $\tilde{\omega} \in \mathfrak{A}^*$. Since the maps $(A, B) \mapsto \tilde{\omega}(AB)$ and $(A, B) \mapsto \tilde{\omega}(A)\tilde{\omega}(B)$, $\forall A, B \in \mathfrak{A}$, agree on the dense subset $\mathfrak{A}^\infty \times \mathfrak{A}^\infty$, it holds $\tilde{\omega}(AB) = \tilde{\omega}(A)\tilde{\omega}(B)$. It descends that $\tilde{\omega} \in \sigma(\mathfrak{A})$ and ev is surjective.

It holds that ev is a continuous bijection. One must identify M with X . Under $\mathfrak{A}^\infty \cong C^\infty(M)$, the evaluation map

$$\text{ev}^M : M \rightarrow \sigma(\mathfrak{A}^\infty), \quad p \mapsto \text{ev}_p^M, \quad \text{ev}_p^M(f) = f(p) \quad \forall f \in \mathfrak{A}^\infty,$$

is a bijection [Lan17, Proposition C.21]. Continuity follows since the evaluation $\text{ev}_p^M(f) = f(p)$ acts on smooth, hence continuous, functions $f \in \mathfrak{A}^\infty$. Since M is compact and $\sigma(\mathfrak{A}^\infty)$ is Hausdorff, ev^M is a homeomorphism by [Mun00, Theorem 26.6]:

$$M \cong \sigma(\mathfrak{A}^\infty).$$

Since $\mathfrak{A}$ is unital, X is compact by Gelfand Theorem Corollary 1.6.30. Also, since $\sigma(\mathfrak{A}^\infty)$ is Hausdorff. The continuous bijection ev is likewise a homeomorphism via [Mun00, Theorem 26.6]:

$$X \cong \sigma(\mathfrak{A}^\infty).$$

Composing the two homeomorphisms yields $X \cong M$. Transporting the smooth atlas of M along this homeomorphism equips X with a smooth structure, making it a smooth manifold diffeomorphic to M . $\square$

Having discussed the limitations and possible extension of the smooth subalgebra $\mathfrak{A}^\infty$, we can collect the components identified so far into a single, self-contained structure.

Definition 1.6.35 (Classical Algebraic System). *A classical mechanical system is completely characterized by the triplet $(\mathfrak{A}, \mathfrak{A}^\infty, \{\cdot, \cdot\})$, where*

1. $\mathfrak{A} \cong C^0(X)$ is a commutative unital C^* -algebra, as in Definition 1.6.13, determining the topological state space X , its Gelfand spectrum.
2. $(\mathfrak{A}^\infty \subset \mathfrak{A}, \{\cdot, \cdot\})$ is a smooth dense Poisson subalgebra, as in Definition 1.6.32, inducing the differential and dynamical structure.

The last step consists of showing that this is sufficient to recover the complete structure of a symplectic manifold. This can be done invoking the representation theory of Poisson algebras. The complete explanation of this theory is outside the scope of this work, however we can refer the reader to the representation theorem presented in [Lan98, Theorem 2.6.7]. The key idea is to use the Poisson bracket defined on the smooth dense subalgebra to endow the smooth manifold M with a dynamical structure. The theorem establishes that irreducible representations of the Poisson subalgebra, presented in Definition 1.6.32, correspond to realizations of the algebra on spaces where the Poisson brackets become strictly non-degenerate. Geometrically, this implies evaluating the global Poisson structure of M on invariant submanifolds that inherit a well-defined symplectic form. By restricting the dynamics to these irreducible geometric domains, one exactly recovers the full symplectic phase space formulation of Hamiltonian mechanics without additional postulates.

Physical Observables and Dynamics

Before ending this chapter, we clarify the physical interpretation of the algebras introduced so far. We start with a unital commutative C^* -algebra $\mathfrak{A}$, per Definition 1.6.11. Using Gelfand Theorem 1.10, one can reconstruct the topological space X . The selfadjoint sector $\mathfrak{A}_{sa}$, already seen in Definition 1.6.17, isolates the measurable physical quantities. The smooth dense Poisson subalgebra $\mathfrak{A}^\infty$ of Definition 1.6.32 is required to formulate Hamiltonian mechanics. These topological, algebraic, and differential requirements combine as follows.

- The continuous measurable observables form the selfadjoint sector $\mathfrak{A}_{sa} = \{f \in \mathfrak{A} \mid f^* = f\}$. This space suffices to define states, probability measures, and continuous symmetries, but it lacks a Poisson bracket and so cannot generate differential dynamics.
- Hamiltonian mechanics further requires smoothness to formulate the equations of motion, so the Hamiltonian H must belong to the smooth selfadjoint sector $\mathfrak{A}_{sa}^\infty := \mathfrak{A}^\infty \cap \mathfrak{A}_{sa}$. This guarantees that the Hamiltonian vector field $X_H = -\{H, \cdot\}$ generates a well-defined, strictly real one-parameter group of $*$ -automorphisms $\alpha_t : \mathfrak{A}^\infty \rightarrow \mathfrak{A}^\infty$.

A non-selfadjoint or non-smooth Hamiltonian would produce complex flows or exit the domain of the Poisson bracket, yielding unphysical dynamics. Restricting dynamical generators to the selfadjoint part of $\mathfrak{A}^\infty$, preserves both the real structure of physical observables and the symplectic geometry of the underlying phase space.

1.7 Example I: Classical Spin System

In this section, we apply the algebraic tools developed in the first chapter to the classical spin system. This example provides an interesting test case for our framework, since it is a non-trivial Poisson algebra whose reduction yields a compact curved manifold, the sphere. This allows every abstract construction introduced so far, from the Gelfand spectrum to the Poisson bracket, to be verified explicitly on a concrete and geometrically intuitive example. Working through this case in detail also illustrates how symmetries of a physical system translate, via Noether's Theorem, into conserved quantities within the same algebraic language.

The Poisson Algebra and Gelfand's Theorem

Consider the polynomial algebra $\mathfrak{A}$, as in Definition 1.6.1, generated by three elements x_1, x_2, x_3 , equipped with the $\mathfrak{so}(3)$ -Poisson bracket

$$\{x_i, x_j\} = \epsilon_{ijk} x_k, \quad i, j, k \in \{1, 2, 3\},$$

where ϵ_{ijk} is the Levi-Civita symbol. This bracket is skew-symmetric and it satisfies the Jacobi identity by virtue of the Lie algebra structure. A direct computation reveals the existence of a Casimir function

$$C = x_1^2 + x_2^2 + x_3^2, \quad \{C, A\} = 0 \quad \forall A \in \mathfrak{A}.$$

Since $\mathfrak{A}$ is defined abstractly, we must first recover a geometric space from its algebraic structure. In order to fix $C = 1$, we identify $A \sim A'$ whenever $A - A' = Q(C - 1)$ for some $A, A', Q \in \mathfrak{A}$. Denote the resulting quotient by $\mathfrak{A}_1$. This identification respects the bracket because C is a Casimir operator that acts as a scalar through the Poisson brackets. It descends that $\{A, B\} - \{A', B\} = (C - 1)\{Q, B\}$ for $B \in \mathfrak{A}$, which again vanishes when descending to the quotient. It descends that $\mathfrak{A}_1$ inherits a well-defined Poisson brackets.

The points of $\mathfrak{A}_1$ are the multiplicative linear functionals $\omega : \mathfrak{A}_1 \rightarrow \mathbb{R}$ in the dual space $\mathfrak{A}_1^*$. We call their collection X . Since x_1, x_2, x_3 generates $\mathfrak{A}_1$, each $\omega \in X$ is fixed by the triple $(\omega(x_1), \omega(x_2), \omega(x_3)) \in \mathbb{R}^3$. Multiplicativity and $C = 1$ in $\mathfrak{A}_1$ force this triple onto $\mathbb{S}^2$. In contrast, every point of $\mathbb{S}^2$ defines such a functional by evaluation. Hence,

$$X \cong \mathbb{S}^2.$$

Completing $\mathfrak{A}_1$ in the uniform norm on X yields a commutative C^* -algebra isomorphic to $C^0(\mathbb{S}^2)$, consistent with Gelfand's Theorem 1.10. Moreover, the brackets descend with it, making $\mathbb{S}^2$ a natural candidate for a symplectic manifold, seen in Definition 1.3.8.

Construction of the Symplectic Manifold

We now equip $\mathbb{S}^2$ with coordinates and explicitly construct the symplectic form. In Cartesian coordinates, the symplectic form in $\mathbb{R}^3$ reads

$$\omega = x_1 dx_2 \wedge dx_3 + x_2 dx_3 \wedge dx_1 + x_3 dx_1 \wedge dx_2.$$

We restrict the form to $\mathbb{S}^2$ using the standard spherical coordinates

$$x_1 = \sin \theta \cos \phi, \quad x_2 = \sin \theta \sin \phi, \quad x_3 = \cos \theta,$$

where $\theta \in (0, \pi)$ and $\phi \in (0, 2\pi)$. Now we must compute the bracket $\{\theta, \phi\}$. Using the chain rule and computing the partial derivatives we get

$$\{x_1, x_2\} = \sin \theta \cos \theta \{\theta, \phi\}.$$

Equating this to $x_3 = \cos \theta$ yields

$$\boxed{\{\theta, \phi\} = \frac{1}{\sin \theta}.} \quad (1.17)$$

This formula is valid away from the poles. Its extension to $\sin \theta = 0$ is given by the Cartesian expression, ensuring global non-degeneracy. Given that the manifold is two-dimensional, to compute the symplectic form in spherical coordinates, we invert the commutator found in Equation (1.17) to obtain

$$\boxed{\omega = \sin \theta d\theta \wedge d\phi.} \quad (1.18)$$

This form is closed $d\omega = 0$ and non-degenerate, thus $(\mathbb{S}^2, \omega)$ is a symplectic manifold as in Definition 1.3.8.

Hamiltonian Action and the Moment Map

The group $G = SO(3)$ acts on $\mathbb{S}^2$ by rotations. For each $\xi \in \mathfrak{so}(3) \cong \mathbb{R}^3$, the infinitesimal generator, seen in Definition 1.5.4, is given by

$$X_\xi(x) = x \times \xi,$$

where $\times$ denotes the standard cross product. We seek a moment map $\mu : \mathbb{S}^2 \rightarrow \mathfrak{so}(3)^*$, as in Definition 1.5.7, satisfying the relation

$$d\langle \mu(x), \xi \rangle = \iota_{X_\xi} \omega, \quad \forall \xi \in \mathfrak{so}(3).$$

Identifying $\mathfrak{so}(3)^* \cong \mathbb{R}^3$ via the dot product, we test the candidate $\mu(x) = -x$. Given $\xi = e_1 = (1, 0, 0)$, then $X_{e_1} = (0, x_3, -x_2)$ and contracting ω with X_{e_1} yields

$$\begin{aligned} \iota_{X_{e_1}} \omega &= \iota_{(0, x_3, -x_2)} (x_1 dx_2 \wedge dx_3 + x_2 dx_3 \wedge dx_1 + x_3 dx_1 \wedge dx_2) \\ &= x_1(x_3 dx_3 + x_2 dx_2) - (x_2^2 + x_3^2) dx_1. \end{aligned}$$

Using the constraint $x_2 dx_2 + x_3 dx_3 = -x_1 dx_1$, given by the restriction on the sphere, we get

$$\iota_{X_{e_1}} \omega = -dx_1.$$

Thus, $d\langle \mu, e_1 \rangle = d(-x_1)$, which implies

$$\boxed{\mu(x) = -x.} \quad (1.19)$$

Physically, this is the negative of the position vector, which corresponds to the angular momentum vector, up to a sign convention.

Noether's Theorem

The moment map constructed above is the algebraic counterpart of a conserved quantity associated with a continuous symmetry, in the sense of Noether's Theorem 1.9. We now illustrate this correspondence through the example of the precession of a classical system that is governed by the same underlying symplectic structure on the sphere. Consider the Hamiltonian

$$H = x_3,$$

which is invariant under rotations about the z -axis. The symmetry generator is $\xi = e_3 = (0, 0, 1)$. The associated Noether invariant is

$$\langle \mu, \xi \rangle = \langle -x, e_3 \rangle = -x_3.$$

As calculated in Equation (1.19), Noether's theorem guarantees x_3 is conserved. In fact, the equations of motion are

$$\begin{aligned}\dot{x}_1 &= \{x_1, H\} = -x_2, \\ \dot{x}_2 &= \{x_2, H\} = x_1, \\ \dot{x}_3 &= \{x_3, H\} = 0.\end{aligned}$$

Solving yields a uniform precession around the z -axis:

$$x_1(t) + ix_2(t) = e^{it}(x_1(0) + ix_2(0)), \quad x_3(t) = x_3(0).$$

Chapter 2

Algebraic Construction of Quantum Mechanics

Nobody takes seriously quantum mechanics.

Adán Cabello Quintero

Quantum phenomena force a redefinition of physical reality and thus require a new descriptive language. In fact, words like 'to measure' are used but their operational semantics shifts. This mirrors the adaptation of Newtonian terms like time, mass, and energy in special relativity. At the same time, some syntactically correct propositions lose physical meaning. An example in special relativity is the assertion that two events occurred simultaneously at different places.

In order to find this new framework, we start by recalling that a physical theory partitions the world into the system and its environment. The framework described here treats only the system while the environment remains classical. This duality is underlined by Niels Bohr [Sch91].

However far the [quantum] phenomena transcend the scope of classical physical explanation, the account of all evidence must be expressed in classical terms. The argument is simply that by the word 'experiment' we refer to a situation where we can tell others what we have done and what we have learned and that, therefore, the account of the experimental arrangement and the results of the observations must be expressed in unambiguous language with suitable application of the terminology of classical physics.

Bohr used to reinforce this point with pseudorealistic sketches of robust clocks and rigid supports. It must be highlighted that he did not claim different physical laws for microscopic and macroscopic systems. He insisted on distinct modes of description between system and measurement apparatus. This approach is nonetheless *unsatisfactory*. Motivated by these considerations, the main aim of the present work is to show the common algebraic structure underlying these two theories understanding the common grounds and the differences.

2.1 General Mathematical Description of a Quantum Theory

This section shows that the same algebraic structure seen in the Hamiltonian case is forced on us directly by the operational meaning of measurement, without assuming a phase space and without assuming that the resulting algebra is commutative. A system is specified solely by the states in which it can be prepared and the observables that can be measured on it. Starting from this minimal data, we build the observable set together with the norm and algebraic structure that let the observable set be embedded in an associative C^* -algebra. We will see that commutativity not used in the classical case is needed to encode the features of quantum mechanics.

We seek this new mode of description by analysing the procedures used in experimental physics. In fact, two categories arise, preparations and tests. In this approach, these are primitive undefined notions of quantum theory, analogous to points and lines in Euclidean geometry. In particular, a preparation is a

completely specified and unambiguous experimental procedure. A measurement is an apparatus interacting with the system where a system property affects an apparatus property, and generally, this can disturb other properties of the system. That occurs even in classical physics, as shown in [Per02, Section 12-2]. However, in that framework we assume the measured property exists prior to the interaction. Instead quantum physics rejects this premise, as shown in [Per02, Section 1.5]. However, this epistemological discussion lies outside our present scope. We refer the reader to [Jam74] for a philosophical discussion about this topic.

Having set preparations and tests, we can now turn to define a 'quantum system'. Following N. Bohr [Boh35], a complete description of a quantum system necessarily includes the measurements to be performed, as these determine the relevant physical quantities. A quantum system thus comprises a physical system, a set of preparations, and a set of measurements. The equivalence of distinct macroscopic procedures giving rise to the same physical object, such as a polarized photon or a free hydrogen atom, induces the notion of a quantum state, characterized by the probability distribution of measurement outcomes arising from repeated identical preparations. Likewise, an observable is defined as an equivalence class of measurements, where two measurements are identified if they yield identical outcome statistics across all the given preparations. This formal framework is developed further in the subsequent sections, where the property that an equivalence class uniquely determines the physical quantity is formalized as Separation in Definition 2.1.4.

The Observable Set and States

An observable is operationally tied to a concrete measuring apparatus, namely a device that, applied to any preparation of the system, returns a real number. Two elementary manipulations of an apparatus carry a physical meaning. Rescaling the readout by a real factor λ , such as relabelling the pointer scale, produces a new observable λA from A . Replacing every reading by its n -th power produces a new observable A^n , obtained by squaring the scale, cubing it, and so on, and this operation inherits the elementary law of exponents, $A^m A^n = A^{m+n}$, directly from the arithmetic of real numbers. In particular, A^0 is the trivial observable whose outcome is always 1, independently of the preparation. A quantum system is specified jointly by a set S of states and by the set of observables obtainable in this way, which we now formalize.

Definition 2.1.1 (Observable set). *We call $\mathcal{O}$ an observable set if it is a set equipped with the operations*

$$(\lambda, A) \mapsto \lambda A, \quad (n, A) \mapsto A^n, \quad A^m A^n = A^{m+n}, \quad A^0 \equiv 1,$$

for $\lambda \in \mathbb{R}$, $m, n \in \mathbb{N}$, $A \in \mathcal{O}$.

An observable is called *positive* if every possible outcome of its measurement is a nonnegative number. Operationally, this is realized when the observable is obtained by squaring another one, for instance by taking a squared pointer scale, whose readings are nonnegative.

Definition 2.1.2 (Positive observable). *Let $A \in \mathcal{O}$ with $\mathcal{O}$ denoting an observable set as in Definition 2.1.1, A is positive if $A = B^2$ for some $B \in \mathcal{O}$.*

Consider the set of preparations given with a quantum system. Through the procedure described earlier, the equivalence classes of identical preparations are called *states*, denoted by ω . The expectation value $\omega(A) \in \mathbb{R}$ is obtained as the limit of sample means of replicated measurements of A on identically prepared copies of the system. Since sample means are linear in the individual readings and intrinsically non-negative when evaluating squares of outcomes, we elevate these operational facts to defining axioms for the states.

Definition 2.1.3 (State). *A state ω on the observable set $\mathcal{O}$, as per Definition 2.1.1, is a real linear functional $\omega : \mathcal{O} \rightarrow \mathbb{R}$ satisfying two properties:*

- *Normalization:* $\omega(1) = 1$.
- *Positivity:* $\omega(B^2) \geq 0$ for every $B \in \mathcal{O}$.

Having defined states, we now present the state space S , which, in complete analogy with the classical case, is convex.

Definition 2.1.4 (State space). *The state space S of a system is a convex set of states which is assumed to be separating for $\mathcal{O}$. To wit, for any two observables $A, B \in \mathcal{O}$, if $\omega(A) = \omega(B)$ for all $\omega \in S$, then $A = B$.*

The separating property mathematically implements the operational completeness of the state space. The physical content of an observable is entirely exhausted by its expectations across all possible preparations. Two states that agree on every observable are identified as one and the same state. This mutual duality is symmetric.

Remark 2.1.5. Definition 2.1.3 guarantees by construction that every state evaluates non-negatively on the algebraic squares of observables. However, the converse implication, namely that state-positivity guarantees the existence of an exact square root within $\mathcal{O}$, fails. Establishing this strict equivalence requires endowing the observable set with a topological structure, such as norm completeness within a C^* -algebra. In this setting the continuous functional calculus ensures that such roots are always well defined.

Every concrete apparatus has an inevitable scale bound, for instance a bounded pointer range, independent of the state being probed. Consequently, the outcomes of measuring a fixed observable A , across all possible states of Definition 2.1.3, form a bounded set of real numbers, and it is natural to record this bound as a norm.

Definition 2.1.6 (Norm on $\mathcal{O}$). *Let $\mathcal{O}$ be the observable set defined in Definition 2.1.1 and let S be the associated state space as in Definition 2.1.4. We can endow $\mathcal{O}$ with a norm $\|\cdot\| : \mathcal{O} \rightarrow \mathbb{R}$ defined as*

$$\|A\| := \sup_{\omega \in S} |\omega(A)| < \infty.$$

This norm is an additional structure layered on the bare set of Definition 2.1.1. Note that we continue to write $\mathcal{O}$ for the resulting normed set, rather than introduce new notation at each enrichment, and will do so consistently from here on.

By homogeneity of the states, $\|\lambda A\| = |\lambda| \|A\|$. Since the states separate the observables, $\|A\| = 0$ implies $A = 0$. We can also prove that using only positivity of the states applied to $\|A\| \mathbb{1} \pm A$ and to $(\|A\| \mathbb{1} \pm A)^2$, the norm can be linked to the squaring operation already available on $\mathcal{O}$.

Proposition 2.1.7 (Compatibility of norm and squares). *Let $\mathcal{O}$ be the observable set defined in Definition 2.1.6. For every $A \in \mathcal{O}$, $\|A\|^2 = \|A^2\|$.*

The complete proof of this statement can be found in [Str05, p.18]. Operationally, the sum of two observables is meaningful whenever there is a third observable whose expectation always equals the sum of the two expectations, for instance kinetic and potential energy. Postulating the existence of such sums for arbitrary pairs turns $\mathcal{O}$ into a real vector space, on which $\|\cdot\|$ is subadditive. Its completion is a real Banach space, still denoted $\mathcal{O}$.

Definition 2.1.8 (Linear extension of $\mathcal{O}$). *Let $\mathcal{O}$ be an observable set as in Definition 2.1.6 and let S be the associated state space as in Definition 2.1.4. For any observables $A, B, C \in \mathcal{O}$, if the identity $\omega(C) = \omega(A) + \omega(B)$ holds for all states $\omega \in S$, we define the sum as $C = A + B$.*

The linear extension of $\mathcal{O}$ is postulated by requiring that such sums, alongside their induced powers $(A+B)^n$, exist for all arbitrary pairs within $\mathcal{O}$. Furthermore the states in S must act upon these extended elements as positive linear functionals.

The structural consequence of this operational hypothesis is that $\mathcal{O}$ acquires the algebraic structure of a real vector space. Furthermore, this linear framework is compatible with the previously defined topological properties. Specifically, the norm becomes subadditive, guaranteeing that

$$\|A + B\| \leq \|A\| + \|B\|, \quad \|\omega(A)\| \leq \|A\|,$$

for all $A, B \in \mathcal{O}$ and all $\omega \in S$. Consequently, $\mathcal{O}$ constitutes a normed real vector space. Taking the norm completion of this space yields a real Banach space, providing the necessary topological completeness for the subsequent algebraic development. This is a strictly larger structure that we continue to denote $\mathcal{O}$.

The Jordan product

So far $\mathcal{O}$ carries only the Banach space structure just introduced, together with the squaring of a single observable. To reach an associative algebra, and eventually the full apparatus of operator theory, a product combining two different observables is needed. Nothing in the operational setting singles out a product AB between two *arbitrary* observables, since only squares, and hence sums of squares, carry a direct experimental meaning. The symmetric combination built from the operations already available, namely squares and sums, is the following.

Definition 2.1.9 (Jordan product). *Let $\mathcal{O}$ be an observable set, as in Definition 2.1.1. The Jordan product is a commutative operation between two observables $A, B \in \mathcal{O}$ defined via*

$$A \circ B := \frac{1}{2}((A+B)^2 - A^2 - B^2) = B \circ A.$$

Remark 2.1.10. As with the earlier enrichments, we do not introduce new notation for this product. For polynomials of a single observable A , the Jordan Product $\circ$ coincides with the ordinary algebraic product, so that $A \circ A^n = A^{n+1}$.

At this stage, $\circ$ is not known to be distributive over the sum already available on $\mathcal{O}$. This property follows, however, from a mild extra assumption, homogeneity of $\circ$, itself a direct extension of the homogeneity already required on scalars.

Proposition 2.1.11 (Homogeneity implies distributivity). *Let $\mathcal{O}$ be an observable set with a Jordan product $\circ$ as in Definition 2.1.9. If $\circ$ is homogeneous, namely $A \circ (\lambda B) = \lambda(A \circ B)$ for $\lambda \in \mathbb{R}$, then $\circ$ is distributive, $A \circ (B + C) = A \circ B + A \circ C$.*

This is proved in [Str05, p.20]. Distributivity alone says nothing about the norm of $A \circ B$. Only combining it with the positivity of the states and Proposition 2.1.7 enforces a bound on $\|A \circ B\|$, as shown in the following proposition.

Proposition 2.1.12 (Norm bound on the Jordan product). *Let $\mathcal{O}$ be an observable set with Jordan product $\circ$ as in Definition 2.1.9. Then $\|A \circ B\| \leq \|A\| \|B\|$ for all $A, B \in \mathcal{O}$.*

Proof. For every state ω and every $\lambda \in \mathbb{R}$, positivity of $(A + \lambda B)^2$ yields

$$0 \leq \omega(A^2) + 2\lambda \omega(A \circ B) + \lambda^2 \omega(B^2) \leq \|A\|^2 + 2\lambda \omega(A \circ B) + \lambda^2 \|B\|^2.$$

A nonnegative discriminant for this quadratic in λ forces $\omega(A \circ B)^2 \leq \|A\|^2 \|B\|^2$ for every ω . This yields $\|A \circ B\| \leq \|A\| \|B\|$. $\square$

One algebraic property of a Jordan algebra remains open, it is called weak associativity

$$A^2 \circ (B \circ A) = (A^2 \circ B) \circ A.$$

This key identity ensures power associativity, guaranteeing that arbitrary polynomials of a single observable are unambiguously defined. Algebraically, this provides the main structure required to establish the continuous functional calculus. The commutativity, distributivity, weak associativity, and norm properties established above are, taken together, exactly the defining axioms of a single algebraic structure. Unlike the earlier enrichments, we now give this structure its own name and record its axioms once and for all in the following definition.

Definition 2.1.13 (Jordan–Banach algebra). *A Jordan–Banach algebra $\mathcal{O}$ is a real Banach space as in Definition 1.6.9, equipped with a commutative, distributive, weakly associative product $\circ$ such that*

$$\|A \circ B\| \leq \|A\| \|B\|, \quad \|A\|^2 = \|A^2\|, \quad \|A^2\| \leq \|A^2 + B^2\| \quad \forall A, B \in \mathcal{O}.$$

Such a structure is also called a JB algebra.

Embedding a JB Algebra into an Associative Algebra

The Jordan product alone is, in general, neither associative nor known to arise from an ordinary matrix-like product. It is technically convenient, and possible for every physically reasonable observable set, to realize $\mathcal{O}$ concretely inside an associative C^* -algebra, so that $\circ$ becomes the symmetrization of the usual product.

Definition 2.1.14 (Embedding of a JB algebra in a C^* -algebra). *A Jordan–Banach algebra $\mathcal{O}$, as in Definition 2.1.13, is embedded in a C^* -algebra $\mathfrak{A}$ if there exists an isometric linear map from $\mathcal{O}$ into $\mathfrak{A}$ such that the image of $\mathcal{O}$ generates $\mathfrak{A}$ as a C^* -algebra. In addition, for all $A, B \in \mathcal{O}$*

1. $A^* = A$ in $\mathfrak{A}$, so that $\mathcal{O}$ is identified with a closed real subspace of the selfadjoint elements $\mathfrak{A}_{sa}$ as per Definition 1.6.5,
2. the Jordan product corresponds to the symmetrized associative product, $A \circ B = \frac{1}{2}(AB + BA)$.

Not every Jordan–Banach algebra admits such an embedding, so the property deserves its own name. Jordan algebras for which the embedding exists are singled out below, since it is precisely this class that will later be identified with the selfadjoint part of a genuine C^* -algebra of observables.

Definition 2.1.15 (JC algebra). *A JB algebra admitting an embedding as in Definition 2.1.14 is called a JC algebra. A Jordan algebra of observables admitting such an embedding is called special, and exceptional otherwise.*

Remark 2.1.16. Exceptional Jordan algebras exist but are classified by an essentially unique finite-dimensional pathological building block, too limited to accommodate the observable algebra of a realistic physical system. We therefore restrict attention throughout to special Jordan algebras of observables, JC algebras, which generate an associative C^* -algebra $\mathfrak{A}$.

The embedding of Definition 2.1.14 realizes $\mathcal{O}$ inside a real associative algebra, but a genuine C^* -algebra is by convention a complex object. To hop from the real Jordan algebra $\mathcal{O}$ to its complex counterpart, one first complexifies, doubling the real dimension while keeping every operation already defined.

Definition 2.1.17 (Complexification). *The complexification of a JC algebra $\mathcal{O}$ as in Definition 2.1.15 is $\mathcal{O}^{\mathbb{C}} = \{A + iB \mid A, B \in \mathcal{O}\}$, with product extended by distributivity,*

$$(A + iB) \circ (C + iD) = (A \circ C - B \circ D) + i(A \circ D + B \circ C),$$

involution $(A + iB)^ = A - iB$, and states extended by linearity, $\omega(A + iB) = \omega(A) + i\omega(B)$.*

One checks directly that $\omega((A + iB) \circ (A - iB)) = \omega(A^2 + B^2) \geq 0$ and $\|A + iB\| = \|A - iB\|$. This yields $\|C\| = \|C^*\|$ for all $C \in \mathcal{O}^{\mathbb{C}}$. The complexified structure already carries the positivity and symmetry expected of a C^* -algebra. The complexified, embedded algebra $\mathfrak{A}$ generated by $\mathcal{O}$ is thus a unital C^* -algebra with identity $\mathbb{1}$. Its subset of selfadjoint elements, which coincides with the original Jordan algebra, is precisely $\mathcal{O}$, and $\mathfrak{A}$ is generated by $\mathcal{O}$.

Lie–Jordan algebras and the associative product

The embedding just constructed makes a second bilinear operation available on the ambient algebra $\mathfrak{A}$ besides $\circ$, the commutator $[A, B] := AB - BA$. The commutator alone does not preserve selfadjointness. In fact, fixing a constant $\hbar > 0$, we can rescale the commutator and physically identify it with the reduced Planck constant. This rescaling preserves selfadjointness and it is the missing concept needed to recover the full associative product from $\mathcal{O}$.

Proposition 2.1.18 (Selfadjointness of the rescaled commutator). *Let $\mathcal{O}$ be a JC algebra as in Definition 2.1.15 and let $A, B \in \mathcal{O}$. Then $[A, B]^* = -[A, B]$, and consequently*

$$\{A, B\} := \frac{1}{i\hbar}[A, B],$$

satisfies $\{A, B\}^ = \{A, B\}$, so that $\{A, B\} \in \mathcal{O}$. The bracket $\{\cdot, \cdot\} : \mathcal{O} \times \mathcal{O} \rightarrow \mathcal{O}$ is thus unambiguously defined on selfadjoint elements.*

Proof. Since $A^* = A$, $B^* = B$, $[A, B]^* = (AB - BA)^* = B^*A^* - A^*B^* = BA - AB = -[A, B]$. Hence $\{A, B\}^* = \left(\frac{1}{i\hbar}[A, B]\right)^* = \frac{1}{-i\hbar}[A, B]^* = \frac{1}{-i\hbar}(-[A, B]) = \{A, B\}$. $\square$

Equivalently, $[A, B] = i\hbar\{A, B\}$. Note that $\{\cdot, \cdot\}$ plays the role of the quantum-mechanical Poisson bracket, and reality of $\{\cdot, \cdot\}$ is exactly what makes the following axioms consistent with the operational structure already established on $\mathcal{O}$.

Definition 2.1.19 (Real Lie–Jordan algebra). *A Jordan–Banach algebra $(\mathcal{O}, \circ)$, as in Definition 2.1.13, equipped with a Lie bracket $\{\cdot, \cdot\}$ is a Lie–Jordan algebra if, there exists $c \in \mathbb{R}$ such that*

$$(A \circ B) \circ C - A \circ (B \circ C) = \frac{c}{4}\{\{A, C\}, B\}, \quad \{A, B \circ C\} = \{A, B\} \circ C + \{A, C\} \circ B,$$

for all $A, B, C \in \mathcal{O}$.

Note that, on physical grounds, one sets $c = \hbar^2$, consistently with $\{\cdot, \cdot\} = \frac{1}{i\hbar}[\cdot, \cdot]$ above. The two products of $(\mathcal{O}, \circ, \{\cdot, \cdot\})$ are not independent data once the embedding of Definition 2.1.14 is in place. They recombine into the single associative product of the ambient algebra $\mathfrak{A}$, in the precise sense made explicit by the following identity.

Proposition 2.1.20 (Recovery of the associative product). *Let $\mathcal{O}$ be a JC algebra as in Definition 2.1.15. For all $A, B \in \mathcal{O}$,*

$$AB = A \circ B + \frac{i\hbar}{2}\{A, B\}. \quad (2.1)$$

Proof. A direct computation shows that $A \circ B + \frac{i\hbar}{2}\{A, B\} = \frac{1}{2}(AB + BA) + \frac{i\hbar}{2} \cdot \frac{1}{i\hbar}(AB - BA) = \frac{1}{2}(AB + BA) + \frac{1}{2}(AB - BA) = AB$. $\square$

This identity can be reversed, and this is the precise sense in which a Lie–Jordan algebra is associated with a C^* -algebra. Starting from the abstract axioms of Definition 2.1.19 alone, without presupposing an ambient associative algebra, one can reconstruct such an algebra by declaring the right-hand side of the identity above to be the product.

Theorem 2.1 (From Lie–Jordan algebras to C^* -algebras). *Let $(\mathcal{O}, \circ, \{\cdot, \cdot\})$ be a real Lie–Jordan algebra, as in Definition 2.1.19, with $c = \hbar^2$. On $\mathcal{O}^{\mathbb{C}} := \mathbb{C} \otimes \mathcal{O}$, with involution $(a \otimes A)^* := \bar{a} \otimes A$, the $\mathbb{C}$ -linear extension of $AB := A \circ B + \frac{i\hbar}{2}\{A, B\}$ is an associative product, and the completion of $\mathcal{O}^{\mathbb{C}}$ with respect to its natural norm is a C^* -algebra $\mathfrak{A}$ whose selfadjoint part is $\mathcal{O}$.*

A complete proof of this theorem is provided by [Mor18, p.807].

Remark 2.1.21. The algebra $\mathfrak{A}$ produced this way is the same pair obtained from the JC-algebra embedding of Definition 2.1.14, and it is exactly the algebra $\mathfrak{A}$ of the quantum system of Definition 2.1.22 in the next subsection.

Before stating the formal definition of a quantum system, we summarize the algebraic reconstruction developed in this section. The starting point consists of the operational procedures of preparations and tests, which define the states and the observables. The observable set is structured using rescaling and squaring operations. States are introduced as normalized positive linear functionals, representing the expectation values of measurements. By recording the bounds of these expectations, a norm is defined, and the observable set is completed into a real Banach space. The symmetric Jordan product is introduced to combine observables using only sums and squares. The addition of a real Lie bracket, corresponding to the rescaled commutator, turns the space into a Lie–Jordan algebra. Through Theorem 2.1, these products generate an associative product on the complexified space. This procedure yields a C^* -algebra whose selfadjoint elements coincide with the initial observable set, establishing the mathematical requirements for the axiomatic definition.

The C^* -algebraic definition of a Quantum System

The operational construction of Section 2.1 leads, for every physically reasonable observable set, to a unital C^* -algebra $\mathfrak{A}$ together with a set of normalized positive linear functionals on it. This is the same kind of pair that describes a classical system, except that commutativity of $\mathfrak{A}$ is no longer assumed. We adopt this pair as the axiomatic definition of a quantum system, and this section collects the additional facts, automatic continuity of states, the spectrum, and separation by states, needed to use the definition as a working framework rather than just formal statement.

Definition 2.1.22 (Quantum system). *A quantum system is defined by the following two sets of data.*

1. *A unital C^* -algebra $\mathfrak{A}$ of observables, its selfadjoint elements being the operationally accessible observables $\mathcal{O}$ of Definition 2.1.1.*
2. *A set S of states, normalized positive linear functionals on $\mathfrak{A}$ satisfying $\omega(\mathbb{1}) = 1$ and $\omega(A^*A) \geq 0$ for all $A \in \mathfrak{A}$, such that S is full. To wit, it separates the elements of $\mathfrak{A}$, so that $\omega(A) = \omega(B)$ for all $\omega \in S$ forces $A = B$. Conversely $\mathfrak{A}$ separates S .*

A classical system is recovered as the special case in which $\mathfrak{A}$ is abelian, as shown in the previous chapter 1.6. Before this pair can serve as a working framework, it must be checked that positivity alone already forces the continuity that would otherwise need to be postulated separately. The following proposition establishes exactly this automatic regularity.

Proposition 2.1.23 (Automatic continuity and positivity). *Any positive linear functional $\omega : \mathfrak{A} \rightarrow \mathbb{C}$ on a unital C^* -algebra $\mathfrak{A}$, as in Definition 1.6.13, is automatically continuous in the norm topology,*

$$|\omega(A)| \leq \|A\| \omega(\mathbb{1}), \quad \forall A \in \mathfrak{A},$$

*and $\omega(A) \geq 0$ for all $\omega \in S$ if and only if $A = B^*B$ for some $B \in \mathfrak{A}$.*

This is proved in [BR81, Proposition 2.3.11]. The spectrum of an observable records, algebraically, the physical values it can legitimately yield upon measurement. We recall its formal definition for a general element of a C^* -algebra $\mathfrak{A}$.

Definition 2.1.24 (Spectrum). *Let $\mathfrak{A}$ be a C^* -algebra in the sense of Definition 1.6.11. The spectrum of $A \in \mathfrak{A}$ is defined as the set $\sigma(A) = \{\lambda \in \mathbb{C} \mid \lambda\mathbb{1} - A \text{ has no two-sided inverse in } \mathfrak{A}\}$.*

The proposition below will show that every value in the spectrum of an observable is attained as an expectation value, but this attainment result holds only for a specific class of elements. We therefore isolate this class first, restricting attention to elements that commute with their own adjoint.

Definition 2.1.25 (Normal elements). *An element $A \in \mathfrak{A}$ is called normal if $AA^* = A^*A$.*

Every physical observable $A \in \mathcal{O}$ is selfadjoint by definition, and therefore inherently normal. As we will see in the next sections, this framework establishes an isomorphism between the C^* -subalgebra generated by A and the algebra of continuous functions on its spectrum, $C(\sigma(A))$. Operationally, this structural correspondence dictates that evaluating the observable behaves exactly like sampling points from $\sigma(A)$. Consequently, the elements of the spectrum represent the only possible deterministic outcomes of a measurement. For normal elements, the functional calculus guarantees that every spectral value corresponds to a sharp physical preparation.

Proposition 2.1.26 (Attainment of spectral values). *Let $\mathfrak{A}$ be a C^* -algebra in the sense of Definition 1.6.11. If $A \in \mathfrak{A}$ is normal as in Definition 2.1.25 and $\lambda \in \sigma(A)$, there exists a state $\omega \in S$ such that $\omega(A) = \lambda$.*

This Proposition is proved in [Mur90, Theorem 3.3.6].

Remark 2.1.27. The attainment proposition mathematically crystallizes the physical interpretation of the spectrum. Operationally, $\sigma(A)$ constitutes the set of values that a measurement of A will yield with zero variance when the system is prepared in a suitable, dispersion-free state. For a generic state ω , the expectation value $\omega(A)$ is not necessarily a sharp reading but rather a statistical average over the spectrum, thereby falling within the convex hull of $\sigma(A)$.

Corollary 2.1.28. *The state space S of Definition 2.1.4 separates all normal elements of $\mathfrak{A}$, and consequently all of $\mathfrak{A}$. Indeed, any generic element $A \in \mathfrak{A}$ can be uniquely decomposed into a complex linear combination of two selfadjoint, and thus normal, elements via the Cartesian decomposition $A = \frac{1}{2}(A + A^*) - \frac{i}{2}(iA - iA^*)$.*

Example 2.1.29 (Continuous spectrum). Let $\mathcal{H} = L^2([0, 1], dx)$ and let $\mathfrak{A} = \mathcal{B}(\mathcal{H})$ be the full algebra of bounded operators on $\mathcal{H}$, unital and noncommutative. Define the position observable $Q \in \mathfrak{A}$ by $(Qf)(x) := xf(x)$ for $f \in \mathcal{H}$. Since $x \in [0, 1]$, Q is bounded and selfadjoint, hence normal as in Definition 2.1.25. For $\lambda \notin [0, 1]$, the function $x \mapsto (\lambda - x)^{-1}$ is bounded and continuous on $[0, 1]$, so multiplication by it provides a two-sided inverse of $\lambda\mathbb{1} - Q$ in $\mathfrak{A}$. Conversely, for $\lambda \in [0, 1]$ no such bounded inverse exists, since $(\lambda - x)^{-1}$ is unbounded near $x = \lambda$. Hence $\sigma(Q) = [0, 1]$ is a continuous spectrum.

2.2 An Introduction to the Hilbert Space Formalism

The Hilbert space formalism of quantum mechanics was developed through the Dirac-Von Neumann axioms. In particular, quantum states are vectors in a Hilbert space, and observables are operators acting on it. Although we will not develop this framework here, we present the key concepts to later understand how algebraic reconstruction links to the Hilbert space formalism.

The discussion begins with the inner product and the completion of a pre-Hilbert space into a full Hilbert space, which provides the necessary framework for the rest of the study. From there, we introduce both bounded and unbounded operators, carefully distinguishing them by their domains. In the case of bounded operators, we can construct their adjoint using the Riesz representation theorem as the key tool. A particular point of emphasis is the difference between symmetric and Selfadjoint unbounded operators, as this distinction affects their spectral properties and domain intricacies. We conclude by introducing cyclic vectors, a concept included to lay the groundwork for the GNS theorem developed in the later sections.

Inner Product Spaces and Hilbert Spaces

We start with a complex vector space endowed with an inner product that yields a notion of length, distance, and orthogonality. A vector space with an inner product is called a pre-Hilbert space. If the normed space is complete under its induced metric, then it is a Hilbert space.

Definition 2.2.1 (Hermitian Inner Product). *Let V be a complex vector space. A Hermitian inner product on V is a map $(\cdot, \cdot) : V \times V \rightarrow \mathbb{C}$ satisfying the following properties for all $v, w, w' \in V$ and $\alpha, \beta \in \mathbb{C}$.*

1. $(v, v) \geq 0$.
2. $(v, v) = 0$ if and only if $v = 0$.
3. $(v, \alpha w + \beta w') = \alpha(v, w) + \beta(v, w')$.
4. $(v, w) = \overline{(w, v)}$.

Note that we adopt the physics convention of linearity in the second argument, so that property 4 combined with property 3 makes the inner product conjugate-linear in the first argument.

Remark 2.2.2. Some texts use the opposite convention, linear in the first argument and conjugate-linear in the second. The two conventions differ only by conjugation. We keep the physics convention throughout. The inner product induces a norm through its square root and makes the complex vector space normed.

Proposition 2.2.3 (The Inner Product Induces a Norm). *Let V be a complex vector space equipped with a Hermitian inner product as per Definition 2.2.1. The map $\|\cdot\| : V \rightarrow \mathbb{R}$ defined by $\|v\| = \sqrt{(v, v)}$ is a norm on V .*

A proof of this statement can be found in [Mor18, Proposition 3.3].

Example 2.2.4. The space $\mathbb{C}^n$ with $(v, w) = \sum_{k=1}^n \bar{v}_k w_k$ is a finite-dimensional pre-Hilbert space. The space ℓ^2 of square-summable sequences, with the analogous inner product, is an infinite-dimensional example used throughout this work.

Before completeness enters the picture, it is useful to isolate the algebraic content of an inner product space on its own. This intermediate structure will serve as the domain on which the completion procedure of Theorem 2.2 acts.

Definition 2.2.5 (Pre-Hilbert Space). *A complex vector space equipped with a Hermitian inner product as per Definition 2.2.1 is called a pre-Hilbert space. With the induced norm of Proposition 2.2.3, it is a normed space.*

We can topologically complete a pre-Hilbert space and obtain the central structure used in quantum mechanics.

Definition 2.2.6 (Hilbert Space). *A Hilbert space $\mathcal{H}$ is a pre-Hilbert space as per Definition 2.2.5 that is complete with respect to its induced norm.*

Example 2.2.7. The space $C([0, 1])$ with $(f, h) = \int_0^1 \overline{f(x)}h(x)dx$ is a pre-Hilbert space that is not complete, since a sequence of continuous functions approximating a jump discontinuity is Cauchy without a continuous limit. Both $\mathbb{C}^n$ and ℓ^2 are complete, hence genuine Hilbert spaces.

The preceding example demonstrates that pre-Hilbert spaces need not be complete. However, completeness is a requirement for many constructions in functional analysis and quantum mechanics. In analogy with the completion of the rationals to the reals, every pre-Hilbert space admits a completion obtained by adjoining the limit points of all Cauchy sequences. To formulate this procedure rigorously, we introduce the notions of dense subsets and linear isometries. The former characterizes how the original space is embedded into its extension, while the latter guarantees that the norm and inner-product structures are preserved under this embedding.

Definition 2.2.8 (Dense Subset). *Let V be a normed space. A subset $W \subseteq V$ is dense in V if for every $v \in V$ and for every $\varepsilon > 0$ there exists $w \in W$ with $\|v - w\| < \varepsilon$.*

Both notions will enter the statement of the completion theorem below. Density describes how the original space sits inside its completion, while a linear isometry is precisely the kind of map under which such an embedding must be realized, since it preserves the norm and inner-product structure intact.

Definition 2.2.9 (Linear Isometry). *Let V, V' be normed spaces. A linear map $J : V \rightarrow V'$ is a linear isometry if $\|Jv\|_{V'} = \|v\|_V$ for every $v \in V$.*

We can show that every pre-Hilbert space embeds densely into a Hilbert space, unique up to unitary isomorphism.

Theorem 2.2 (Completion of a Pre-Hilbert Space). *Let V be a pre-Hilbert space as per Definition 2.2.5. There exists a Hilbert space $\mathcal{H}$ as per Definition 2.2.6 and an injective linear isometry $J : V \rightarrow \mathcal{H}$ as per Definition 2.2.9 such that JV is dense in $\mathcal{H}$ as per Definition 2.2.8. The completion is unique up to unitary isomorphism, and*

$$(Jv, Jw)_{\mathcal{H}} = (v, w)_V, \quad (2.2)$$

for all $v, w \in V$.

A complete proof of this statement is found in [Mor18, Theorem 3.11].

Bounded Operators on Hilbert Spaces

In finite-dimensional linear algebra, every linear map is continuous. However, on normed spaces, continuity becomes a non-trivial property, the fundamental equivalence between continuity and boundedness makes the latter a convenient algebraic proxy for analytic regularity. This equivalence leads to the operator norm, which turns the space of operators into a metric setting and it allows us to quantify their size. Then, we will specialize to continuous linear functionals, whose collection forms the dual space. At the end, since many operators of interest are initially defined only on a proper dense subspace, we determine whether they can be extended to the whole space. The bounded extension theorem supplies this, guaranteeing a unique continuous extension whenever the operator is bounded on its dense domain. We start with defining boundedness.

Definition 2.2.10 (Operator Norm and Boundedness). *Let X, Y be normed spaces and let $T \in \mathcal{L}(X, Y)$ with $\mathcal{L}(X, Y)$ the space of linear maps between X and Y . The operator T is bounded if there exists $K \geq 0$ such that*

$$\|Tv\|_Y \leq K\|v\|_X, \quad (2.3)$$

for every $v \in X$. Thus the operator norm is $\|T\| = \sup_{v \neq 0} \frac{\|Tv\|_Y}{\|v\|_X}$.

The set of bounded linear operators from X to Y is denoted $\mathcal{B}(X, Y)$. While continuity is inherently a local property at a given point, linearity of an operator forces this local behaviour to propagate globally. Furthermore, a particularly convenient metric for quantifying continuity is the operator norm, which leads to the following fundamental equivalence.

Theorem 2.3 (Continuity if and only if Boundedness). *Let $T : X \rightarrow Y$ be linear, with X, Y normed spaces. The following statements are equivalent.*

1. *T is continuous at $0 \in X$.*
2. *T is continuous everywhere on X .*
3. *T is bounded as per Definition 2.2.10.*

This theorem is proved in [Mor18, Theorem 3.11].

Example 2.2.11. On ℓ^2 , the right shift $(Sv)_1 = 0$, $(Sv)_{k+1} = v_k$ satisfies $\|Sv\| = \|v\|$, so S is bounded with $\|S\| = 1$. On the subspace of finitely supported sequences, $(Dv)_k = kv_k$ is linear but unbounded, since $\|Dv\| = n\|v\|$ for v the n -th standard basis vector.

Having characterized bounded operators between arbitrary normed spaces, we now focus on functionals. These objects lie in dual spaces and provide a coordinate-like description of infinite-dimensional spaces.

Definition 2.2.12 (Algebraic Dual). *Let X be a complex vector space. The algebraic dual of X is $X^* = \mathcal{L}(X, \mathbb{C})$.*

The algebraic dual imposes no regularity on the functionals it contains, and in infinite dimensions most of them fail to be continuous. Restricting attention to the bounded, hence continuous, functionals singles out the subspace that will actually be useful in the Hilbert space setting.

Definition 2.2.13 (Topological Dual). *Let X be a normed space. The topological dual of X is $X' = \mathcal{B}(X, \mathbb{C}) \subseteq X^*$, the set of bounded linear functionals as per Definition 2.2.12 and Definition 2.2.10.*

Note also that by Theorem 2.3, X' is the space of continuous linear functionals on X . In addition, we show that a bounded operator defined on a dense subspace extends uniquely to the whole space.

Proposition 2.2.14 (Extension of Bounded Operators). *Let X be a normed space, let Y be a Banach space as in Definition 1.6.9, and let $W \subseteq X$ be a dense subspace as per Definition 2.2.8. If $T \in \mathcal{B}(W, Y)$ as per Definition 2.2.10, there exists a unique $\tilde{T} \in \mathcal{B}(X, Y)$ such that $\tilde{T}|_W = T$ and $\|\tilde{T}\| = \|T\|$.*

A complete proof can be found in [Mor18, Proposition 2.47]. Now, we turn to a more strict class of bounded operators that will be used later.

Definition 2.2.15 (Compact operators). *An operator $A \in \mathcal{B}(\mathcal{H})$, as per Definition 2.2.10, is compact if it is a norm limit of finite-rank operators.*

It can be showed that compact operators form a closed two-sided ideal $\mathfrak{B}_0(\mathcal{H}) \subset \mathcal{B}(\mathcal{H})$. In addition, in finite dimension $\mathfrak{B}_0(\mathcal{H}) = \mathcal{B}(\mathcal{H})$. A more detailed discussion can be found in [Mor18, Chapter 4].

Orthogonal Decomposition of an Hilbert space

Two vectors in a Hilbert space are orthogonal if their inner product vanishes. Every closed subspace splits $\mathcal{H}$ into two orthogonal pieces, which yields the Riesz representation theorem for continuous linear functionals.

Definition 2.2.16 (Orthogonal Complement). *Let $\mathcal{H}$ be a Hilbert space as per Definition 2.2.6 and let $K \subseteq \mathcal{H}$. The orthogonal complement of K is*

$$K^\perp = \{v \in \mathcal{H} \mid (v, w) = 0 \text{ for every } w \in K\}. \quad (2.4)$$

The orthogonal complement of a closed subspace K splits $\mathcal{H}$ in two components, namely K and $K^\perp$.

Theorem 2.4 (Orthogonal Decomposition). *Let $\mathcal{H}$ be a Hilbert space as per Definition 2.2.6 and let $K \subseteq \mathcal{H}$ be a closed linear subspace. Every $v \in \mathcal{H}$ admits a unique decomposition $v = p + q$, with $p \in K$ and $q \in K^\perp$ as per Definition 2.2.16. Consequently $\mathcal{H} = K \oplus K^\perp$, and $K^\perp$ is nontrivial whenever K is a proper subspace of $\mathcal{H}$.*

The proof of this theorem is available in [Mor18, Theorem 3.13]. We recall a particular class of operators that project any vector to a particular subspace of $\mathcal{H}$. Among all these operators, we will refer to the orthogonal ones with just the term projector.

Definition 2.2.17 (Projector). *Let $\mathcal{H}$ be a Hilbert space as in Definition 2.2.6. A projector on $\mathcal{H}$, also called an orthogonal projection, is a bounded operator $P \in \mathcal{B}(\mathcal{H})$ satisfying*

$$P^2 = P, \quad P^* = P. \quad (2.5)$$

Rank one projectors arise from a single unit vector, rather than from a higher dimensional closed subspace. Hence, we state here the definition that will be used later.

Definition 2.2.18 (One-dimensional projector). *Let $\mathcal{H}$ be a Hilbert space as in Definition 2.2.6 and let $\Psi \in \mathcal{H}$ be a unit vector, $\|\Psi\| = 1$. The one-dimensional projector, or rank-one projector, associated to Ψ is the projector P_Ψ , as per Definition 2.2.17, defined by*

$$P_\Psi \Phi := (\Psi, \Phi) \Psi, \quad \forall \Phi \in \mathcal{H}. \quad (2.6)$$

Its range $P_\Psi \mathcal{H}$ is the one dimensional subspace $\mathbb{C}\Psi$ spanned by Ψ .

Remark 2.2.19. The operator P_Ψ of Definition 2.2.18 is indeed a projector, as per Definition 2.2.17. It is self-adjoint, since

$$(\Phi_1, P_\Psi \Phi_2) = (\Psi, \Phi_2)(\Phi_1, \Psi) = (\Phi_1, \Psi)(\Psi, \Phi_2) = (P_\Psi \Phi_1, \Phi_2), \quad \forall \Phi_1, \Phi_2 \in \mathcal{H}, \quad (2.7)$$

using conjugate-linearity of the inner product in its first argument together with $\overline{(\Psi, \Phi_1)} = (\Phi_1, \Psi)$. It is idempotent, since $\|\Psi\|^2 = (\Psi, \Psi) = 1$ gives

$$P_\Psi^2 \Phi = (\Psi, \Phi) P_\Psi \Psi = (\Psi, \Phi)(\Psi, \Psi) \Psi = (\Psi, \Phi) \Psi = P_\Psi \Phi, \quad \forall \Phi \in \mathcal{H}. \quad (2.8)$$

Continuing our discussion, a continuous linear functional F has kernel $\text{Ker } F$, a closed hyperplane. Its orthogonal complement produces the vector representing F . This is the content of the Riesz Theorem.

Theorem 2.5 (Riesz Representation Theorem). *Let $\mathcal{H}$ be a Hilbert space as per Definition 2.2.6 and let $\mathcal{H}'$ be its topological dual as per Definition 2.2.13. For every $F \in \mathcal{H}'$ there exists a unique vector $v_F \in \mathcal{H}$ such that*

$$Fw = (v_F, w), \quad (2.9)$$

for every $w \in \mathcal{H}$. The map $F \mapsto v_F$ is a conjugate-linear isometry from $\mathcal{H}'$ onto $\mathcal{H}$.

This is proved in [Mor18, Theorem 3.16]. We propose an example showing an application of Riesz Theorem 2.5.

Example 2.2.20. On $\mathcal{H} = \ell^2$, fix $a \in \ell^2$ and define $Fw = \sum_{k=1}^{\infty} \bar{a}_k w_k$. Then F is a bounded linear functional with $\|F\| = \|a\|$, and $Fw = (a, w)$, so the Riesz vector representing F is a itself.

The Riesz theorem yields the adjoint of a bounded operator, defined in the next section.

The Adjoint Operation and Unbounded Operators

The adjoint of a bounded operator follows from the Riesz representation Theorem 2.5, applied to the functional $\psi \mapsto (\varphi, T\psi)$ for fixed φ . Although we will not show the proof, the existence of T^* follows directly from the Riesz representation theorem. In fact, it guarantees that the bounded functional $\psi \mapsto (\varphi, T\psi)$ is uniquely represented as $(T^*\varphi, \psi)$ for some vector $T^*\varphi$. This unique representability is precisely what ensures the existence and well definedness of the adjoint operator T^* .

Proposition 2.2.21 (Adjoint of Bounded Operators). *Let $\mathcal{H}_1, \mathcal{H}_2$ be Hilbert spaces as per Definition 2.2.6 and let $T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$ as per Definition 2.2.10. There exists a unique operator $T^* : \mathcal{H}_2 \rightarrow \mathcal{H}_1$ such that*

$$(\varphi, T\psi)_{\mathcal{H}_2} = (T^*\varphi, \psi)_{\mathcal{H}_1}, \quad (2.10)$$

for every $\psi \in \mathcal{H}_1$ and $\varphi \in \mathcal{H}_2$. Moreover $T^* \in \mathcal{B}(\mathcal{H}_2, \mathcal{H}_1)$ and $\|T^*\| = \|T\|$.

This statement is proved in [Mor18, Proposition 3.38]. Comparing T with T^* singles out several classes of operators, generalizing normal, Hermitian, and unitary matrices.

Definition 2.2.22 (Special Classes of Bounded Operators). *Let $\mathcal{H}$ be a Hilbert space as per Definition 2.2.6 and let $T \in \mathcal{B}(\mathcal{H})$ as per Definition 2.2.10, with T^* as per Proposition 2.2.21. The operator T is called*

1. *normal if $TT^* = T^*T$,*
2. *selfadjoint if $T = T^*$,*
3. *isometric if $\|T\psi\| = \|\psi\|$ for every $\psi \in \mathcal{H}$,*
4. *unitary if it is isometric and surjective,*
5. *positive, written $T \geq 0$, if $(\psi, T\psi) \geq 0$ for every $\psi \in \mathcal{H}$.*

Remark 2.2.23. Having set Definition 2.2.22, observe the following.

- T is isometric when $T^*T = I$, with I the identity on $\mathcal{H}$.
- T is unitary when $T^*T = TT^* = I$.
- Every positive operator is selfadjoint, and admits a unique positive square root.

Example 2.2.24. The shift S of Example 2.2.11 is isometric but not unitary, since no sequence with first entry 1 lies in its range.

Bounded operators are defined on the whole space. Position and momentum operators, for example, are not since they satisfy no bound $\|T\psi\| \leq K\|\psi\|$ on their domain. We define such operators as unbounded operators which are densely defined. We will see that we can extend the concept of adjoint also to this other class of operators.

Definition 2.2.25 (Unbounded Operator). *Let $\mathcal{H}$ be a Hilbert space as per Definition 2.2.6. An unbounded operator on $\mathcal{H}$ is a linear map $T : D(T) \rightarrow \mathcal{H}$, where $D(T)$ is a subspace of $\mathcal{H}$ on which no bound as per Definition 2.2.10 is assumed. The graph of T is*

$$G(T) = \{(\psi, T\psi) \in \mathcal{H} \times \mathcal{H} \mid \psi \in D(T)\}. \quad (2.11)$$

The unique existence of the adjoint of an unbounded operator requires $D(T)$ to be dense.

Definition 2.2.26 (Adjoint of Unbounded Operators). *Let $T : D(T) \rightarrow \mathcal{H}$ be an unbounded operator as per Definition 2.2.25 with $D(T)$ dense in $\mathcal{H}$ as per Definition 2.2.8. The adjoint T^* has domain*

$$D(T^*) = \{\varphi \in \mathcal{H} \mid \exists \omega_\varphi \in \mathcal{H} \text{ such that } (\varphi, T\psi) = (\omega_\varphi, \psi) \forall \psi \in D(T)\}, \quad (2.12)$$

and for $\varphi \in D(T^*)$ the value $T^*\varphi$ is set equal to ω_φ .

Comparing T with T^* yields the unbounded analogues of the bounded case presented in Proposition 2.2.21.

Definition 2.2.27 (Hermitian Operator). *Let $T : D(T) \rightarrow \mathcal{H}$ be an unbounded operator as per Definition 2.2.25. The operator T is Hermitian if*

$$(\varphi, T\psi) = (T\varphi, \psi), \quad (2.13)$$

for all $\varphi, \psi \in D(T)$.

Hermiticity alone says nothing about the size of the domain $D(T)$, which may fail to be dense. Adding this density requirement singles out the class for which the adjoint itself is guaranteed to be well defined, as the following definition makes precise.

Definition 2.2.28 (Symmetric Operator). *A Hermitian operator T as per Definition 2.2.27 whose domain $D(T)$ is dense in $\mathcal{H}$ as per Definition 2.2.8 is called symmetric.*

Symmetry makes T^* well defined.

Definition 2.2.29 (Selfadjoint Operator). *A symmetric operator T as per Definition 2.2.28 is selfadjoint if $D(T) = D(T^*)$ and $T = T^*$ on this common domain, with T^* as per Definition 2.2.26.*

A symmetric operator may still be enlarged to a Selfadjoint one.

Definition 2.2.30 (Essentially Selfadjoint Operator). *A symmetric operator T as per Definition 2.2.28 is essentially selfadjoint if $D(T^*)$ is dense in $\mathcal{H}$ as per Definition 2.2.8 and $T^{**} = T^*$.*

The distinction between symmetric, selfadjoint, and essentially selfadjoint operators concerned the interplay of T with its own adjoint. A further classification, familiar from the bounded case, focuses on whether T and T^* commute with each other.

Definition 2.2.31 (Normal Operator, Unbounded Case). *An unbounded operator T as per Definition 2.2.25 with dense domain is normal if $T^*T = TT^*$ on the common domain where both sides are defined.*

Another consequence of the Riesz representation theorem is the Hellinger–Toeplitz theorem. It shows that any everywhere-defined symmetric operator on a Hilbert space is automatically bounded. This theorem thus justifies why the unbounded operators that arise in quantum mechanics, such as position and momentum, can never be defined on the entire Hilbert space but only on dense, proper subspaces.

Theorem 2.6 (Hellinger–Toeplitz Theorem). *Let $\mathcal{H}$ be a Hilbert space as per Definition 2.2.6 and let $T : \mathcal{H} \rightarrow \mathcal{H}$ be a Hermitian operator as per Definition 2.2.27 defined on the whole space $\mathcal{H}$. Then T is bounded and selfadjoint as per Definition 2.2.10 and Definition 2.2.29.*

A complete proof can be found in [Mor18, Theorem 5.14]. Symmetric operators that are not selfadjoint exist only on proper dense subspaces.

Example 2.2.32. On $\mathcal{H} = L^2(\mathbb{R})$, $(Q\psi)(x) = x\psi(x)$ is Hermitian on $D(Q) = \{\psi \in L^2(\mathbb{R}) \mid x\psi(x) \in L^2(\mathbb{R})\}$, which is dense. Hence Q is symmetric, and in fact Selfadjoint, since Q^* has the same domain and action.

Within this framework, we introduce the notion of a cyclic vectors. They provide a structural bridge between the abstract operator and its spectral realization. A vector $\psi \in \mathcal{H}$ is defined cyclic for an operator T if the linear span of $\{T^n\psi \mid n \geq 0\}$ is dense in $\mathcal{H}$. For unbounded operators, the existence of cyclic vectors guarantees that T is unitarily equivalent to a multiplication operator on an L^2 -space. Thus, although an unbounded operator cannot be defined everywhere, a single cyclic vector captures all its spectral data.

Definition 2.2.33 (Cyclic Vector for an Operator). *Let $\mathcal{H}$ be a Hilbert space as per Definition 2.2.6 and let $T \in \mathcal{B}(\mathcal{H})$ as per Definition 2.2.10. A vector $v \in \mathcal{H}$ is cyclic for T if the linear span of $\{T^n v \mid n \geq 0\}$ is dense in $\mathcal{H}$ as per Definition 2.2.8.*

A single cyclic vector need not exist for an arbitrary bounded operator, but the notion becomes considerably more flexible once a whole family of operators is allowed to act jointly.

Definition 2.2.34 (Cyclic Vector for a Family of Operators). Let $\mathcal{H}$ be a Hilbert space as per Definition 2.2.6 and let $\mathcal{F} \subseteq \mathcal{B}(\mathcal{H})$ be a collection of bounded operators as per Definition 2.2.10. A vector $v \in \mathcal{H}$ is cyclic for $\mathcal{F}$ if the linear span of $\{Av \mid A \in \mathcal{F}\}$ is dense in $\mathcal{H}$ as per Definition 2.2.8. Taking $\mathcal{F} = \{T^n \mid n \geq 0\}$ one recovers Definition 2.2.33.

Example 2.2.35. On $\mathcal{H} = L^2([0, 1])$, with $(T\psi)(x) = x\psi(x)$, the constant function $v = 1$ is cyclic for T as per Definition 2.2.33, since $\{T^n v \mid n \geq 0\}$ spans the polynomials, which is dense in $L^2([0, 1])$ by the Weierstrass approximation theorem.

2.3 The Gelfand-Naimark-Segal Construction

The operational construction presented in Section 2.1 led to the conclusion that quantum observables can be embedded inside a unital non-commutative C^* -algebra. However, such an object carries no Hilbert space with it, so it is not obvious how this new formalism connects with the Dirac-Von Neumann axiomatization. The Gelfand-Naimark-Segal construction shows that every state on a C^* -algebra $\mathfrak{A}$ determines a Hilbert space $\mathcal{H}_\omega$ and a representation π_ω of $\mathfrak{A}$ as bounded operators $\mathcal{B}(\mathcal{H}_\omega)$, in which the state itself is reconstructed as the expectation value in a particular cyclic vector. This can be seen as the non-commutative counterpart of the Gelfand construction used in Chapter 1 for the classical case.

Before stating the definition of a representation, we isolate the class of maps between $*$ -algebras that preserve all the algebraic relations.

Definition 2.3.1 ($*$ -Homomorphism, $*$ -isomorphism, $*$ -automorphism). Let $\mathfrak{A}$ and $\mathfrak{B}$ be two $*$ -algebras with identity, as per Definition 1.6.2. A $*$ -homomorphism $\pi : \mathfrak{A} \rightarrow \mathfrak{B}$ is a map preserving all the algebraic relations of the $*$ -algebra structure: It is linear, multiplicative and $*$ -preserving, that is $\pi(A^*) = \pi(A)^*$ for every $A \in \mathfrak{A}$. A bijective $*$ -homomorphism is called a $*$ -isomorphism, and a $*$ -isomorphism of $\mathfrak{A}$ onto itself is called a $*$ -automorphism.

A $*$ -homomorphism gains importance once the target algebra $\mathfrak{B}$ is taken to be that of bounded operators on a Hilbert space.

Definition 2.3.2 (Representation, faithful and irreducible representation). Let $\mathfrak{A}$ be a C^* -algebra, as per Definition 1.6.11, and let $\mathcal{H}$ be a Hilbert space as in Definition 2.2.6. A representation π of $\mathfrak{A}$ on $\mathcal{H}$ is a $*$ -homomorphism, as per Definition 2.3.1, of $\mathfrak{A}$ into the C^* -algebra $\mathcal{B}(\mathcal{H})$ of bounded operators on $\mathcal{H}$. The representation π is called faithful when $\ker \pi = \{0\}$, and it is called irreducible when the only closed subspaces of $\mathcal{H}$ invariant under $\pi(\mathfrak{A})$ are $\{0\}$ and $\mathcal{H}$ itself.

Having established representations, we return to the notion of cyclic vectors, introduced in Definition 2.2.34. In fact, note that such vectors serves as a physical reference state from which the entire spectrum of accessible states is generated through the action of the observable algebra. Imposing cyclicity disconnects mathematical subspaces not physically accessible, as shown in the next remark.

Remark 2.3.3 (Cyclic vectors). Recalling Definition 2.2.34, a vector $\Psi \in \mathcal{H}$ is called cyclic for a representation π when $\pi(\mathfrak{A})\Psi$ is dense in $\mathcal{H}$. In an irreducible representation, every non-zero vector is cyclic. In fact, the closure of $\pi(\mathfrak{A})\Psi$ is, by construction, a closed subspace invariant under $\pi(\mathfrak{A})$. Therefore, irreducibility forces it to coincide either with $\{0\}$ or with the whole of $\mathcal{H}$, but it cannot be $\{0\}$ because it contains at least $\Psi = \pi(1)\Psi$.

A representation π on a Hilbert space $\mathcal{H}$ together with a chosen cyclic vector Ψ is denoted by the triple $(\mathcal{H}, \pi, \Psi)$. Before presenting the key result of this section, we show that a $*$ -homomorphism can only decrease norms, becoming norm preserving when it is injective.

Proposition 2.3.4. Let $\mathfrak{A}$ and $\mathfrak{B}$ be two unital C^* -algebras as in Definition 1.6.13 and let $\pi : \mathfrak{A} \rightarrow \mathfrak{B}$ be a $*$ -homomorphism as in Definition 2.3.1. Then

$$\|\pi(A)\|_{\mathfrak{B}} \leq \|A\|_{\mathfrak{A}}, \quad \forall A \in \mathfrak{A}. \quad (2.14)$$

If π is a $*$ -isomorphism, equality holds.

The proof of this statement can be found in [Str05, Proposition 2.2.3]. The norm bound stated above indicates that any representation of a C^* -algebra must be realized by bounded operators. The theorem stated next, due to Gelfand and Naimark and later reformulated by Segal, shows that this quest always succeeds.

Theorem 2.7 (Gelfand-Naimark-Segal). *Let $\mathfrak{A}$ be a unital C^* -algebra, as in Definition 1.6.13, and let ω be a state on $\mathfrak{A}$, as per Definition 2.1.3. There exist a Hilbert space $\mathcal{H}_\omega$ and a representation $\pi_\omega : \mathfrak{A} \rightarrow \mathcal{B}(\mathcal{H}_\omega)$, as per Definition 2.3.2, such that*

- i) $\mathcal{H}_\omega$ contains a cyclic vector Ψ_ω , as per Definition 2.2.34,
- ii) $\omega(A) = (\Psi_\omega, \pi_\omega(A)\Psi_\omega)$ for every $A \in \mathfrak{A}$,
- iii) every other representation $(\mathcal{H}, \pi, \Psi)$ such that

$$\omega(A) = (\Psi, \pi(A)\Psi), \quad \forall A \in \mathfrak{A}, \quad (2.15)$$

is unitarily equivalent to π_ω : There exists an isometry $U : \mathcal{H}_\pi \rightarrow \mathcal{H}_\omega$ such that

$$U\pi(A)U^{-1} = \pi_\omega(A), \quad U\Psi = \Psi_\omega. \quad (2.16)$$

Proof. Part i) The state ω turns the algebra $\mathfrak{A}$, regarded as a vector space, into a space with a semidefinite inner product, setting

$$(A, B) := \omega(A^*B), \quad A, B \in \mathfrak{A}. \quad (2.17)$$

Positivity of ω yields $(A, A) = \omega(A^*A) \geq 0$ for every $A \in \mathfrak{A}$, while sesquilinearity of Equation (2.17) follows directly from antilinearity of ω and of the involution.

Consider the set

$$J := \{A \in \mathfrak{A} \mid \omega(B^*A) = 0, \forall B \in \mathfrak{A}\}. \quad (2.18)$$

By definition, $A \in J$ when A is orthogonal to the whole of $\mathfrak{A}$ with respect to $(\cdot, \cdot)$, and taking $B = A$ shows in particular that $(A, A) = \omega(A^*A) = 0$. That makes J the set of vanishing vectors of the semidefinite inner product as in Equation (2.17). We claim that J is a left ideal of $\mathfrak{A}$.

Recall that a left ideal of an algebra $\mathfrak{A}$ is a linear subspace $J \subseteq \mathfrak{A}$ that absorbs left multiplication by an arbitrary element of the algebra $\mathfrak{A}J \subseteq J$. That means that $CA \in J \forall C \in \mathfrak{A}$ and $\forall A \in J$. To verify this property for the set in Equation (2.18), take $B, C \in \mathfrak{A}$ and $A \in J$. The Cauchy-Schwarz inequality associated with the semidefinite inner product from Equation (2.17) yields

$$|\omega(B^*(CA))| = |(B, CA)| = |(C^*B, A)| \leq (C^*B, C^*B)^{1/2}(A, A)^{1/2} = 0, \quad (2.19)$$

where the second equality regroups $\omega(B^*CA) = \omega((C^*B)^*A)$, while the right-hand side vanishes because $(A, A) = 0$, as $A \in J$. Since $B \in \mathfrak{A}$ is arbitrary, this shows $CA \in J$, which proves $\mathfrak{A}J \subseteq J$. Being a left ideal is the algebraic property that allows multiplication by a fixed element of $\mathfrak{A}$ to be defined consistently on equivalence classes modulo J .

We can therefore consider the quotient vector space $\mathfrak{A}/J$, whose elements are the equivalence classes $[A] = \{A+B : B \in J\}$, for $A \in \mathfrak{A}$. The inner product from Equation (2.17) descends to a well defined, strictly positive inner product on $\mathfrak{A}/J$. That is, because the value $\omega(A^*B)$ is unchanged when A or B is modified by an element of J , since J is the set of vanishing vectors identified above. It is strictly positive because a class $[A]$ with $([A], [A]) = 0$ has $A \in J$ by definition, that is, $[A] = [0]$. The completion of $\mathfrak{A}/J$ with respect to the norm induced by this inner product is a Hilbert space, denoted $\mathcal{H}_\omega$.

To each $A \in \mathfrak{A}$ we associate an operator denoted by $\pi_\omega(A)$ acting on $\mathcal{H}_\omega$, defined on the dense subspace $\mathfrak{A}/J$ by left multiplication,

$$\pi_\omega(A)[B] := [AB], \quad [B] \in \mathfrak{A}/J. \quad (2.20)$$

If $[B] = [C]$, that is, $B - C \in J$, then $A(B - C) = AB - AC$ belongs to J as well, because J absorbs left multiplication by A . Hence $[AB] = [AC]$.

The operator $\pi_\omega(A)$ is bounded on $\mathfrak{A}/J$, with norm controlled by $\|A\|$. In fact, for every $B \in \mathfrak{A}$,

$$\|\pi_\omega(A)[B]\|^2 = ([AB], [AB]) = \omega(B^*A^*AB) =: \omega_B(A^*A), \quad (2.21)$$

where $\omega_B(X) := \omega(B^*XB)$. The functional ω_B is itself positive and therefore continuous, as per [Str05, Prop. 1.6.3, Appendix C], and it satisfies $\omega_B(\mathbb{1}) = \omega(B^*B) = \|[B]\|^2$. Hence continuity of ω_B at the identity yields

$$\omega_B(A^*A) \leq \omega_B(\mathbb{1}) \|A\|^2 = \|[B]\|^2 \|A\|^2. \quad (2.22)$$

Therefore $\|\pi_\omega(A)[B]\| \leq \|A\| \|[B]\|$ for every $[B] \in \mathfrak{A}/J$. Hence $\pi_\omega(A)$ extends by continuity to a bounded operator on the whole of $\mathcal{H}_\omega$ by Theorem 2.2.14, still denoted $\pi_\omega(A)$, with

$$\|\pi_\omega(A)\| \leq \|A\|. \quad (2.23)$$

Linearity of π_ω in A is immediate from Equation (2.20) and linearity of the product in $\mathfrak{A}$. Multiplicativity follows from associativity of the product, since

$$\pi_\omega(A)\pi_\omega(A')[B] = \pi_\omega(A)[A'B] = [AA'B] = \pi_\omega(AA')[B],$$

for all $A, A', B \in \mathfrak{A}$. The identity $\pi_\omega(A)^*[B] = \pi_\omega(A^*)[B]$ holds because $([AB], [C]) = \omega(B^*A^*C) = ([B], [A^*C])$ for all $B, C \in \mathfrak{A}$. Together with Equation (2.23), this shows that π_ω is a representation of $\mathfrak{A}$ on $\mathcal{H}_\omega$, as per Definition 2.3.2.

We now identify the vector $\Psi_\omega := [\mathbb{1}] \in \mathcal{H}_\omega$. By Equation (2.20), $\pi_\omega(A)\Psi_\omega = [A\mathbb{1}] = [A]$ for every $A \in \mathfrak{A}$, so the orbit $\pi_\omega(\mathfrak{A})\Psi_\omega$ coincides with the whole of $\mathfrak{A}/J$, which is dense in $\mathcal{H}_\omega$ by construction: Ψ_ω is a cyclic vector, proving part i).

Part ii) It follows from the same computation together with Equation (2.17) that

$$(\Psi_\omega, \pi_\omega(A)\Psi_\omega) = ([\mathbb{1}], [A]) = \omega(\mathbb{1}^*A) = \omega(A). \quad (2.24)$$

Part iii) Let π be a representation on a Hilbert space $\mathcal{H}_\pi$ with a cyclic vector Ψ satisfying Equation (2.15). Define a map on the dense subspace $\pi_\omega(\mathfrak{A})\Psi_\omega = \mathfrak{A}/J$ of $\mathcal{H}_\omega$ by

$$U^{-1}\pi_\omega(A)\Psi_\omega := \pi(A)\Psi, \quad A \in \mathfrak{A}. \quad (2.25)$$

This assignment is well defined on $\mathfrak{A}/J$. If $[A] = [A']$, that is, $A - A' \in J$, then in particular $\omega((A - A')^*(A - A')) = 0$. Taking $B = A - A'$ in Equation (2.18), and applying Equation (2.15) to the element $(A - A')^*(A - A') \in \mathfrak{A}$ yields

$$\|\pi(A)\Psi - \pi(A')\Psi\|^2 = (\Psi, \pi((A - A')^*(A - A'))\Psi) = \omega((A - A')^*(A - A')) = 0, \quad (2.26)$$

so $\pi(A)\Psi = \pi(A')\Psi$, as required. The same computation, applied to A^*A rather than to a difference, shows that U^{-1} is norm preserving on $\mathfrak{A}/J$,

$$\|\pi(A)\Psi\|^2 = (\Psi, \pi(A^*A)\Psi) = \omega(A^*A) = ([A], [A]) = \|A\|^2. \quad (2.27)$$

Since Ψ is cyclic for π , the range of U^{-1} is dense in $\mathcal{H}_\pi$, and U^{-1} is an isometry defined on a dense subspace of $\mathcal{H}_\omega$. Therefore it extends uniquely to a unitary map $\mathcal{H}_\omega \rightarrow \mathcal{H}_\pi$, whose inverse is the required isometry $U : \mathcal{H}_\pi \rightarrow \mathcal{H}_\omega$. Equation (2.16) holds by construction on the dense subspace $\pi_\omega(\mathfrak{A})\Psi_\omega$, since

$$U^{-1}\pi_\omega(A')\pi_\omega(A)\Psi_\omega = U^{-1}\pi_\omega(A'A)\Psi_\omega = \pi(A'A)\Psi = \pi(A')\pi(A)\Psi = \pi(A')U^{-1}\pi_\omega(A)\Psi_\omega, \quad (2.28)$$

for all $A, A' \in \mathfrak{A}$. Furthermore it extends by continuity to the whole of $\mathcal{H}_\omega$. Taking $A = \mathbb{1}$ in the definition of U^{-1} yields $U^{-1}\Psi_\omega = \Psi$, that is, $U\Psi = \Psi_\omega$, which completes the proof. $\square$

Remark 2.3.5 (Significance of the GNS construction). The GNS construction reduces the existence of Hilbert space representations of a C^* -algebra to that of states on it. That is guaranteed in general by [Str05, Prop. 1.6.4, Appendix C]. In addition, GNS Theorem 2.7 shows the dependence of two postulates of the Dirac-von Neumann axiomatization, namely that observables are represented by operators and that states are represented by vectors in a Hilbert space.

This theorem shows that cyclic representations are the basic building blocks of the representation theory of a C^* -algebra. In addition, the GNS construction of Theorem 2.7 is what produces it practically.

Remark 2.3.6 (Brief summary of the GNS construction). Given a state ω on a C^* -algebra $\mathfrak{A}$, as per Definition 2.1.3, the proof of Theorem 2.7 produces the pair $(\mathcal{H}_\omega, \pi_\omega)$ through the following steps.

1. Define the sesquilinear form $(A, B) := \omega(A^*B)$ on $\mathfrak{A}$, positive semidefinite because ω is a state, as per Equation (2.17).
2. Its null space $J = \{A \in \mathfrak{A} : \omega(B^*A) = 0, \forall B \in \mathfrak{A}\}$, as per Equation (2.18), coincides with the set $\{A \in \mathfrak{A} : \omega(A^*A) = 0\}$ on account of the Cauchy-Schwarz inequality for $(\cdot, \cdot)$. Furthermore it is a left ideal of $\mathfrak{A}$.
3. The form $(\cdot, \cdot)$ descends to a inner product on the quotient $\mathfrak{A}/J$, whose completion is the Hilbert space $\mathcal{H}_\omega$.
4. The representation π_ω is defined on the dense subspace $\mathfrak{A}/J \subset \mathcal{H}_\omega$ by left multiplication, $\pi_\omega(A)[B] := [AB]$, as per Equation (2.20). It is extended to all of $\mathcal{H}_\omega$ by continuity.
5. The cyclic vector is $\Psi_\omega := [1]$, and it reproduces the state, $(\Psi_\omega, \pi_\omega(A)\Psi_\omega) = \omega(A)$ for every $A \in \mathfrak{A}$.

Part iii) of Theorem 2.7 compares an arbitrary representation with the specific GNS representation of the state it reproduces. The same argument, applied twice, yields a direct comparison between any two cyclic representations that reproduce the same expectation values on their respective cyclic vectors.

Corollary 2.3.7. *Let π_1, π_2 be representations of a C^* -algebra $\mathfrak{A}$ on Hilbert spaces $\mathcal{H}_1, \mathcal{H}_2$, as per Definition 2.3.2, with cyclic vectors $\Psi_1 \in \mathcal{H}_1, \Psi_2 \in \mathcal{H}_2$ respectively, as in Definition 2.2.34. If*

$$(\Psi_1, \pi_1(A)\Psi_1) = (\Psi_2, \pi_2(A)\Psi_2), \quad \forall A \in \mathfrak{A}, \quad (2.29)$$

then π_1 and π_2 are equivalent, as per Definition 1.2.23.

Proof. Denote by ω the common state $A \mapsto (\Psi_1, \pi_1(A)\Psi_1) = (\Psi_2, \pi_2(A)\Psi_2)$. By Theorem 2.7 iii), applied once to π_1 and once to π_2 , both representations are equivalent to the GNS representation π_ω defined by ω . Composing the two unitaries obtained this way yields a unitary between $\mathcal{H}_1$ and $\mathcal{H}_2$ intertwining π_1 and π_2 , so π_1 and π_2 are equivalent. $\square$

Pure and Mixed States

The construction above produces many more states than the one it started from. In fact, every vector of $\mathcal{H}_\omega$, not only the distinguished cyclic vector Ψ_ω , determines a state on $\mathfrak{A}$ through the representation π_ω . This observation is specified by the following result.

Proposition 2.3.8. *Let ω be a state on a C^* -algebra $\mathfrak{A}$, as in Definition 2.1.3, and let $(\mathcal{H}_\omega, \pi_\omega, \Psi_\omega)$ be its GNS representation, as per Theorem 2.7. For every $\Phi \in \mathcal{H}_\omega$, the functional $\varphi : \mathfrak{A} \rightarrow \mathbb{C}$ defined via*

$$\varphi(A) := (\Phi, \pi_\omega(A)\Phi), \quad (2.30)$$

where $A \in \mathfrak{A}$, is a positive linear functional on $\mathfrak{A}$.

Proof. The functional φ is positive because $\varphi(A^*A) = \|\pi_\omega(A)\Phi\|^2 \geq 0$ for every $A \in \mathfrak{A}$, and it is linear because π_ω is linear together with linearity of the inner product in its second argument. $\square$

The next definition fixes the terminology used throughout the section to refer to such states.

Definition 2.3.9 (State vector). *Let ω be a state on a C^* -algebra $\mathfrak{A}$, as in Definition 2.1.3, and let $(\mathcal{H}_\omega, \pi_\omega, \Psi_\omega)$ be its GNS representation, as per Theorem 2.7. A vector $\Phi \in \mathcal{H}_\omega$ which defines a state as in Proposition 2.3.8 is called a state vector.*

Remark 2.3.10 (Physical Interpretation of State vectors). Since, by the GNS construction, vectors of the form $\pi_\omega(A)\Psi_\omega$, for $A \in \mathfrak{A}$, are dense in $\mathcal{H}_\omega$, the state vectors obtained by acting with $\pi_\omega(\mathfrak{A})$ on Ψ_ω have the physical interpretation of states reached from ω through a physically realizable operation on the observables. If $\Phi \in \mathcal{H}_\omega$ is itself cyclic for π_ω , the representation $(\mathcal{H}_\omega, \pi_\omega, \Phi)$ is unitarily equivalent, as per Theorem 2.7 iii), to the GNS representation defined by φ .

In addition, distinct states on $\mathfrak{A}$ can be combined into new ones, simply by taking convex combinations of them. This operation identifies, among all states, those that describe a definite physical situation rather than a statistical combination of several ones.

Definition 2.3.11 (Mixture and pure state). Given two states ω_1, ω_2 on a C^* -algebra $\mathfrak{A}$, as in Definition 2.1.3, the convex combination

$$\omega := \lambda\omega_1 + (1 - \lambda)\omega_2, \quad 0 < \lambda < 1, \quad (2.31)$$

is again a state on $\mathfrak{A}$, called a mixture of ω_1 and ω_2 , or a mixed state. A state ω is called a pure state when it cannot be written as a convex combination Equation (2.31) of other states. The set of all pure states of a C^* -algebra $\mathfrak{A}$ is denoted as $\mathcal{P}(\mathfrak{A})$.

These definitions correspond to those of the classical case expressed in Definition 1.6.20. We can state an equivalent definition of pure states using the following operation.

Definition 2.3.12. A state ω on a C^* -algebra $\mathfrak{A}$, as in Definition 2.1.3, is said to majorize a positive linear functional ϕ on $\mathfrak{A}$ when $\omega - \phi$ is itself a positive linear functional.

Using Definition 2.3.12, we can state equivalently that a state ω is called pure if the only positive linear functionals majorized by ω are of the form $\lambda\omega$, with $0 < \lambda < 1$.

The dichotomy between pure and mixed states has a counterpart at the level of GNS representations. In fact, purity of the state that generates the representation controls whether the representation itself is irreducible, as per Definition 2.3.2.

Proposition 2.3.13. The GNS representation, as per Theorem 2.7, defined by a state ω on a C^* -algebra $\mathfrak{A}$ is irreducible if and only if ω is a pure state, as per Definition 2.3.11.

For a complete proof see [Str05, Appendix D, Proposition 2.6.2]. A mixed state can therefore never be represented by a state vector, as per Definition 2.3.9, of an irreducible representation.

Mixed states admit a concrete description inside an irreducible representation. The next result specifies this construction. It shows that it always yields a positive linear functional on $\mathfrak{A}$.

Proposition 2.3.14. Let π be an irreducible representation of a C^* -algebra $\mathfrak{A}$ on a Hilbert space $\mathcal{H}$ as in Definition 2.3.2. Given orthonormal vectors $\Psi_i \in \mathcal{H}$, $i = 1, \dots, N$, with $P_i \in \mathcal{B}(\mathcal{H})$ the corresponding one-dimensional orthogonal projections as in Definition 2.2.18, let

$$\rho := \sum_{i=1}^N \lambda_i P_i, \quad \text{with } \lambda_i \geq 0, \quad \sum_{i=1}^N \lambda_i = 1. \quad (2.32)$$

Then the functional

$$\omega_\rho(A) := \text{Tr}(\rho \pi(A)), \quad A \in \mathfrak{A}, \quad (2.33)$$

defines a state on $\mathfrak{A}$.

Proof. Complete the orthonormal set $\{\Psi_i\}_{i=1}^N$ to a complete orthonormal set $\{\Psi_n\}_{n=1}^d$ of $\mathcal{H}$, with $d \in \overline{\mathbb{R}}$ being the dimensions of $\mathcal{H}$. Expanding the trace in this basis, and using $\rho\Psi_n = \lambda_n\Psi_n$ for $n \leq N$ and $\rho\Psi_n = 0$ for $n > N$, we get

$$\omega_\rho(A) = \sum_{n=1}^d (\Psi_n, \rho \pi(A) \Psi_n) = \sum_{i=1}^N \lambda_i (\Psi_i, \pi(A) \Psi_i) = \sum_{i=1}^N \lambda_i \omega_i(A), \quad (2.34)$$

where $\omega_i(A) := (\Psi_i, \pi(A) \Psi_i)$ is, as per Definition 2.3.9, the state vector associated to Ψ_i . Being a convex combination of the states ω_i , the functional ω_ρ is itself a positive linear functional, and $\omega_\rho(\mathbb{1}) = \sum_i \lambda_i \omega_i(\mathbb{1}) = \sum_i \lambda_i = 1$: ω_ρ is a state on $\mathfrak{A}$. $\square$

The operator ρ used in the previous construction is subject to the two conditions in Equation (2.32). These admit a characterization that does not refer to any particular choice of orthonormal vectors, stated in the next definition.

Definition 2.3.15 (Density matrix, folium of a representation). *An operator $\rho : \mathcal{H} \rightarrow \mathcal{H}$ on a Hilbert space $\mathcal{H}$, as in Equation (2.32), is called a density matrix when it is a positive trace class operator with $\text{Tr } \rho = 1$. Given a representation π of a C^* -algebra $\mathfrak{A}$ on $\mathcal{H}$, the set of states of the form $\omega_\rho(A) = \text{Tr}(\rho \pi(A))$, for ρ a density matrix on $\mathcal{H}$, is called the folium of the representation π .*

Remark 2.3.16. When ρ is a one-dimensional projection, the state ω_ρ of Proposition 2.3.14 coincides with the state vector, as per Definition 2.3.9, associated to the corresponding unit vector. Therefore, state vectors form a special case of the folium of Definition 2.3.15. When $N > 1$, ω_ρ is instead a mixed state, as per Definition 2.3.11, and generally every mixed state can be described by a density matrix.

The notion of faithfulness, already introduced at the level of representations in Definition 2.3.2, has a natural counterpart for states themselves. The two notions are directly related through the GNS construction.

Definition 2.3.17 (Faithful state). *A state ω on a C^* -algebra $\mathfrak{A}$ is called faithful when $\omega(A^*A) > 0$ for every non-zero $A \in \mathfrak{A}$.*

Remark 2.3.18. The GNS representation, as per Theorem 2.7, defined by a faithful state is itself faithful, as per Definition 2.3.2. Indeed, if $\pi_\omega(A) = 0$ then, in particular, $\pi_\omega(A)\Psi_\omega = [A] = [0]$, that is, A belongs to the ideal J of Equation (2.18), so $\omega(A^*A) = 0$, taking $B = A$ in Equation (2.18). Faithfulness of ω then forces $A = 0$, so $\ker \pi_\omega = \{0\}$.

Gelfand-Naimark Theorem for Observables

GNS representation, constructed in Theorem 2.7, can fail to be faithful. In fact, elements of $\mathfrak{A}$ that act differently as abstract algebra elements may be represented by the same, or even by the zero, operator. This would cause a failure of the separation property we presented in Definition 2.1.4. Recovering a faithful realization of $\mathfrak{A}$ as an algebra of operators therefore requires working not with a single state but with a sufficiently large family of them at once. This is the content of the Gelfand-Naimark theorem, which shows that every C^* -algebra is isomorphic to an algebra of bounded operators on some Hilbert space.

Before proving that theorem, we need to state an equivalent definition of a separating state space. Recalling Definition 2.1.4, a state set $\mathcal{S}$ is said to be *separating* when $\omega(A) = \omega(B)$ for all states $\omega \in \mathcal{S}$ implies $A = B$. By linearity and the fundamental properties of the states, this implies that $\omega(A^*A) = 0$ across all states if and only if $A = 0$. Hence, we can state an alternative operational definition more suitable for this framework.

Definition 2.3.19 (Separating property for a family of states). *Given a C^* -algebra $\mathfrak{A}$ as in Definition 1.6.11. A family of states $\mathcal{F}$ is said to separate the algebra if, for every non-zero $A \in \mathfrak{A}$, there exists $\omega \in \mathcal{F}$ such that the represented operator $\pi_\omega(A)$ is not the zero operator.*

Using GNS Theorem 2.7, the operator $\pi_\omega(A)$ vanishes on $\mathcal{H}_\omega$ if and only if the element A lies in the Gelfand left ideal J defined in Equation (2.18), which is the condition $\omega(A^*A) = 0$. Consequently, the two definitions coincide. We are ready to present the non-commutative counterpart of Theorem 1.10.

Theorem 2.8 (Gelfand-Naimark). *Every C^* -algebra $\mathfrak{A}$, as in Definition 1.6.11, is isomorphic to an algebra of bounded operators on a Hilbert space, as per Definition 2.2.6.*

Proof. Using [Str05, Prop. 1.6.5, Appendix C] we can state the existence of a family $\mathcal{F}$ of separating states for every C^* -algebra. That is the state set described in Definition 2.1.22. Fix one such family $\mathcal{F}$, and consider the direct sum of the corresponding GNS representations,

$$\mathcal{H} := \bigoplus_{\omega \in \mathcal{F}} \mathcal{H}_\omega, \quad \pi := \bigoplus_{\omega \in \mathcal{F}} \pi_\omega. \quad (2.35)$$

Hence, $\mathcal{H}$ is the space of families $x = \{x_\omega\}_{\omega \in \mathcal{F}}$, with $x_\omega \in \mathcal{H}_\omega$ for every ω , such that $\sum_\omega \|x_\omega\|^2 < \infty$, equipped with the inner product $(\{x_\omega\}, \{y_\omega\}) := \sum_\omega (x_\omega, y_\omega)$, and π acts on such a family term by term, $[\pi(A)x]_\omega := \pi_\omega(A)x_\omega$.

The inner product on $\mathcal{H}$ is well defined and $\mathcal{H}$ is complete, by construction. In particular, convergence of $\sum_\omega (x_\omega, y_\omega)$ follows from the Cauchy-Schwarz inequality on each summand together with $\sum_\omega \|x_\omega\|^2 < \infty$ and $\sum_\omega \|y_\omega\|^2 < \infty$, and completeness of $\mathcal{H}$ is inherited coordinatewise from the completeness of each $\mathcal{H}_\omega$. The operator $\pi(A)$ is bounded on $\mathcal{H}$, since Proposition 2.3.4 yields $\|\pi_\omega(A)\| \leq \|A\|$ for every $\omega \in \mathcal{F}$, that is

$$\|\pi(A)x\|^2 = \sum_{\omega \in \mathcal{F}} \|\pi_\omega(A)x_\omega\|^2 \leq \|A\|^2 \sum_{\omega \in \mathcal{F}} \|x_\omega\|^2 = \|A\|^2 \|x\|^2. \quad (2.36)$$

Furthermore π inherits linearity, multiplicativity, and the $*$ -preserving property from each π_ω , hence π is a representation of $\mathfrak{A}$ on $\mathcal{H}$, as per Definition 2.3.2.

Since $\mathcal{F}$ separates $\mathfrak{A}$, for every non-zero $A \in \mathfrak{A}$ there exists $\omega \in \mathcal{F}$ with $\pi_\omega(A) \neq 0$, hence $\pi(A) \neq 0$ and the representation π is faithful, $\ker \pi = \{0\}$, as per Definition 2.3.2, hence π is a $*$ -isomorphism of $\mathfrak{A}$ onto the C^* -algebra $\pi(\mathfrak{A}) \subseteq \mathcal{B}(\mathcal{H})$. Applying Proposition 2.3.4 to the $*$ -isomorphism π yields $\|\pi(A)\| = \|A\|$ for every $A \in \mathfrak{A}$. This implies that π is an isometric $*$ -isomorphism of $\mathfrak{A}$ onto an algebra of bounded operators on the Hilbert space $\mathcal{H}$. $\square$

Remark 2.3.20 (Recovering the commutative case). When $\mathfrak{A}$ is commutative, Theorem 2.8 can be specialized to recover the Gelfand characterization of Chapter 1. For a commutative C^* -algebra, irreducible representations coincide with the GNS representations defined by pure states, as per Proposition 2.3.13, and pure states on a commutative C^* -algebra are multiplicative, as per [Str05, Prop. 2.6.3]. Hence each such representation is one-dimensional, $\pi_\omega(A) = \omega(A)\mathbb{1} \ \forall A \in \mathfrak{A}$. Given a family $\mathcal{F}$ of pairwise inequivalent irreducible representations, it coincides with the Gelfand spectrum of $\mathfrak{A}$, as per Definition 1.6.24, and the faithful representation

$$\pi(A) := \bigoplus_{\omega \in \mathcal{F}} \pi_\omega(A) \quad \forall A \in \mathfrak{A}$$

is given by the collection $\{\omega(A) \mid \omega \in \mathcal{F}\}$ for every $A \in \mathfrak{A}$. That is because the correspondent continuous function of the observable A is given by $\omega(A)$, call it $\tilde{A}$. The norm identity of Theorem 2.8 then reads $\|\tilde{A}\|_\infty = \sup_\omega |\tilde{A}(\omega)| = \sup_\omega |\omega(A)| = \|A\|$, and $\mathcal{F}$ is a compact Hausdorff space in the weak- $*$ topology, by the Alaoglu-Banach theorem, as per [RS03, Theorem IV.21], with each $\tilde{A}$ continuous on it. This shows the structural difference between the commutative and the non-commutative case already announced in Section 2.1.

2.4 The Weyl Algebra and its Representations

In the previous chapter, we characterized a commutative C^* -algebra as the correct algebraic model for a classical particle. The present section is devoted to find the algebra that embeds the canonical commutation relations between position and momentum in quantum mechanics. A naive answer, generating an algebra directly from the operators q and p , turns out to be incompatible with the very notion of a C^* -algebra. In fact, we show that no finite norm can be assigned to either operator once the commutation relations hold. Weyl's resolution replaces q and p with the bounded unitary exponentials that they formally generate. The resulting Weyl C^* -algebra becomes the correct object on which to invoke the noncommutative Gelfand-Naimark Theorem 2.8. In addition, von Neumann uniqueness theorem then shows that this representation, if it is regular, is unique up to unitary equivalence, so that the quantum particle possesses a single state space. The section closes by exhibiting the Schrödinger representation on $L^2(\mathbb{R})$. The following work in this section is based on [Str05, Chapter 3].

The Heisenberg relations and the boundedness obstruction

A classical particle in one space dimension is described by the algebra generated by the position q and the momentum p . The natural quantum analogue is obtained by requiring that the same two generators satisfy the following Heisenberg commutation relations.

$$[q, p] = i, \quad [q, q] = 0, \quad [p, p] = 0, \quad (2.37)$$

Here, the value of $\hbar$ has been set to one by a suitable choice of units for the sake of simplicity. Note that the discussion is restricted to a single space dimension for definiteness. The word *generated*, however, admits more than one interpretation, and this reading is specified the first task of the section.

Definition 2.4.1 (Heisenberg Lie algebra). *The Heisenberg Lie algebra is the complex vector space spanned by q , p and the identity element $\mathbb{1}$, equipped with the Lie bracket determined by the commutation relations Equation (2.37) together with $[q, \mathbb{1}] = [p, \mathbb{1}] = 0$.*

The Lie bracket alone does not specify how q and p act as operators. The associative completion obtained in this way, rather than the Lie algebra of Definition 2.4.1 itself, is the structure usually meant with Heisenberg algebra. We need to examine the relations of this object to the operational framework of Section 2.1.

Definition 2.4.2 (Heisenberg algebra). *The Heisenberg algebra is the associative algebra generated, through sums and products, by q and p , with commutation relations fixed by Equation (2.37).*

The Heisenberg algebra of Definition 2.4.2 does not fit into the operational framework of Section 2.1. The reason why is that q and p cannot be realized as bounded, or selfadjoint, elements of a C^* -algebra. The proposition below specifies this obstruction and quantifies how badly q and p fail to admit a finite C^* -norm.

Proposition 2.4.3. *Let q and p satisfy the Heisenberg commutation relations Equation (2.37). If a C^* -norm could be assigned to q and p , then*

$$\|q\| \|p\| \geq \frac{n}{2}, \quad \forall n \in \mathbb{N}. \quad (2.38)$$

Hence $\|q\|$ and $\|p\|$ cannot both be finite.

Proof. The commutation relations Equation (2.37) entail, by induction on n ,

$$[p, q^n] = -i n q^{n-1}. \quad (2.39)$$

If a C^* -norm existed on the algebra generated by q and p , submultiplicativity of the norm applied to Equation (2.39) would yield

$$n \|q^{n-1}\| \leq \|p q^n\| + \|q^n p\| \leq 2 \|p\| \|q\| \|q^{n-1}\|. \quad (2.40)$$

Since $\|q^{n-1}\| = \|q\|^{n-1}$ by the C^* -identity and since it cannot vanish for any n . We are contradicting Equation (2.37). Dividing by $\|q^{n-1}\|$ therefore yields Equation (2.38) for every $n \in \mathbb{N}$, and letting n grow without bound shows that $\|q\|$ and $\|p\|$ cannot both be finite. $\square$

Remark 2.4.4. The mathematical obstruction identified in Proposition 2.4.3 has a direct operational counterpart. Since any experimental apparatus is subject to scale bounds, what is actually measured is never q or p themselves rather bounded functions of them. Among this we include the position within the region accessible to the apparatus and the momentum within an interval set by the available energy. A formulation built on the Heisenberg algebra of Definition 2.4.2 therefore involves an extrapolation beyond what is operationally accessible in the sense of Section 2.1. Such an extrapolation that carries no physical consequence yet forces a change of strategy at the algebraic level.

Proposition 2.4.3 is, in fact, an elementary instance of a considerably more general fact. This result makes no reference to the specific value of the commutator between q and p and holds for an arbitrary pair of bounded operators.

Theorem 2.9 (Wintner's Obstruction Theorem). *Let $\mathcal{H}$ be a Hilbert space, as in Definition 2.2.6, and let $A, B \in \mathcal{B}(\mathcal{H})$. Then*

$$AB - BA \neq c \mathbb{1}, \quad \forall c \in \mathbb{C} \setminus \{0\}. \quad (2.41)$$

The result was first proved by Wintner, with an independent proof given shortly afterwards by Wielandt. The proof is not reproduced here, and the reader is referred to [Mor18, Proposition 11.32]. Taking $A = q$ and $B = p$ in Theorem 2.9 recovers the impossibility already established by direct computation in Proposition 2.4.3.

The Weyl Operators and the Weyl Algebra

Weyl's proposal replaces the problematic extrapolation with a family of operators that are bounded by construction, namely the formal exponentials of q and p . In addition, it is on these that a C^* -algebra can be built.

Remark 2.4.5. Note that from now on, $s \in \mathbb{N}$ will denote a finite number of dimensions.

Definition 2.4.6 (Weyl operators). *Given q and p satisfying the Heisenberg commutation relations Equation (2.37) on $\mathbb{R}^s$, the Weyl operators are the bounded formal functions*

$$U(\alpha) = e^{i\alpha q}, \quad V(\beta) = e^{i\beta p}, \quad \alpha, \beta \in \mathbb{R}^s. \quad (2.42)$$

The algebraic properties of the Weyl operators can be read off from their formal relation to q and p , through an identity that will be used again in the proof of the von Neumann uniqueness theorem below.

Lemma 2.4.7 (Baker–Hausdorff formula). *Let A, B be densely defined operators on a Hilbert space, as in Definition 2.2.25, with a common dense domain on which $A, B, A + B$ and their exponentials can be freely applied. Suppose that $C := [A, B]$ commutes with both A and B . Then*

$$e^A e^B = e^{A+B+C/2}. \quad (2.43)$$

A complete proof of this lemma can be found in [Str05, p.61]. Equation (2.43) applies to q and p as soon as the commutator between the two exponents is a scalar. That is the situation encompassed by the Heisenberg relations (2.37).

Proposition 2.4.8 (Weyl relations). *The Weyl operators of Definition 2.4.6 satisfy*

$$U(\alpha)V(\beta) = V(\beta)U(\alpha)e^{-i\alpha\beta}, \quad (2.44)$$

$$U(\alpha)U(\alpha') = U(\alpha + \alpha'), \quad V(\beta)V(\beta') = V(\beta + \beta'). \quad (2.45)$$

Proof. The element $[i\alpha q, i\beta p] = -i\alpha\beta$ is a scalar multiple of the identity by Equation (2.37), hence commutes with both $i\alpha q$ and $i\beta p$, and Lemma 2.4.7 applies to the pair $A = i\alpha q, B = i\beta p$. It yields

$$U(\alpha)V(\beta) = e^{i\alpha q + i\beta p - i\alpha\beta/2}. \quad (2.46)$$

Exchanging the roles of A and B yields, by the same argument,

$$V(\beta)U(\alpha) = e^{i\alpha q + i\beta p + i\alpha\beta/2} = U(\alpha)V(\beta)e^{i\alpha\beta}, \quad (2.47)$$

which is Equation (2.44). Equations (2.45) follow from Equation (2.37), since q commutes with itself and p with itself. Hence no correction term of the form $[A, B]/2$ appears in Lemma 2.4.7 when both operators belong to the same one-parameter family. $\square$

The relations of Proposition 2.4.8, referred to as the Weyl form of the canonical commutation relations, no longer make reference to unbounded operators. It can therefore be taken as the defining relations of an abstract algebra.

Definition 2.4.9 (Weyl algebra). *The Weyl algebra $\mathfrak{A}_W$ is the abstract algebra generated, through complex linear combinations and products, by abstract elements $U(\alpha), V(\beta)$ with $\alpha, \beta \in \mathbb{R}$, subject to the relations Equation (2.44) and Equation (2.45).*

Remark 2.4.10. A natural involution on $\mathfrak{A}_W$ is fixed by declaring $U(\alpha)^* = U(-\alpha)$ and $V(\beta)^* = V(-\beta)$, matching the selfadjointness that q and p would carry at the formal level. The Weyl relations then force $U(\alpha)$ and $V(\beta)$ to be unitary, and consequently every monomial in U and V has unit norm as well, since the Weyl relations reduce any such monomial to a single product $U(\alpha)V(\beta)$ up to a phase.

Assigning a norm to the remaining elements of $\mathfrak{A}_W$, so as to obtain a C^* -algebra, is a nontrivial step, shown in the following result.

Proposition 2.4.11. *There exists a unique C^* -norm on $\mathfrak{A}_W$, and the completion of $\mathfrak{A}_W$ with respect to it is a C^* -algebra.*

This proposition is not proved here, and we refer the reader to [BR81, Theorem 5.2.8]. With a unique C^* -norm available on $\mathfrak{A}_W$, its completion can now be regarded as a C^* -algebra. The next definition fixes the notation for this completed algebra, the object to which the noncommutative Gelfand-Naimark theorem will be applied below.

Definition 2.4.12 (Weyl C^* -algebra). *The completion of $\mathfrak{A}_W$ with respect to the norm of Proposition 2.4.11 is a C^* -algebra, still denoted $\mathfrak{A}_W$, called the Weyl C^* -algebra.*

The quantum particle is defined as the quantum system characterized by the Weyl C^* -algebra $\mathfrak{A}_W$.

Remark 2.4.13. The states of the quantum particle are identified with the representations of $\mathfrak{A}_W$, as the states of the classical particle were identified in Section 1.6 with the representations of the corresponding commutative algebra. What Theorem 2.8 does not by itself provide is uniqueness, since a C^* -algebra can admit inequivalent representations, and it is this question that occupies the remainder of the section.

Before turning to uniqueness, it is worth recording the structure implicit in the Weyl relations, since it offers an equivalent description of representations of $\mathfrak{A}_W$.

Definition 2.4.14 (Heisenberg Lie group). *The Heisenberg Lie group is the Lie group whose elements are triples $(\alpha, \beta, \lambda) \in \mathbb{R}^s \times \mathbb{R}^s \times \mathbb{R}$, with group law*

$$(\alpha, \beta, \lambda)(\alpha', \beta', \lambda') = \left(\alpha + \alpha', \beta + \beta', \lambda + \lambda' + \frac{1}{2}(\alpha' \beta - \alpha \beta') \right). \quad (2.48)$$

Remark 2.4.15. The group law Equation (2.48) is obtained from the identification $(\alpha, \beta, \lambda) \leftrightarrow \exp i(\alpha q + \beta p + \lambda \mathbb{1})$ together with the Weyl relations of Proposition 2.4.8. It exhibits the Heisenberg Lie group as the Lie group associated, through the exponential map, with the Heisenberg Lie algebra of Definition 2.4.1. Determining the representations of the Weyl C^* -algebra $\mathfrak{A}_W$ is therefore equivalent to determining the unitary representations of the Heisenberg Lie group.

Regularity and the von Neumann Uniqueness Theorem

Theorem 2.8 guarantees the existence at least one representation of $\mathfrak{A}_W$ on a Hilbert space. Yet it carries no informations about how many inequivalent ones exist. An infinite multiplicity of inequivalent representations would mean that the algebraic characterization of the quantum particle given in Definition 2.4.12 fails to determine its physics uniquely. That is this possibility that the regularity condition introduced below is planned to rule out.

Definition 2.4.16 (Regular representation). *A representation π of the Weyl algebra $\mathfrak{A}_W$, as in Definition 2.4.9, on a separable Hilbert space $\mathcal{H}$, as in Definition 2.2.6, is regular if $\pi(U(\alpha))$ and $\pi(V(\beta))$ are strongly continuous one-parameter groups of unitary operators in α and β , respectively as per Definition 1.2.28.*

Regularity is useful because strongly continuous one-parameter unitary groups admit self-adjoint generators, by the following theorem.

Theorem 2.10 (Stone's theorem). *Let $t \mapsto U(t)$ be a strongly continuous one-parameter group of unitary operators on a Hilbert space $\mathcal{H}$, as per Definition 1.2.28. Then there exists a unique selfadjoint operator A on $\mathcal{H}$, generally unbounded and densely defined as per Definition 2.2.25, such that*

$$U(t) = e^{itA}, \quad \forall t \in \mathbb{R}. \quad (2.49)$$

Conversely, every self-adjoint operator A on $\mathcal{H}$ generates in this way a strongly continuous one-parameter unitary group.

The operator A is called the generator of U . The proof combines the spectral theorem for self-adjoint operators with the properties of strongly continuous unitary groups. The reader is referred to [Mor18, Theorem 9.33] for a complete textbook treatment.

Remark 2.4.17. Although regularity is formally defined by requiring the strong continuity of the one-parameter unitary groups $\pi(U(\alpha))$ and $\pi(V(\beta))$, it constitutes a mild condition that is satisfied in non pathological physical models. For families of unitary operators, strong continuity is equivalent to the weak one. This is a consequence of the following identity.

$$\|(U(t) - 1)\Psi\|^2 = 2\|\Psi\|^2 - 2\operatorname{Re}(\Psi, U(t)\Psi). \quad (2.50)$$

That demonstrates that the strong limit exists provided the real part of the transition amplitude $(\Psi, U(t)\Psi)$ approaches $\|\Psi\|^2$ as $t \rightarrow 0$. That is the weak continuity condition. Furthermore, on a separable Hilbert space, weak continuity is strictly equivalent to weak measurability of the scalar mappings $t \mapsto (\Psi, U(t)\Psi)$. Therefore, verifying regularity reduces to checking the Lebesgue measurability of these matrix elements $(\Psi, U(t)\Psi)$.

A key physical consequence of regularity concerns the unbounded observables. By Stone's theorem, strong continuity is the hypothesis for one-parameter unitary groups to admit unique self-adjoint generators on the Hilbert space $\mathcal{H}$. These generators, which are unbounded, correspond to the momentum and position operators, p and q , within the representation π . Regularity also shows that these operators share a common, dense, invariant domain. Therefore, this condition provides the framework necessary to reconstruct the foundational unbounded observables of the quantum particle.

Having established regularity, we can state the next theorem to classify representations.

Theorem 2.11 (von Neumann uniqueness theorem). *All the regular irreducible representations of the Weyl C^* -algebra $\mathfrak{A}_W$ are unitarily equivalent.*

A complete proof of this statement can be found in [Str05, Theorem 3.2.2].

Remark 2.4.18 (The Fock state and unbounded generators). The constructive argument of the proof identifies a distinguished state on $\mathfrak{A}_W$, known as the *Fock state* ω_F . It is unambiguously defined by its expectation values on the generating elements of the algebra in the following way

$$\omega_F(e^{i\alpha\beta/2} U(\alpha) V(\beta)) = e^{-(|\alpha|^2 + |\beta|^2)/4}. \quad (2.51)$$

The GNS construction associated with ω_F yields the *Fock representation*, and its corresponding cyclic vector Ψ_0 is called the *Fock vector*. By exploiting the regularity of the representation, differentiating Equation (2.51) at $\alpha = \beta = 0$ shows that Ψ_0 satisfies

$$(q + ip)\Psi_0 = 0.$$

This is an unbounded equation obtained from a statement that only ever involved bounded operators, providing an instance of the mechanism described earlier. The unbounded operators q and p act on Ψ_0 although they are not themselves elements of $\mathfrak{A}_W$. Their action is recovered by differentiating the bounded representation along the directions that regularity keeps under control.

The Schrödinger representation

Theorem 2.11 reduces the classification of regular irreducible representations to the exhibition of a single example. Historically, the standard choice is the following.

Definition 2.4.19 (Schrödinger representation). *The Schrödinger representation π_S of the Weyl C^* -algebra $\mathfrak{A}_W$, as per Definition 2.4.9, acts on the Hilbert space $\mathcal{H} = L^2(\mathbb{R}^s, d^s x)$ by*

$$(U(\alpha)\psi)(x) = e^{i\alpha x} \psi(x), \quad (V(\beta)\psi)(x) = \psi(x + \beta), \quad \psi \in \mathcal{H}. \quad (2.52)$$

The definition would be of little use if it did not satisfy the two hypotheses of Theorem 2.11, regularity and irreducibility, and both properties are verified from Equation (2.52).

Proposition 2.4.20. *The Schrödinger representation of Definition 2.4.19 is a regular, irreducible representation of $\mathfrak{A}_W$.*

A full proof of this proposition can be found in [Str05, p.65].

Remark 2.4.21. The physical states of a quantum particle are, in this representation, the vectors of $L^2(\mathbb{R}^s)$, usually called Schrödinger wave functions. By Stone's Theorem 2.10, the strong continuity of $U(\alpha)$ and $V(\beta)$ established in Proposition 2.4.20 guarantees the existence, as strong limits on a dense domain, of the selfadjoint generators

$$(q\psi)(x) = x\psi(x), \quad \psi \in D_q = \{\psi \in L^2(\mathbb{R}) \mid x\psi \in L^2(\mathbb{R})\}, \quad (2.53)$$

$$(p\psi)(x) = -i \frac{d\psi}{dx}(x), \quad \psi \in D_p = \{\psi \in L^2(\mathbb{R}) \mid \psi' \in L^2(\mathbb{R})\}, \quad (2.54)$$

The derivative in Equation (2.54) is understood in the weak sense. The Schwartz space $\mathcal{S} := \mathcal{S}(\mathbb{R}^s)$ of smooth functions of fast decrease is a common dense domain of essential selfadjointness for q and p , following [RS03, Chapter VIII, Section 2]. The Heisenberg relations Equation (2.37) hold on $\mathcal{S}$.

Equations (2.53) and (2.54) are the concrete, unbounded realization of the operators that Proposition 2.4.3 excluded from the Weyl C^* -algebra $\mathfrak{A}_W$ from the outset. In fact, theorem 2.10 provides a reconstruction of q and p as the infinitesimal generators, as per Definition 1.5.4, of the bounded one-parameter groups $\pi_S(U(\alpha))$ and $\pi_S(V(\beta))$.

2.5 The Dirac-von Neumann Axioms Revisited

The reconstruction carried out in Section 2.1 and Section 2.3 started from purely operational data, namely preparations and tests. Consequently, we show the embedding of Definition 2.1.14 into a unital C^* -algebra $\mathfrak{A}$. Historically, however, quantum mechanics reached this level of rigor along the opposite route. Heisenberg's matrix mechanics of 1925 was cast by Paul Dirac into a first systematic formulation set directly at the level of a Hilbert space. Von Neumann later refined it into the set of postulates that most textbooks still adopt today. Therefore, it is worth recording this historical formulation and matching each of its postulates against the results already established above. The comparison shows that every Dirac-von Neumann axiom finds a counterpart among the theorems proved earlier in the chapter. Following [Str05, Sect. 3.6], we state the axioms of the Hilbert space formulation of quantum mechanics.

Definition 2.5.1 (Dirac-von Neumann axioms). *The Dirac-von Neumann formulation of quantum mechanics rests on the following postulates:*

- I. *States.* The states of a quantum system are described by density matrices $\rho = \sum_i \lambda_i P_i$, as in Definition 2.3.15, with $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$, each is a P_i one-dimensional projections, in a separable complex Hilbert space $\mathcal{H}$, as in Definition 2.2.6.
- II. *Observables.* The observables of a quantum system are described by the set of bounded self-adjoint operators in $\mathcal{H}$, as in Definition 2.2.22.
- III. *Expectations.* If a state ω is described by the vector $\Psi_\omega \in \mathcal{H}$, the expectation value of an observable A is the Hilbert space matrix element $\omega(A) = (\Psi_\omega, A\Psi_\omega)$, or more generally $\omega(A) = \text{Tr}(\rho_\omega A)$ for a state described by the density matrix ρ_ω .
- IV. *Canonical quantization.* For a system with finitely many degrees of freedom, the canonical coordinates and momenta satisfy $[q_i, q_j] = 0 = [p_i, p_j]$ and $[q_i, p_j] = i\hbar\delta_{ij}$, as in Equation (2.37).
- V. *Schrödinger representation.* The commutation relations of Axiom IV are represented on $\mathcal{H} = L^2(\mathbb{R}^s, dx)$ by $(q_i\psi)(x) = x_i\psi(x)$ and $(p_j\psi)(x) = -i\hbar \partial\psi/\partial x_j(x)$.

The five constructions that follow take up each axiom in the order in which Definition 2.5.1 lists them. We compare it against what already developed in this chapter.

Recovering Axiom I. The state space S of Definition 2.1.4, together with the identification of pure and mixed states of Definition 2.3.11, plays the role of Axiom I. A pure state is represented, through the GNS construction of Theorem 2.7, by a unit vector in the Hilbert space $\mathcal{H}_\omega$, since Proposition 2.3.13 identifies purity of ω with irreducibility of the associated representation π_ω . A mixed state is instead represented by a density matrix on $\mathcal{H}_\omega$ as per Definition 2.3.15, and Proposition 2.3.14 shows that every such density matrix defines a state. Separability of $\mathcal{H}_\omega$, required by Axiom I, is not automatic in this construction, and the next subsection discusses the circumstances under which it can fail. However, in most of physical systems, this hypothesis can be verified.

Recovering Axiom II The embedding of Definition 2.1.14 identifies the observable set $\mathcal{O}$ of Definition 2.1.1 with the selfadjoint sector $\mathfrak{A}_{sa}$ of the ambient C^* -algebra, matching Axiom II directly. Boundedness of the observables, absent from Dirac's original formulation according to [Str05, Sect. 3.6], is not an additional postulate in the present reconstruction. It descends from Definition 2.1.6 that assigns a finite norm to every observable. In fact, in the previous subsection, we showed that even unbounded operators like position and momentum can be represented as bounded operators using Weyl's algebra of Definition 2.4.9.

Hopping from an abstract state on a C^* -algebra to a Hilbert space matrix element, postulated outright in Axiom III, is instead a theorem of the operational construction.

Corollary 2.5.2 (Expectation values from the GNS construction). *Let ω be a state on a C^* -algebra $\mathfrak{A}$ as per Definition 2.1.3, and let $(\mathcal{H}_\omega, \pi_\omega, \Psi_\omega)$ be its GNS representation as per Theorem 2.7. For every $A \in \mathfrak{A}$,*

$$\omega(A) = (\Psi_\omega, \pi_\omega(A)\Psi_\omega). \quad (2.55)$$

More generally, for every density matrix ρ on $\mathcal{H}_\omega$ as per Definition 2.3.15, the state ω_ρ of Proposition 2.3.14 satisfies $\omega_\rho(A) = \text{Tr}(\rho \pi_\omega(A))$.

Proof. Equation (2.55) is part ii) of Theorem 2.7. The general statement is Equation (2.33) of Proposition 2.3.14, applied to the representation π_ω . $\square$

The last two axioms stand apart from the first three. The axioms I to III describe general mathematical structures common to every quantum system while IV and V refer to one specific realization. The remaining two remarks take up this starting from the commutation relations imposed by Axiom IV.

Recovering Axiom IV. The commutation relations of Axiom IV are Equation (2.37) of Definition 2.4.1. They identify a single instance among the class of Definition 2.1.22. That is the quantum particle whose algebra of observables is the Weyl C^* -algebra $\mathfrak{A}_W$ of Definition 2.4.12.

Recovering Axiom V. The Schrödinger representation of Axiom V coincides with Definition 2.4.19. Axiom V in Definition 2.5.1, however, postulates the use of this representation. However, in the algebraic reconstruction, the von Neumann uniqueness Theorem 2.11, shows that it is, up to unitary equivalence, the only regular irreducible representation of $\mathfrak{A}_W$. Note that the restriction to a finite number s of dimensions, introduced ahead of Definition 2.4.6, is an hypothesis of the uniqueness theorem itself. In the next subsection, it is discussed what becomes of Axiom V once this hypothesis is dropped.

Limits of the GNS Construction and the Infinite-Dimensional Case

The construction in Section 2.5 isolated the hypothesis on which the whole uniqueness statement of Theorem 2.11 rests. This restriction to a finite number of degrees of freedom is the boundary between the general postulates I to III and the special postulates IV and V. Note that theories like quantum field theory, quantum statistical mechanics, and quantum many-body systems all require infinitely many degrees of freedom. In fact, the regime in which the Schrödinger representation of Definition 2.4.19 ceases to be available, see [Str05, Sect. 3.6].

Remark 2.5.3 (Inequivalent representations). The GNS construction of Theorem 2.7 remains valid without modification for the algebra of a field theory or of an infinite spin chain. In the infinite-dimensional analogue of the Weyl algebra of Definition 2.4.9, regular irreducible representations as per Definition 2.4.16 are no longer unitarily equivalent to one another. In particular, different equilibrium temperatures or different values of a mass parameter, correspond to physically distinct situations rather than to a single Hilbert space picture.

A second, more technical mismatch concerns the separability required by Axiom I itself. Remark 2.5 already presented this point. It is worth stating how the algebraic construction can fail to prove it.

Remark 2.5.4 (Separability). When $\mathfrak{A}$ is itself separable in the norm topology, the GNS space $\mathcal{H}_\omega$ inherits separability from the quotient $\mathfrak{A}/J$ used in the proof of Theorem 2.7. Hence, a single physically relevant state always leads to a Hilbert space compatible with Axiom I. Theorem 2.8, however, assembles the faithful representation as the direct sum $\mathcal{H} = \bigoplus_{\omega \in \mathcal{F}} \mathcal{H}_\omega$ over an entire separating family $\mathcal{F}$ of states. When inequivalent representations proliferate as in Remark 2.5.3, no countable subfamily of $\mathcal{F}$ typically suffices to separate $\mathfrak{A}$. The resulting direct sum $\mathcal{H}$ is then non separable.

None of this affects the classical construction of Chapter 1 in the same way, since a commutative algebra of continuous functions carries no analogue of an inequivalent representation once its Gelfand spectrum is fixed.

2.6 A Comparison between Classical and Quantum Algebraic Reconstructions

The two classical and quantum constructions share the same starting vocabulary, algebra, states, norm, and representation, yet the order in which these structures arise is different between the two cases. We compare the two constructions, first at the level of the foundational objects and then at the level of the representation theorems that recover a geometric or Hilbert space picture. Particular attention is paid to the bracket. In fact, Chapter 1 had to postulate the Poisson bracket as a structure external to the C^* -algebra, while the present chapter obtained a Lie bracket directly from noncommutativity of the associative product.

Foundational objects

The first difference between the two constructions concerns the order in which the algebraic operations arise. Chapter 1 starts from a fully formed associative commutative product. Differently, the present chapter has to build one from more primitive operations, as the following remark specifies.

Remark 2.6.1 (The associative product, given or constructed). The classical continuous algebra of Definition 1.6.15 is an associative commutative algebra of functions under pointwise multiplication. The observable set $\mathcal{O}$ of Definition 2.1.1, in contrast, starts with only rescaling and squaring, and an associative product becomes available only at the end of the whole setting.

The norm, instead, is postulated as part of the function algebra on one side becomes. On the other, it is a consequence of the positivity already required of the states.

Remark 2.6.2 (The norm, given or derived). The supremum norm of Equation (1.13) is inherited directly from the function algebra $C^0(X; \mathbb{C})$ underlying Definition 1.6.15. The norm of Definition 2.1.6, by contrast, is derived from the operational bound on measurement outcomes across all states. Furthermore, the defining property of a C^* -algebra, postulated for the classical case, is a consequence of positivity of the states for the quantum case as shown in Proposition 2.1.7.

Both constructions require their states to separate the algebra, yet the topological and the purely algebraic route to this property follows a completely different path.

Remark 2.6.3 (States and the separating property). Definition 2.1.3 characterizes a state, on either side, by positivity and normalization on an already given algebra. In the classical setting, the classical observable algebra $\mathfrak{A}_{sa}$ separates the points of the Gelfand spectrum X as a consequence of Urysohn's lemma, once X is known to be a compact Hausdorff space. In the quantum picture, there is no topological space available yet at the point where separation is needed. Therefore, Definition 2.1.4 postulates the separating property

as part of the definition of the state space S . It is only afterward, through the family $\mathcal{F}$ of Theorem 2.8, that a faithful representation, and hence an indirect geometric picture, is obtained.

The most concerning difference between the construction is given by the appearance of Lie bracket and commutator. The following proposition isolates the reason why a bracket cannot be recovered from a commutative algebra alone.

Proposition 2.6.4 (The bracket vanishes on a commutative algebra). *Let $\mathfrak{A}$ be a commutative C^* -algebra embedding a Jordan-Banach algebra $\mathcal{O}$ as per Definition 2.1.14. Then the bracket $\{A, B\} = \frac{1}{i\hbar}[A, B]$ defined for $A, B \in \mathcal{O}$ vanishes identically.*

Proof. Since $\mathfrak{A}$ is commutative, $[A, B] = AB - BA = 0$ for every $A, B \in \mathfrak{A}$, and in particular for every $A, B \in \mathcal{O} \subset \mathfrak{A}_{sa}$. Hence $\{A, B\} = 0$. $\square$

Remark 2.6.5 (The Poisson bracket, imposed or free). Since $C^0(X; \mathbb{C})$ is commutative, the natural candidate for a bracket built from the commutator is identically zero. Poisson bracket has to be supplied as a new piece of data, namely the smooth dense Poisson subalgebra $\mathfrak{A}^\infty$ of Definition 1.6.32. Hence, the choice of $\mathfrak{A}^\infty$ constitutes the external input of the classical reconstruction, as stated after Definition 1.6.32.

The quantum case follows the opposite path. As soon as $\mathcal{O}$ is realized inside a noncommutative associative algebra through Definition 2.1.14, the commutator $[A, B]$ is generally non-zero. This construction constitutes the way to repair to the innatural choice made in the classical case and to understand that all the structures derived from the Lie algebra follow from noncommutativity.

Representation theorems

Remark 2.3.20 already identifies the classical construction as a special case of the quantum one rather than an analogous procedure. The proposition below specifies this relation.

Proposition 2.6.6 (The classical Gelfand-Naimark theorem as a commutative corollary). *Let $\mathfrak{A}$ be an commutative unital C^* -algebra as per Definition 1.6.13, and let $\mathcal{F}$ be a separating family of pure states on $\mathfrak{A}$ as per Definition 1.6.20 and Proposition 1.6.27. The faithful representation $\pi = \bigoplus_{\omega \in \mathcal{F}} \pi_\omega$ of Theorem 2.8 is unitarily equivalent to the Gelfand isomorphism of Theorem 1.10, under the identification of $\mathcal{F}$ with the Gelfand spectrum $\sigma(\mathfrak{A})$ of Definition 1.6.24.*

For a proof of this statement, see Remark 2.3.20. Consequently $\pi(A)$ coincides with the family $\{\omega(A)\}_{\omega \in \mathcal{F}}$. This relies on results already established, yet its content deserves to be observed independently of the formal statement.

Remark 2.6.7 (Points versus Hilbert spaces). Proposition 2.6.6 shows that the two representation theorems are a single Theorem 2.8, one of them specialized to the commutative case. What changes between the two cases is the meaning attached to a pure state. On one hand, it assumes the meaning of a character as in Definition 1.6.24, evaluating every observable to a real number. On the other hand, it represents a full irreducible representation on a generally infinite-dimensional Hilbert space, per Proposition 2.3.13. Every value $\omega(A)$ that a classical observable takes at a point of X becomes, once commutativity is dropped, an operator $\pi_\omega(A)$ acting on $\mathcal{H}_\omega$, and it is this passage from numbers to operators that the whole of Section 2.3 was devoted to constructing.

Remark 2.6.8 (Topology, compactness versus completeness). Turning the Gelfand spectrum $\sigma(\mathfrak{A})$ into a compact Hausdorff space, as required by Theorem 1.10, requires the weak-* topology of Definition 1.6.28 together with the Banach-Alaoglu compactness of Proposition 1.6.29. No analogous step appears in the construction of $\mathcal{H}_\omega$, since the inner product of Equation (2.17) already supplies a metric topology on the quotient $\mathfrak{A}/J$, completed directly into a Hilbert space by the same procedure as Theorem 2.2.

Table 2.1 collects every point of comparison drawn in this section into a single overview.

The comparison collected in Table 2.1 can be summarized in a single statement. Hopping from the commutative to the noncommutative case replaces points of a topological space with Hilbert spaces carrying a representation. It also replaces evaluation of a function at a point with the action of an operator on a cyclic vector. Furthermore, it replaces an externally imposed Poisson bracket, confined to a dense subalgebra,

Aspect	Classical reconstruction, Chapter 1	Quantum reconstruction, present chapter
Associative product	Given from the start, pointwise multiplication on $C^0(X; \mathbb{C})$, Definition 1.6.15	Constructed step by step, from squares to the Jordan product of Definition 2.1.9, to full associativity through the embedding of Definition 2.1.14
Norm	Given, the supremum norm of Equation (1.13)	Derived from the states, $\ A\ ^2 = \ A^2\ $ proved in Proposition 2.1.7
Bracket	Imposed externally, smooth dense Poisson subalgebra of Definition 1.6.32, discontinuous in the uniform norm as per Remark 1.6.33	Intrinsic, $\{A, B\} = \frac{1}{i\hbar}[A, B]$ as per Definition 2.1.19, defined on all of $\mathcal{O}$ and proved via Theorem 2.1
Separation of states	Via Urysohn's lemma, once X is known to be compact Hausdorff	Postulated directly as part of Definition 2.1.4
Representation theorem	Gelfand-Naimark theorem, Theorem 1.10, giving $\mathfrak{A} \cong C_0(X)$	Noncommutative Gelfand-Naimark theorem, Theorem 2.8, via the GNS construction of Theorem 2.7
Topology required	Weak-* topology and Banach-Alaoglu compactness, Definition 1.6.28 and Proposition 1.6.29	None beyond the Hilbert space completion of Theorem 2.2

Table 2.1: Summary of the comparison between the classical reconstruction of Chapter 1 and the quantum reconstruction of the present chapter.

with a Lie bracket intrinsic to the whole algebra, as shown in Remark 2.6.5. Every new piece of mathematics required for the noncommutative case, the Jordan product, the GNS construction, and the Lie-Jordan identities of Theorem 2.1, exists to compensate for the loss of commutativity and a given geometric space.

2.7 Example II: Quantum Spin System

This example illustrates the GNS construction of Section 2.3 and the noncommutative Gelfand-Naimark Theorem. The quantum spin is generated by three operators satisfying the $\mathfrak{su}(2)$ commutation relations. In contrast with the classical spin of Section 1.7 where the same Lie algebra was realised as a Poisson bracket on the sphere. The comparison shows how the same algebraic relations yield a symplectic manifold in the commutative case and a Hilbert space of states in the noncommutative one.

The Pauli Algebra and the GNS Representation

The quantum spin system is modeled by the complex unital C^* -algebra $\mathfrak{A}$, as in Definition 1.6.13, generated by three selfadjoint elements J_1, J_2, J_3 and equipped with the usual C^* -norm and involution. These three generators are required to satisfy the $\mathfrak{su}(2)$ commutation relations

$$[J_i, J_j] = i \epsilon_{ijk} J_k, \quad i, j, k \in \{1, 2, 3\}. \quad (2.56)$$

This is the same Lie algebra that generated the classical spin system of Section 1.7 through a Poisson bracket rather than through a commutator. A spin-1/2 system is identified among all the representations of these relations by fixing the value of the Casimir operator $C := J_1^2 + J_2^2 + J_3^2$ to $\frac{3}{4}\mathbb{1}$. Passing to the quotient algebra $\mathfrak{A}_0 := \mathfrak{A}/(C - \frac{3}{4}\mathbb{1})$ then restricts attention to this sector alone, as the analogous restriction did in the classical case of Section 1.7. The quotient $\mathfrak{A}_0$ remains a unital C^* -algebra as in Definition 1.6.13. A state ω_0 on $\mathfrak{A}_0$, as in Definition 2.1.3, is fixed by prescribing its values on the identity and on products of at most two generators

$$\omega_0(\mathbb{1}) = 1, \quad \omega_0(J_i) = 0, \quad \omega_0(J_i J_j) = \frac{1}{4} \delta_{ij}, \quad (2.57)$$

extended to the whole of $\mathfrak{A}_0$ by linearity. The coefficient $\frac{1}{4}$ appearing in Equation (2.57) is fixed by the Casimir relation itself. In fact, applying ω_0 to the Casimir identity $\sum_i J_i^2 = \frac{3}{4}\mathbb{1}$ and using isotropy yields

$3\omega_0(J_i^2) = \frac{3}{4}$. It descends that $\omega_0(J_i^2) = \frac{1}{4}$ for every i . The resulting functional is positive and normalized, hence it defines a state on $\mathfrak{A}_0$ by Definition 2.1.3. Because the three directions J_1, J_2, J_3 are indistinguishable for ω_0 , it is the unique $SO(3)$ invariant state on $\mathfrak{A}_0$. Physically, this symmetry is what identifies ω_0 as the state of complete ignorance about the orientation of the spin.

We carry out this construction explicitly to represent concretely ω_0 . It is worth following Remark 2.3.6 in order to keep the space of abstract observables $\mathfrak{A}_0$ on one side and the Hilbert space of state vectors that the construction produces on the other.

1. The first step defines on $\mathfrak{A}_0$ itself the sesquilinear form $(A, B) := \omega_0(A^*B)$, as in Equation (2.17), which is positive semidefinite because ω_0 is a state.
2. The second step isolates the null space of this form, $J := \{A \in \mathfrak{A}_0 \mid \omega_0(B^*A) = 0 \text{ for every } B \in \mathfrak{A}_0\}$, as in Equation (2.18), which coincides, by the Cauchy-Schwarz inequality for $(\cdot, \cdot)$, with the smaller looking set $\{A \in \mathfrak{A}_0 \mid \omega_0(A^*A) = 0\}$ and is a left ideal of $\mathfrak{A}_0$. We identify $\mathfrak{A}_0$ concretely below, showing that this ideal will vanish $J = \{0\}$, so that the algebraic state ω_0 turns out to be faithful.
3. Since $J = \{0\}$, the form $(\cdot, \cdot)$ is already a inner product on the whole of $\mathfrak{A}_0$, and no further quotient is needed before completing it. The *Hilbert space of state vectors* $\mathcal{H}_{\omega_0}$ is obtained by completing $\mathfrak{A}_0$ with respect to the norm induced by this inner product. Note that an element $B \in \mathfrak{A}_0$, which up to this point has only played the role of an abstract observable, is now reinterpreted as a state vector $[B] \in \mathcal{H}_{\omega_0}$. The same mathematical object, an element of $\mathfrak{A}_0$, is thus made to carry two different physical meanings depending on which of the two spaces, $\mathfrak{A}_0$ or $\mathcal{H}_{\omega_0}$, it is regarded as belonging to.
4. The fourth step supplies the *space of physical operators* $\mathcal{B}(\mathcal{H}_{\omega_0})$, the algebra of bounded linear operators on $\mathcal{H}_{\omega_0}$. The representation $\pi_{\omega_0} : \mathfrak{A}_0 \rightarrow \mathcal{B}(\mathcal{H}_{\omega_0})$ of Theorem 2.7 is defined by left multiplication. It is through π_{ω_0} that an abstract observable $A \in \mathfrak{A}_0$ is instantiated as the concrete physical operator $\pi_{\omega_0}(A)$. This operator acts on a state vector $[B] \in \mathcal{H}_{\omega_0}$ by

$$\pi_{\omega_0}(A)[B] := [AB], \quad (2.58)$$

as in Equation (2.20). That shows in a single formula the two roles just distinguished, A on the left of the product enters as the observable being represented, while B inside the bracket enters as the state vector it acts on.

5. The fifth and last step identifies the cyclic vector, as per Definition 2.2.34, with $\Psi_{\omega_0} := [1]$. The state vector corresponding to the identity element of $\mathfrak{A}_0$. This vector closes the construction by reproducing the original algebraic state through the Hilbert space inner product,

$$(\Psi_{\omega_0}, \pi_{\omega_0}(A)\Psi_{\omega_0}) = \omega_0(A), \quad \forall A \in \mathfrak{A}_0. \quad (2.59)$$

Hence that the whole of ω_0 is recovered as an expectation value in Ψ_{ω_0} .

The Gelfand-Naimark Theorem and the Operator Algebra

Determining the dimension of $\mathfrak{A}_0$ requires classifying the representations of the $\mathfrak{su}(2)$ Equation (2.56) that are compatible with the fixed value of the Casimir element, $C = \frac{3}{4}1$. Representation theory of $\mathfrak{su}(2)$ associates the two-dimensional irreducible representation the following

$$\pi_{1/2}(J_i) = \frac{1}{2} \sigma_i. \quad (2.60)$$

In which σ_i denote the Pauli matrices. Fixing the Casimir at $\frac{3}{4}$ forces every $*$ -representation of Equation (2.56) to be built out of copies of this single representation alone. This identification of representations translates into an identification of algebras showing that $\mathfrak{A}_0 \cong M_2(\mathbb{C})$. The full argument behind this identification is developed in [Hal03, Section 4.4].

Since the identity together with the three Pauli matrices form a basis of $M_2(\mathbb{C})$, the image of $\pi_{1/2}$ already exhausts the whole of $M_2(\mathbb{C})$. The commutant of $M_2(\mathbb{C})$ inside $\mathcal{B}(\mathbb{C}^2)$ consists only of the scalar multiples of

the identity. Schur's Lemma 1.2.26 states that a representation is irreducible when its commutant reduces to scalars in this way. Combining the two facts shows that $\pi_{1/2}$ is irreducible. Irreducibility of $\pi_{1/2}$ has an immediate consequence for the states it defines. By Proposition 2.3.13, every vector state

$$\omega_\psi(A) = (\psi, \pi_{1/2}(A)\psi), \quad \forall \psi \in \mathbb{C}^2, \quad (2.61)$$

is pure, and $\pi_{1/2}$ is, up to unitary equivalence, the unique irreducible representation of $\mathfrak{A}_0$.

Having established an irreducible representation of $\mathfrak{su}(2)$, we continue by write the state ω_0 analytically. The basis $\{\mathbb{1}, \sigma_1, \sigma_2, \sigma_3\}$ of $M_2(\mathbb{C})$ satisfies $\text{Tr}(\mathbb{1}) = 2$, $\text{Tr}(\sigma_i) = 0$, and $\text{Tr}(\sigma_i \sigma_j) = 2\delta_{ij}$. These are the values of Equation (2.57), up to a factor $\frac{1}{2}$, used to define ω_0 , and by linearity the two functionals coincide on the whole of $\mathfrak{A}_0$,

$$\omega_0(A) = \frac{1}{2} \text{Tr}(A), \quad \forall A \in \mathfrak{A}_0. \quad (2.62)$$

The trace of a strictly positive operator on $\mathbb{C}^2$ is itself strictly positive, that is $\omega_0(A^*A) = \frac{1}{2} \text{Tr}(A^*A) > 0$ for every non-zero $A \in \mathfrak{A}_0$. Therefore, the state ω_0 is a faithful state by Definition 2.3.17. This confirms the vanishing of the left kernel anticipated in the second step above, $J = \{0\}$. The construction above then yields

$$\mathcal{H}_{\omega_0} \cong \mathfrak{A}_0 \cong \mathbb{C}^4. \quad (2.63)$$

This identification of $\mathcal{H}_{\omega_0}$ with $\mathbb{C}^4$ has to be read with care, since the four-dimensional space $\mathbb{C}^4$ appearing here does not describe a new physical system with four levels. It is instead $M_2(\mathbb{C})$ of all linear operators on $\mathbb{C}^2$, now regarded as a Hilbert space of state vectors rather than as an algebra of observables because it includes also density matrices, as in Definition 2.3.15.

Faithfulness of ω_0 has another consequence, it makes the representation π_{ω_0} itself injective. Consequently, Proposition 2.3.4 shows that an injective representation is isometric. That is the property that the Gelfand-Naimark Theorem of Section 2.3 guarantees in general. Making π_{ω_0} explicit is the remaining task of this subsection.

To achieve that, we first recall the canonical vector space isomorphism $M_2(\mathbb{C}) \cong \mathbb{C}^2 \otimes \mathbb{C}^2$. Let $\{e_1, e_2\} \in \mathbb{C}^2$ be the standard orthonormal basis of $\mathbb{C}^2$, and let $E_{ij} \in M_2(\mathbb{C})$ denote the matrix units, defined by their action $E_{ij}e_k = \delta_{jk}e_i$. The isomorphism $M_2(\mathbb{C}) \cong \mathbb{C}^2 \otimes \mathbb{C}^2$ is fixed by the linear extension of the map $E_{ij} \mapsto e_i \otimes e_j$. Then, every abstract observable of $\mathfrak{A}_0$ acquires, once identified with its matrix representation, a corresponding vector in $\mathbb{C}^2 \otimes \mathbb{C}^2$.

Under the isomorphism just fixed, left multiplication acts only on the first tensor factor. For any observable $A \in \mathfrak{A}_0$, with matrix representation $\pi_{1/2}(A) \in M_2(\mathbb{C})$,

$$\pi_{1/2}(A)E_{ij} \longleftrightarrow (\pi_{1/2}(A) \otimes \mathbb{1}_{\mathbb{C}^2})(e_i \otimes e_j). \quad (2.64)$$

By contrast, right multiplication by an arbitrary matrix $B \in M_2(\mathbb{C})$ acts only on the second factor,

$$E_{ij}B \longleftrightarrow (\mathbb{1}_{\mathbb{C}^2} \otimes B^T)(e_i \otimes e_j), \quad (2.65)$$

where B^T denotes the transpose of B with respect to the basis $\{e_1, e_2\}$. We observe that the first one carries the physical degrees of freedom of the spin. In fact, it is on this factor alone that the observable algebra acts through π_{ω_0} . Given that, for every $B \in M_2(\mathbb{C})$, the operators $\mathbb{1}_{\mathbb{C}^2} \otimes B^T$ commute with every operator $\pi_{1/2}(A) \otimes \mathbb{1}_{\mathbb{C}^2}$ with $A \in \mathfrak{A}_0$, the second factor is what generates the commutant of the observable algebra inside $\mathcal{B}(\mathcal{H}_{\omega_0})$. Therefore, we can write the representation π_{ω_0} in the following way

$$\pi_{\omega_0}(A) = \pi_{1/2}(A) \otimes \mathbb{1}_{\mathbb{C}^2}, \quad A \in \mathfrak{A}_0. \quad (2.66)$$

The cyclic vector, as in Definition 2.2.34, is identified above as $\Psi_{\omega_0} = [\mathbb{1}]$. The identity matrix decomposes as $\mathbb{1} = E_{11} + E_{22}$. Under the map shown earlier, we can express the cyclic vector as

$$\Psi_{\omega_0} \longleftrightarrow \frac{1}{\sqrt{2}}(e_1 \otimes e_1 + e_2 \otimes e_2). \quad (2.67)$$

This vector embodies the mechanism of purification. The state ω_0 , which on $\mathbb{C}^2$ alone requires the density matrix $\rho = \frac{1}{2}\mathbb{1}$ as in Equation (2.33), is here represented instead as a vector state in the enlarged space $\mathbb{C}^2 \otimes \mathbb{C}^2$. That is engineered so that its expectation value on every physical observable $\pi_{1/2}(A) \otimes \mathbb{1}_{\mathbb{C}^2}$ reproduces the evaluation already dictated by the original density matrix $\rho = \frac{1}{2}\mathbb{1}$.

The Bloch Sphere and Mixed States

The pure states of the system are the rays of $\mathbb{C}^2$. That is the equivalence classes of unit vectors modulo multiplication by a phase. The space of such rays is the complex projective line $\mathbb{P}\mathbb{C}^1$, subject of work in the next Chapter. Each vector $\psi \in \mathbb{C}^2$ identifies a ray $[\psi] \in \mathbb{P}\mathbb{C}^1$. That is parametrised by the expectation values of the Pauli matrices given by $\mathbf{r} = (\psi, \boldsymbol{\sigma}\psi) \in \mathbb{C}^3$, which satisfy $|\mathbf{r}| = 1$ to determine the ray uniquely. Therefore $\mathbb{P}\mathbb{C}^1 \cong \mathbb{S}^2$, and this identification is called the Bloch sphere. States on this projective space are defined via

$$\omega_\psi(A) = (\psi, \pi_{1/2}(A)\psi), \quad \forall \psi \in \mathbb{C}^2, \|\psi\| = 1. \quad (2.68)$$

More generally, every state of $\mathfrak{A}_0$ is

$$\omega_\rho(A) = \text{Tr}(\rho \pi_{1/2}(A)), \quad \rho = \frac{1}{2}(\mathbb{1} + \mathbf{r} \cdot \boldsymbol{\sigma}), \quad |\mathbf{r}| \leq 1, \quad (2.69)$$

and its space is called the Bloch ball. Pure states are identified with points at the boundary $|\mathbf{r}| = 1$, where ω_0 is the centre $\mathbf{r} = 0$ representing the maximally mixed state.

By Theorem 2.7, every state, pure or mixed, has a *cyclic* GNS vector. What changes with $\mathbf{r}$ is the irreducibility of the representation for Proposition 2.3.13. For $|\mathbf{r}| < 1$, ρ is invertible and ω_ρ is faithful, so the argument used above for ω_0 applies. Only at $|\mathbf{r}| = 1$ does it collapse to the irreducible $\pi_{1/2}$ on $\mathbb{C}^2$.

Comparison with the Classical Spin System

The classical spin of Section 1.7, generated by x_1, x_2, x_3 with $\{x_i, x_j\} = \epsilon_{ijk}x_k$ yields the commutative algebra $C(\mathbb{S}^2)$. Its Gelfand spectrum, which was the sphere itself, is a continuum of pure states. The quantum spin uses the same Lie algebra noncommutatively admitting a *unique* irreducible representation $\pi_{1/2}$ up to unitary equivalence. Its pure states reproduce the same sphere $\mathbb{S}^2$. But the full quantum state space, the Bloch ball, is strictly richer. In fact, it contains mixed states with no counterpart in $C(\mathbb{S}^2)$. The GNS construction and the Gelfand-Naimark Theorem are what let the same $\mathfrak{su}(2)$ algebra generate a symplectic manifold commutatively and a Hilbert space noncommutatively.

These extra features arise from the difference in the *geometry* of the two state spaces. The classical state space, representing probability measures on $\mathbb{S}^2$, as show in Section 1.6, is a generally simplex. Moreover, every classical mixed state decomposes uniquely into pure ones. However, the Bloch ball is strictly convex, and this uniqueness fails completely. This makes ω_0 at its centre an equal mixture of ω_ψ and $\omega_{\psi^\perp}$ for *every* unit vector $\psi \in \mathbb{C}^2$. A classical mixed state can always be read as an ignorance about which pure state actually obtains. Instead, a generic quantum mixed state cannot be given such a reading at all. It is this loss of a unique "ignorance interpretation" that foundational frameworks take as the operational signature of nonclassicality.

The next Example 3.8 will continue this formulation leading the discussion to time evolution, symmetries and Quantum Noether's Theorem 3.11 that will be presented in the next chapter.

Chapter 3

Recovery of Hamiltonian Structures in the Projective State Space

The correspondence between the quantum and classical theories lies not so much in the limiting agreement when $\hbar \rightarrow 0$ as in the fact that the mathematical operations on the two theories obey in many cases the same laws.

Paul A. M. Dirac, [Dir25]

The previous chapter showed an algebraic reformulation of quantum mechanics that only addressed the time-independent axiomatics. We ask how a quantum system evolves and how its continuous symmetries manifest. Instead of postulating a separate equation of motion or invoking group representations on a Hilbert space, we derive both the dynamics and the symmetry principles from the geometry as in Hamiltonian mechanics. The starting point is the set of pure states, which becomes a projective Hilbert space. This manifold carries a natural geometric structure that combines symplectic, Riemannian, and complex structures in a mutually compatible way called a Kähler structure.

The symplectic form provides the Poisson bracket that generates time evolution. Given the energy operator, its mean value becomes a smooth function on the state space, and the associated Hamiltonian vector field defines a flow whose integral curves are the solutions of the Schrödinger equation. The Riemannian metric captures the statistical and probabilistic aspects of the theory. In fact, it induces a notion of distance between pure states. Moreover, the geodesic separation between two neighbouring states coincides with the transition probability predicted by the Born rule. At the end, continuous symmetries of the system are implemented as flows generated by such admissible functions. Geometry guarantees the existence of a momentum map assigning to each symmetry generator a conserved quantity. This construction yields a geometric version of Noether's theorem. The entire structure thus follows from the intrinsic geometry of the state manifold, providing a fully geometrical formulation of quantum mechanics.

3.1 A Brief Introduction to Kähler Geometry

The mathematical framework used in this chapter is a generalization of the one used in Chapter 1. We see that differential manifolds can be stated substituting open sets of $\mathbb{R}^n$ as the image of charts of the maximal atlas. In fact, to generalize differential geometry to complex numbers, we need an operation that mimics the complex unit, called a complex structure. Together with a particular metric and a manifold, this will create the key structure of the chapter, the Kähler manifold. This mathematical concept is rich enough to accommodate the whole framework of quantum mechanics, as we will see in the next sections.

We start by modeling a smooth manifold on a Hilbert space, as per Definition 2.2.6. Although this statement follows Definition 1.1.4, we still need to ensure compatibility with the structures already imposed on the Hilbert space.

Definition 3.1.1 (Hilbertian manifold). A real smooth Hilbertian manifold $\mathcal{M}$ modeled on a real Hilbert space $\mathcal{H}$, as per Definition 2.2.6, is a smooth manifold, as per Definition 1.1.4, whose maximal atlas $\mathfrak{A} = \{(U_\alpha, \phi_\alpha)\}_{\alpha \in I}$, as in Definition 1.1.2, satisfies the following conditions.

1. For every $\alpha \in I$, the image $\phi_\alpha(U_\alpha)$ is an open subset of $\mathcal{H}$.
2. For every $\alpha, \beta \in I$ such that $U_\alpha \cap U_\beta \neq \emptyset$, the transition map $\phi_\beta \circ \phi_\alpha^{-1}$ is of class C^∞ in the sense of the Fréchet derivative.

The almost complex structures and Riemannian metrics of an almost Kähler manifold act as sections of spaces of multilinear operators on the tangent space. Fixing notation for these spaces here avoids repeating it at every subsequent definition.

Definition 3.1.2 (Families of multilinear maps). Let $\mathcal{M}$ be a real smooth Hilbertian manifold, as per Definition 3.1.1. For each $x \in \mathcal{M}$, consider the Banach spaces of bounded linear operators on $T_x\mathcal{M}$, of bounded linear maps $T_x^*\mathcal{M} \rightarrow T_x\mathcal{M}$, and of bounded n -linear forms on $T_x\mathcal{M}$. The disjoint unions of these spaces over all $x \in \mathcal{M}$ are denoted, respectively, by

$$\mathcal{B}(T\mathcal{M}) := \bigsqcup_{x \in \mathcal{M}} \mathcal{B}(T_x\mathcal{M}), \quad \mathcal{B}(T^*\mathcal{M}, T\mathcal{M}) := \bigsqcup_{x \in \mathcal{M}} \mathcal{B}(T_x^*\mathcal{M}, T_x\mathcal{M}), \quad \mathcal{B}_n(T\mathcal{M}, \mathbb{R}) := \bigsqcup_{x \in \mathcal{M}} \mathcal{B}_n(T_x\mathcal{M}, \mathbb{R}).$$

With this notation available, the almost complex structures needed to define an almost Kähler manifold can be introduced as smooth sections of $\mathcal{B}(T\mathcal{M})$ squaring to minus the identity.

Definition 3.1.3 (Almost complex structure). Let $\mathcal{M}$ be a real smooth Hilbert manifold, as per Definition 3.1.1. An almost complex structure on $\mathcal{M}$ is a smooth section $J : \mathcal{M} \rightarrow \mathcal{B}(T\mathcal{M})$, such that for every $x \in \mathcal{M}$, $J(x) = J_x \in \mathcal{B}(T\mathcal{M})$ satisfies

$$J_x \circ J_x = -\mathbb{1}_{T_x\mathcal{M}}.$$

Consequently, the map J endows each real tangent space with the algebraic structure of a complex vector space playing the role of the imaginary unit.

Example 3.1.4. On $\mathbb{R}^{2n} \cong \mathbb{C}^n$ with real coordinates $(x^1, \dots, x^n, y^1, \dots, y^n)$ and $z^j = x^j + iy^j$, the linear map defined pointwise by $J\partial_{x^j} = \partial_{y^j}$, $J\partial_{y^j} = -\partial_{x^j}$ satisfies $J^2 = -\mathbb{1}$. It descends that it is an almost complex structure as per Definition 3.1.3. It is the endomorphism induced by multiplication by i on $\mathbb{C}^n$ under the identification $T_z\mathbb{C}^n \cong \mathbb{C}^n$. Its integrability, in the sense of Definition 3.1.5 comes from the trivial holomorphic atlas of $\mathbb{C}^n$.

Not every almost complex structure comes from an underlying complex manifold structure. The obstruction is measured by a torsion tensor, and the case in which it vanishes deserves its own name because it is the case relevant to Kähler geometry.

Definition 3.1.5 (Integrable almost complex structure). Let $\mathcal{M}$ be a real smooth Hilbert manifold, as per Definition 3.1.1. An almost complex structure J as per Definition 3.1.3 is integrable if its torsion vanishes. That is, if for every open set $\mathcal{U} \subset \mathcal{M}$ and every pair of smooth vector fields X, Y on $\mathcal{U}$,

$$[JX, JY] - [X, Y] - J[X, JY] - J[JX, Y] = 0.$$

Having set the complex structure, the second needed ingredient is a metric.

Definition 3.1.6 (Riemannian metric). Let $\mathcal{M}$ be a real smooth Hilbertian manifold, as per Definition 3.1.1. A Riemannian metric on $\mathcal{M}$ is a smooth section $g : \mathcal{M} \rightarrow \mathcal{B}_2(T\mathcal{M}, \mathbb{R})$, as in Definition 3.1.2, such that, for each $x \in \mathcal{M}$, the evaluation $g_x : T_x\mathcal{M} \times T_x\mathcal{M} \rightarrow \mathbb{R}$ is a symmetric, and positive-definite bilinear map.

Remark 3.1.7 (Local coordinate expression and line element). In a local coordinate chart (U, x) with real coordinates $(x^1, \dots, x^n)$, the Riemannian metric g is a symmetric bilinear map on vectors. It is expressed via the tensor product as

$$g = \sum_{i,j=1}^n g_{ij} dx^i \otimes dx^j, \tag{3.1}$$

where $g_{ij} := g\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right)$ are the smooth metric components and $\{dx^i\}$ constitutes the dual basis of the cotangent space. In the literature, the metric is also conventionally denoted by the infinitesimal line element

$$ds^2 = \sum_{i,j=1}^n g_{ij} dx^i dx^j. \quad (3.2)$$

In this notation, the juxtaposition $dx^i dx^j$ is a formal abbreviation for the symmetric tensor product $\frac{1}{2}(dx^i \otimes dx^j + dx^j \otimes dx^i)$. Consequently, expressions such as dx^2 denote the rank-two tensor $dx \otimes dx$. When evaluated on a tangent vector field, this tensor returns the square of the vector's spatial projection along the associated coordinate axis.

Having set all the structures needed, we are ready to enrich a Hilbertian manifold with an almost complex structure and a Riemannian metric. This generates an almost Kähler manifold.

Definition 3.1.8 (Almost Kähler manifold). *An almost Kähler manifold is a triple $(\mathcal{M}, J, g)$, where $\mathcal{M}$ is a real smooth Hilbertian manifold, as per Definition 3.1.1, J is an almost complex structure as per Definition 3.1.3, while g is a Riemann metric, as in Definition 3.1.6 such that*

- (i) g is J -invariant, that is, $g_x(J_x X_x, J_x Y_x) = g_x(X_x, Y_x)$ for all $x \in \mathcal{M}$, $X_x, Y_x \in T_x \mathcal{M}$.
- (ii) The fundamental form defined via $\omega_x(X_x, Y_x) := g_x(J_x X_x, Y_x)$ is closed.

Remark 3.1.9. Definition 3.1.8 intertwines Riemannian, complex, and symplectic structures.

- Condition (i) makes J a metric isometry, yielding an almost Hermitian structure.
- Condition (ii) ensures that the fundamental 2-form $\omega = g(J \cdot, \cdot)$ is closed and hence non-degenerate, elevating it to a symplectic form. Thus an almost Kähler manifold is a symplectic manifold, as in Definition 1.3.8, with compatible J and metric g .

Physically, the triple (g, J, ω) unifies Hamiltonian mechanics, via ω , quantum state distances via g , and complex phase via J . The qualifier “almost” reflects that J need not be integrable, i.e., its torsion tensor need not vanish.

Once the almost complex structure of an almost Kähler manifold is required to be integrable in the sense of Definition 3.1.5, the manifold acquires an underlying holomorphic structure. It deserves the sharper name of Kähler manifold, which is the setting in which the projective Hilbert space of a quantum system will be placed.

Definition 3.1.10 (Kähler manifold). *An almost Kähler manifold $(\mathcal{M}, J, g)$ as per Definition 3.1.8 is called a Kähler manifold if its almost complex structure J is integrable, as per Definition 3.1.5.*

The fundamental form ω and the metric g of an almost Kähler manifold each induce an isomorphism from the cotangent to the tangent space at every point, and it is these two musical isomorphisms, rather than ω and g themselves, that will enter into the definition of the Poisson and Riemann brackets.

Definition 3.1.11 (Musical isomorphisms of an almost Kähler manifold). *Let $(\mathcal{M}, J, g)$ be an almost Kähler manifold as per Definition 3.1.8, with fundamental form ω . The musical isomorphisms $I_x, G_x : T_x^* \mathcal{M} \rightarrow T_x \mathcal{M}$, for $x \in \mathcal{M}$, are the topological-linear isomorphisms defined implicitly by*

$$\omega_x(I_x \alpha_x, X_x) = \langle \alpha_x, X_x \rangle, \quad g_x(G_x \alpha_x, X_x) = \langle \alpha_x, X_x \rangle, \quad (3.3)$$

for $\alpha_x \in T_x^* \mathcal{M}$, $X_x \in T_x \mathcal{M}$. The corresponding smooth sections of $\mathcal{B}(T^* \mathcal{M}, T \mathcal{M})$ are denoted I, G .

Remark 3.1.12. The two musical isomorphisms of Definition 3.1.11 are related by $G = J \circ I$. Indeed, for $\alpha_x \in T_x^* \mathcal{M}$ and $X_x \in T_x \mathcal{M}$, the definitions of ω and I_x yield

$$g_x(J_x I_x \alpha_x, X_x) = \omega_x(I_x \alpha_x, X_x) = \langle \alpha_x, X_x \rangle = g_x(G_x \alpha_x, X_x), \quad (3.4)$$

and non-degeneracy of g_x entails that $J_x I_x \alpha_x = G_x \alpha_x$ for every α_x , that is, $G = J \circ I$.

With the musical isomorphisms in place, two bilinear brackets between smooth functions can be defined, one recovering the ordinary symplectic Poisson bracket and the other, depending only on the metric, providing the dispersion structure that will later encode the probabilistic content of quantum mechanics.

Definition 3.1.13 (Poisson and Riemann brackets). *Let $(\mathcal{M}, J, g)$ be an almost Kähler manifold as per Definition 3.1.8, and let $f, l \in C^\infty(\mathcal{M}, \mathbb{R})$. The Poisson bracket $\{f, l\} \in C^\infty(\mathcal{M}, \mathbb{R})$ and the Riemann bracket $((f, l)) \in C^\infty(\mathcal{M}, \mathbb{R})$ are defined as*

$$\{f, l\} := \langle df, I dl \rangle, \quad ((f, l)) := \langle df, G dl \rangle. \quad (3.5)$$

Here I, G are the musical isomorphisms of Definition 3.1.11, while $\langle \cdot, \cdot \rangle$ is the inner product between forms and vectors. By bilinearity, both extend to smooth complex-valued functions on $\mathcal{M}$.

Example 3.1.14. On $\mathbb{R}^2$, identified with $\mathbb{C}$ via $z = x + iy$, define a 2-form, as in Definition 1.1.17 via $\omega = dx \wedge dy$. In addition, define a Riemannian metric, as in Definition 3.1.6 as $g = dx^2 + dy^2$. Moreover, let J be a complex structure, as in Definition 3.1.3 via $J\partial_x = \partial_y$.

For the coordinate functions $x, y \in C^\infty(\mathbb{R}^2, \mathbb{R})$, the differentials are dx and dy . By Definition 3.1.11, $I dy = \partial_x$ and $G dy = \partial_y$. Hence $\{x, y\} = \langle dx, \partial_x \rangle = 1$ recovers the canonical Poisson bracket, while $((x, y)) = \langle dx, \partial_y \rangle = 0$, since dx and dy are g -orthogonal. The two coordinate functions are canonically conjugate but independent.

Remark 3.1.15. The Poisson bracket of Definition 3.1.13 is the standard symplectic bracket associated with ω as in Definition 1.4.7, while the Riemann bracket, depending only on g , is not a symplectic invariant. Using $G = J \circ I$, as per Remark 3.1.12, together with the J -invariance of ω and g ,

$$\{f, l\} = \omega(Idf, Idl) = \omega(Gdf, Gdl), \quad ((f, l)) = g(Gdf, Gdl) = g(Idf, Idl). \quad (3.6)$$

The Poisson and Riemann brackets combine into a single complex-valued bracket. Together with the ordinary pointwise product they generate two further bilinear products on smooth functions, $\circ_\nu$ and $*_\nu$ products. They will be the geometric counterpart of the anticommutator and of the operator product once the manifold is specialized to $\mathbb{P}\mathcal{H}$.

Definition 3.1.16 (Kähler bracket, $\circ_\nu$ and $*_\nu$ products). *Let $(\mathcal{M}, J, g)$ be an almost Kähler manifold as per Definition 3.1.8, and let $f, l \in C^\infty(\mathcal{M}, \mathbb{C})$. The Kähler bracket between f and l is*

$$\langle f, l \rangle := ((f, l)) + i\{f, l\}, \quad (3.7)$$

where $\{\cdot, \cdot\}$ and $((\cdot, \cdot))$ are the brackets of Definition 3.1.13. For $\nu \in \mathbb{R}$, the $\circ_\nu$ product and the $*_\nu$ product between f and l are the smooth complex functions

$$f \circ_\nu l := \frac{1}{2} \nu ((f, l)) + fl, \quad f *_\nu l := \frac{1}{2} \nu \langle f, l \rangle + fl. \quad (3.8)$$

Remark 3.1.17. Equipped with the Riemann bracket, or alternatively with the $\circ_\nu$ product, $C^\infty(\mathcal{M}, \mathbb{R})$, respectively $C^\infty(\mathcal{M}, \mathbb{C})$, is a real, respectively complex, commutative, nonassociative algebra. The Kähler bracket and the $*_\nu$ product provide two further complex, nonassociative algebra structures on $C^\infty(\mathcal{M}, \mathbb{C})$. With either of these two products and with complex conjugation as involution, $C^\infty(\mathcal{M}, \mathbb{C})$ is an involutive algebra, as per [CMP90, p. 2892].

The four products introduced in Definition 3.1.16 are not independent. Each can be recovered from any other two, a fact that will be used repeatedly to move between the symplectic and the Jordan-like descriptions of the same algebraic structure.

Proposition 3.1.18. *Let $(\mathcal{M}, J, g)$ be an almost Kähler manifold as per Definition 3.1.8, and let $f, l \in C^\infty(\mathcal{M}, \mathbb{C})$. Then, with $\circ_\nu, *_\nu$ and $\{\cdot, \cdot\}$ as per Definitions 3.1.13 and 3.1.16,*

$$f *_\nu l = f \circ_\nu l + \frac{i}{2} \nu \langle f, l \rangle, \quad (3.9)$$

$$f \circ_\nu l = \frac{1}{2} (f *_\nu l + l *_\nu f), \quad (3.10)$$

$$\{f, l\} = \frac{1}{i\nu} (f *_\nu l - l *_\nu f) \quad (3.11)$$

The proof of this statement can be found in [CMP90, Equations (2.10)–(2.12)]. Among the smooth functions on an almost Kähler manifold, a distinguished class is identified by the requirement that its Hamiltonian flow preserve the metric, and not only the symplectic form. This is the class that will be identified with the observables of a quantum system.

Definition 3.1.19 (Kählerian function). *Let $(\mathcal{M}, J, g)$ be an almost Kähler manifold as per Definition 3.1.8, let $f \in C^\infty(\mathcal{M}, \mathbb{R})$, and let $X_f := Idf$, with I as per Definition 3.1.11. The function f is Kählerian if $\mathcal{L}_{X_f}g = 0$, where $\mathcal{L}$ denotes the Lie derivative, as in Definition 1.1.30. If $f \in C^\infty(\mathcal{M}, \mathbb{C})$, f is Kählerian if $\operatorname{Re}f$ and $\operatorname{Im}f$ are. The set of Kählerian functions is denoted $\mathcal{K}(\mathcal{M}, \mathbb{R})$, respectively $\mathcal{K}(\mathcal{M}, \mathbb{C})$.*

Remark 3.1.20. The vector field $X_f = Idf$ of Definition 3.1.19 is the Hamiltonian vector field of f , and the condition $\mathcal{L}_{X_f}g = 0$ states that the Hamiltonian flow of f preserves the metric g . Since ω is closed and $\iota_{X_f}\omega = df$ by Definition 1.4.1, Cartan's formula, Equation (1.2), yields

$$\mathcal{L}_{X_f}\omega = d\iota_{X_f}\omega + \iota_{X_f}d\omega = d(df) = 0.$$

The flow automatically preserves the fundamental form. Moreover, from $\mathcal{L}_{X_f}g = 0$ it follows that $\mathcal{L}_{X_f}J = 0$, since J is determined by ω and g through Definition 3.1.8. Therefore, if f is Kählerian, its Hamiltonian flow preserves the whole Kähler structure.

This closure is what will let the observables of a quantum system, once identified with Kählerian functions, be combined without leaving the class of admissible dynamical generators.

Proposition 3.1.21. *Let $\mathcal{K}(\mathcal{M}, \mathbb{R})$ as in Definition 3.1.19, respectively $\mathcal{K}(\mathcal{M}, \mathbb{C})$, is a Lie subalgebra of $C^\infty(\mathcal{M}, \mathbb{R})$, respectively $C^\infty(\mathcal{M}, \mathbb{C})$, with respect to the Poisson bracket of Definition 3.1.13.*

This statement is proved in [CMP90, p.2892]

Example 3.1.22 (Kählerian functions and the $\ast_\nu$ product). Continuing the flat space setup on $\mathbb{R}^2$ with $\omega = dx \wedge dy$ and $g = dx^2 + dy^2$. Consider the Hamiltonian function of the isotropic harmonic oscillator, $H(x, y) = \frac{1}{2}(x^2 + y^2)$. Its differential is $dH = xdx + ydy$. Applying the musical isomorphism I , the associated Hamiltonian vector field is $X_H = I dH = y\partial_x - x\partial_y$. Evaluating the Lie derivative of the metric tensor with respect to it yields

$$\mathcal{L}_{X_H}g = \mathcal{L}_{y\partial_x - x\partial_y}(dx \otimes dx + dy \otimes dy) = 0. \quad (3.12)$$

Hence, H is a Kählerian function as per Definition 3.1.19.

Furthermore, the geometric framework algebraically composes functions via the Kähler bracket. Using the vanishing Riemann bracket $((x, y)) = 0$ and the canonical Poisson bracket $\{x, y\} = 1$ established previously, Definition 3.1.16 yields the exact evaluation of the $\ast_\nu$ product

$$x \ast_\nu y = xy + \frac{\nu}{2}(((x, y)) + i\{x, y\}) = xy + \frac{i\nu}{2}. \quad (3.13)$$

Fixing $\nu = \hbar$, this geometric product reconstructs the associative, non-commutative, algebraic product of the quantum position and momentum observables for a flat phase space.

Complex manifolds and the Dolbeault operators

The definitions and theorems of the preceding sections can appear abstract when viewed through the lens of pure linear algebra on the Hilbert space $\mathcal{H}$. To ground these concepts and to show the way for explicit computations, it is essential to examine the local coordinate picture. The key simplification offered by Kähler geometry is the reduction of a complex, multi-component geometric object to a single, real-valued scalar function called the Kähler potential. We begin by setting the definition of a complex manifold which is the extension of a real one to complex numbers.

Definition 3.1.23 (Complex manifold). *A complex manifold of complex dimension n is a smooth manifold $\mathcal{M}$, as in Definition 1.1.4, of real dimension $2n$ together with an atlas of charts $\varphi_\alpha : U_\alpha \rightarrow \mathbb{C}^n$, $\mathcal{M} = \bigcup_\alpha U_\alpha$ such that every transition map $\varphi_\beta \circ \varphi_\alpha^{-1}$, between open subsets of $\mathbb{C}^n$, is holomorphic. Such atlas is called a holomorphic atlas for $\mathcal{M}$.*

A holomorphic atlas as per Definition 3.1.23 determines an integrable, almost complex structure on $\mathcal{M}$, obtained by transporting the multiplication by i on $\mathbb{C}^n$ through each chart. The converse, that every integrable almost complex structure arises this way, is the content of the Newlander-Nirenberg theorem, recalled here without proof.

Theorem 3.1 (Newlander-Nirenberg). *Let $\mathcal{M}$ be a smooth manifold of real dimension $2n$ equipped with an integrable almost complex structure J , as per Definition 3.1.5. Then $\mathcal{M}$ admits a holomorphic atlas, as per Definition 3.1.23, inducing J .*

This theorem is proved in [Mor12, Theorem 7.4]. Fix a complex manifold $\mathcal{M}$ of complex dimension n as per Definition 3.1.23, and a local holomorphic chart (U, φ) with complex coordinates $z = (z^1, \dots, z^n) \in \mathbb{C}^n$, so that $z^j = x^j + iy^j$. Here $(x^1, \dots, x^n, y^1, \dots, y^n)$ are the corresponding local real coordinates on U . The complexified cotangent space at each point of U splits according to the holomorphic and antiholomorphic differentials. This splitting is what allows differential forms to be assigned a bidegree.

Definition 3.1.24 (Forms of type (p, q)). *Let (U, φ) be a local holomorphic chart on a complex manifold $\mathcal{M}$, as per Definition 3.1.23, with complex coordinates $z = (z^1, \dots, z^n)$. The subspaces $\Lambda^{1,0}$ and $\Lambda^{0,1}$ of the complexified cotangent space $T^*\mathcal{M} \otimes \mathbb{C}$ at each point of U are the spans of $\{dz^1, \dots, dz^n\}$ and of $\{d\bar{z}^1, \dots, d\bar{z}^n\}$, respectively. A differential form on U is of type (p, q) if, at every point, it lies in*

$$\Lambda^{p,q} := \underbrace{\Lambda^{1,0} \wedge \dots \wedge \Lambda^{1,0}}_p \wedge \underbrace{\Lambda^{0,1} \wedge \dots \wedge \Lambda^{0,1}}_q. \quad (3.14)$$

The decomposition of forms by type, as per Definition 3.1.24, is compatible with the exterior derivative, and this compatibility is what makes it possible to split d into a holomorphic and an antiholomorphic piece.

Definition 3.1.25 (Dolbeault operators). *Let (U, φ) be a local holomorphic chart on a complex manifold $\mathcal{M}$, as per Definition 3.1.23, with complex coordinates $z = (z^1, \dots, z^n)$. On $C^\infty(U, \mathbb{C})$, the Dolbeault operators $\partial, \bar{\partial}$ are the linear differential operators defined locally by*

$$\partial f := \sum_{j=1}^n \frac{\partial f}{\partial z^j} dz^j, \quad \bar{\partial} f := \sum_{j=1}^n \frac{\partial f}{\partial \bar{z}^j} d\bar{z}^j, \quad (3.15)$$

extended to forms of type (p, q) , as per Definition 3.1.24, by $d = \partial + \bar{\partial}$, with ∂ raising p by one and $\bar{\partial}$ raising q by one.

Remark 3.1.26. The nilpotency of the exterior derivative, $d^2 = 0$, together with the splitting $d = \partial + \bar{\partial}$ of Definition 3.1.25, yields $0 = d^2 = \partial^2 + \bar{\partial}^2 + (\partial \bar{\partial} + \bar{\partial} \partial)$. Since ∂^2 , $\bar{\partial}^2$ and $\partial \bar{\partial} + \bar{\partial} \partial$ raise the bidegree (p, q) of a form to $(p+2, q)$, $(p, q+2)$ and $(p+1, q+1)$ respectively, these three subspaces intersect trivially. Consequently, each term vanishes separately:

$$\partial^2 = 0, \quad \bar{\partial}^2 = 0, \quad \partial \bar{\partial} + \bar{\partial} \partial = 0. \quad (3.16)$$

A closed real $(1, 1)$ -form on a complex manifold, such as the fundamental form of a Kähler manifold, admits a local description in terms of the Dolbeault operators just introduced. This is the analytic feature underlying the notion of Kähler potential.

Theorem 3.2 (Local $\partial \bar{\partial}$ -lemma). *Let ω be a real, closed differential form of type $(1, 1)$, as per Definition 3.1.24, on a complex manifold $\mathcal{M}$, as per Definition 3.1.23. For every point $p \in \mathcal{M}$, there exists a contractible local holomorphic chart (U, z) containing p and a smooth real-valued function $K \in C^\infty(U, \mathbb{R})$ such that, on U ,*

$$\omega = i \partial \bar{\partial} K \quad (3.17)$$

For the proof, we will refer the reader to [Bal06, Theorem 4.17]. This theorem introduces a key function used to compute the symplectic form on a complex manifold.

Definition 3.1.27 (Kähler potential). *Let ω be a real, closed $(1, 1)$ -form on a complex manifold $\mathcal{M}$, as per Definition 3.1.23, and let (U, z) be a local holomorphic chart. A smooth function $K \in C^\infty(U, \mathbb{R})$ such that $\omega = i \partial \bar{\partial} K$ on U , as provided by Theorem 3.2, is called a local Kähler potential for ω on (U, z) .*

Remark 3.1.28. The local Kähler potential of Definition 3.1.27 is not unique. In fact, if K and K' are both local Kähler potentials for ω on (U, z) , then $\partial\bar{\partial}(K - K') = 0$ on U , so that $K - K'$ coincides with the real part of a holomorphic function. Conversely, adding to K the real part of any holomorphic function on U produces another local Kähler potential for ω .

Once a local Kähler potential is available, the whole Riemannian metric tensor of a Kähler manifold as in Definition 3.1.10, restricted to a finite-dimensional holomorphic chart, is determined by the mixed second-order derivatives of that single scalar function.

Theorem 3.3 (Local coordinate expression of the metric). *Let $(\mathcal{M}, J, g)$ be a Kähler manifold, as per Definition 3.1.10, of finite complex dimension n , and let K be a local Kähler potential, as per Definition 3.1.27, for the fundamental form ω on a chart (U, z) . The components of the metric tensor g in the coordinates $(z^1, \dots, z^n)$ are given by*

$$g_{j\bar{k}} = \frac{\partial^2 K}{\partial z^j \partial \bar{z}^k}. \quad (3.18)$$

The resulting Hermitian matrix $(g_{j\bar{k}})$ is positive definite everywhere on U .

This statement is not proved here. For further readings, see [Mor12, p.58].

Remark 3.1.29. The positive definiteness recorded in Theorem 3.3, in the coordinates $(z^1, \dots, z^n)$, is equivalent to the requirement $g(X, X) > 0$ for every nonzero tangent vector X , which is already part of g being a Riemannian metric on $\mathcal{M}$ as per Definition 3.1.8.

Local Expression of Curvature

The same local potential that determines the metric also determines its curvature. This geometrical object measures the local deviation from a flat Euclidean space. Since the metric components $g_{j\bar{k}}$ are defined by the second derivatives of the Kähler potential K , the local obstruction to flattening the metric relies on the second derivatives of $g_{j\bar{k}}$. Consequently, the Riemann curvature tensor is algebraically encoded within the fourth-order mixed partial derivatives of the single scalar function K .

Definition 3.1.30 (Riemann curvature via Kähler potential). *Let $(\mathcal{M}, J, g)$ be a Kähler manifold, as in Definition 3.1.10 with local holomorphic coordinates $(z^1, \dots, z^n)$ and Kähler potential K , as in Definition 3.1.27. The components of the Riemann curvature tensor $R_{j\bar{k}m\bar{l}}$ evaluated on holomorphic planes are uniquely determined by the metric components $g_{j\bar{k}} = \partial_j \partial_{\bar{k}} K$.*

In a holomorphic chart centered at a point, the Kähler potential can always be chosen so that the metric components satisfy $g(\partial_j, \partial_{\bar{k}})|_0 = \delta_{jk}$ and all first derivatives vanish at the origin. This is equivalent to the freedom to add the real part of a holomorphic function to K . In such adapted coordinates, the curvature tensor depends on the fourth derivatives of K , with no lower-order interference.

At any local origin $z = 0$, where the first derivatives of the metric vanish, $\partial_m g_{j\bar{k}}|_0 = 0$, the curvature tensor components reduce to

$$R_{j\bar{k}m\bar{l}} := -\frac{\partial^2 g_{j\bar{k}}}{\partial z^m \partial \bar{z}^l} = -\frac{\partial^4 K}{\partial z^j \partial \bar{z}^k \partial z^m \partial \bar{z}^l}.$$

Contracting $R_{j\bar{k}m\bar{l}}$ with X and $\bar{X}$, normalized by $g(X, X)$, yields the holomorphic sectional curvature. That is the curvature along the complex curve tangent to X . This scalar will characterise $\mathbb{PH}$ in Theorem 3.5 once shown constant.

Definition 3.1.31 (Holomorphic sectional curvature). *Let $(\mathcal{M}, J, g)$ be a Kähler manifold, as in Definition 3.1.10 with Riemann curvature as in Definition 3.1.30. Given a tangent vector $X = \sum v^j \partial_{z^j}$, the holomorphic sectional curvature $H(X)$ is the normalized restriction of the curvature tensor along the complex line generated by X :*

$$H(X) := \frac{\sum R_{j\bar{k}m\bar{l}} v^j \bar{v}^k v^m \bar{v}^l}{2(\sum g_{j\bar{k}} v^j \bar{v}^k)^2}. \quad (3.19)$$

Before curving the manifold, it is worth checking that the formula returns the answer expected of the flat model, which carries no curvature at all.

Example 3.1.32 (Flat space). On $\mathbb{C}^n$ with the flat Kähler potential $K(z, \bar{z}) = \|z\|^2$, the metric components are $g_{j\bar{k}} = \partial_j \partial_{\bar{k}} K = \delta_{jk}$, constant on all of $\mathbb{C}^n$. Every derivative of $g_{j\bar{k}}$ vanishes, so $R_{j\bar{k}m\bar{l}} = 0$ and $H(X) = 0$ for every tangent vector X , as expected of Euclidean space regarded as a Kähler manifold.

The local Kähler potential thus determines the entire geometry, metric and curvature alike. The non-vanishing terms at the fourth order of the Taylor expansion of K provide the exact fundamental obstruction to mapping the manifold globally to a Euclidean space. This algebraic reduction establishes the explicit starting point for evaluating the constant holomorphic sectional curvature of the projective Hilbert space in Proposition 3.3.8 below.

3.2 The Projective Hilbert Space as a Kähler Manifold

We specialize the general Kähler framework to the manifold of physical interest, namely a projective Hilbert space. This space is obtained by identifying non vanishing vectors of a complex Hilbert space which differ by complex scaling. We show that the quotient is the pure state space of the algebra of compact operators. Equipped with a holomorphic atlas, it acquires a complex structure, while the underlying inner product naturally induces both a Riemannian metric and a compatible symplectic form. These two structures, together with the underlying complex structure, form the Fubini–Study geometry, which is the canonical Kähler structure of the pure state space. Subsequently, we derive their explicit local coordinate expressions. This is achieved introducing a global Kähler potential on the chart, as outlined in the previous section. We start by constructing the projective Hilbert space.

Definition 3.2.1 (Space of unit norm representatives). *Let $\mathcal{H}$ be a Hilbert space, as in Definition 2.2.6. The space of unit norm representatives is*

$$\mathbb{S}\mathcal{H} := \{\Psi \in \mathcal{H} \mid (\Psi, \Psi) = 1\}.$$

To construct the state manifold, the Hilbert space is quotiented by an equivalence relation that identifies vectors differing only by a non-zero complex scalar. This eliminates the unobservable scale and phase.

Definition 3.2.2 (Projective Hilbert space). *Let $\mathcal{H}$ be a Hilbert space. On $\mathcal{H}^* := \mathcal{H} \setminus \{0\}$ set $\Psi \sim \Phi$ if $\exists z \in \mathbb{C} \setminus \{0\}$ such that $\Phi = z\Psi$. The projective Hilbert space is*

$$\mathbb{P}\mathcal{H} := \mathcal{H}^* / \sim,$$

with $[\Psi]$ the equivalence class of $\Psi \in \mathcal{H}^*$.

The quotient operation on the full Hilbert space restricts to a phase quotient on the unit sphere, as shown in the next proposition.

Proposition 3.2.3. *Let $\mathbb{S}\mathcal{H}$ be the space of unit norm representatives as in Definition 3.2.1. Restricting the identification in Definition 3.2.2, $\sim$, to $\mathbb{S}\mathcal{H}$ yields $[\Psi] \mapsto \tau(\Psi)$. This is a bijection $\mathbb{P}\mathcal{H} \rightarrow \mathbb{S}\mathcal{H}/U(1)$.*

This statement is proved in [Lan98, Section 2.5]. To connect the geometric quotient space $\mathbb{P}\mathcal{H}$ to the operational framework of Chapter 2, the pure state space of an arbitrary C^* -algebra must be mapped to projective Hilbert spaces. The GNS construction establishes that the total pure state space decomposes into a disjoint union of sectors, proving that studying a single $\mathbb{P}\mathcal{H}$ is sufficient to capture the geometry of any superselection sector.

Theorem 3.4 (Pure state decomposition). *The pure state space $\mathcal{P}(\mathfrak{A})$ of a C^* -algebra $\mathfrak{A}$ as in Definition 2.3.11 is a disjoint union $\mathcal{P}(\mathfrak{A}) = \bigcup_{\alpha} \mathbb{P}\mathcal{H}_{\alpha}$, where $\mathcal{H}_{\alpha}$ is the Hilbert space of the irreducible GNS representation Theorem 2.7, of a state in $\mathbb{P}\mathcal{H}_{\alpha}$. States in the same $\mathbb{P}\mathcal{H}_{\alpha}$ are equivalent, states in different $\mathbb{P}\mathcal{H}_{\alpha}$ are inequivalent, and each inclusion $\mathbb{P}\mathcal{H}_{\alpha} \hookrightarrow \mathcal{P}(\mathfrak{A})$ is continuous.*

This decomposition is proved in [Lan98, Theorem 2.5.4]. In the specific case where the observable algebra consists of the compact operators, the pure state space reduces to a single sector.

Proposition 3.2.4. *Let $\mathbb{P}\mathcal{H}$ be a projective Hilbert space as in Definition 3.2.2. Under the bijection specified in Proposition 3.2.3, $\mathbb{P}\mathcal{H}$ is identified with the pure state space of $\mathfrak{B}_0(\mathcal{H})$.*

The identification with the pure states of $\mathfrak{B}_0(\mathcal{H})$ follows from the correspondence between vector states and rank-one projections [Lan98, Corollary 2.5.3]. Having established $\mathbb{P}\mathcal{H}$ as the physical state space, it must be rewritten as a Hilbertian manifold, as per Definition 3.1.1. This requires defining a maximal atlas of charts on a Hilbert space. These are constructed by projecting a neighborhood of a state onto the orthogonal complement of its representative vector.

Definition 3.2.5 (Local chart of $\mathbb{P}\mathcal{H}$). *Let $\mathbb{P}\mathcal{H}$ be a projective Hilbert space as in Definition 3.2.2. For $\psi \in \mathbb{P}\mathcal{H}$ and for $\Psi \in \mathcal{SH}$ a representative, $N_\psi := \{\varphi \in \mathbb{P}\mathcal{H} \mid (\Psi, \Phi) \neq 0\}$ with Φ any representative of φ , set $F_\psi : N_\psi \rightarrow \mathcal{H}$ as*

$$F_\psi(\varphi) := \frac{\Phi}{(\Psi, \Phi)} - \Psi. \quad (3.20)$$

The structural properties of the map F_ψ guarantee that it provides a valid coordinate system modeled on the closed subspace $\Psi^\perp$.

Proposition 3.2.6. *Let $\mathbb{P}\mathcal{H}$ be a projective Hilbert space as in Definition 3.2.2. For $\psi \in \mathbb{P}\mathcal{H}$, let F_ψ be the map and N_ψ be the set as per Definition 3.2.5. It holds that*

- F_ψ is well defined,
- $F_\psi(\psi) = 0$,
- the image of F_ψ is $\Psi^\perp$,
- $F_\psi : N_\psi \rightarrow \Psi^\perp$ is a homeomorphism.

Hence, $(N_\psi, F_\psi, \Psi^\perp)$ is a local chart of $\mathbb{P}\mathcal{H}$, as in Definition 1.1.2.

Proof. The element $\Phi/(\Psi, \Phi) \in \mathbb{P}\mathcal{H}$ is invariant under $\Phi \mapsto z\Phi$, $z \neq 0$. Hence, F_ψ is well defined, and $F_\psi(\psi) = \Psi/(\Psi, \Psi) - \Psi = 0$. In addition, F_ψ takes values in $\Psi^\perp$ because $(\Psi, F_\psi(\varphi)) = (\Psi, \Phi)/(\Psi, \Phi) - 1 = 0$. The map $\chi \mapsto \tau(\Psi + \chi)$, $\chi \in \Psi^\perp$, is a well-defined two-sided continuous inverse, since $\Psi + \chi \neq 0$ and since both maps are built from continuous algebraic operations on the domain $(\Psi, \Phi) \neq 0$. $\square$

Definition 3.2.5 can be used together with its properties in Proposition 3.2.6 to build an atlas on $\mathbb{P}\mathcal{H}$, to endow it with the structure of a Hilbertian manifold. This is made explicit in the next proposition.

Proposition 3.2.7 (Holomorphic atlas). *The collection of charts $\{(N_\psi, F_\psi, \Psi^\perp)\}_{[\Psi] \in \mathbb{P}\mathcal{H}}$ endows $\mathbb{P}\mathcal{H}$ with the structure of a Hilbertian manifold, as in Definition 3.1.1.*

The proof of this proposition is brought by [Lan98, Section 2.5]. The assignment of this differential structure induces a manifold topology on $\mathbb{P}\mathcal{H}$. To be physically consistent with the algebraic formulation, this manifold topology must coincide with the weak-* one, which governs the convergence of expectation values.

Remark 3.2.8 (Operational versus geometric topologies). The weak-* topology on the state space, as per Definition 1.6.28, is defined as the following. A sequence of states ω_α converges to ω if and only if $\omega_\alpha(A) \rightarrow \omega(A)$ for every observable $A \in \mathfrak{B}_0(\mathcal{H})$. This topology measures statistical indistinguishability. Conversely, the manifold topology induced by the holomorphic atlas of Proposition 3.2.7 measures geometric proximity, determined by the norm distance of representative vectors in the underlying Hilbert space. We will see in Section 3.4 that these two notions coincide.

Proposition 3.2.9. *The manifold topology on $\mathbb{P}\mathcal{H}$ coincides with the weak*-topologies induced by the dual space embeddings $\mathbb{P}\mathcal{H} \subset \mathfrak{B}_0(\mathcal{H})^*$ and $\mathbb{P}\mathcal{H} \subset \mathcal{B}(\mathcal{H})^*$, as per Definition 1.6.28.*

The proof of this statement is given in [Lan98, Proposition 2.5.2].

Remark 3.2.10 (Physical consistency of the manifold structure). The topological equivalence established in Proposition 3.2.9 guarantees that smooth curves in the projective manifold correspond to continuous variations in the expectation values of all physical observables. Without this identification, the differential geometric structures introduced subsequently, would be operationally meaningless with respect to the algebraic state space.

Corollary 3.2.11. *The pure state space of $\mathfrak{B}_0(\mathcal{H})$ equipped with the weak* topology is homeomorphic to $\mathbb{P}\mathcal{H}$ equipped with the manifold topology.*

Proof. Combine the bijection of Proposition 3.2.3 with the topological equivalence of Proposition 3.2.9. $\square$

This formulation establishes the topological and differentiable equivalence between the algebraic pure state space and the geometric manifold $\mathbb{P}\mathcal{H}$. Proceeding from this, the canonical metric and symplectic structures are introduced in the subsequent section.

Explicit Kähler structure

Using Proposition 3.2.6, we make explicit the construction of the Kähler manifold onto the projective Hilbert space. Let $\psi \in \mathbb{P}\mathcal{H}$ and let $\Psi \in \mathcal{SH}$ be a normalized representative, as shown in Proposition 3.2.3. By Proposition 3.2.6, the holomorphic tangent space $T_\psi \mathbb{P}\mathcal{H}$ is canonically isomorphic to the complex subspace $\Psi^\perp \subset \mathcal{H}$. To construct the Riemannian and symplectic structures, $T_\psi \mathbb{P}\mathcal{H}$ must be modeled as a real tangent space. This is achieved through the realification of $\Psi^\perp$.

The realification procedure restricts the field of scalars of $\Psi^\perp$ from $\mathbb{C}$ to $\mathbb{R}$. Consequently, the original complex scalar multiplication by the imaginary unit i is reinterpreted as a real-linear endomorphism on the realified space.

$$J : \Psi^\perp \rightarrow \Psi^\perp, \quad JV := iV,$$

satisfying $J^2 = -\mathbb{1}$. This is an almost complex structure on $\mathbb{P}\mathcal{H}$ as per Definition 3.1.3, integrable as per Definition 3.1.5 since it is induced by the holomorphic atlas of Proposition 3.2.7.

The Hilbert space inner product $(\cdot, \cdot)$ restricts to a Hermitian counterpart on $\Psi^\perp$. Its real and imaginary parts,

$$g_0(U, V) := \operatorname{Re}(U, V), \quad \omega_0(U, V) := \operatorname{Im}(U, V), \quad (3.21)$$

define a Riemannian metric g_0 on $\mathbb{P}\mathcal{H}$, as per Definition 3.1.6, and a real bilinear form ω_0 , with $(U, V) = g_0(U, V) + i\omega_0(U, V)$. Since scalar products are conjugate-linear in the first argument, $(iU, iV) = (U, V)$. It descends that $g_0(JU, JV) = g_0(U, V)$. This agrees with condition (i) of Definition 3.1.8. Evaluating instead on (U, JV) ,

$$(U, JV) = (U, iV) = i(U, V) = -\omega_0(U, V) + ig_0(U, V),$$

and comparing real parts with $(U, V) = g_0(U, V) + i\omega_0(U, V)$ yields the compatibility identity

$$\omega_0(U, V) = g_0(JU, V). \quad (3.22)$$

Closedness of ω_0 , that is condition (ii) of Definition 3.1.8, is verified below through its local expression via a Kähler potential. As a consequence, $(\mathbb{P}\mathcal{H}, J, g_0)$ is a Kähler manifold as per Definition 3.1.10.

This triple $(\mathbb{P}\mathcal{H}, J, g_0)$ constitutes the unscaled Fubini–Study structure of the projective Hilbert space. It is canonically induced by the linear inner product of $\mathcal{H}$ without introducing external parameters. However, g_0 and ω_0 are dimensionless mathematical objects. To correctly encode physical dynamics and quantum statistical dispersion, this canonical structure must be uniformly scaled by a real parameter.

Local coordinate expression via the Kähler potential

The function $\Phi \mapsto \|\Phi\|^2$ on $\mathcal{H}^*$ is the natural invariant of the quotient defining $\mathbb{P}\mathcal{H}$. We note that for $\lambda \in \mathbb{C} \setminus \{0\}$,

$$\ln \|\lambda\Phi\|^2 = \ln \lambda + \ln \bar{\lambda} + \ln \|\Phi\|^2.$$

In addition, $\partial \bar{\partial}$ annihilates the holomorphic term $\ln \lambda$ and the antiholomorphic term $\ln \bar{\lambda}$. It descends that

$$\partial \bar{\partial} \ln \|\lambda\Phi\|^2 = \partial \bar{\partial} \ln \|\Phi\|^2.$$

Hence $K_0 := \frac{1}{2} \ln \|\Phi\|^2$ descends to a local Kähler potential, as per Definition 3.1.27, unique up to the real part of a holomorphic function by Remark 3.1.28.

We apply Theorem 3.3 on the chart $N_\psi \simeq \Psi^\perp \simeq \mathbb{C}^n$ of Definition 3.2.5. A nearby state is represented by $\Phi = \Psi + \sum_{m=1}^n z^m e_m$, with $\{e_m\}$ an orthonormal basis of $\Psi^\perp$ and $z = (z^1, \dots, z^n) \in \mathbb{C}^n$. Since Ψ is normalized and orthogonal to each e_m ,

$$\|\Phi\|^2 = 1 + \|z\|^2.$$

The local Kähler potential is therefore $K_0(z, \bar{z}) = \frac{1}{2} \ln(1 + \|z\|^2)$, and

$$\frac{\partial K_0}{\partial z^j} = \frac{\bar{z}^j}{2(1 + \|z\|^2)}, \quad g_{0,j\bar{k}} = \frac{\partial^2 K_0}{\partial z^j \partial \bar{z}^k} = \frac{\delta_{jk}(1 + \|z\|^2) - \bar{z}^j z^k}{2(1 + \|z\|^2)^2}.$$

At $z = 0$ this yields $g_{0,j\bar{k}} = \delta_{jk}/2$, matching g_0 as per Equation (3.21) evaluated on the orthonormal basis $\{e_m\}$

$$g_0(\partial_j, \partial_k) = 2\operatorname{Re}(g_{0,j\bar{k}}) = 2g_{0,j\bar{k}} = \delta_{j\bar{k}}.$$

The fundamental form is $\omega_0 = i \sum_{j,k} g_{0,j\bar{k}} dz^j \wedge d\bar{z}^k$, which is closed by Remark 3.1.26, completing the proof that $(\mathbb{P}\mathcal{H}, J, g_0)$ is Kähler.

3.3 Schrödinger Equation as an Hamiltonian Flow

The Kähler structure (J, g_0) of Section 3.2 is fixed only up to an overall positive rescaling of g_0 . In fact, replacing g_0 by λg_0 , for any $\lambda > 0$, again satisfies Definition 3.1.8, since J -invariance and closedness of the fundamental form survive positive rescaling. This section fixes λ by requiring the resulting Hamiltonian flow to reproduce the Schrödinger equation. It shows that the value obtained is the one already implicit, on the algebraic side, in the convention $\frac{1}{i\hbar}[A, B]$ fixed for the Lie–Jordan algebra of Definition 2.1.19.

We begin by recalling the map $\pi : \mathcal{H} \setminus \{0\} \rightarrow \mathbb{P}\mathcal{H}$, $x \mapsto [x]$, the quotient map defining $\mathbb{P}\mathcal{H}$ as in Definition 3.2.2.

Definition 3.3.1 (Mean value function). *Given a Hilbert space $\mathcal{H}$, as in Definition 2.2.6, and a bounded operator $A \in \mathcal{B}(\mathcal{H})$, the mean value function of A is*

$$\langle A \rangle : \mathbb{P}\mathcal{H} \rightarrow \mathbb{C}, \quad \langle A \rangle([x]) := \frac{(x, Ax)}{\|x\|^2}.$$

If A in Definition 3.3.1 is selfadjoint, then $\langle A \rangle$ is real valued. By Stone's Theorem 2.10, a selfadjoint operator $H \in \mathcal{B}(\mathcal{H})$ generates the strongly continuous one-parameter unitary group $U_t = e^{iatH}$ for some $\alpha \in \mathbb{R}$. The physical postulate of Schrödinger dynamics fixes this constant to $\alpha = -1/\hbar$ and identifies H with the energy operator, so that a state $[x] \in \mathbb{P}\mathcal{H}$ evolves along

$$\gamma(t) := [U_t x] = [e^{-itH/\hbar} x], \quad t \in \mathbb{R}.$$

This postulated curve already carries the physical constant $\hbar$, while the Kähler structure $(J, \lambda g_0)$ of Section 3.2 does not. Its normalization $\lambda > 0$ is left free by Definition 3.1.8. We determine λ by requiring γ to be the integral curve of the Hamiltonian vector field $X_{\langle H \rangle} := I d\langle H \rangle$ of Definition 3.1.19, with I the musical isomorphism of Definition 3.1.11. Equivalently, we ask for which λ the postulated flow γ coincides with the flow generated via

$$\iota_\gamma \omega = d\langle H \rangle,$$

by the function $\langle H \rangle$ on $(\mathbb{P}\mathcal{H}, \lambda g_0)$.

Let $v \in T_{[x]}\mathbb{P}\mathcal{H}$ be a tangent vector. By Proposition 3.2.6, v is uniquely represented by $\eta \in x^\perp \subset \mathcal{H}$, for $x \in \mathcal{S}\mathcal{H}$ a normalized representative of $[x]$. We first compute the right-hand side $d\langle H \rangle_{[x]}(v)$ for generic v , then match it against the left-hand side produced by γ itself.

To compute $d\langle H \rangle_{[x]}(v)$, evaluate the directional derivative of the mean value function along the auxiliary linear path $\sigma(s) := x + s\eta$, whose projection to $\mathbb{P}\mathcal{H}$ passes through $[x]$ at $s = 0$ with velocity v . The evaluation of the mean value function along this path yields

$$\langle H \rangle([\sigma(s)]) = \frac{(x + s\eta, H(x + s\eta))}{\|x + s\eta\|^2}.$$

Expanding the inner products in the numerator and denominator by sesquilinearity produces

$$\langle H \rangle([\sigma(s)]) = \frac{(x, Hx) + s(\eta, Hx) + s(x, H\eta) + s^2(\eta, H\eta)}{(x, x) + s(\eta, x) + s(x, \eta) + s^2(\eta, \eta)}.$$

Imposing $(x, x) = 1$ and $(\eta, x) = (x, \eta) = 0$, the denominator simplifies to $1 + s^2\|\eta\|^2$, and the quotient rule at $s = 0$ isolates the linear coefficients of the numerator alone:

$$d\langle H \rangle_{[x]}(v) = (\eta, Hx) + (x, H\eta) = 2\operatorname{Re}(\eta, Hx).$$

Turning to the left-hand side, differentiating $\gamma(t) = [e^{-itH/\hbar}x]$ at $t = 0$ gives the representative velocity

$$\eta_H := \dot{\gamma}(0) = -\frac{i}{\hbar}Hx.$$

Although η_H need not lie in $x^\perp$, its component along x pairs to zero against every $\eta \in x^\perp$ under ω_0 , since

$$\omega_0(cx, \eta) = \operatorname{Im}(\bar{c}(x, \eta)) = 0,$$

whenever $(x, \eta) = 0$. Hence η_H is the tangent vector it represents in $T_{[x]}\mathbb{P}\mathcal{H} \cong x^\perp$. Using $\omega = \lambda\omega_0$, $\omega_0(U, V) = \operatorname{Im}(U, V)$, and conjugate-linearity of $(\cdot, \cdot)$ in the first argument, we have that

$$\omega_{[x]}(\eta_H, \eta) = \lambda \operatorname{Im}\left(-\frac{i}{\hbar}Hx, \eta\right) = \frac{\lambda}{\hbar} \operatorname{Im}(i(Hx, \eta)) = \frac{\lambda}{\hbar} \operatorname{Re}(Hx, \eta) = \frac{\lambda}{\hbar} \operatorname{Re}(\eta, Hx).$$

The curve γ is the integral curve of $X_{\langle H \rangle}$ when $\iota_{\eta_H}\omega = d\langle H \rangle_{[x]}$ as functionals on $x^\perp$, that is, when $\omega_{[x]}(\eta_H, \eta) = d\langle H \rangle_{[x]}(v)$ for every $\eta \in x^\perp$. Comparing the two expressions above,

$$\frac{\lambda}{\hbar} \operatorname{Re}(\eta, Hx) = 2\operatorname{Re}(\eta, Hx) \quad \forall \eta \in x^\perp,$$

forces

$$\lambda = 2\hbar. \tag{3.23}$$

Since $[x] \in \mathbb{P}\mathcal{H}$ was arbitrary, Equation (3.23) is the unique normalization for which the physically postulated Schrödinger flow coincides, at every point of $\mathbb{P}\mathcal{H}$. Equation (3.23) fixes the free normalization of Section 3.2. From here on,

$$g := 2\hbar g_0, \quad \omega := 2\hbar \omega_0, \tag{3.24}$$

denote the physical Fubini–Study metric and symplectic form of $\mathbb{P}\mathcal{H}$, with I, G the associated musical isomorphisms of Definition 3.1.11. By linearity the local Kähler potential and the metric of Section 3.2 rescale to

$$K = \hbar \ln(1 + \|z\|^2), \quad g_{j\bar{k}} = 2\hbar g_{0,j\bar{k}} = \hbar \frac{\delta_{jk}(1 + \|z\|^2) - \bar{z}^j z^k}{(1 + \|z\|^2)^2}.$$

This construction can be summarized in the following proposition.

Proposition 3.3.2 (Hamiltonian flow equals Schrödinger flow). *Let $H \in \mathcal{B}(\mathcal{H})$ be selfadjoint. The Hamiltonian flow of $\langle H \rangle$ with respect to ω is*

$$\Phi_t([x]) = [e^{-itH/\hbar}x], \quad t \in \mathbb{R},$$

which satisfies the Schrödinger equation $i\hbar \frac{d}{dt}(e^{-itH/\hbar}x) = He^{-itH/\hbar}x$.

Remark 3.3.3. The value $\lambda = 2\hbar$ in (3.24) is the only normalization compatible with the Stone evolution $e^{-itH/\hbar}$. If the flow had been stated as $e^{-itH/\hbar'}$ for some $\hbar' \neq \hbar$, the computation above would have fixed $\lambda = 2\hbar'$ instead.

This same normalization is what identifies the geometric $\ast_\nu$ product at $\nu = \hbar$ with the associative product $AB = A \circ B + \frac{i\hbar}{2}\{A, B\}$ of the Lie–Jordan algebra of Definition 2.1.19. The complete discussion of this correspondence will be presented in Corollary 3.5.5. Proposition 3.3.4 captures the Lie structure isomorphism.

Proposition 3.3.4 (Poisson bracket and commutator). *For any pair of selfadjoint operators $A, B \in \mathcal{B}(\mathcal{H})$,*

$$\{\langle A \rangle, \langle B \rangle\} = \langle \{A, B\} \rangle = \left\langle \frac{1}{i\hbar}[A, B] \right\rangle.$$

Proof. By Definitions 3.1.13 and 3.1.19, $\{f, l\} = df(X_l)$. Using the representative of $X_{\langle B \rangle}$ found above,

$$\{\langle A \rangle, \langle B \rangle\}([x]) = d\langle A \rangle_{[x]}(X_{\langle B \rangle}([x])) = 2\operatorname{Re}\left(-\frac{i}{\hbar}Bx, Ax\right) = -\frac{2}{\hbar}\operatorname{Im}(Bx, Ax).$$

Since A, B are selfadjoint,

$$(x, [A, B]x) = (Ax, Bx) - (Bx, Ax) = -2i \operatorname{Im}(Bx, Ax),$$

which yields $\left\langle \frac{1}{i\hbar}[A, B] \right\rangle([x]) = \frac{1}{i\hbar}(x, [A, B]x) = -\frac{2}{\hbar}\operatorname{Im}(Bx, Ax)$. It agrees with $\{\langle A \rangle, \langle B \rangle\}([x])$. $\square$

Remark 3.3.5. The map $A \mapsto \langle A \rangle$ is thus a Lie algebra homomorphism from the selfadjoint part of $\mathcal{B}(\mathcal{H})$, with bracket $\frac{1}{i\hbar}[\cdot, \cdot]$, to $C^\infty(\mathbb{P}\mathcal{H}, \mathbb{R})$, with Poisson bracket $\{\cdot, \cdot\}$ with no extra numerical factor once the normalization of Equation (3.23) is in place.

Geometry of the flow

The flow of Proposition 3.3.2 is generated by the unitaries $U_t = e^{-itH/\hbar}$. Since each unitary preserves the Hilbert space inner product, its projectivization preserves g and ω , so each Φ_t is a Kähler isomorphism of $(\mathbb{P}\mathcal{H}, \omega, g)$.

Proposition 3.3.6 (Kähler isomorphisms and their generators). *Let $\mathcal{H}$ be a Hilbert space as in Definition 2.2.6, and let $\mathbb{P}\mathcal{H}$ be its projective space as in Definition 3.2.2. It holds the following.*

1. *Unitary operators on $\mathcal{H}$ induce Kähler isomorphisms of $\mathbb{P}\mathcal{H}$, and conversely every Kähler isomorphism is of this form, uniquely up to phase.*
2. *Selfadjoint operators generate one-parameter groups of Kähler isomorphisms via $\Phi_t([x]) = [e^{-itH/\hbar}x]$. Conversely every such group is generated by a unique H up to additive constant.*

Proof. The first statement follows because unitaries preserve the inner product, hence their projectivization preserves both ω and g . In addition, any Kähler automorphism of $\mathbb{P}\mathcal{H}$ lifts to a unitary or antiunitary operator on $\mathcal{H}$ by Wigner’s Theorem, [Mor18, Theorem 12.11]. Restricting to continuous paths through the identity selects the unitary lifts.

For the second statement, $e^{-itH/\hbar}$ is a strongly continuous one-parameter unitary group by Stone’s Theorem 2.10, so its projectivization is a continuous group of Kähler isomorphisms. Conversely, any such group lifts to a strongly continuous one-parameter unitary group U_t , up to phases, whose selfadjoint generator H satisfies $U_t = e^{-itH/\hbar}$ by Stone’s Theorem 2.10, unique up to an additive real constant, which does not affect the projectivized flow. $\square$

Remark 3.3.7. A quantum system is thus a Hamiltonian system. In particular, it is a symplectic manifold $(\mathbb{P}\mathcal{H}, \omega)$ together with a distinguished one-parameter group of symplectic, indeed Kähler, isomorphisms generated by the Hamiltonian function $\langle H \rangle$.

Curvature of the State Manifold

We now specialize the holomorphic sectional curvature of Definition 3.1.31 to the projective Hilbert space. A direct computation will show that in the case of $\mathbb{P}\mathcal{H}$ it is constant and positive. The general fact is stated by Hawley-Igusa Theorem 3.5 showing that projective spaces are the only possible choice for quantum mechanics.

Proposition 3.3.8 (Curvature of the projective state space). *Let $\mathbb{P}\mathcal{H}$ be the projective Hilbert space, as in Definition 3.2.2, endowed with the Fubini–Study metric g_λ generated by the scaled local Kähler potential, as in Definition 3.1.27, via*

$$K_\lambda(z, \bar{z}) := \frac{\lambda}{2} \ln(1 + \|z\|^2),$$

where $\lambda > 0$. The holomorphic sectional curvature of $\mathbb{P}\mathcal{H}$, as in Definition 3.1.31, is constant being equal to $2/\lambda$.

The core proof of this proposition is found in [BŽ17, Section 4.5., p.120]. The rescaling fixed in Equation (3.23) was chosen to reproduce the Schrödinger flow, but it has an equally direct geometric consequence for the curvature computed in Proposition 3.3.8.

Corollary 3.3.9 (The curvature of $\mathbb{P}\mathcal{H}$ fixes $\hbar$). *The physical Kähler structure of Equation (3.24) has constant holomorphic sectional curvature $1/\hbar$.*

Proof. The local Kähler potential $K = \hbar \ln(1 + \|z\|^2)$, obtained by rescaling as in Equation (3.24), is the potential K_λ of Proposition 3.3.8 for $\lambda = 2\hbar$, whose holomorphic sectional curvature is $2/\lambda = 1/\hbar$. $\square$

This result can be generalized to any projective space. The central concept presented in the following theorem is the uniqueness, up to isomorphisms, of the projective Hilbert space formalism. The key importance of this next theorem will become clear during the constructions in the next sections.

Theorem 3.5 (Hawley–Igusa). *Up to isomorphism, projective spaces are the only connected, simply connected, complete Kähler manifolds of constant, positive holomorphic sectional curvature.*

The proof of this theorem is presented in [KN96, Theorem 7.9].

3.4 Riemannian Metric and Uncertainty Relations

While the symplectic form ω rules the Hamiltonian dynamics of the quantum system, as discussed in Section 3.3, the Riemannian metric g encodes its statistical structure. The metric allows for the calculation of dispersion, distance, and transition probability on the state manifold.

Following Definition 3.1.13 applied to the Kähler manifold $(\mathbb{P}\mathcal{H}, \omega, g)$, the state space carries a symmetric Riemann bracket. For any two smooth functions $f, k \in C^\infty(\mathbb{P}\mathcal{H}, \mathbb{R})$, this bracket is defined by $((f, k)) = \langle df, Gdk \rangle$, where G is the musical isomorphism associated with the Fubini–Study metric g . We use the notation $((\cdot, \cdot))$, regardless whether the arguments are general smooth functions on $\mathbb{P}\mathcal{H}$ or mean value functions of quantum observables.

In the next lemma, we show that exploiting the linear structure of the ambient Hilbert space $\mathcal{H}$ helps us to compute the intrinsic Kähler structures. We demonstrate how the Riemannian metric and the symplectic form on the tangent space of $\mathbb{P}\mathcal{H}$ can be recovered from the standard Hilbert inner product by projecting out the unphysical scale and phase degrees of freedom.

Lemma 3.4.1 (Metric and form via ambient representatives). *Let $\mathbb{P}\mathcal{H}$ be a projective Hilbert space as in Definition 3.2.2. Let $\psi = [\Psi] \in \mathbb{P}\mathcal{H}$, $\|\Psi\| = 1$, and let $\eta, \eta' \in \mathcal{H}$ be arbitrary. Define $\eta_\psi := \eta - (\Psi, \eta)\Psi \in \Psi^\perp$ for the component of η tangent to $\mathbb{P}\mathcal{H}$ at ψ , under the identification $T_\psi \mathbb{P}\mathcal{H} \cong \Psi^\perp$ of Proposition 3.2.6. Then*

$$(\eta_\psi, \eta'_\psi) = (\eta, \eta') - (\eta, \Psi)(\Psi, \eta'). \quad (3.25)$$

and, by Equation (3.24),

$$g(\eta_\psi, \eta'_\psi) = 2\hbar \operatorname{Re}[(\eta, \eta') - (\eta, \Psi)(\Psi, \eta')], \quad \omega(\eta_\psi, \eta'_\psi) = 2\hbar \operatorname{Im}[(\eta, \eta') - (\eta, \Psi)(\Psi, \eta')]. \quad (3.26)$$

Proof. Since $\|\Psi\| = 1$, η_Ψ is the orthogonal projection of η onto $\Psi^\perp$, and likewise for η'_Ψ . Denoting by $a := (\Psi, \eta)$, $b := (\Psi, \eta')$, expanding by conjugate-linearity in the first argument yields

$$(\eta_\Psi, \eta'_\Psi) = (\eta - a\Psi, \eta' - b\Psi) = (\eta, \eta') - b(\eta, \Psi) - \bar{a}b + \bar{a}b = (\eta, \eta') - (\eta, \Psi)(\Psi, \eta'),$$

which is Equation (3.25). Equation (3.26) follows applying Equation (3.24). $\square$

Remark 3.4.2. In particular, for $A \in \mathcal{B}(\mathcal{H})$ selfadjoint, $x = \Psi \in \mathbb{S}\mathcal{H}$ normalized, and $\eta := -\frac{i}{\hbar}Ax$, one has $(\Psi, \eta) = -\frac{i}{\hbar}\langle A \rangle([x])$ and $\eta_\Psi = -\frac{i}{\hbar}(A - \langle A \rangle)x$. This recovers the vector field $X_{\langle A \rangle}([x])$ of Section 3.3.

The following proposition establishes how the Jordan product, as in Definition 2.1.9, between two selfadjoint operators is connected with the Riemannian structure of $\mathbb{P}\mathcal{H}$, requiring no reference to the underlying linear Hilbert space.

Proposition 3.4.3 (Intrinsic Jordan decomposition). *Let $\mathbb{P}\mathcal{H}$ be a projective Hilbert space as in Definition 3.2.2. Let $A, K \in \mathcal{B}(\mathcal{H})$ be selfadjoint operators, as in Definition 2.2.22, and let $f = \langle A \rangle$ and $k = \langle K \rangle$ be their corresponding mean value functions on $\mathbb{P}\mathcal{H}$, as in Definition 3.3.1. The mean value of their Jordan product $A \circ K$ is uniquely determined by the Riemann bracket on $\mathbb{P}\mathcal{H}$ via the identity*

$$\langle A \circ K \rangle = \frac{\hbar}{2}((f, k)) + f k. \quad (3.27)$$

Proof. Fix $[x] \in \mathbb{P}\mathcal{H}$ and a normalized representative $\Psi = x \in \mathbb{S}\mathcal{H}$. By the remark following Lemma 3.4.1, X_f and X_k are represented, under $T_{[x]}\mathbb{P}\mathcal{H} \cong x^\perp$, by the projections of $\eta := -\frac{i}{\hbar}Ax$ and $\eta' := -\frac{i}{\hbar}Kx$. Kähler compatibility, as in Remark 3.1.15, yields $((f, k)) = g(X_f, X_k) = g(\eta_x, \eta'_x)$. Hence Lemma 3.4.1 applies. Using $(\eta, x) = \frac{i}{\hbar}\langle A \rangle([x])$, $(x, \eta') = -\frac{i}{\hbar}\langle K \rangle([x])$, selfadjointness of A, K , and

$$\operatorname{Re}(Ax, Kx) = \left(x, \frac{1}{2}(AK + KA)x\right) = \langle A \circ K \rangle([x]),$$

we have that

$$((f, k))_\omega([x]) = 2\hbar \operatorname{Re} \left[\frac{1}{\hbar^2}(Ax, Kx) - \frac{1}{\hbar^2}\langle A \rangle([x])\langle K \rangle([x]) \right] = \frac{2}{\hbar}[\langle A \circ K \rangle([x]) - f([x])k([x])],$$

which is Equation (3.27). Equation (3.28) is the case $K = A$, since $A \circ A = A^2$. $\square$

The dispersions of two observable functions are controlled jointly, through the metric and the symplectic form, by the Kähler counterpart of the Robertson–Schrödinger uncertainty relation.

Definition 3.4.4 (Dispersion). *Let $f \in \mathcal{K}(\mathbb{P}\mathcal{H}, \mathbb{R})$ be a Kählerian function, as per Definition 3.1.19, and let $p \in \mathbb{P}\mathcal{H}$. The dispersion of f at p is*

$$\Delta f(p) := \sqrt{\hbar/2} \sqrt{((f, f))(p)}. \quad (3.28)$$

If $f = \langle A \rangle$ for a selfadjoint $A \in \mathcal{B}(\mathcal{H})$, $\Delta f(p)$ coincides with the usual quantum-mechanical standard deviation of A in the state p , by Remark 3.4.5 below.

Remark 3.4.5. Setting $A = K$ and $f = k = \langle A \rangle$ in Proposition 3.4.3 yields $\langle A^2 \rangle = \frac{\hbar}{2}((f, f)) + f^2$, that is,

$$((f, f)) = \frac{2}{\hbar}(\langle A^2 \rangle - \langle A \rangle^2) = \frac{2}{\hbar}(\Delta f)^2, \quad (3.29)$$

consistently with Definition 3.4.4. While $((f, k))$ alone does not represent in general the expectation value of a single observable, it serves as the geometric formulation of the quantum variance between f and k at any given state $p \in \mathbb{P}\mathcal{H}$.

This geometric interpretation of variances allows to formulate the most general uncertainty relations without invoking the linear structure of $\mathcal{H}$.

Theorem 3.6 (Robertson–Schrödinger relation, geometric form). *Let $\mathbb{P}\mathcal{H}$ be a projective Hilbert space as in Definition 3.2.2. For any pair of observable functions $f, k \in \mathcal{K}(\mathbb{P}\mathcal{H}, \mathbb{R})$, as in Definition 3.5.1, their statistical dispersions evaluated at any state $p \in \mathbb{P}\mathcal{H}$ satisfy the bound:*

$$(\Delta f)(\Delta k) \geq \frac{\hbar}{2} \sqrt{\{f, k\}^2 + ((f, k))^2}. \quad (3.30)$$

Proof. Fix $[x] \in \mathbb{P}\mathcal{H}$. Since $X_f, X_k \in x^\perp$ are already tangent, Equation (3.24) yields

$$((f, k)) + i\{f, k\} = g(X_f, X_k) + i\omega(X_f, X_k) = 2\hbar(X_f, X_k).$$

By Equation (3.28), $\|X_f\|^2 = (X_f, X_f) = \frac{1}{2\hbar}((f, f)) = \frac{(\Delta f)^2}{\hbar^2}$, and likewise for k . The Cauchy–Schwarz inequality on $\mathcal{H}$ then yields

$$\sqrt{((f, k))^2 + \{f, k\}^2} = 2\hbar |(X_f, X_k)| \leq 2\hbar \|X_f\| \|X_k\| = \frac{2}{\hbar}(\Delta f)(\Delta k),$$

which is the stated inequality. $\square$

Remark 3.4.6. The weaker Heisenberg relation $(\Delta f)(\Delta k) \geq \frac{\hbar}{2} |\{f, k\}|$ follows dropping the Riemann bracket term. Equality in Theorem 3.6 holds at $[x] \in \mathbb{P}\mathcal{H}$ if and only if X_f and X_k are proportional over $\mathbb{C}$ there, $X_f = aX_k + bJX_k$ for some $a, b \in \mathbb{R}$. The extreme cases $b = 0$ and $a = 0$ correspond, respectively, to the vanishing of the Poisson bracket and of the covariance $((f, k))$ at $[x]$.

Example 3.4.7. For $f = \langle \sigma_x \rangle$, $k = \langle \sigma_y \rangle$ lying on the Bloch sphere, given the state $[x] = [(1, 0)^T]$ one computes $\{f, k\} = \frac{2}{\hbar} \langle \sigma_z \rangle = \frac{2}{\hbar}$ and $((f, k)) = 0$. Hence Theorem 3.6 yields $(\Delta f)(\Delta k) \geq 1$. This bound is saturated at this state since $\langle \sigma_x \rangle = \langle \sigma_y \rangle = 0$ force $\Delta f = \Delta k = 1$, matching the standard spin uncertainty relation.

A related special case, isolating the variance of a single Kählerian function against its own Hamiltonian flow, is worth recording explicitly.

Example 3.4.8 (Anandan–Aharonov relation). For a bounded Hamiltonian $H \in \mathcal{B}(\mathcal{H})$ with observable function $h = \langle H \rangle \in \mathcal{K}(\mathbb{P}\mathcal{H}, \mathbb{R})$, the energy variance takes the form $(\Delta h)^2 = \frac{\hbar}{2}((h, h))$. By the Kähler compatibility outlined in Remark 3.1.15, $((h, h)) = g(X_h, X_h)$. Thus, the energy uncertainty Δh coincides, up to a multiplicative constant, with the Riemannian length of the Hamiltonian vector field generating time evolution. The energy dispersion dictates the Riemannian speed at which the state traverses the projective manifold $\mathbb{P}\mathcal{H}$ during unitary dynamics.

Geodesics and Riemannian Distance

The Riemannian metric g studied so far entails the infinitesimal statistical structure of $\mathbb{P}\mathcal{H}$. That encodes the local dispersion and variance of observables within the tangent space. Instead, capturing the macroscopic distinguishability between quantum states, we must integrate this structure across the manifold. The global physical role of the metric is thus expressed through the distance it induces. The extremal curves that minimize this geometric length are called geodesics, which we define as follows.

Definition 3.4.9 (Geodesic). *Let $(\mathcal{M}, J, g)$ be a Kähler manifold, as per Definition 3.1.10. For a piecewise C^1 curve $\gamma : [a, b] \rightarrow \mathcal{M}$, its length is*

$$L(\gamma) := \int_a^b \sqrt{g(\dot{\gamma}(t), \dot{\gamma}(t))} dt. \quad (3.31)$$

A curve $\gamma : [a, b] \rightarrow \mathcal{M}$, parametrized with respect to arc length, is a geodesic if every $t_0 \in [a, b]$ admits a neighborhood $[t_0 - \varepsilon, t_0 + \varepsilon] \cap [a, b]$ for each $\varepsilon \in \mathbb{R}^+$ on which γ minimizes L among piecewise C^1 curves with the same endpoints.

Geodesics are the Riemannian analogue of straight lines, and their lengths define a genuine distance function on $\mathcal{M}$, compatible with its manifold topology.

Definition 3.4.10 (Riemannian distance). *Let $(\mathcal{M}, J, g)$ be a Kähler manifold, as per Definition 3.1.10. The Riemannian distance between $p, q \in \mathcal{M}$ is*

$$d_g(p, q) := \inf_{\gamma} L(\gamma), \quad (3.32)$$

where the infimum runs over all piecewise C^1 curves $\gamma : [a, b] \rightarrow \mathcal{M}$ with $\gamma(a) = p$, $\gamma(b) = q$, while L is as in Definition 3.4.9.

Remark 3.4.11. Note that, in Definition 3.4.10, if the manifold $\mathcal{M}$ is connected, d_g is a metric inducing the manifold topology as per Proposition 3.2.7. Moreover, when attained, the infimum is realized by a geodesic of Definition 3.4.9.

Among submanifolds of $\mathcal{M}$, those on which the ambient and intrinsic geodesics coincide play a distinguished role. For these, the intrinsic distance agrees with the distance measured in the ambient manifold.

Definition 3.4.12 (Totally geodesic submanifold). *Let $(\mathcal{M}, J, g)$ be a Kähler manifold, as per Definition 3.1.10, and let $N \subset \mathcal{M}$ be a submanifold, equipped with the induced metric $g|_N$. N is totally geodesic if every geodesic of $(N, g|_N)$, as per Definition 3.4.9, is also one of $(\mathcal{M}, g)$.*

Remark 3.4.13. If $N \subset \mathcal{M}$ is totally geodesic as per Definition 3.4.12 and $\mathcal{M}$ is connected, the Riemannian distance $d_{g|_N}$ induced on N by Definition 3.4.10 coincides with the restriction of d_g to $N \times N$.

Transition Probabilities

Beyond its algebraic role in defining the quantum variance of specific observable functions, the Riemannian metric g has an operational meaning independent of any fixed physical quantity. In fact, it quantifies the statistical distinguishability of infinitesimally close pure states. When evaluating a continuous parameterization of the state space, the metric tensor yields the quantum Fisher information. That sets the geometric limit on how nearby states can be resolved by any experimental measurement.

Theorem 3.7 (Provost–Vallee). *Let $\mathcal{H}$ be a Hilbert space as in Definition 2.2.6 and $\mathbb{P}\mathcal{H}$ be its projective space as per Definition 3.2.2. Let $\theta \in \mathbb{R} \mapsto x(\theta) \in \mathcal{H} \setminus \{0\}$ be a smooth curve, and $\psi(\theta) := [x(\theta)] \in \mathbb{P}\mathcal{H}$. Setting the velocity $\dot{x} := \partial_{\theta} x$ yields*

$$g\left(\frac{d\psi}{d\theta}, \frac{d\psi}{d\theta}\right) = 2\hbar \left(\frac{(\dot{x}, \dot{x})}{(x, x)} - \frac{(x, \dot{x})(\dot{x}, x)}{(x, x)^2} \right). \quad (3.33)$$

This result is not proved here. The reader is referred to the original article by Provost and Vallee in [PV80, Section 2].

Remark 3.4.14. The scalar quantity

$$\mathcal{F}(\theta) := 4 \left(\frac{(\dot{x}, \dot{x})}{(x, x)} - \frac{(x, \dot{x})(\dot{x}, x)}{(x, x)^2} \right) \quad x = x(\theta) \quad (3.34)$$

is the one-dimensional quantum Fisher information of the family of pure states $\psi(\theta)$ with respect to the parameter θ . Theorem 3.7 identifies it with the pulled-back metric, $g(\dot{\psi}, \dot{\psi}) = \frac{\hbar}{2} \mathcal{F}(\theta)$. The Riemannian line element of $\mathbb{P}\mathcal{H}$ is, up to the multiplicative constant $\hbar/2$, the statistical distance between infinitesimally separated pure states, measuring the maximum achievable precision in estimating the physical parameter θ .

We clarify its meaning through the next example.

Example 3.4.15 (Fisher Information of a Rotation). Let $\mathcal{H} = \mathbb{C}^2$ and consider a single-parameter family of pure states defined by the normalized vector $x(\theta) = (\cos(\theta/2), \sin(\theta/2))^T$. This corresponds to a one-parameter unitary rotation of the state vector. The parameter derivative is

$$\dot{x}(\theta) = \frac{1}{2} \begin{pmatrix} -\sin(\theta/2) \\ \cos(\theta/2) \end{pmatrix}. \quad (3.35)$$

The inner products evaluate to $\langle \dot{x}, \dot{x} \rangle = \frac{1}{4}(\sin^2(\theta/2) + \cos^2(\theta/2)) = \frac{1}{4}$ and $\langle x, \dot{x} \rangle = -\frac{1}{2} \cos(\theta/2) \sin(\theta/2) + \frac{1}{2} \sin(\theta/2) \cos(\theta/2) = 0$. Replacing these into the quantum Fisher information formula yields

$$\mathcal{F}(\theta) = 4 \left(\frac{1}{4} - 0 \right) = 1. \quad (3.36)$$

The pulled-back Fubini–Study metric yields to $g\left(\frac{d\psi}{d\theta}, \frac{d\psi}{d\theta}\right) = \frac{\hbar}{2}$. The quantum Fisher information being strictly constant and equal to 1 demonstrates that the parameter θ acts directly as the geodesic arc length on the Bloch sphere up to a scaling factor, meaning the statistical distinguishability of the state accumulates uniformly with respect to θ .

The Geometrical Born Rule

The metric g extends beyond infinitesimal statistical dispersion to govern the macroscopic distinguishability of quantum states. Integrating the metric distance across the manifold yields the postulate of quantum measurement.

Theorem 3.8 (Geometrical Born Rule). *Let $d_g : \mathbb{P}\mathcal{H} \times \mathbb{P}\mathcal{H} \rightarrow \mathbb{R}$ denote the Riemannian distance, as per Definition 3.4.10, induced by the Fubini–Study metric g on $\mathbb{P}\mathcal{H}$. For any two pure states $[x], [y] \in \mathbb{P}\mathcal{H}$, the quantum mechanical transition probability $P([x], [y])$ satisfies*

$$P([x], [y]) := \frac{|(x, y)|^2}{\|x\|^2 \|y\|^2} = \cos^2 \left(\frac{d_g([x], [y])}{\sqrt{2\hbar}} \right). \quad (3.37)$$

We will not present the proof of this statement. For further readings, we refer to [AS97, Theorem II.2].

Example 3.4.16 (Qubit). For $\mathcal{H} = \mathbb{C}^2$, the projective space $\mathbb{P}\mathcal{H}$ is identified with $\mathbb{CP}^1$, the Bloch sphere of radius $\sqrt{\hbar/2}$ in the Fubini–Study metric g of (3.24). Antipodal points on the sphere correspond to orthogonal states $[x], [y]$, for which $P([x], [y]) = |(x, y)|^2 / (\|x\|^2 \|y\|^2) = 0$. By Theorem 3.8 this forces $d_g([x], [y]) = \pi\sqrt{\hbar/2}$, the diameter of the sphere, matching the maximal geodesic distance recorded in the remark following Theorem 3.5. Conversely, identical states sit at $d_g = 0$, where $P = 1$. Hence the cosine squared function interpolates continuously between certainty and mutual exclusivity as the geodesic distance ranges over $[0, \pi\sqrt{\hbar/2}]$.

Remark 3.4.17. The metric g completely determines the probabilistic structure of quantum mechanics. Orthogonal states, representing mutually exclusive measurement outcomes, correspond to points separated by the maximum geodesic distance $d_g = \pi\sqrt{\hbar/2}$.

3.5 Observable Functions on Kähler manifolds

In classical mechanics, the space of physical observables is identified with the entire commutative algebra of smooth real-valued functions on the symplectic phase space. In the quantum setting, by contrast, the constraints set by the symplectic structure ω and the Riemannian metric g define the admissible observables. The physical observables are those functions whose Hamiltonian flow respects the full Kähler geometry of the state manifold.

Definition 3.5.1 (Observable function). *A smooth map $f : \mathbb{P}\mathcal{H} \rightarrow \mathbb{R}$ is an observable function if there exists a selfadjoint operator $A \in \mathcal{B}(\mathcal{H})$ such that $f = \langle A \rangle$.*

An equivalent definition can be stated using a dynamical consideration based on Section 3.3. By Proposition 3.3.6, the flow generated by a selfadjoint operator A via the Schrödinger equation acts on $\mathbb{P}\mathcal{H}$ as a continuous one-parameter group of Kähler isomorphisms. Consequently, the Hamiltonian vector field X_f of an observable function $f = \langle A \rangle$ preserves the symplectic form ω and also the Fubini–Study metric g .

Definition 3.5.2 (Killing vector field). *Let $(\mathcal{M}, J, g)$ be a Kähler manifold, as per Definition 3.1.10. A vector field $X \in \mathfrak{X}(\mathcal{M})$ is a Killing vector field if*

$$\mathcal{L}_X g = 0, \quad (3.38)$$

where $\mathcal{L}_X$ denotes the Lie derivative as per Definition 1.1.30.

Hence, we can characterize the observables as the functions which generate a Killing Hamiltonian vector field.

Theorem 3.9 (Geometric characterization of observables). *Given a projective Hilbert space $\mathbb{P}\mathcal{H}$ as per Definition 3.2.2, a smooth function $f \in C^\infty(\mathbb{P}\mathcal{H}, \mathbb{R})$ is an observable function, as in Definition 3.5.1, if and only if its Hamiltonian vector field X_f is a Killing vector field for the Fubini–Study metric g , as per Definition 3.5.2.*

The proof of this statement is shown in [AS97, p.13]. For instance, we present an example of a Killing vector field on the Bloch sphere.

Example 3.5.3. On the Bloch sphere $\mathbb{P}\mathcal{H} \cong \mathbb{CP}^1$ with the Fubini–Study metric g , the one-parameter group $[x] \mapsto [e^{-it\sigma_z/2}x]$ rotates the sphere about its polar axis at constant angular speed. Since rotations preserve the round metric, the generating vector field $X_{\langle\sigma_z\rangle}$ is Killing. This confirms, via Theorem 3.9, that $\langle\sigma_z\rangle$ is an observable function.

Remark 3.5.4 (Symmetry data and finite-dimensionality). A property of Killing vector fields on a connected Kähler manifold is their unique global reconstruction from the initial data consisting of their pointwise value and their first derivative at a single prescribed base point. Through the symplectic correspondence

$$X_f = I \, df.$$

This uniqueness principle entails that any observable function on the projective Hilbert space $\mathbb{P}\mathcal{H}$ is completely determined by its value, its exterior derivative, and its Hessian tensor, all evaluated at an arbitrary reference state $[x] \in \mathbb{P}\mathcal{H}$. Consequently, in the case where $\mathcal{H}$ is finite-dimensional, the vector space of such observables is likewise finite-dimensional. This stands in contrast with the infinite-dimensional structure of the standard algebra of smooth functions $C^\infty(\mathbb{P}\mathcal{H}, \mathbb{R})$, which exhibits no such finite parametrization.

Theorem 3.9 establishes that observables are the Kählerian functions in Definition 3.1.19 of the projective Hilbert space. They act as the generators of the structural symmetries of the state manifold. This geometric constraint allows the non-commutative algebraic operations of $\mathcal{B}(\mathcal{H})$ to be translated into geometric operations on $\mathbb{P}\mathcal{H}$.

Theorem 3.10 (Homomorphism of Lie–Jordan and Kähler algebras). *Let $(\mathcal{O}, \circ, \{\cdot, \cdot\})$ be a real Lie–Jordan algebra, as per Definition 2.1.19. Let $\mathfrak{A}$ be the C^* -algebra generated by $\mathcal{O}$, as per Theorem 2.1, and let $\mathbb{P}\mathcal{H}_\alpha$ be the projective Hilbert space corresponding to a fixed sector of the pure state space $\mathcal{P}(\mathfrak{A})$, as per Theorem 3.4.*

The mean value map $A \mapsto \langle A \rangle$, as per Definition 3.3.1, on $\mathbb{P}\mathcal{H}_\alpha$, is a Lie–Jordan homomorphism between $\mathcal{O}$ and the algebra of Kählerian functions $\mathcal{K}(\mathbb{P}\mathcal{H}_\alpha, \mathbb{R})$, as per Definition 3.1.19. The map is an algebraic-geometric isomorphism if and only if the GNS representation of $\mathfrak{A}$ on the sector α , as per Theorem 2.7, is faithful, as per Definition 2.3.2.

Proof. By Theorem 2.1, the abstract Lie–Jordan algebra $\mathcal{O}$ embeds as the selfadjoint part of the C^* -algebra $\mathfrak{A}$. By Theorem 3.4, the pure state space $\mathcal{P}(\mathfrak{A})$ decomposes into a disjoint union of projective Hilbert spaces $\mathbb{P}\mathcal{H}_\alpha$, each endowed with an intrinsic Kähler structure (J, g, ω) , as per Definition 3.1.10.

For a fixed sector $\mathbb{P}\mathcal{H}_\alpha$, an element $A \in \mathcal{O}$ defines a smooth real-valued expectation function $\langle A \rangle : \mathbb{P}\mathcal{H}_\alpha \rightarrow \mathbb{R}$. The algebraic operations defined on $\mathcal{O}$ correspond to the geometric structures on $\mathbb{P}\mathcal{H}_\alpha$ via the following evaluations:

1. **Lie structure (Symplectic form).** The Lie bracket $\{A, B\}$ in $\mathcal{O}$ corresponds to the Poisson bracket generated by the symplectic form ω , as per Definition 3.1.13. Applying Proposition 3.3.4 yields the relation

$$\{\langle A \rangle, \langle B \rangle\} = \langle \{A, B\} \rangle. \quad (3.39)$$

2. **Jordan structure (Riemannian metric).** The Jordan product $A \circ B$ in $\mathcal{O}$ governs the statistical dispersion structure. This maps to the scaled Riemann bracket generated by the Fubini–Study metric g , as per Definition 3.1.13, isolating the $\circ_{\hbar}$ product by Proposition 3.4.3 and Definition 3.1.16

$$\langle A \circ B \rangle = \langle A \rangle \circ_{\hbar} \langle B \rangle = \frac{\hbar}{2}((\langle A \rangle, \langle B \rangle)) + \langle A \rangle \langle B \rangle. \quad (3.40)$$

3. **Associative product (Kähler form).** The total associative product $AB = A \circ B + \frac{i\hbar}{2}\{A, B\}$, given by Equation (2.1), within the complexified enveloping algebra $\mathfrak{A}$ reconstructs the geometric Kähler product $*_{\hbar}$, as per Definition 3.1.16. Applying the linearity of the expectation value map on the components yields

$$\begin{aligned}\langle AB \rangle &= \langle A \circ B \rangle + \frac{i\hbar}{2}\langle \{A, B\} \rangle \\ &= \left(\frac{\hbar}{2}((\langle A \rangle, \langle B \rangle)) + \langle A \rangle \langle B \rangle \right) + \frac{i\hbar}{2}\{ \langle A \rangle, \langle B \rangle \} \\ &= \langle A \rangle *_{\hbar} \langle B \rangle.\end{aligned}\tag{3.41}$$

This establishes the structural homomorphism from the abstract algebra $\mathcal{O}$ into the geometric algebra on $\mathbb{P}\mathcal{H}_{\alpha}$. The kernel of this map coincides with the kernel of the GNS representation π_{α} associated with the sector α , as per Theorem 2.7. Consequently, the map is an exact isomorphism if and only if the specified sector provides a faithful representation of the observable algebra, as per Definition 2.3.2. $\square$

Before discussing an example, we note the simultaneous presence of the constant $\hbar$ in the geometric $*_{\hbar}$ product and in the rescaling of Equation (3.24). As we have seen, Kähler compatibility forces $\omega = 2\hbar\omega_0$, giving $g = 2\hbar g_0$ and $K_{\hbar} = \hbar \ln(1 + \|z\|^2)$. In addition, the holomorphic sectional curvature becomes $1/\hbar$. Likewise, the Lie–Jordan algebra isomorphism requires $\circ_{\nu}$ and $*_{\nu}$ to be evaluated at $\nu = \hbar$.

Corollary 3.5.5 (Imposition of $\nu = \hbar$). *Let $\mathbb{P}\mathcal{H}$ be a projective Hilbert space as in Definition 3.2.2 and $A, B \in \mathcal{B}(\mathcal{H})$ be selfadjoint operators. If the products, defined in Definition 3.1.16, $((\langle A \rangle, \langle B \rangle))$ and $\{ \langle A \rangle, \langle B \rangle \}$ are non vanishing on $\mathbb{P}\mathcal{H}$, then*

$$\langle A \rangle *_{\nu} \langle B \rangle = \langle AB \rangle \text{ on } \mathbb{P}\mathcal{H} \iff \nu = \hbar.\tag{3.42}$$

Proof. Fix the functions $f = \langle A \rangle$ and $k = \langle B \rangle$. By Equation (2.1), $AB = A \circ B + \frac{i\hbar}{2}\{A, B\}$, linearity of the mean value map yields

$$\langle AB \rangle = \langle A \circ B \rangle + \frac{i\hbar}{2}\langle \{A, B\} \rangle.$$

By Proposition 3.4.3, $\langle A \circ B \rangle = f k + \frac{\hbar}{2}((f, k))$, and by Proposition 3.3.4, $\langle \{A, B\} \rangle = \{f, k\}$. Hence

$$\langle AB \rangle = f k + \frac{\hbar}{2}((f, k)) + \frac{i\hbar}{2}\{f, k\}.\tag{3.43}$$

By Definition 3.1.16, $f *_{\nu} k = f k + \frac{\nu}{2}[(f, k)) + i\{f, k\}]$. Subtracting from Equation (3.43),

$$f *_{\nu} k - \langle AB \rangle = \frac{\nu - \hbar}{2}[(f, k)) + i\{f, k\}].$$

Since $((f, k))$ and $\{f, k\}$ are real-valued, the bracket $((f, k)) + i\{f, k\}$ vanishes at $[x] \in \mathbb{P}\mathcal{H}$ if and only if both terms do. By assumption $f *_{\nu} k = \langle AB \rangle$ on $\mathbb{P}\mathcal{H}$ forces $\nu = \hbar$. Conversely, $\nu = \hbar$ yields equality by Equation (3.43). $\square$

Requiring the Kähler product $*_{\nu}$ to reproduce the associative operator product AB on mean value functions, equivalently, requiring compatibility between the algebraic structure of $\mathcal{B}(\mathcal{H})$ and the Kähler geometry of $\mathbb{P}\mathcal{H}$, forces $\nu = \hbar$. The value of $\hbar$ is thus fixed by the same normalization $\lambda = 2\hbar$ of Equation (3.23) that makes the Hamiltonian flow reproduce the Schrödinger equation.

Example 3.5.6. Let $A = \sigma_x$, $B = \sigma_y$ be Pauli matrices on $\mathcal{H} = \mathbb{C}^2$. Direct computation yields

$$A \circ B = \frac{1}{2}(\sigma_x \sigma_y + \sigma_y \sigma_x) = 0 \quad , \quad \{A, B\} = \frac{1}{i\hbar}[\sigma_x, \sigma_y] = \frac{2}{\hbar}\sigma_z.$$

This entails that $AB = A \circ B + \frac{i\hbar}{2}\{A, B\} = i\sigma_z$, matching the familiar identity $\sigma_x \sigma_y = i\sigma_z$. Under the isomorphism of Theorem 3.10, this same identity is recovered geometrically as $\langle \sigma_x \rangle *_{\hbar} \langle \sigma_y \rangle = \langle i\sigma_z \rangle$ on the Bloch sphere $\mathbb{P}\mathcal{H} \cong \mathbb{CP}^1$, illustrating how the noncommutative operator product becomes the $*_{\hbar}$ product of Definition 3.1.16.

3.6 Quantum Symmetries and the Momentum Map

Continuous symmetries and conservation laws on $\mathbb{P}\mathcal{H}$ adopt classical Hamiltonian nomenclature. The momentum map, Definition 1.5.7, can be translated smoothly, yet the domain imposes stricter constraints. In fact, quantum symmetries must preserve the full Kähler triad (ω, g, J) , whereas classical ones only preserve ω . By Wigner's Theorem, [Mor18, Theorem 12.11], continuous symmetry groups arise from strongly continuous unitary representations on $\mathcal{H}$. We construct the quantum momentum map, assigning an observable to each Lie-algebra generator. Classical divergences appear, non-exact ω requires simple connectivity of $\mathbb{P}\mathcal{H}$ for global map existence, while g -invariance forces Kählerian components. The Quantum Noether Theorem then proves that geometric conservation of these components along the flow is equivalent to the vanishing commutator $[\hat{f}, \hat{H}]$.

Proposition 3.6.1 (Quantum momentum map). *Let $G \subseteq U(\mathcal{H})$ be a connected Lie group, as in Definition 1.2.8, acting on $\mathbb{P}\mathcal{H}$, a projective Hilbert space, via projective unitary transformations, as in Definition 2.2.22. Let $\mathfrak{g}$ be its Lie algebra, as in Definition 1.2.1, consisting of skew-selfadjoint operators on $\mathcal{H}$. There exists a unique momentum map $\mu : \mathbb{P}\mathcal{H} \rightarrow \mathfrak{g}^*$ such that, for every $\xi \in \mathfrak{g}$,*

$$d\langle \mu, \xi \rangle = \iota_{\xi_{\mathbb{P}\mathcal{H}}} \omega,$$

as per Definition 1.5.7, with $\xi_{\mathbb{P}\mathcal{H}}$ the infinitesimal generator of Definition 1.5.4. This is given pointwise by

$$\langle \mu([x]), \xi \rangle = \langle i\hbar\xi \rangle([x]) = \frac{(x, i\hbar\xi x)}{\|x\|^2}, \quad (3.44)$$

where $i\hbar\xi$ is the corresponding selfadjoint generator in $\mathcal{B}(\mathcal{H})$, while $\langle \cdot \rangle$ is the mean value function, as in Definition 3.3.1.

Proof. Let $G \subseteq U(\mathcal{H})$ be a connected Lie group acting on $\mathbb{P}\mathcal{H}$ via projective unitary transformations. The Lie algebra $\mathfrak{g}$ consists of skew-adjoint operators. Hence, $i\hbar\xi$ is selfadjoint for every $\xi \in \mathfrak{g}$. By Remark 1.5.8, the defining condition of the momentum map $d\langle \mu, \xi \rangle = \iota_{\xi_{\mathbb{P}\mathcal{H}}} \omega$ holds if and only if $X_{\langle \mu, \xi \rangle} = \xi_{\mathbb{P}\mathcal{H}}$. By Definition 1.5.4, the infinitesimal generator $\xi_{\mathbb{P}\mathcal{H}}$ induces the one-parameter flow $\Phi_t([x]) = [\exp(t\xi)x]$ on $\mathbb{P}\mathcal{H}$. At the same time, by Proposition 3.3.2, the Hamiltonian vector field associated with the mean value function $\langle H \rangle$ of a selfadjoint operator H generates the flow $\Phi_t([x]) = [\exp(-itH/\hbar)x]$. Setting $H = i\hbar\xi$, the Hamiltonian flow evaluates to $[\exp(t\xi)x]$. This establishes the vector field identity

$$X_{\langle i\hbar\xi \rangle} = \xi_{\mathbb{P}\mathcal{H}}.$$

Therefore, $\langle \mu([x]), \xi \rangle = \langle i\hbar\xi \rangle([x])$ satisfies the definition of the momentum map. $\square$

The triviality of the $U(1)$ phase action on $\mathbb{P}\mathcal{H}$ offers an immediate consistency check for the formula just derived.

Example 3.6.2. Let $G = U(1)$ act on $\mathcal{H}$ by $\Psi \mapsto e^{i\theta}\Psi$, with Lie algebra generator $\xi = i \cdot \mathbb{1}$. Since this action is trivial on $\mathbb{P}\mathcal{H}$, because it fixes every ray $[\Psi]$, the induced vector field $\xi_{\mathbb{P}\mathcal{H}}$ vanishes identically. The momentum map of Proposition 3.6.1 reduces to the constant $\langle \mu([x]), \xi \rangle = \langle i\hbar\xi \rangle([x]) = -\hbar$, consistently with the overall phase of Ψ carrying no physical content on the projective state space.

This construction differs from its classical counterpart in two respects, both rooted in the topology and geometry of $\mathbb{P}\mathcal{H}$.

Remark 3.6.3 (Topological exactness and structural rigidity). The transition from classical phase spaces to $\mathbb{P}\mathcal{H}$ introduces two geometric distinctions regarding the momentum map.

1. *Non-exactness of the symplectic form.* Classical momentum maps are frequently constructed via the exactness of the symplectic form $\omega = -d\theta$, typical of cotangent spaces T^*Q . Conversely, the Fubini-Study form ω on $\mathbb{P}\mathcal{H}$ is closed but not exact. This is because, for $\dim \mathcal{H} < \infty$, $\mathbb{P}\mathcal{H}$ is compact, strictly prohibiting exact symplectic forms. The existence of the global momentum map μ on $\mathbb{P}\mathcal{H}$ relies instead on the property of $\mathbb{P}\mathcal{H}$ being simply connected.

2. *Metric constraints.* In classical mechanics, the component functions $\langle \mu, \xi \rangle$ are smooth real functions generating symplectic vector fields. In this framework, because the flow of $\xi_{\mathbb{P}\mathcal{H}}$ preserves the Riemannian metric g , Theorem 3.9 guarantees that the components of the momentum map are Kählerian functions, or equivalently Killing vector fields. The classical observable corresponding to a continuous symmetry direction ξ is identified with the expectation value of the quantum observable $i\hbar\xi$.

The classical geometric Noether Theorem 1.9, translates directly into the quantum formalism. The structure of $\mathbb{P}\mathcal{H}$ uniquely identifies the vanishing of the classical Poisson bracket with the vanishing of the quantum commutator.

Theorem 3.11 (Quantum Noether Theorem). *Let $\mathbb{P}\mathcal{H}$ be a projective Hilbert space and $H \in \mathcal{B}(\mathcal{H})$ be a selfadjoint Hamiltonian operator governing the dynamics as in Section 3.3. Let $G \subseteq U(\mathcal{H})$ be a connected Lie group, as in Definition 1.2.8, acting on $\mathbb{P}\mathcal{H}$ with momentum map $\mu : \mathbb{P}\mathcal{H} \rightarrow \mathfrak{g}^*$, as in Proposition 3.6.1. The momentum map is a prime integral, meaning $\mu(\Phi_t([x])) = \mu([x])$ for all $t \in \mathbb{R}$, if and only if the algebra of generators commutes with the Hamiltonian*

$$[H, i\hbar\xi] = 0 \quad \forall \xi \in \mathfrak{g}. \quad (3.45)$$

Proof. By Theorem 1.9, μ is conserved along the flow of $X_{\langle H \rangle}$ if and only if the Poisson bracket vanishes. That is $\{\langle H \rangle, \langle \mu, \xi \rangle\} = 0$ for all $\xi \in \mathfrak{g}$. Replacing the explicit form of the quantum momentum map from Proposition 3.6.1, the condition becomes $\{\langle H \rangle, \langle i\hbar\xi \rangle\} = 0$. Applying the algebraic-geometric correspondence of Theorem 3.10, this symplectic bracket maps to the mean value of the operator commutator

$$\{\langle H \rangle, \langle i\hbar\xi \rangle\} = \left\langle \frac{1}{i\hbar} [H, i\hbar\xi] \right\rangle. \quad (3.46)$$

The function $\langle \frac{1}{i\hbar} [H, i\hbar\xi] \rangle$ vanishes identically on $\mathbb{P}\mathcal{H}$ if and only if the corresponding operator $[H, i\hbar\xi]$ is the zero operator in $\mathcal{B}(\mathcal{H})$. $\square$

3.7 Algebraic-Geometric Axiomatization of Quantum Mechanics

Chapter 2 reconstructed quantum mechanics from operational data. It started with a C^* -algebra $\mathfrak{A}$ and a separating state space S , per Definition 2.1.22, whose GNS representation, Theorem 2.7, supplies a Hilbert space. However, neither a dynamics nor a reason for the Hamiltonian structure of the observable algebra is derived there. In fact, nothing in $\mathfrak{A}$ singles out a Poisson bracket among its automorphisms. This chapter closes both gaps geometrically. Theorem 3.4 shows that the pure states of $\mathfrak{A}$ already organize themselves into a disjoint union of projective Hilbert spaces $\mathbb{P}\mathcal{H}_\alpha$. Hence, fixing one sector lets the Kähler manifold $\mathbb{P}\mathcal{H}_\alpha$ of Section 3.2, rather than $\mathfrak{A}$ itself, generate the symplectic form that drives the dynamics and the Riemann bracket that reproduces the Jordan product. We observe that the mean value map $A \mapsto \langle A \rangle$ is an algebra isomorphism by Theorem 3.10. It entails that every geometric statement proved about $\mathbb{P}\mathcal{H}_\alpha$ is simultaneously translated for $\mathfrak{A}$. We also note that the choice of $\mathbb{P}\mathcal{H}_\alpha$, rather than some other Kähler manifold, is forced by Hawley–Igusa, Theorem 3.5.

In this section, we construct an axiomatization of a quantum theory based on the algebraic-geometric framework presented in this chapter and the previous one.

Pure state decomposition and the choice of a sector

By Theorem 3.4, the pure state space of $\mathfrak{A}$ decomposes as

$$\mathcal{P}(\mathfrak{A}) = \bigcup_{\alpha} \mathbb{P}\mathcal{H}_\alpha,$$

indexed by the unitary equivalence classes of irreducible representations of $\mathfrak{A}$, that is, by the, so called, superselection sectors. Every construction from Section 3.2 to Section 3.6 is carried out on a single sector. In fact, fixing α fixes an irreducible GNS Hilbert space $\mathcal{H}_\alpha$ and its projective space $\mathbb{P}\mathcal{H}_\alpha$. It is $\mathbb{P}\mathcal{H}_\alpha$, not $\mathcal{P}(\mathfrak{A})$,

that carries the triad (J, g, ω) of Equation (3.24). For instance, if $\mathfrak{A} = \mathfrak{B}_0(\mathcal{H})$, then Proposition 3.2.3 forces the decomposition to the single term $\mathbb{P}\mathcal{H}_\alpha = \mathbb{P}\mathcal{H}$. For an algebra with different sectors, the construction below applies separately on each $\mathcal{H}_\alpha$. The sectors are merged only at that of $\mathcal{P}(\mathfrak{A})$, not at the level of any single symplectic or Riemannian structure.

Remark 3.7.1 (Vector states and the geometric point of a pure state). For a pure state $\omega \in \mathcal{P}(\mathfrak{A})$ lying in sector $\mathbb{P}\mathcal{H}_\alpha$, the GNS cyclic vector $\Psi_\omega \in \mathcal{H}_\alpha$ determines the point $[\Psi_\omega] \in \mathbb{P}\mathcal{H}_\alpha$. The expectation value can be rewritten geometrically as

$$\omega(A) = \frac{(\Psi_\omega, A\Psi_\omega)}{\|\Psi_\omega\|^2} = \langle A \rangle([\Psi_\omega]), \quad \forall A \in \mathfrak{A}. \quad (3.47)$$

Hence, an algebraic state is the same object as a point of $\mathbb{P}\mathcal{H}_\alpha$, and evaluating the state functional is evaluating the mean value function of Definition 3.3.1.

Physical observables

The point at which the algebraic formalism is transferred to the geometric one is Theorem 3.9. On the algebraic side, an observable is a selfadjoint element $A \in \mathfrak{A}$. Definition 3.5.1 transports this to $\mathbb{P}\mathcal{H}_\alpha$ as any smooth $f = \langle A \rangle$. Theorem 3.9 then characterizes the image of this map without reference back to $\mathcal{H}_\alpha$. At this point, f arises from a selfadjoint operator if and only if its Hamiltonian vector field X_f is a Killing field for g , that is, if and only if f is Kählerian as per Definition 3.1.19. This is where the algebraic notion of observable becomes a differential-geometric one, identified by a metric condition on $\mathbb{P}\mathcal{H}_\alpha$.

Compatibility via the Lie–Jordan–Kähler isomorphism

Once observables are identified with Kählerian functions the mean value map $A \mapsto \langle A \rangle$ is the bridge between the two frameworks. Theorem 3.10 is the compatibility statement. In fact, we show that $A \mapsto \langle A \rangle$ is an isomorphism between the Lie–Jordan algebra $(\mathcal{O}, \circ, \{\cdot, \cdot\})$ of bounded selfadjoint operators on $\mathcal{H}$ and the algebra of Kählerian functions $(\mathcal{K}(\mathbb{P}\mathcal{H}, \mathbb{R}), \circ_{\hbar}, \{\cdot, \cdot\})$. Hence, through this isomorphism, the geometric Poisson bracket of Definition 3.1.13 is, sector by sector, analogous to the algebraic commutator. Likewise the Riemann bracket is the Jordan anticommutator. Complexifying and passing to the norm, Theorem 3.10 upgrades this to an isometric isomorphism of C^* -algebras between $\mathfrak{A}$ restricted to the sector and $\mathcal{K}(\mathbb{P}\mathcal{H}_\alpha, \mathbb{C})$ with the $*_{\hbar}$ product. Every geometric statement proved in this chapter about $(\mathbb{P}\mathcal{H}, \omega, g)$ is, through Theorem 3.10, already valid on $\mathfrak{A}$.

Dynamics

The Kähler structure resolves the first gap identified above. Proposition 3.3.2 shows that the Schrödinger flow can be rewritten as the Hamiltonian one, with respect to ω , of the observable function $\langle H \rangle$. Hence, Schrödinger dynamics is a case of Hamilton's equations on the projective manifold $\mathbb{P}\mathcal{H}$. Proposition 3.3.6 supplies the converse. A continuous one-parameter group of transformations of $\mathbb{P}\mathcal{H}$ preserving the full triad (ω, g, J) is necessarily of Schrödinger form, generated by a selfadjoint operator unique up to an additive constant. Preserving g together with ω is the geometric counterpart of unitary evolution. The algebraic notion of this structural preservation is captured through automorphisms of the algebra. However, the GNS framework presented does not identify a single particular way to define the Schrödinger automorphism. At the symmetry level, Theorem 3.11 uses this requirement to identify the vanishing Poisson bracket $\{\langle H \rangle, \langle \mu, \xi \rangle\}$ with the commutator condition $[H, i\hbar\xi] = 0$, term by term, through Theorem 3.10.

Hawley–Igusa and the constant $\hbar$

Hawley–Igusa Theorem 3.5 is what makes it consistent to read $\hbar$ as a curvature. Among connected, simply connected, complete Kähler manifolds, constant positive holomorphic sectional curvature determines the manifold up to isomorphism. The curvature is consequently an intrinsic invariant of $\mathbb{P}\mathcal{H}_\alpha$, not just the normalization chosen for g . Corollary 3.3.9 evaluates this invariant, for the physical structure of Equation (3.24), to $1/\hbar$, positive for every $\hbar > 0$. Hawley–Igusa Theorem 3.5 guarantees that this is the value

forced once $\mathbb{P}\mathcal{H}_\alpha$ is required to be homogeneously curved. Every other occurrence of $\hbar$ in the chapter is then fixed as a function of this single curvature as shown in Corollary 3.3.9. In fact, the metric normalization has been fixed at $\lambda = 2\hbar$ by Equation (3.23). Moreover, the Hamiltonian flow that reproduce the Stone evolution $e^{-itH/\hbar}$ agrees with this normalization factor.

Once the curvature has been fixed, the value $\nu = \hbar$ identified by Corollary 3.5.5 is the only normalization for which the geometric product $\ast_\nu$ reproduces the associative operator product AB . Hence, Schrödinger's equation and the Lie–Jordan algebra structure of $\mathfrak{A}$ are tied to a single geometric invariant of the state manifold.

Measurement

The Riemannian metric g that fixes the observable algebra also carries the entire probabilistic content of quantum mechanics, at two complementary scales. Infinitesimally, Definition 3.4.4 reads the dispersion of an observable off the Riemann bracket, $\Delta f = \sqrt{\hbar/2} \sqrt{((f, f))}$. Moreover, Theorem 3.6 shows that the Robertson–Schrödinger uncertainty relation represents the Cauchy–Schwarz inequality for g . Hence, uncertainty is a local, metric phenomenon on $\mathbb{P}\mathcal{H}_\alpha$.

Globally, integrating g along curves produces the geodesics of Definition 3.4.9 and the Riemannian distance d_g of Definition 3.4.10. Theorem 3.8 identifies the transition probability $P([x], [y])$ with the function of the Riemannian metric $\cos^2(d_g([x], [y])/\sqrt{2\hbar})$. We showed that the same metric that bounds simultaneous dispersion at a point also measures the macroscopic distinguishability of two states through the length of the geodesic joining them. The Born rule is, in this sense, the large-scale expression of the same Riemannian metric already responsible for the uncertainty relations.

Axiomatics of the Geometric Framework

The original formulation combined geometric postulates with derived theorems. A geometric axiomatization of a quantum theory removes any *a priori* dependence on the linear structure of a Hilbert space or the algebraic properties of an ambient C^* -algebra. The framework rests on four irreducible, independent postulates.

- (G1) **State Space Manifold:** The pure state space is a connected, simply connected, complete Kähler manifold $(\mathcal{M}, J, g)$ of constant holomorphic sectional curvature equal to $1/\hbar$.
- (G2) **Observables:** The physical observables correspond bijectively to the algebra of Kählerian functions $\mathcal{K}(\mathcal{M}, \mathbb{R})$.
- (G3) **Dynamics:** Time evolution is governed by the Hamiltonian flow of a distinguished observable function $h \in \mathcal{K}(\mathcal{M}, \mathbb{R})$.
- (G4) **Measurement:** The transition probability between two pure states $p, q \in \mathcal{M}$ is determined by their Riemannian geodesic distance $d_g(p, q)$ via the geometric Born rule $P(p, q) = \cos^2\left(\frac{d_g(p, q)}{\sqrt{2\hbar}}\right)$.

All other structures established in Sections 3.2–3.6 are mathematical derivations proceeding from these four postulates.

- **Fubini–Study Geometry:** Postulate (G1) forces the manifold $\mathcal{M}$, up to isomorphism, to be a complex projective space $\mathbb{P}\mathcal{H}$ equipped with the scaled Fubini–Study triad (J, g, ω) , by the Hawley–Igusa Theorem 3.5.
- **Algebraic Isomorphism:** Postulate (G2), combined with the intrinsic Poisson and Riemann brackets of the Kähler manifold, structurally generates a Lie–Jordan algebra. Theorem 3.10 derives the exact algebraic-geometric isomorphism with the space of bounded selfadjoint operators.
- **Schrödinger Evolution:** Postulate (G3) guarantees that the dynamics consist of global Kähler isomorphisms. By Proposition 3.3.2, this flow uniquely reconstructs the unitary Stone evolution on the underlying Hilbert space, eliminating the need to postulate the Schrödinger equation.

- **Uncertainty Relations:** The Robertson–Schrödinger bounds derived in Theorem 3.6 follow intrinsically from the metric and symplectic structures fixed by (G1) and (G2), completely independent of the probabilistic postulate (G4).

The unified axiomatic system

The last discussion we present aims to merge the algebraic construction of the previous chapter with the geometric one proposed above. Hence, we introduce the notion of algebraic-geometric quantum system into a minimal set of axioms.

Definition 3.7.2 (Algebraic-geometric quantum system). *An algebraic-geometric quantum system is a triple $(\mathfrak{A}, S, H)$ defined by the following.*

- (i) **Algebraic core:** $(\mathfrak{A}, S)$ is a quantum system, as per Definition 2.1.22.
- (ii) **Kählerian dynamics:** On each superselection sector, identified via the theorems of Section 3.2 with a projective Hilbert space $\mathbb{P}\mathcal{H}_\omega$, time evolution acts as the Hamiltonian flow, with respect to the Kähler structure normalized to constant holomorphic sectional curvature $1/\hbar$, of the mean value function $\langle H \rangle$ of a distinguished selfadjoint element $H \in \mathfrak{A}_{sa}$.
- (iii) **Born rule:** Transition probabilities between two pure states p, q of $\mathfrak{A}$ lying in a common sector, represented as points of $\mathbb{P}\mathcal{H}_\omega$, are fixed by the geometric Born rule

$$P(p, q) = \cos^2 \left(\frac{d_g(p, q)}{\sqrt{2\hbar}} \right),$$

as per Theorem 3.8, where d_g denotes the Riemannian distance induced by the Kähler structure fixed in Axiom (ii).

Remark 3.7.3 (Axiom (i) carries no independent content). The previous formulation of Axiom (i) restated, inside the axiom itself, the defining properties of a quantum system, unitality of $\mathfrak{A}$ and separation of $\mathfrak{A}$ by S . Axiom (i) above only names the pair $(\mathfrak{A}, S)$ as an instance of Definition 2.1.22, leaving every further physical requirement to Axioms (ii) and (iii).

Remark 3.7.4 (Axiom (ii) imposes sector decomposition, the observable isomorphism, and dynamical evolution). As we have seen, Theorem 3.4 already forces $\mathcal{P}(\mathfrak{A})$ to split as a disjoint union of projective Hilbert spaces, each equipped with the canonical Fubini–Study triad of Section 3.2, for every C^* -algebra satisfying Axiom (i). What Axiom (i) leaves open is the overall scale of the metric on each sector, and fixing that scale is the entire content of Axiom (ii). By the Hawley–Igusa Theorem 3.5, a connected, simply connected, complete Kähler manifold of constant positive holomorphic sectional curvature is, up to isomorphism, a projective space at that curvature, so prescribing the value $1/\hbar$ singles out, through Corollary 3.3.9, the physical normalization of Equation (3.24) without any further postulate. Axiom (ii) is therefore the point at which $\hbar$ enters the theory once and for all, as the reciprocal of an intrinsic curvature of the state manifold rather than as a separately imposed physical constant.

Once the curvature is fixed, the observable isomorphism follows as a theorem. Theorem 3.10 yields, on every sector, a Lie–Jordan isomorphism between $\mathfrak{A}_{sa}$ and the algebra of Kählerian functions, with no further choice beyond the curvature normalization already imposed.

The dynamical equivalence follows in the same way. Proposition 3.3.2 identifies the Hamiltonian flow of $\langle H \rangle$ with the Stone evolution $[e^{-itH/\hbar}x]$, and Proposition 3.3.6 shows conversely that every one-parameter group of Kähler isomorphisms of the state manifold arises this way, from a selfadjoint generator unique up to an additive constant. Axiom (ii) therefore already entails, through these two propositions, that the flow it postulates coincides with the automorphism $A \mapsto e^{itH/\hbar}Ae^{-itH/\hbar}$ on $\mathfrak{A}$.

Remark 3.7.5 (Axiom (iii) fixes the probabilistic structure). Transition probabilities are governed by the geodesic distance d_g through the cosine-squared law of Theorem 3.8. Its role, however, differs from that of Axioms (i) and (ii). Those two fix the algebraic and geometric structure of the theory, while Axiom (iii) only prescribes how the geometry they already determine is to be read operationally.

Remark 3.7.6 (The inherited constant $c = \hbar^2$ of the Lie–Jordan algebra). Definition 2.1.19 fixes the real Lie–Jordan algebra $(\mathcal{O}, \circ, \{\cdot, \cdot\})$ underlying $\mathfrak{A}_{sa}$ through a single associator constant $c \in \mathbb{R}$, matched on physical grounds to $c = \hbar^2$ so that $\{\cdot, \cdot\} = \frac{1}{i\hbar}[\cdot, \cdot]$. Under Definition 3.7.2, this identification is no longer an independent piece of physical input. Once Axiom (ii) fixes the curvature of the state manifold to $1/\hbar$, the isomorphism of Theorem 3.10 transports, sector by sector, the intrinsic Poisson and Riemann brackets of the Kähler manifold onto $\mathfrak{A}_{sa}$. In addition, Corollary 3.5.5 shows that the resulting associative product coincides with $AB = A \circ B + \frac{i\hbar}{2}\{A, B\}$ at $\nu = \hbar$. The value $c = \hbar^2$ is therefore canonically induced, through Theorem 3.10.

Having established the features of this axiomatization, we turn to the limitations of this construction. In particular, mixed states are considered in Chapter 3. To include them it is needed to extend this work to more complex manifolds called stratified spaces or passing through purification as shown in Section 2.7.

Remark 3.7.7 (Purification and stratified spaces). The space of density operators does not constitute a global smooth manifold. It forms a stratified space where the dimension of the local tangent space depends on the rank of the density matrix. Differential geometric structures, such as the symplectic form and the Riemannian metric, become singular across boundaries where the matrix rank changes.

The Gelfand–Naimark–Segal (GNS) construction circumvents this dimensional discontinuity via algebraic purification. By Theorem 2.7, any state ω on the observable algebra $\mathfrak{A}$, including mixed states, induces a Hilbert space $\mathcal{H}_\omega$ containing a cyclic vector Ψ_ω . Although Proposition 2.3.13 dictates that the representation π_ω is reducible when ω is mixed, the operational state is identically recovered as a pure vector state evaluation: $\omega(A) = (\Psi_\omega, \pi_\omega(A)\Psi_\omega)$. This maps the singular mixed state to the ray $[\Psi_\omega]$ in the corresponding projective space $\mathbb{P}\mathcal{H}_\omega$. Since $\mathbb{P}\mathcal{H}_\omega$ is a connected, complete Kähler manifold regardless of the reducibility of π_ω , the Kähler geometry is globally recovered, bypassing the rank-induced singularities of the density operator space. For a classification of these stratified manifolds, see [BŽ17].

3.8 Example III: Dynamics and Symmetries on the Bloch Sphere

In this example we focus once more on the quantum spin system of Section 2.7, there reconstructed algebraically through the GNS representation of the $\mathfrak{su}(2)$ -algebra generated by J_1, J_2, J_3 . The same Pauli algebra and the same Bloch ball of Equation (2.69) reappear here as the projective Hilbert space $\mathbb{P}\mathcal{H}$ for $\mathcal{H} = \mathbb{C}^2$, now carrying the Kähler structure of Section 3.2. This finite-dimensional instance of the projective Hilbert space provides a concrete test case for the geometric apparatus developed in this chapter, from the Kähler structure of Section 3.2 to the momentum map of Section 3.6. Working through this example in detail also illustrates, on the same $\mathfrak{su}(2)$ Lie algebra already treated classically in Section 1.7, how symmetries of a quantum system translate, through the Quantum Noether Theorem 3.11, into conserved quantities within the geometric language of $\mathbb{P}\mathcal{H}$.

Construction of the Kähler Manifold

First, we equip $\mathbb{P}\mathcal{H}(\mathbb{C}^2)$ with coordinates and explicitly construct its Kähler structure. Let $z = (z_0, z_1)^T \in \mathbb{C}^2 \setminus \{0\}$ denote a pure state vector, and let U_0 be the standard affine chart defined by the condition $z_0 \neq 0$. Introducing the local holomorphic coordinate $w := z_1/z_0 \in \mathbb{C}$, the state is represented on U_0 by the unnormalized vector $\Phi = (1, w)^T \in \mathbb{C}^2$, of squared norm $\|\Phi\|^2 = 1 + w\bar{w}$. By Equation (3.24), the local Kähler potential on U_0 is

$$K(w, \bar{w}) = \hbar \ln(1 + w\bar{w}). \quad (3.48)$$

Differentiating Equation (3.48) as per Theorem 3.3 yields the metric component

$$g_{w\bar{w}} = \partial_{\bar{w}} \partial_w K = \frac{\hbar}{(1 + |w|^2)^2}, \quad (3.49)$$

positive on all of U_0 . It descends that the fundamental form

$$\omega = i g_{w\bar{w}} dw \wedge d\bar{w} = \frac{i\hbar}{(1 + |w|^2)^2} dw \wedge d\bar{w} \quad (3.50)$$

is nondegenerate. Since $\mathbb{P}\mathcal{H}(\mathbb{C}^2)$ has real dimension two, ω is a top-degree form on U_0 . It descends that $d\omega = 0$ identically, so that, together with Equation (3.49), $(\mathbb{P}\mathcal{H}(\mathbb{C}^2), J, g)$ is a Kähler manifold as per Definition 3.1.10, in agreement with the general construction of Section 3.2.

The Bloch vector components $\mathbf{r} = (r_1, r_2, r_3) \in \mathbb{R}^3$, identified on the pure state boundary of the Bloch ball with the mean value functions $r_i := \langle \sigma_i \rangle$, constitute a set of global smooth functions on $\mathbb{P}\mathcal{H}(\mathbb{C}^2)$. On the local chart U_0 with coordinate $w = z_1/z_0$ and normalized representative $\Psi = (1 + |w|^2)^{-1/2}(1, w)^T$, they evaluate to the smooth functions

$$r_1 = \frac{w + \bar{w}}{1 + |w|^2}, \quad r_2 = i \frac{\bar{w} - w}{1 + |w|^2}, \quad r_3 = \frac{1 - |w|^2}{1 + |w|^2}. \quad (3.51)$$

The symplectic form $\omega = i g_{w\bar{w}} dw \wedge d\bar{w}$ defined in Equation (3.50) induces a Poisson bracket on U_0 . For smooth functions $f, k \in C^\infty(U_0, \mathbb{R})$, the Poisson bracket is generated by the inverse of the symplectic matrix

$$\{f, k\} = \frac{1}{i g_{w\bar{w}}} \left(\frac{\partial f}{\partial w} \frac{\partial k}{\partial \bar{w}} - \frac{\partial f}{\partial \bar{w}} \frac{\partial k}{\partial w} \right) = \frac{(1 + |w|^2)^2}{i\hbar} \left(\frac{\partial f}{\partial w} \frac{\partial k}{\partial \bar{w}} - \frac{\partial f}{\partial \bar{w}} \frac{\partial k}{\partial w} \right). \quad (3.52)$$

Evaluating the partial derivatives of the mean value functions r_1 and r_2 yields

$$\frac{\partial r_1}{\partial w} = \frac{1 - \bar{w}^2}{(1 + |w|^2)^2}, \quad \frac{\partial r_1}{\partial \bar{w}} = \frac{1 - w^2}{(1 + |w|^2)^2}, \quad (3.53)$$

$$\frac{\partial r_2}{\partial w} = -i \frac{1 + \bar{w}^2}{(1 + |w|^2)^2}, \quad \frac{\partial r_2}{\partial \bar{w}} = i \frac{1 + w^2}{(1 + |w|^2)^2}. \quad (3.54)$$

Substitution into the Poisson bracket exhibits the algebraic cancellation:

$$\begin{aligned} \{r_1, r_2\} &= \frac{(1 + |w|^2)^2}{i\hbar} \left[\left(\frac{1 - \bar{w}^2}{(1 + |w|^2)^2} \right) \left(i \frac{1 + w^2}{(1 + |w|^2)^2} \right) - \left(\frac{1 - w^2}{(1 + |w|^2)^2} \right) \left(-i \frac{1 + \bar{w}^2}{(1 + |w|^2)^2} \right) \right] \\ &= \frac{1}{i\hbar(1 + |w|^2)^2} \left[i(1 + w^2 - \bar{w}^2 - |w|^4) + i(1 + \bar{w}^2 - w^2 - |w|^4) \right] \\ &= \frac{2(1 - |w|^4)}{\hbar(1 + |w|^2)^2} = \frac{2}{\hbar} \frac{1 - |w|^2}{1 + |w|^2}. \end{aligned} \quad (3.55)$$

Comparing this scalar field with the structural definition of r_3 establishes the geometric derivation of the algebraic commutator, $\{r_1, r_2\} = \frac{2}{\hbar} r_3$. By cyclic permutation of the coordinate basis, this reproduces the complete $\mathfrak{so}(3)$ -Poisson bracket algebra

$$\boxed{\{r_i, r_j\} = \left\langle \frac{1}{i\hbar} [\sigma_i, \sigma_j] \right\rangle = \frac{2}{\hbar} \epsilon_{ijk} r_k.} \quad (3.56)$$

Hamiltonian Action and the Moment Map

The group $G = SU(2)$ acts on $\mathbb{P}\mathcal{H}(\mathbb{C}^2)$ through projective unitary transformations, playing the role that $SO(3)$ played for the classical sphere of Section 1.7. Its Lie algebra $\mathfrak{su}(2)$ is spanned by the skew-adjoint generators $\xi_j := -\frac{i}{2} \sigma_j$, normalized so that $i\hbar \xi_j = \frac{\hbar}{2} \sigma_j$ is the physical spin operator. By Proposition 3.6.1, the induced action is strongly Hamiltonian, and its momentum map $\mu : \mathbb{P}\mathcal{H}(\mathbb{C}^2) \rightarrow \mathfrak{su}(2)^*$ is given pointwise by

$$\langle \mu([\psi]), \xi_j \rangle = \langle i\hbar \xi_j, [\psi] \rangle = \frac{\hbar}{2} r_j([\psi]).$$

Identifying $\mathfrak{su}(2)^* \cong \mathbb{R}^3$ through the three components $\langle \mu, \xi_j \rangle$, $j = 1, 2, 3$, this yields

$$\boxed{\mu = \frac{\hbar}{2} \mathbf{r}.} \quad (3.57)$$

The momentum map thus corresponds to the quantum spin angular momentum vector, the $\hbar/2$ -rescaled counterpart of the classical moment map $\mu_{\text{cl}}(\mathbf{r}) = -\mathbf{r}$ of Equation (1.19).

Noether's Theorem

The momentum map constructed above is the geometric counterpart of a conserved quantity associated with a continuous symmetry, in the sense of the Quantum Noether Theorem 3.11. We illustrate this correspondence through the precession of a spin placed in a uniform magnetic field, governed by the same Kähler structure fixed above.

Consider the Hamiltonian $H = \frac{\hbar\omega}{2}\sigma_3$, describing a spin in a uniform magnetic field along the z -axis and invariant under rotations about it. The symmetry generator is $\xi_3 = -\frac{i}{2}\sigma_3$. The associated Noether invariant is

$$\langle \mu, \xi_3 \rangle = \frac{\hbar}{2} r_3.$$

As computed in Equation (3.57), and since $[H, i\hbar\xi_3] = \left[\frac{\hbar\omega}{2}\sigma_3, \frac{\hbar}{2}\sigma_3\right] = 0$, the Quantum Noether Theorem 3.11 guarantees that r_3 is conserved. In fact, writing $h := \langle H \rangle = \frac{\hbar\omega}{2}r_3$, Equation (3.56) together with $\dot{f} = \{f, h\}$, descending from the proof of Proposition 3.3.4, yields the equations of motion for the Bloch vector along the Hamiltonian flow of h :

$$\begin{aligned}\dot{r}_1 &= \{r_1, h\} = -\omega r_2, \\ \dot{r}_2 &= \{r_2, h\} = \omega r_1, \\ \dot{r}_3 &= \{r_3, h\} = 0.\end{aligned}$$

This yields a uniform precession about the z -axis:

$$r_1(t) + ir_2(t) = e^{i\omega t}(r_1(0) + ir_2(0)), \quad r_3(t) = r_3(0).$$

Conclusion

In this thesis, we presented an algebraic reconstruction of classical and quantum mechanics along with their subsequent geometric unification. In the first chapter, we rebuilt classical phase space from a commutative C^* -algebra via Gelfand-Naimark, supplemented by a Poisson bracket to endow it with a symplectic structure. In the second chapter, we reconstructed quantum mechanics from a non-commutative algebra using the GNS construction to obtain Hilbert space representations. Hence, the Poisson bracket was replaced by a Lie bracket, establishing a formal correspondence with the classical framework. In the third chapter, we joined the algebraic and geometric formulations by identifying the GNS-derived pure state space with projective Hilbert space. Its intrinsic Kähler structure generates Schrödinger dynamics from its symplectic form. Moreover, it encodes Born-rule probabilities through its metric, while expressing symmetries as conserved charges via a geometric moment map.

From this work, we can conclude the following key facts. In algebraic classical mechanics, once the topological state space is fixed, it is needed a choice of the smooth subalgebra, as in Definition 1.6.32. Instead, in the algebraic quantum picture, we observe that through GNS Theorem 2.7 we recover the Dirac-Von Neumann formalism and a Lie algebra using only operational structures. At the end, once the theories are rewritten geometrically, we can conclude the quantum algebraic axiomatization using a classical geometrical analogy because we showed that same geometrical objects represents the same physical processes as in the classical case.

Limitations of this framework discussed in Section 2.5 and Remark 3.7.7 can be overcome by extending the formalism to infinite dimensional Hilbert spaces as in [BR02] and mixed states in a future work. This thesis can also lead to tackling the choice problem of the smooth structure in Chapter 1. In fact, one can use the quantum Kählerian algebra in Definition 3.1.19 on the projective space to identify a proper smooth Lie algebra, using it as a choice of a Poisson algebra of smooth functions. Another future work can focus on finding an operational interpretation and a subsequent ontology for this new algebraic-geometric framework.

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Questo ambiente folle del collegio mi ha portato anche a conoscere persone normali, se così possono essere definite, con cui ho stretto un meraviglioso rapporto di amicizia. Ringrazio di cuore Viking per quello che sei. Sei la persona più disponibile, premurosa e gentile che io conosca oltre ad essere veramente un genio in quello che fai. Ringrazio moltissimo Ste per essere riuscito ad andare oltre le apparenze anche quando tutti guardavano dall'altra parte.

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Un Caloroso Abbraccio a Reggio

Ringrazio di cuore Iari, ci conosciamo ormai da otto anni e dopo averne passate di ogni al liceo ancora ci fermiamo a parlare per ore davanti a una birra. Ti confesso che moltissime discussioni che abbiamo fatto davanti a una birra mi hanno portato a lunghe riflessioni, che si parlasse di amore o fisica, esempio Section 3.8. Spero che tu trovi la tua strada perchè con la testa che hai puoi puntare molto più in alto di quanto credi. Ringrazio anche di cuore Jack, un'amicizia forse già premeditata dai nostri nonni. I ricordi insieme sono infiniti e ogni volta che ci ripenso ricado in un mondo senza pensieri e leggero, dove basta una cazzata per farci ridere tutto il pomeriggio.

Un Caloroso Abbraccio al Lido degli Estensi

22.30 Collins

Persona che arriverà alle 23:20

Ringrazio Gio, Manu, Bob, PJ, Max, Carvi, Marti, Marco, Bons, Zanna e Gioac poi ancora Alessia, Jacopo, Brego, Tia, Zanna, Ferro, GESU', Fragio, Noa, Andre, Luca, Jack, Cremmy, Lu e Cecco per tutti questi anni. Vi ringrazio per aver chiamato la Guardia Costiera quella fatidica notte. Vi ringrazio per le caraffe al Collins e le piadine di King Kong. Vi ringrazio per tutte le "partite" di bocce in spiaggia e di un altro gioco in mare... vabbè. Vi ringrazio per ogni Ferragosto e anche per ogni notte di "massì proviamo questo nuovo locale". Vi ringrazio per avermi curato da ogni problema che avessi, per avermi fatto distrarre ogni volta che ero stressato e per essermi stati vicino nei veri momenti cupi.

Ogni anno, d'estate, una ventina di persone da tutto il Nord Italia scappano dalle loro routine quotidiane per passare il più lungo periodo di tempo possibile in un piccolo lido sulla costa romagnola. Se ci penso per un momento ha dell'incredibile, eppure, abbiamo costruito una magia che dura da vent'anni e forse durerà per tutta la vita.